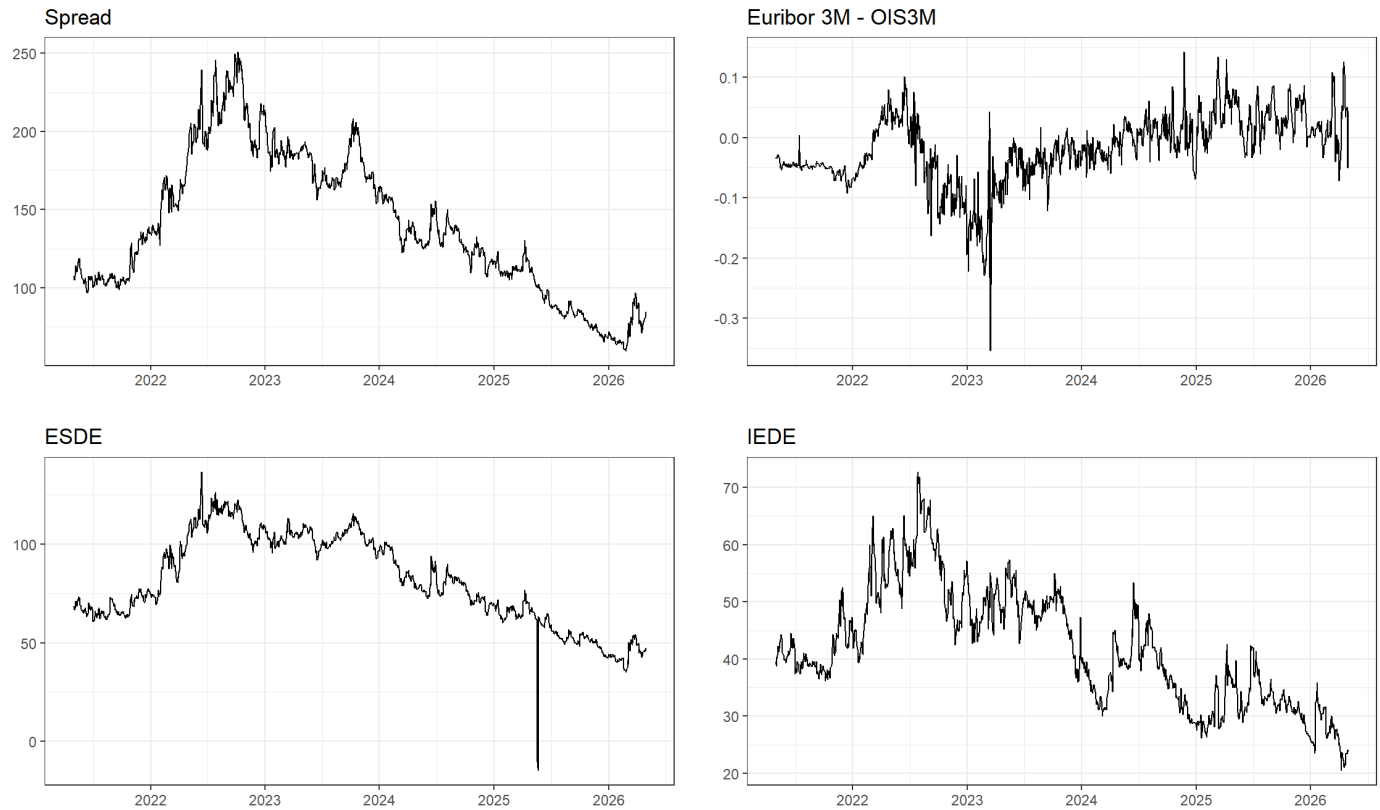
Spread: ARMA-GARCH Model

2026-04-30

Data setup

FVR FF ## Load data

```
spread <- read_excel("spread_daily_30_04.xlsx")
ois3m <- read_excel("OIS3M_30_04.xlsx")
euribor3m <- read_excel("euribor3M_30_04.xlsx")
ESDE <- read_excel("SPREAD_ES_30_04.xlsx")
IEDE <- read_excel("SPREAD_IE_30_04.xlsx")

data <- spread |>
  left_join(ois3m, by = "Date") |>
  left_join(euribor3m, by = "Date") |>
  left_join(ESDE, by = "Date")|>
  left_join(IEDE, by = "Date")

data <- data[-1, ]
data <- data |>
  mutate(across(-Date, as.numeric))

data <- data |> rename(spread = names(data)[2],
                       ois3m = names(data)[3],
                       euribor3m = names(data)[4],
                       ESDE = names(data)[5],
                       IEDE=names(data)[6])

data$spread <- as.numeric(data$spread) * -1
data$ESDE <- as.numeric(data$ESDE) * -1
data$IEDE <- as.numeric(data$IEDE) * -1
data$eurois3m <-data$euribor3m - data$ois3m
data <- data |> dplyr::select(Date,eurois3m, ,ESDE, IEDE,spread)

# 3. GESTIONE NA OTTIMIZZATA (Sostituisce na.omit)
# Trasformiamo in data.frame per usare le funzioni di zoo in modo vettoriale
data_filled <- data %>%
  arrange(Date) %>%
  # Riempie i buchi interni (es. festività sfasate tra paesi)
  mutate(across(-Date, ~ na.approx(., na.rm = FALSE))) %>%
  # Riempie i NA rimanenti all'inizio e alla fine (trascinamento valori)
  mutate(across(-Date, ~ na.locf(., na.rm = FALSE))) %>%
  mutate(across(-Date, ~ na.locf(., fromLast = TRUE, na.rm = FALSE)))
data <- xts(data_filled[-1], order.by = as.Date(data_filled$Date))
```



Data preparation

Train/Test

Use only the last `year` years of data, then split.

```
n_test  <- round(nrow(data) * 0.2)
n_train <- nrow(data) - n_test

train <- data[1:n_train]
test  <- data[(n_train + 1):nrow(data)]

cat("\nTraining observations:", n_train)
```

```
##
## Training observations: 1020
```

```
cat("\nTest observations:", n_test)
```

```
##
## Test observations: 255
```

```
cat("\nTrain period:", as.character(index(train)[1]), "->", as.character(index(train)[nrow(trai
n)]))
```

```
##
## Train period: 2021-04-30 -> 2025-04-25
```

```
cat("\nTest period:", as.character(index(test)[1]), "->", as.character(index(test)[nrow(tes
t)]),"\n")
```

```
##
## Test period: 2025-04-28 -> 2026-04-30
```

Stationarity

Check if raw data are stationary around a drift with ADF test, use `ur.df()` , then first-difference and check again

```
check_stat <- function(series, type = c("adf", "kpss")) {
  # To run both just don't select the type
  df_list <- list()
  # ADF test
  if ("adf" %in% type) {
     for (test in c("none", "drift", "trend")) {
      adf <- ur.df(series, type = test, selectlags = "AIC")
      row <- data.frame(
        test_type = paste("ADF: ", test),
        teststat = adf@teststat[1],
        cval = adf@cval[1, "5pct"],
        result = ifelse(adf@teststat[1] < adf@cval[1, "5pct"], "Stationary", "Non-Stationary")
      )
      df_list[[length(df_list) + 1]] <- row
    }
  }
  # KPSS test
  if ("kpss" %in% type) {
    for (test in c("mu", "tau")) {
      kpss <- ur.kpss(series, type = test, lags = "short")
      row <- data.frame(
        test_type = paste("KPSS: ", ifelse(test == "mu", "const", "trend")),
        teststat = kpss@teststat,
        cval = kpss@cval[1, "5pct"],
        result = ifelse(kpss@teststat < kpss@cval[1, "5pct"], "Stationary", "Non-Stationary")
      )
      df_list[[length(df_list) + 1]] <- row
    }
  }
  df <- bind_rows(df_list)
  return(df)
}

cat("\nRaw data:\n")
```

```
##
## Raw data:
```

```
for (name in names(train)){
  test_stat <- check_stat(train[, name], type = "adf")
  cat(name, ":\n")
  print(test_stat)
}
```

```
## eurois3m :
##     test_type  teststat  cval     result
## 1  ADF:  none -4.705673 -1.95 Stationary
## 2 ADF:  drift -5.325281 -2.86 Stationary
## 3 ADF:  trend -5.868160 -3.41 Stationary
## ESDE :
##     test_type   teststat  cval         result
## 1  ADF:  none -0.3693214 -1.95 Non-Stationary
## 2 ADF:  drift -1.8374715 -2.86 Non-Stationary
## 3 ADF:  trend -1.9561826 -3.41 Non-Stationary
## IEDE :
##     test_type   teststat  cval         result
## 1  ADF:  none -0.5954626 -1.95 Non-Stationary
## 2 ADF:  drift -2.7378828 -2.86 Non-Stationary
## 3 ADF:  trend -3.3330593 -3.41 Non-Stationary
## spread :
##     test_type   teststat  cval         result
## 1  ADF:  none -0.3615813 -1.95 Non-Stationary
## 2 ADF:  drift -1.7444294 -2.86 Non-Stationary
## 3 ADF:  trend -1.9214057 -3.41 Non-Stationary
```

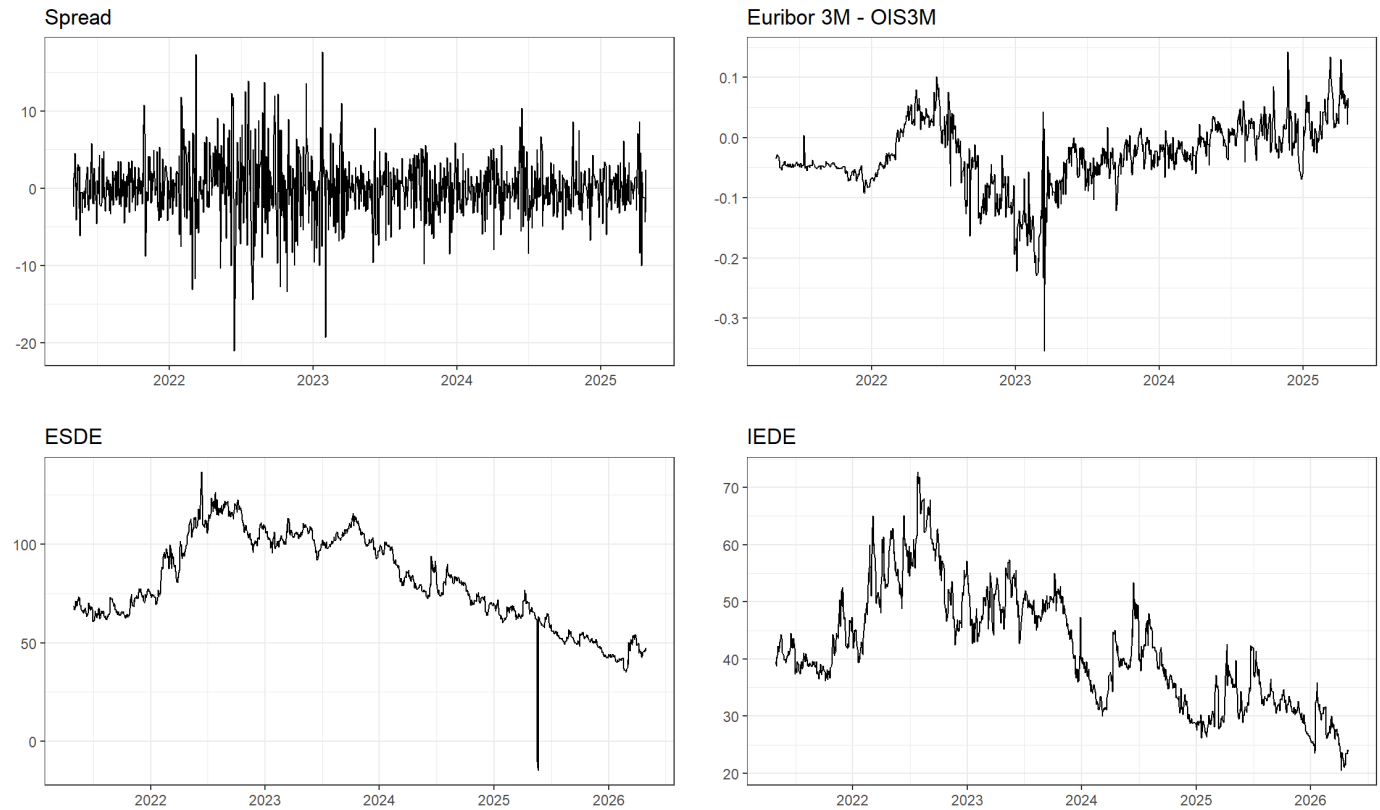
```
var_dif = c("spread","ESDE","IEDE")
d_data = data
d_data[, var_dif] = d_data[, var_dif] |> diff()
d_data = d_data |> na.omit()
d_train <- d_data[1:(n_train - 1)]
d_test  <- d_data[n_train:nrow(d_data)]

cat("\nTransformed data:\n")
```

```
##
## Transformed data:
```

```
for (name in names(d_train)){
  test_stat <- check_stat(d_train[, name], type = "adf")
  cat(name, ":\n")
  print(test_stat)
}
```

```
## eurois3m :
##     test_type  teststat  cval     result
## 1  ADF:  none -4.704066 -1.95 Stationary
## 2 ADF:  drift -5.322679 -2.86 Stationary
## 3 ADF:  trend -5.866651 -3.41 Stationary
## ESDE :
##     test_type  teststat  cval     result
## 1  ADF:  none -22.50426 -1.95 Stationary
## 2 ADF:  drift -22.49318 -2.86 Stationary
## 3 ADF:  trend -22.52778 -3.41 Stationary
## IEDE :
##     test_type  teststat  cval     result
## 1  ADF:  none -23.05358 -1.95 Stationary
## 2 ADF:  drift -23.04239 -2.86 Stationary
## 3 ADF:  trend -23.03723 -3.41 Stationary
## spread :
##     test_type  teststat  cval     result
## 1  ADF:  none -22.60203 -1.95 Stationary
## 2 ADF:  drift -22.59097 -2.86 Stationary
## 3 ADF:  trend -22.63347 -3.41 Stationary
```

- `spread` series showed non-stationary behaviour, it does says “trend stationary” but looking at the series plot we can see that the supposed trend isn’t linear, for semplicity we decided to difference it.
- We differenced `eur1m` becasu, even if it pass the test, if not differenced it will cause problems in the model, the differenced series works better, also looking at the plot we can see that its stationarity is kinda false since it doesn’t look like it.
- On the other side, and `eurois3m` is perfectly stationary.



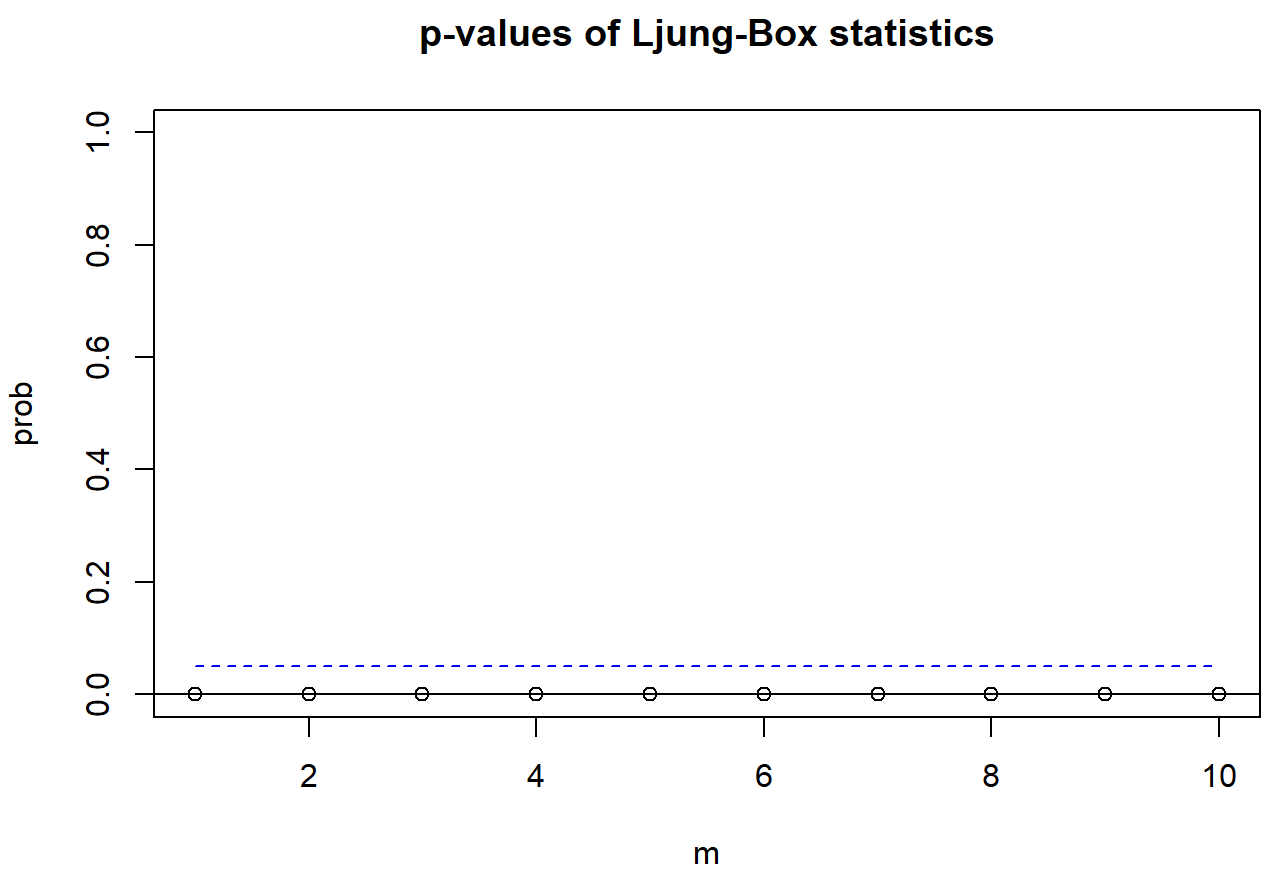
VAR

Estimation

Check if there's autocorrelation between variables

```
mq(d_train, lag = 10)
```

```
## Ljung-Box Statistics:
##         m       Q(m)     df    p-value
##  [1,]     1       796     16        0
##  [2,]     2      1535     32        0
##  [3,]     3      2246     48        0
##  [4,]     4      2884     64        0
##  [5,]     5      3524     80        0
##  [6,]     6      4124     96        0
##  [7,]     7      4698    112        0
##  [8,]     8      5273    128        0
##  [9,]     9      5831    144        0
## [10,]    10      6372    160        0
```



Test shows there's autocorrelation present across variables

Check if variables granger-cause `spread`

```
var_names = names(d_train)

print(var_names)
```

```
## [1] "eurois3m" "ESDE"     "IEDE"     "spread"
```

```
for (v in var_names[var_names != "spread"]) {
  cat("\n", v, "\n")
  print(lmtest::grangertest(d_train[,v], d_train[,"spread"], order = 5))
}
```

```
##
##  eurois3m
## Granger causality test
##
## Model 1: d_train[, "spread"] ~ Lags(d_train[, "spread"], 1:5) + Lags(d_train[, v], 1:5)
## Model 2: d_train[, "spread"] ~ Lags(d_train[, "spread"], 1:5)
##   Res.Df Df      F   Pr(>F)
## 1   1003
## 2   1008 -5 3.1266 0.008303 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##  ESDE
## Granger causality test
##
## Model 1: d_train[, "spread"] ~ Lags(d_train[, "spread"], 1:5) + Lags(d_train[, v], 1:5)
## Model 2: d_train[, "spread"] ~ Lags(d_train[, "spread"], 1:5)
##   Res.Df Df      F  Pr(>F)
## 1   1003
## 2   1008 -5 2.8595 0.01428 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##  IEDE
## Granger causality test
##
## Model 1: d_train[, "spread"] ~ Lags(d_train[, "spread"], 1:5) + Lags(d_train[, v], 1:5)
## Model 2: d_train[, "spread"] ~ Lags(d_train[, "spread"], 1:5)
##   Res.Df Df      F   Pr(>F)
## 1   1003
## 2   1008 -5 3.2708 0.006174 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Both granger-cause `spread`

Find the optimal lag

```
selection = vars::VARselect(y = d_train, lag.max = 10,
                            type = "const")

print(t(selection$criteria))
```

```
##       AIC(n)     HQ(n)     SC(n)     FPE(n)
## 1  -3.582560 -3.545536 -3.485103 0.02780443
## 2  -3.714793 -3.648149 -3.539370 0.02436052
## 3  -3.732518 -3.636254 -3.479129 0.02393264
## 4  -3.722557 -3.596674 -3.391202 0.02417239
## 5  -3.736838 -3.581336 -3.327518 0.02382989
## 6  -3.721927 -3.536804 -3.234641 0.02418831
## 7  -3.705839 -3.491098 -3.140588 0.02458114
## 8  -3.701915 -3.457554 -3.058698 0.02467853
## 9  -3.699643 -3.425662 -2.978460 0.02473561
## 10 -3.676486 -3.372886 -2.877337 0.02531629
```

```
print(selection$selection)
```

```
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      5      2      2      5
```

Following the AIC we will estimate a VAR(5).

```
model = vars::VAR(d_train, lag=5, type = "const")
summary(model)
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: eurois3m, ESDE, IEDE, spread
## Deterministic variables: const
## Sample size: 1014
## Log Likelihood: -3768.334
## Roots of the characteristic polynomial:
## 0.9734 0.6356 0.6356 0.629 0.629 0.6142 0.6142 0.593 0.593 0.5877 0.5877 0.5823 0.5823 0.5577
0.5577 0.5525 0.5525 0.5157 0.5157 0.259
## Call:
## vars::VAR(y = d_train, type = "const", lag.max = 5)
##
##
## Estimation results for equation eurois3m:
## =========================================
## eurois3m = eurois3m.l1 + ESDE.l1 + IEDE.l1 + spread.l1 + eurois3m.l2 + ESDE.l2 + IEDE.l2 + sp
read.l2 + eurois3m.l3 + ESDE.l3 + IEDE.l3 + spread.l3 + eurois3m.l4 + ESDE.l4 + IEDE.l4 + sprea
d.l4 + eurois3m.l5 + ESDE.l5 + IEDE.l5 + spread.l5 + const
##
##               Estimate Std. Error t value Pr(>|t|)
## eurois3m.l1  5.122e-01  3.132e-02  16.353  < 2e-16 ***
## ESDE.l1     -2.134e-03  7.441e-04  -2.868  0.00422 **
## IEDE.l1     -4.176e-04  5.684e-04  -0.735  0.46270
## spread.l1    7.416e-04  3.619e-04   2.049  0.04072 *
## eurois3m.l2  2.424e-01  3.525e-02   6.876 1.08e-11 ***
## ESDE.l2      1.271e-03  7.464e-04   1.703  0.08879 .
## IEDE.l2      8.365e-04  5.693e-04   1.469  0.14209
## spread.l2   -5.358e-04  3.622e-04  -1.479  0.13933
## eurois3m.l3  7.510e-02  3.618e-02   2.076  0.03818 *
## ESDE.l3     -4.696e-04  7.459e-04  -0.630  0.52912
## IEDE.l3      3.582e-04  5.677e-04   0.631  0.52826
## spread.l3   -2.035e-04  3.628e-04  -0.561  0.57495
## eurois3m.l4 -4.514e-02  3.536e-02  -1.277  0.20194
## ESDE.l4      3.809e-05  7.473e-04   0.051  0.95935
## IEDE.l4     -1.056e-03  5.675e-04  -1.861  0.06309 .
## spread.l4    1.384e-04  3.627e-04   0.382  0.70280
## eurois3m.l5  1.633e-01  3.145e-02   5.191 2.53e-07 ***
## ESDE.l5      1.176e-04  7.453e-04   0.158  0.87461
## IEDE.l5      1.075e-03  5.671e-04   1.896  0.05824 .
## spread.l5   -3.279e-04  3.632e-04  -0.903  0.36678
## const       -1.426e-03  9.005e-04  -1.584  0.11358
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.02471 on 993 degrees of freedom
## Multiple R-Squared: 0.8097,  Adjusted R-squared: 0.8059
## F-statistic: 211.2 on 20 and 993 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation ESDE:
```

```
## ======================================
## ESDE = eurois3m.l1 + ESDE.l1 + IEDE.l1 + spread.l1 + eurois3m.l2 + ESDE.l2 + IEDE.l2 + sprea
d.l2 + eurois3m.l3 + ESDE.l3 + IEDE.l3 + spread.l3 + eurois3m.l4 + ESDE.l4 + IEDE.l4 + spread.l4
+ eurois3m.l5 + ESDE.l5 + IEDE.l5 + spread.l5 + const
##
##               Estimate Std. Error t value Pr(>|t|)
## eurois3m.l1 -4.440e+00  2.534e+00  -1.752  0.08010 .
## ESDE.l1      4.770e-02  6.022e-02   0.792  0.42849
## IEDE.l1      9.550e-02  4.600e-02   2.076  0.03813 *
## spread.l1   -3.654e-02  2.929e-02  -1.248  0.21247
## eurois3m.l2  8.828e+00  2.853e+00   3.095  0.00203 **
## ESDE.l2     -4.006e-02  6.040e-02  -0.663  0.50729
## IEDE.l2     -7.389e-03  4.607e-02  -0.160  0.87261
## spread.l2    2.404e-02  2.931e-02   0.820  0.41222
## eurois3m.l3  8.837e-01  2.928e+00   0.302  0.76282
## ESDE.l3     -1.018e-01  6.036e-02  -1.686  0.09218 .
## IEDE.l3     -1.011e-01  4.594e-02  -2.200  0.02806 *
## spread.l3    3.147e-02  2.936e-02   1.072  0.28397
## eurois3m.l4 -2.100e+00  2.861e+00  -0.734  0.46318
## ESDE.l4     -4.583e-02  6.047e-02  -0.758  0.44874
## IEDE.l4      5.079e-03  4.592e-02   0.111  0.91195
## spread.l4    5.000e-04  2.935e-02   0.017  0.98641
## eurois3m.l5 -3.021e+00  2.545e+00  -1.187  0.23548
## ESDE.l5      4.784e-02  6.031e-02   0.793  0.42789
## IEDE.l5      7.013e-02  4.589e-02   1.528  0.12679
## spread.l5   -4.568e-02  2.939e-02  -1.554  0.12044
## const       -6.695e-05  7.287e-02  -0.001  0.99927
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 2 on 993 degrees of freedom
## Multiple R-Squared: 0.04262, Adjusted R-squared: 0.02334
## F-statistic:  2.21 on 20 and 993 DF,  p-value: 0.001682
##
##
## Estimation results for equation IEDE:
## ======================================
## IEDE = eurois3m.l1 + ESDE.l1 + IEDE.l1 + spread.l1 + eurois3m.l2 + ESDE.l2 + IEDE.l2 + sprea
d.l2 + eurois3m.l3 + ESDE.l3 + IEDE.l3 + spread.l3 + eurois3m.l4 + ESDE.l4 + IEDE.l4 + spread.l4
+ eurois3m.l5 + ESDE.l5 + IEDE.l5 + spread.l5 + const
##
##              Estimate Std. Error t value Pr(>|t|)
## eurois3m.l1  0.895840   2.107022   0.425    0.671
## ESDE.l1      0.012163   0.050061   0.243    0.808
## IEDE.l1     -0.034763   0.038242  -0.909    0.364
## spread.l1   -0.006817   0.024348  -0.280    0.780
## eurois3m.l2  2.678009   2.371576   1.129    0.259
## ESDE.l2     -0.040328   0.050212  -0.803    0.422
## IEDE.l2     -0.023112   0.038303  -0.603    0.546
## spread.l2    0.034922   0.024366   1.433    0.152
## eurois3m.l3 -3.279113   2.433926  -1.347    0.178
```

```
## ESDE.l3     -0.004777   0.050184  -0.095    0.924
## IEDE.l3     -0.079340   0.038191  -2.077    0.038 *
## spread.l3    0.034034   0.024406   1.394    0.163
## eurois3m.l4  2.481888   2.378543   1.043    0.297
## ESDE.l4      0.003466   0.050274   0.069    0.945
## IEDE.l4     -0.047841   0.038179  -1.253    0.210
## spread.l4   -0.019007   0.024397  -0.779    0.436
## eurois3m.l5 -2.413172   2.115803  -1.141    0.254
## ESDE.l5      0.016340   0.050140   0.326    0.745
## IEDE.l5     -0.012164   0.038153  -0.319    0.750
## spread.l5    0.004328   0.024431   0.177    0.859
## const        0.004079   0.060583   0.067    0.946
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 1.663 on 993 degrees of freedom
## Multiple R-Squared: 0.01882, Adjusted R-squared: -0.0009375
## F-statistic: 0.9526 on 20 and 993 DF,  p-value: 0.5192
##
##
## Estimation results for equation spread:
## =======================================
## spread = eurois3m.l1 + ESDE.l1 + IEDE.l1 + spread.l1 + eurois3m.l2 + ESDE.l2 + IEDE.l2 + spre
ad.l2 + eurois3m.l3 + ESDE.l3 + IEDE.l3 + spread.l3 + eurois3m.l4 + ESDE.l4 + IEDE.l4 + spread.l
4 + eurois3m.l5 + ESDE.l5 + IEDE.l5 + spread.l5 + const
##
##              Estimate Std. Error t value Pr(>|t|)
## eurois3m.l1 -9.677215   4.910266  -1.971  0.04902 *
## ESDE.l1      0.082337   0.116663   0.706  0.48050
## IEDE.l1      0.155535   0.089119   1.745  0.08125 .
## spread.l1   -0.043272   0.056741  -0.763  0.44586
## eurois3m.l2 17.736998   5.526789   3.209  0.00137 **
## ESDE.l2     -0.093133   0.117016  -0.796  0.42628
## IEDE.l2      0.008580   0.089262   0.096  0.92344
## spread.l2    0.035520   0.056783   0.626  0.53176
## eurois3m.l3  3.379682   5.672093   0.596  0.55142
## ESDE.l3     -0.196053   0.116951  -1.676  0.09398 .
## IEDE.l3     -0.156783   0.089002  -1.762  0.07845 .
## spread.l3    0.059148   0.056876   1.040  0.29862
## eurois3m.l4 -5.595937   5.543026  -1.010  0.31296
## ESDE.l4      0.001091   0.117160   0.009  0.99257
## IEDE.l4     -0.018632   0.088974  -0.209  0.83417
## spread.l4   -0.051721   0.056857  -0.910  0.36322
## eurois3m.l5 -4.461656   4.930728  -0.905  0.36575
## ESDE.l5      0.179654   0.116848   1.538  0.12449
## IEDE.l5      0.098330   0.088912   1.106  0.26903
## spread.l5   -0.120268   0.056935  -2.112  0.03490 *
## const        0.041235   0.141184   0.292  0.77030
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
##
## Residual standard error: 3.874 on 993 degrees of freedom
## Multiple R-Squared: 0.04214, Adjusted R-squared: 0.02285
## F-statistic: 2.184 on 20 and 993 DF,  p-value: 0.00196
##
##
##
## Covariance matrix of residuals:
##           eurois3m    ESDE     IEDE    spread
## eurois3m 0.0006107 0.00295 0.002212  0.002703
## ESDE     0.0029503 3.99914 1.862131  6.423807
## IEDE     0.0022124 1.86213 2.763969  3.075176
## spread   0.0027030 6.42381 3.075176 15.010838
##
## Correlation matrix of residuals:
##          eurois3m   ESDE    IEDE  spread
## eurois3m  1.00000 0.0597 0.05385 0.02823
## ESDE      0.05970 1.0000 0.56009 0.82910
## IEDE      0.05385 0.5601 1.00000 0.47742
## spread    0.02823 0.8291 0.47742 1.00000
```

Now we simplify the model eliminating non-significant parameters, with t-ratio < 1.64 $(\alpha = 10\%)$

```
model_s = vars::restrict(model, method = "ser", thresh = 1.64)
fixed_p = model_s$p
fixed_params = model_s$restrictions
summary(model_s)
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: eurois3m, ESDE, IEDE, spread
## Deterministic variables: const
## Sample size: 1014
## Log Likelihood: -3798.864
## Roots of the characteristic polynomial:
## 0.9802 0.6803 0.6803 0.6572 0.6572 0.6245 0.6245 0.6104 0.6104 0.5734 0.5734 0.5381 0.5381 0.
5145 0.5145 0.4763 0.4763 0.3444 0.3444 8.827e-17
## Call:
## vars::VAR(y = d_train, type = "const", lag.max = 5)
##
##
## Estimation results for equation eurois3m:
## =========================================
## eurois3m = eurois3m.l1 + ESDE.l1 + spread.l1 + eurois3m.l2 + ESDE.l2 + eurois3m.l3 + ESDE.l3
+ IEDE.l4 + eurois3m.l5 + IEDE.l5
##
##               Estimate Std. Error t value Pr(>|t|)
## eurois3m.l1  0.5065580  0.0308937  16.397  < 2e-16 ***
## ESDE.l1     -0.0022459  0.0006998  -3.209  0.00137 **
## spread.l1    0.0007247  0.0003611   2.007  0.04500 *
## eurois3m.l2  0.2408153  0.0344207   6.996 4.80e-12 ***
## ESDE.l2      0.0007753  0.0003898   1.989  0.04698 *
## eurois3m.l3  0.0694296  0.0342664   2.026  0.04301 *
## ESDE.l3     -0.0006476  0.0003890  -1.665  0.09624 .
## IEDE.l4     -0.0009443  0.0004733  -1.995  0.04631 *
## eurois3m.l5  0.1431071  0.0276984   5.167 2.87e-07 ***
## IEDE.l5      0.0008987  0.0004728   1.901  0.05761 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.02472 on 1004 degrees of freedom
## Multiple R-Squared: 0.8512,  Adjusted R-squared: 0.8497
## F-statistic: 574.4 on 10 and 1004 DF,  p-value: < 2.2e-16
##
##
## Estimation results for equation ESDE:
## =====================================
## ESDE = eurois3m.l1 + IEDE.l1 + eurois3m.l2 + IEDE.l3 + eurois3m.l5
##
##             Estimate Std. Error t value Pr(>|t|)
## eurois3m.l1 -4.76244    2.37278  -2.007 0.045004 *
## IEDE.l1      0.08909    0.03777   2.359 0.018525 *
## eurois3m.l2  8.69731    2.55739   3.401 0.000698 ***
## IEDE.l3     -0.13576    0.03787  -3.585 0.000353 ***
## eurois3m.l5 -3.72344    2.00302  -1.859 0.063330 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
##
## Residual standard error: 1.996 on 1009 degrees of freedom
## Multiple R-Squared: 0.03069, Adjusted R-squared: 0.02589
## F-statistic:  6.39 on 5 and 1009 DF,  p-value: 7.457e-06
##
##
## Estimation results for equation IEDE:
## =====================================
## IEDE = IEDE.l3 + spread.l3 + spread.l4
##
##           Estimate Std. Error t value Pr(>|t|)
## IEDE.l3   -0.07388    0.03549  -2.082   0.0376 *
## spread.l3  0.02750    0.01506   1.826   0.0681 .
## spread.l4 -0.02655    0.01329  -1.998   0.0460 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 1.656 on 1011 degrees of freedom
## Multiple R-Squared: 0.008698,    Adjusted R-squared: 0.005756
## F-statistic: 2.957 on 3 and 1011 DF,  p-value: 0.03155
##
##
## Estimation results for equation spread:
## =======================================
## spread = eurois3m.l1 + IEDE.l1 + eurois3m.l2 + IEDE.l3 + eurois3m.l4 + spread.l4 + ESDE.l5 +
spread.l5
##
##             Estimate Std. Error t value Pr(>|t|)
## eurois3m.l1 -9.68452    4.66177  -2.077 0.038015 *
## IEDE.l1      0.16340    0.07335   2.228 0.026126 *
## eurois3m.l2 18.29642    5.08072   3.601 0.000332 ***
## IEDE.l3     -0.22416    0.07335  -3.056 0.002301 **
## eurois3m.l4 -7.35478    4.30285  -1.709 0.087707 .
## spread.l4   -0.05277    0.03109  -1.697 0.089963 .
## ESDE.l5      0.21752    0.10938   1.989 0.047019 *
## spread.l5   -0.11474    0.05645  -2.033 0.042348 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 3.863 on 1006 degrees of freedom
## Multiple R-Squared: 0.03539, Adjusted R-squared: 0.02772
## F-statistic: 4.613 on 8 and 1006 DF,  p-value: 1.512e-05
##
##
##
## Covariance matrix of residuals:
##           eurois3m    ESDE     IEDE    spread
## eurois3m 0.0006166 0.00297 0.002029  0.002613
## ESDE     0.0029704 4.04897 1.863357  6.471906
## IEDE     0.0020289 1.86336 2.792493  3.086387
```

```
## spread   0.0026129 6.47191 3.086387 15.115210
##
## Correlation matrix of residuals:
##          eurois3m    ESDE   IEDE  spread
## eurois3m  1.00000 0.05945 0.0489 0.02707
## ESDE      0.05945 1.00000 0.5542 0.82728
## IEDE      0.04890 0.55415 1.0000 0.47506
## spread    0.02707 0.82728 0.4751 1.00000
```

```
count_coeff = function(model){
  names = names(model$varresult)
  n = c()
  for (name in names){
    n = c(n, length(model$varresult[[name]]$coefficients))
  }
  sum(n)
}

cat("\nUnrestricted model coeff: ", count_coeff(model),
    " Restricted model coeff: ", count_coeff(model_s), "\n")
```

```
##
## Unrestricted model coeff:  84  Restricted model coeff:  26
```

```
cat("\nUr model SC: ", BIC(model), "R model SC: ", BIC(model_s))
```

```
##
## Ur model SC:  8118.088 R model SC:  7777.692
```

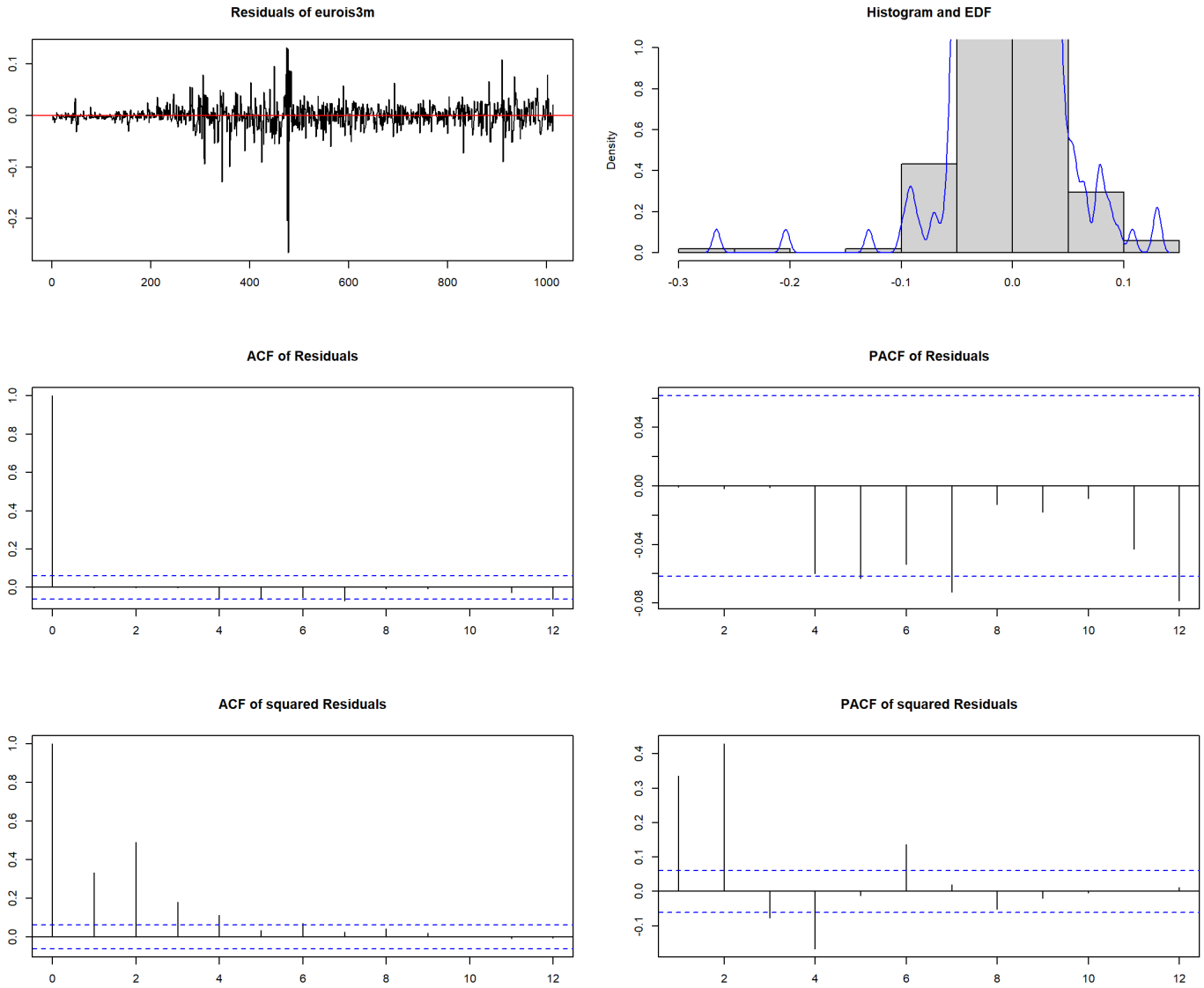
```
cat("\nUr model AIC: ", AIC(model), "R model AIC: ", AIC(model_s))
```

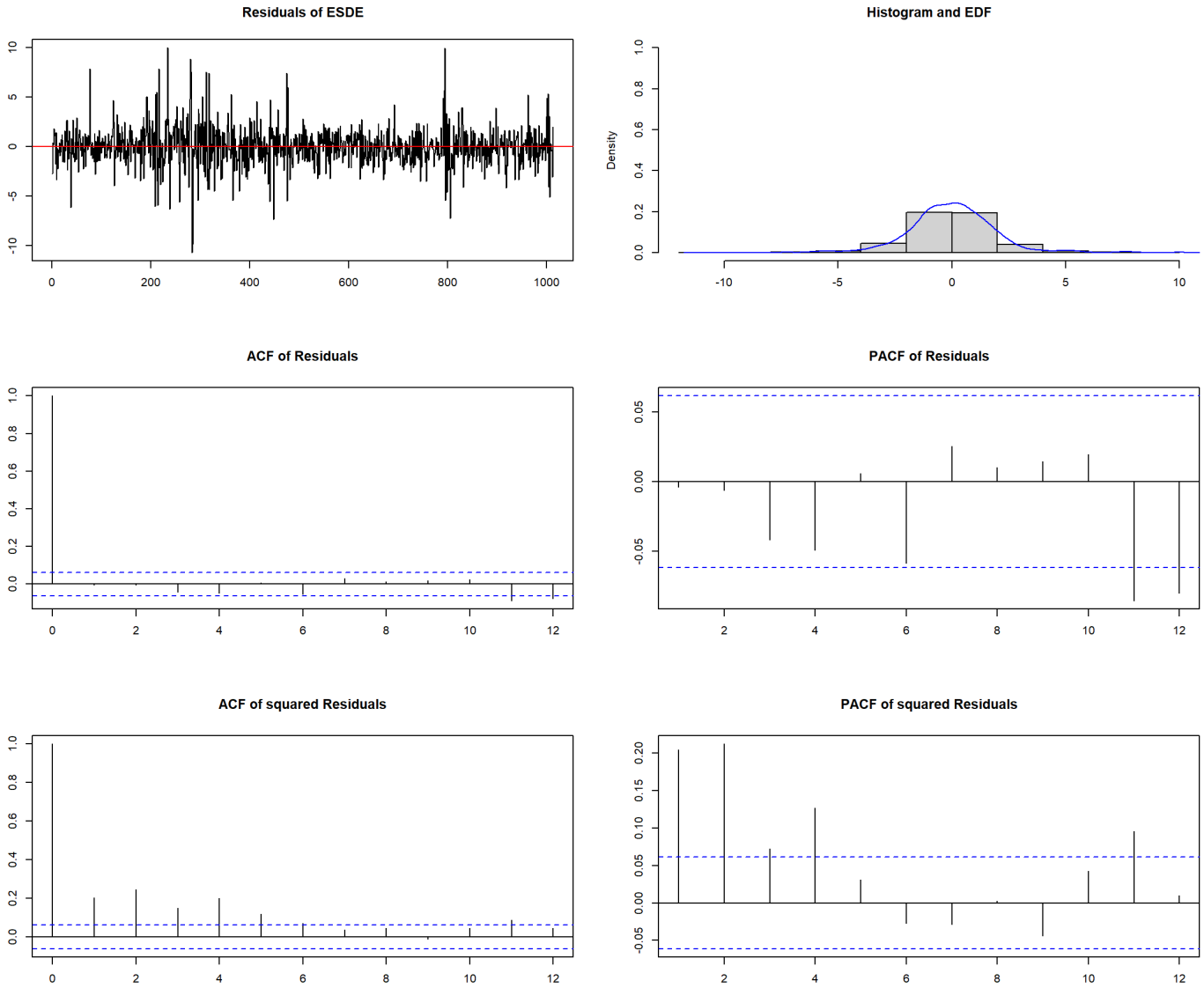
```
##
## Ur model AIC:  7704.668 R model AIC:  7649.728
```

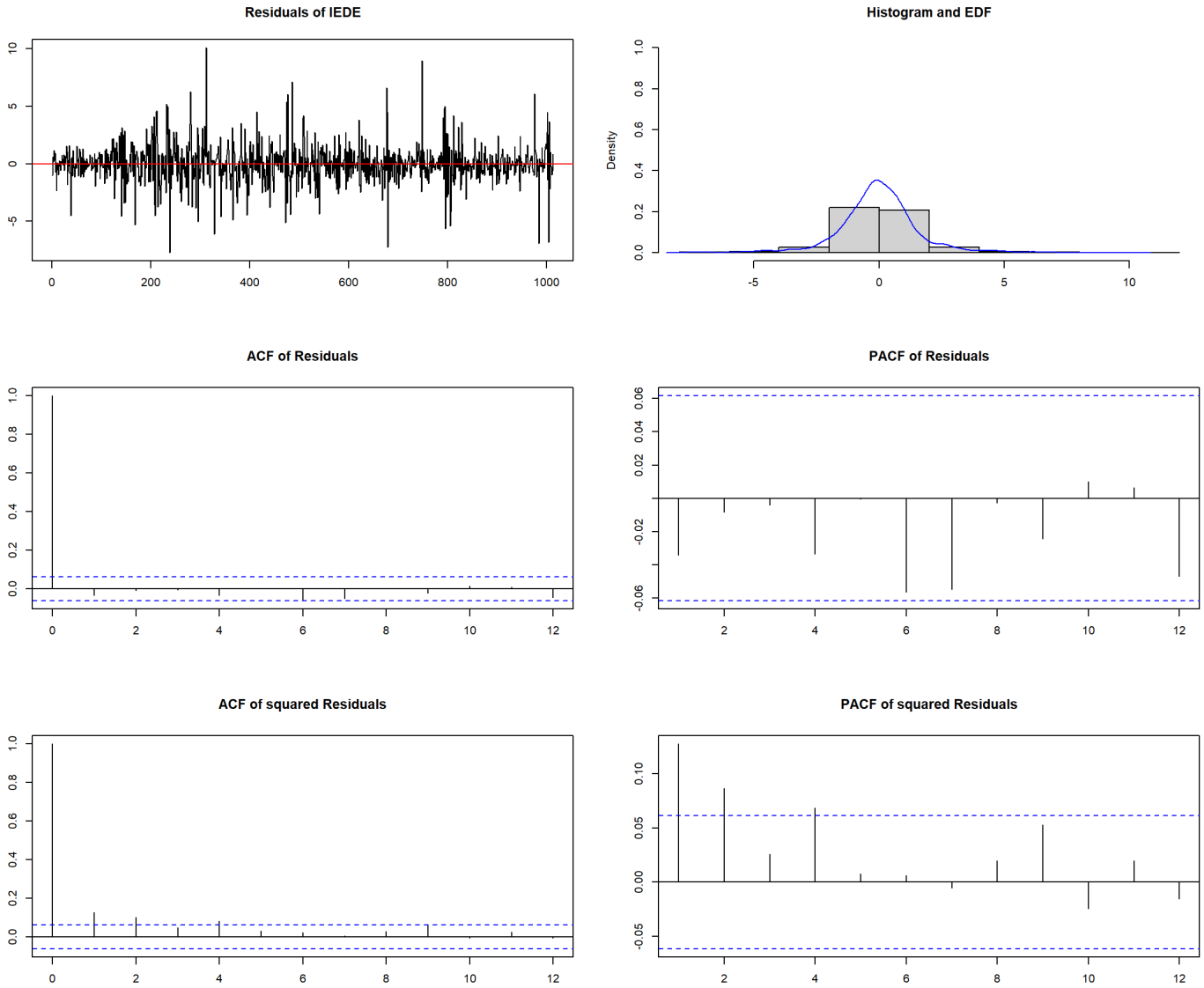
Diagnostic test

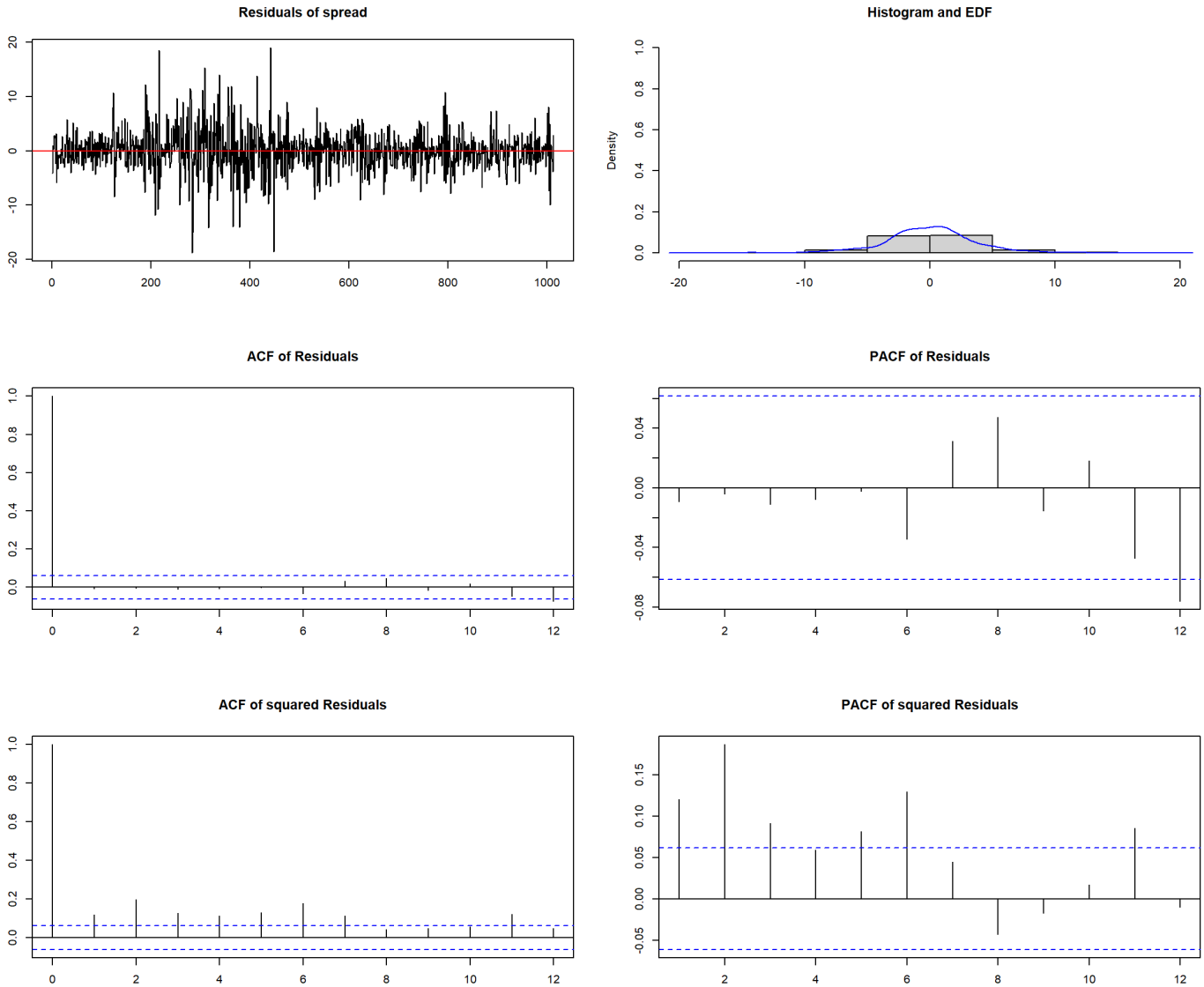
Residuals autocorrelation

```
par(ask = FALSE)
plot(vars::serial.test(model_s, lags.pt = 10, type = "PT.adjusted"))
```


Residuals of eurois3m
Histogram and EDF
Density
ACF of Residuals
PACF of Residuals
ACF of squared Residuals
PACF of squared Residuals


Residuals of ESDE
Histogram and EDF
Density
ACF of Residuals
PACF of Residuals
ACF of squared Residuals
PACF of squared Residuals

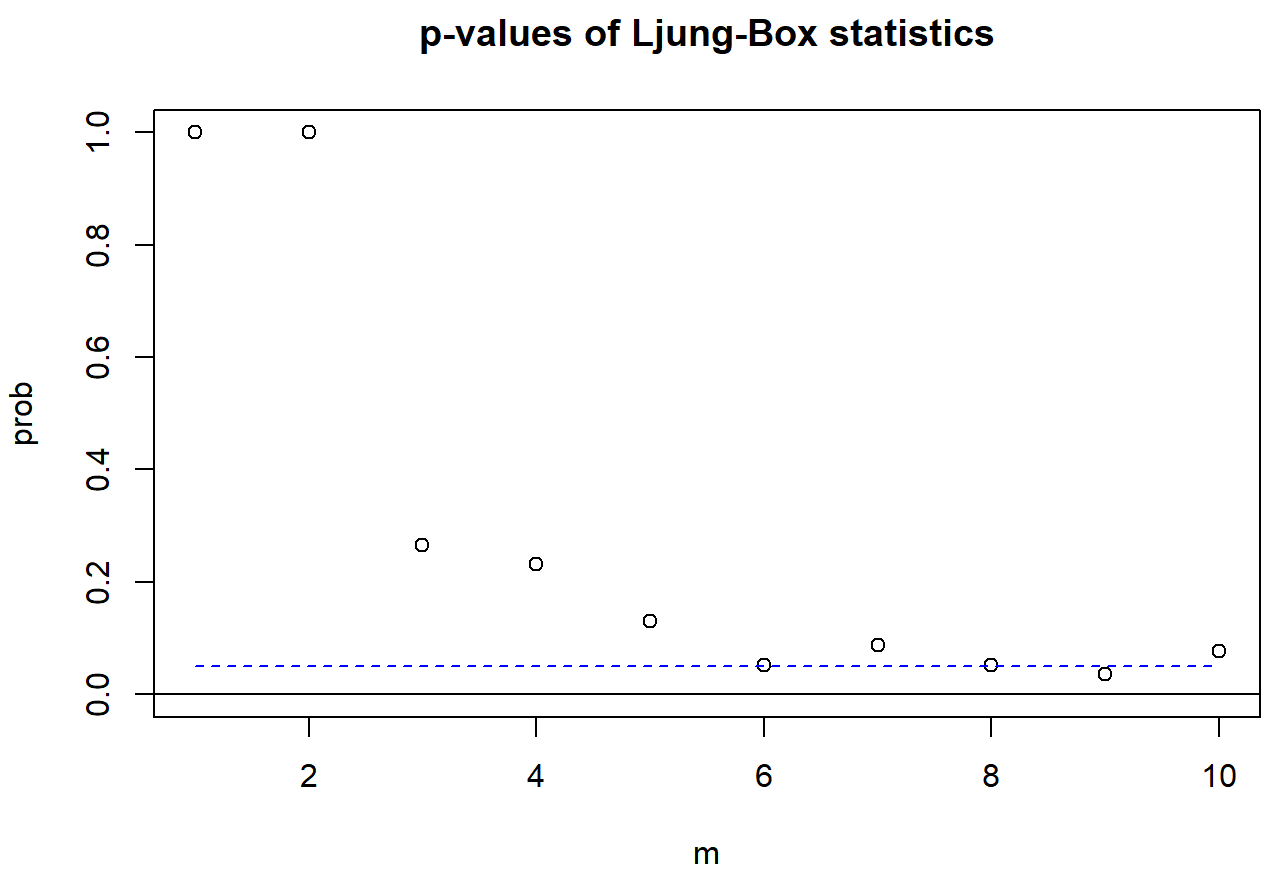

Residuals of IEDE
Histogram and EDF
Density
ACF of Residuals
PACF of Residuals
ACF of squared Residuals
PACF of squared Residuals



Run a Portmanteau test $H_0 : \text{No cross-correlation}$ to test for autocorrelatino in residuals

```
adj_var = (nrow(coef(model_s)$spread) + nrow(coef(model_s)$ESDE) + nrow(coef(model_s)$eurois3m)
+ nrow(coef(model_s)$IEDE))
res = residuals(model_s)
mq(res,lag=10, adj = adj_var)
```

```
## Ljung-Box Statistics:
##          m       Q(m)     df    p-value
##  [1,]   1.00      5.29  -10.00     1.00
##  [2,]   2.00     17.56    6.00     1.00
##  [3,]   3.00     25.70   22.00     0.26
##  [4,]   4.00     44.01   38.00     0.23
##  [5,]   5.00     65.82   54.00     0.13
##  [6,]   6.00     90.29   70.00     0.05
##  [7,]   7.00    104.31   86.00     0.09
##  [8,]   8.00    126.34  102.00     0.05
##  [9,]   9.00    147.20  118.00     0.04
## [10,]  10.00    158.04  134.00     0.08
```



We fail reject H_0, indicating the presence of no autocorrelation in residuals, apart from some rejection in the further away lags

Then we test for ARCH effects

ARCH test

```
vars::arch.test(model_s, lags.single = 16, lags.multi = 5, multivariate.only = FALSE)
```

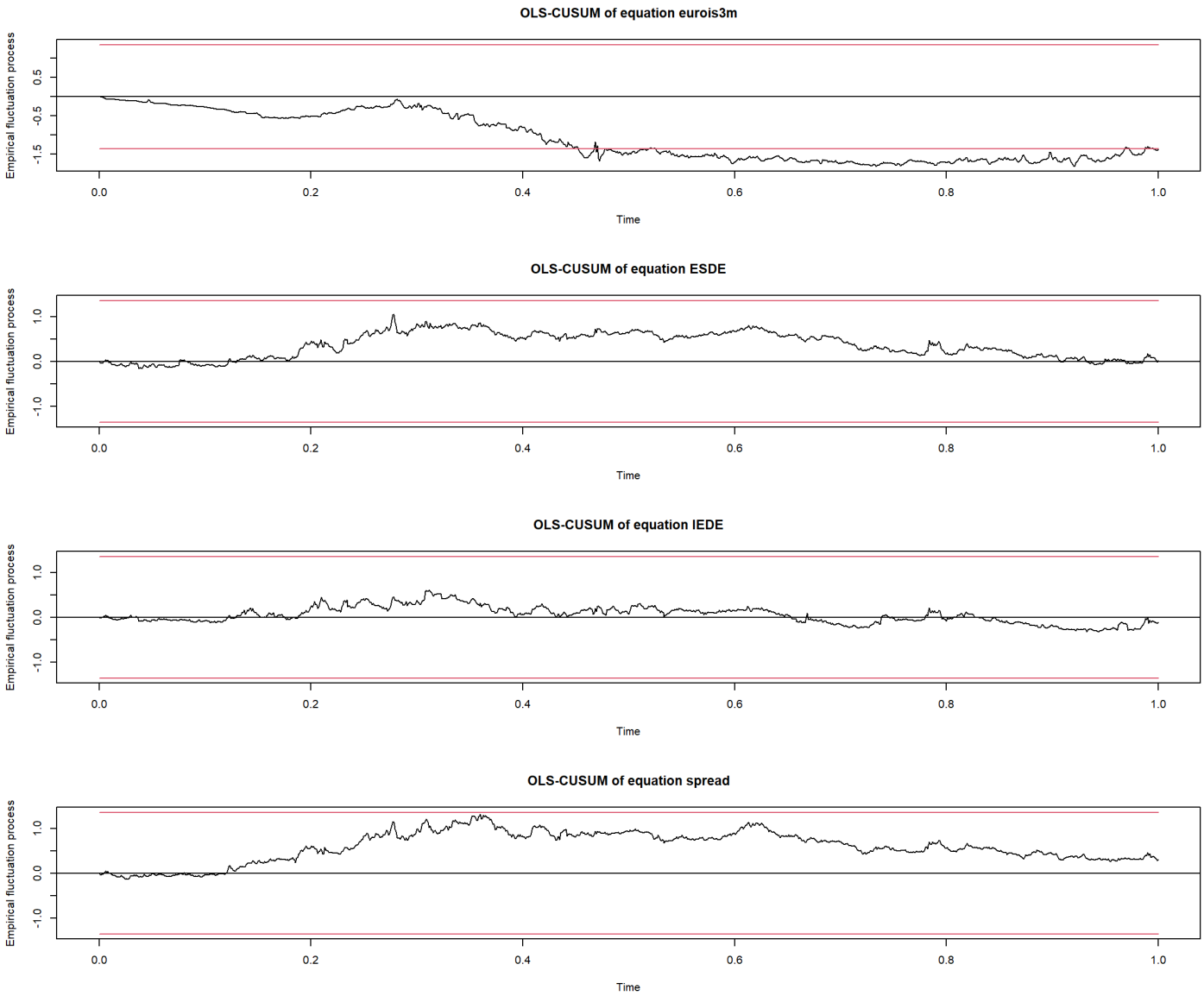
```
## $eurois3m
##
##  ARCH test (univariate)
##
## data:  Residual of eurois3m equation
## Chi-squared = 318.18, df = 16, p-value < 2.2e-16
##
##
## $ESDE
##
##  ARCH test (univariate)
##
## data:  Residual of ESDE equation
## Chi-squared = 119.56, df = 16, p-value < 2.2e-16
##
##
## $IEDE
##
##  ARCH test (univariate)
##
## data:  Residual of IEDE equation
## Chi-squared = 39.721, df = 16, p-value = 0.0008549
##
##
## $spread
##
##  ARCH test (univariate)
##
## data:  Residual of spread equation
## Chi-squared = 93.083, df = 16, p-value = 6.739e-13
##
##
##
##  ARCH (multivariate)
##
## data:  Residuals of VAR object model_s
## Chi-squared = 1905.5, df = 500, p-value < 2.2e-16
```

We strongly reject the null of no ARCH effects, a GARCH modelling is required.

Test for stability in the residuals series with the CUSUM test, if the series cross the 5% significance level the parameters are unstable

Stability test

We run CUSUM test to check for possible structural breaks



All variables residuals lay around 0, even if they slighty drift. There's no evidence of structural breaks

Normality

Check if residuals are normal distributed, if not try to fit them to a t-distribution (more likely).

Use a KS test to test wheter they follow a t-distribution $H_0 : \text{serie follows supposed distribution}$

```
# p_value < 0.05 -> non-normal distributed
vars::normality.test(model_s, multivariate.only = FALSE)
```

```
## $eurois3m
##
##  JB-Test (univariate)
##
## data:  Residual of eurois3m equation
## Chi-squared = 18851, df = 2, p-value < 2.2e-16
##
##
## $ESDE
##
##  JB-Test (univariate)
##
## data:  Residual of ESDE equation
## Chi-squared = 740.99, df = 2, p-value < 2.2e-16
##
##
## $IEDE
##
##  JB-Test (univariate)
##
## data:  Residual of IEDE equation
## Chi-squared = 1068.5, df = 2, p-value < 2.2e-16
##
##
## $spread
##
##  JB-Test (univariate)
##
## data:  Residual of spread equation
## Chi-squared = 429.38, df = 2, p-value < 2.2e-16
##
##
## $JB
##
##  JB-Test (multivariate)
##
## data:  Residuals of VAR object model_s
## Chi-squared = 22486, df = 8, p-value < 2.2e-16
##
##
## $Skewness
##
##  Skewness only (multivariate)
##
## data:  Residuals of VAR object model_s
## Chi-squared = 427.64, df = 4, p-value < 2.2e-16
##
##
## $Kurtosis
##
##  Kurtosis only (multivariate)
##
```

```
## data:  Residuals of VAR object model_s
## Chi-squared = 22058, df = 4, p-value < 2.2e-16
```

```
check_dist <- function(residuals) {
  t = suppressWarnings(MASS::fitdistr(residuals, densfun = "t",
        start = list(m = mean(residuals), s = sd(residuals), df = 5)))  # <- add this
  n = MASS::fitdistr(residuals, densfun = "normal")
  aic = c(AIC(t), AIC(n))
  bic = c(BIC(t), BIC(n))
  df = data.frame(aic, bic, row.names = c("t", "n"))
  df
}

apply(residuals(model_s), 2, check_dist)
```

```
## $eurois3m
##         aic       bic
## t -5034.334 -5019.569
## n -4634.394 -4624.551
##
## $ESDE
##        aic      bic
## t 4120.662 4135.427
## n 4278.429 4288.272
##
## $IEDE
##        aic      bic
## t 3664.806 3679.571
## n 3901.699 3911.542
##
## $spread
##        aic      bic
## t 5491.670 5506.435
## n 5614.108 5623.952
```

We reject the null of the JB test for all the univariate series and the joint multivariate, thus the residuals do not follow a normal distribution.

The subsequent code, consisting in estimating a normal a t distribution on the residuals and then compute their IC, showed that a t-distribution is a better fit than the normal.

Causality

```
vars::causality(model_s, cause = c("eurois3m","ESDE","IEDE"))
```

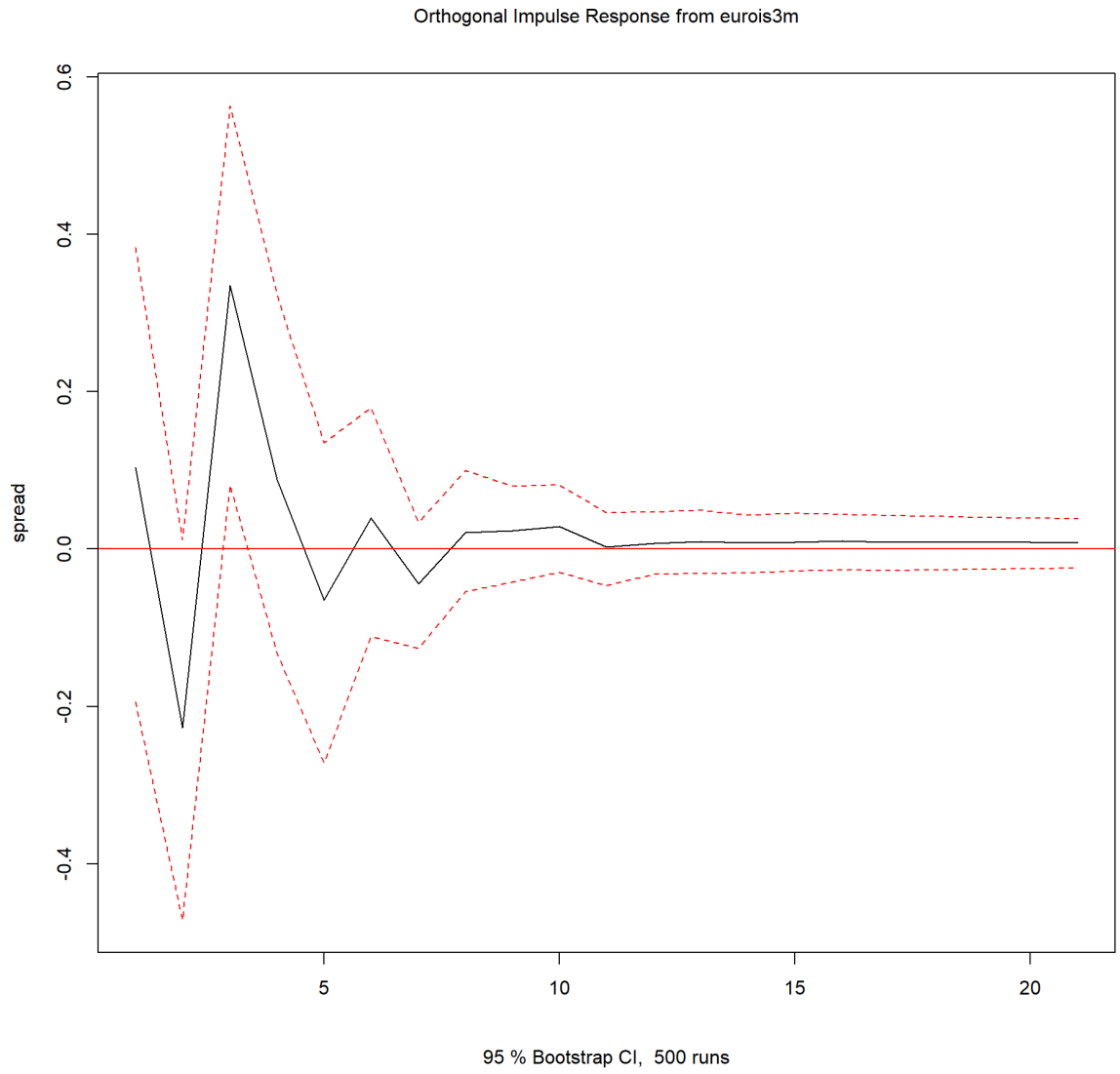
```
## $Granger
##
##  Granger causality H0: eurois3m ESDE IEDE do not Granger-cause spread
##
## data:  VAR object model_s
## F-Test = 1.3105, df1 = 15, df2 = 3972, p-value = 0.1861
##
##
## $Instant
##
##  H0: No instantaneous causality between: eurois3m ESDE IEDE and spread
##
## data:  VAR object model_s
## Chi-squared = 412.31, df = 3, p-value < 2.2e-16
```

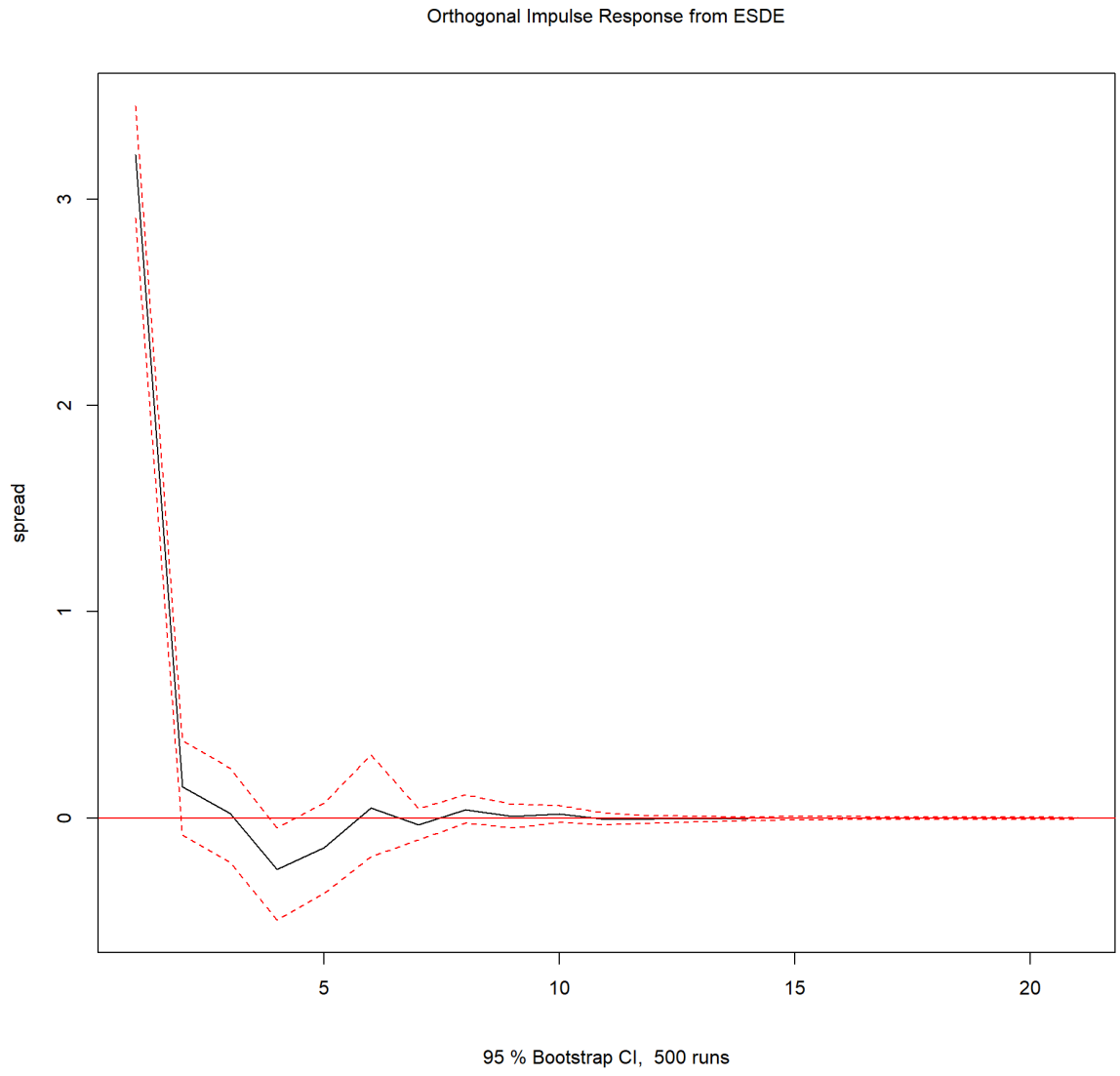
```
vars::causality(model, cause = c( "eurois3m","ESDE","IEDE"))
```

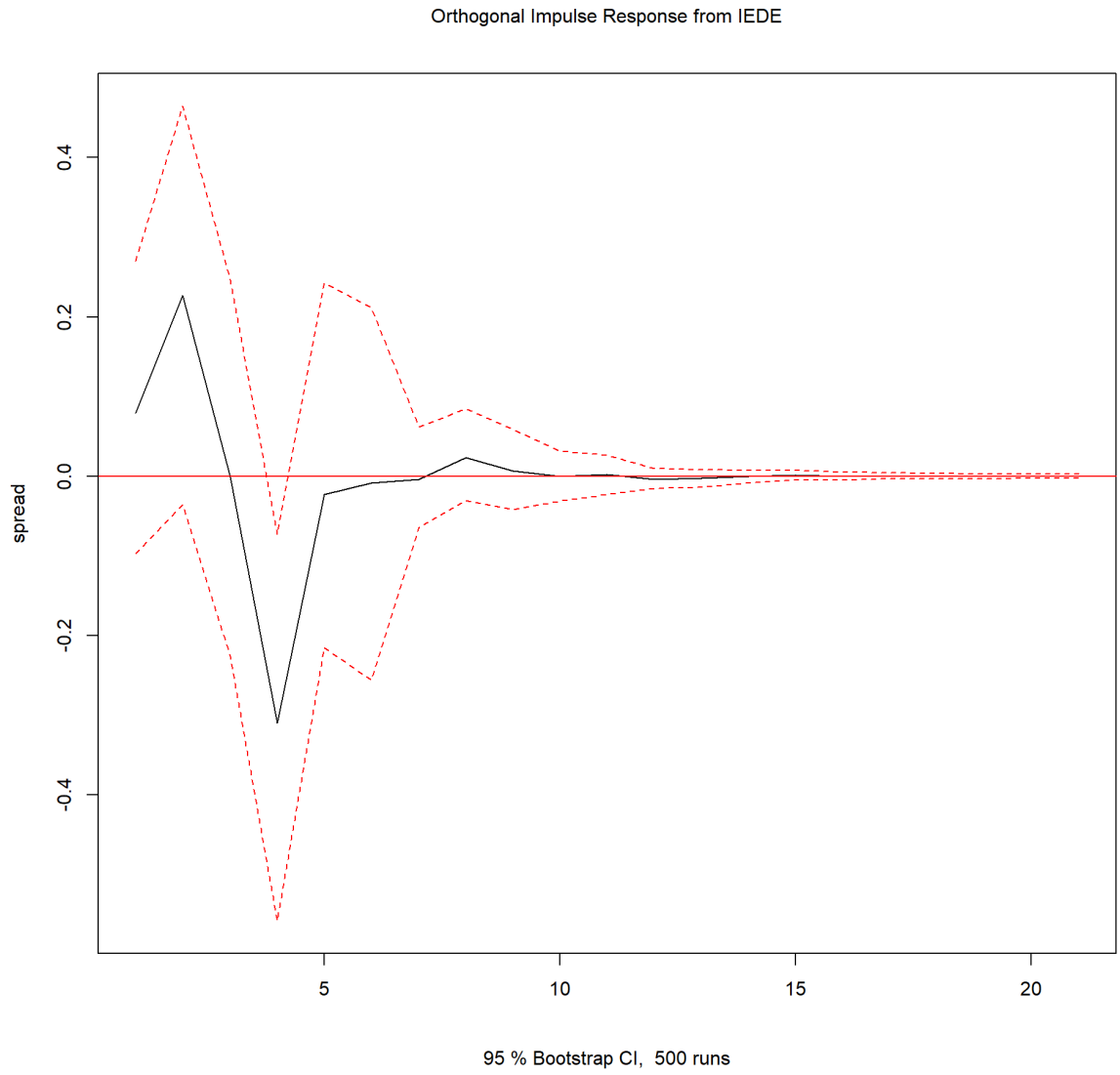
```
## $Granger
##
##  Granger causality H0: eurois3m ESDE IEDE do not Granger-cause spread
##
## data:  VAR object model
## F-Test = 2.4216, df1 = 15, df2 = 3972, p-value = 0.001644
##
##
## $Instant
##
##  H0: No instantaneous causality between: eurois3m ESDE IEDE and spread
##
## data:  VAR object model
## Chi-squared = 413.33, df = 3, p-value < 2.2e-16
```

```
#In the restricted model the granger-causality disappear ???
```

Impulse Response Functions IRF




Orthogonal Impulse Response from ESDE
spread
0
1
2
3
5
10
15
20
95 % Bootstrap CI, 500 runs

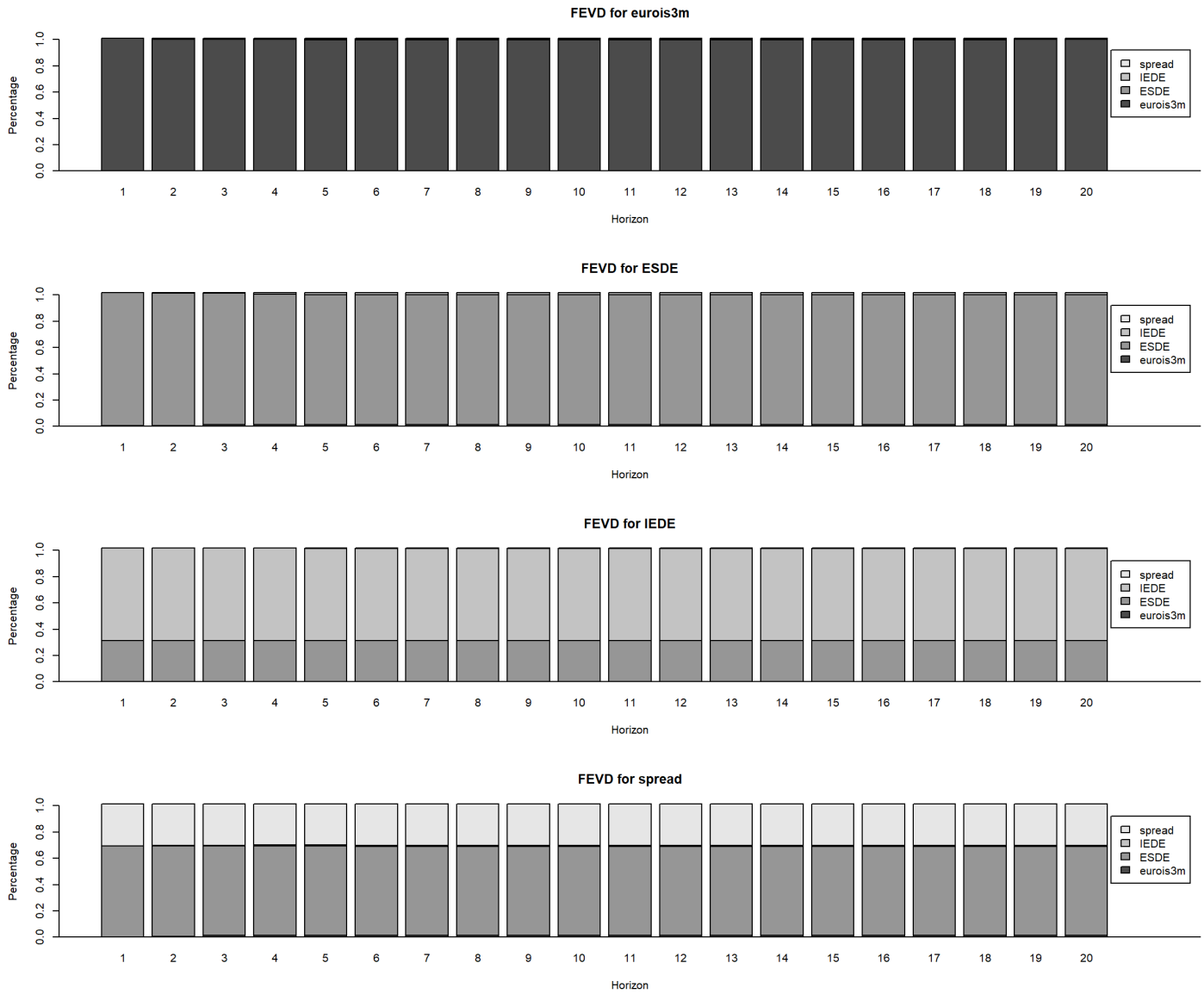



```
## --- ANALISI SIGNIFICATIVITÀ LAG (CI 95%) ---
```

```
## eurois3m     : Significativo ai lag [2]
## ESDE         : Significativo ai lag [0, 3]
## IEDE         : Significativo ai lag [3]
```

```
## ---------------------------------------------
```

Forecast Error Variance Decomposition FEVD



GARCH

For the mulivariate GARCH we use the DCC way

Then i re-run the same tests to see it the model is better.

```
res = residuals(model_s)

uni_spec = ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1,1)),
  mean.model = list(armaOrder = c(0,0), include.mean = FALSE),
  distribution.model = "std"
)

multi_spec = multispec(replicate(ncol(res), uni_spec))

dcc_spec = dccspec(uspec = multi_spec, dccOrder = c(1,1), distribution = "mvt")

dcc_fit = dccfit(dcc_spec, data = res)
```

```
## , , 2021-05-10
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04186737 0.04567508 0.01885192
## ESDE     0.04186737 1.00000000 0.51160462 0.82180371
## IEDE     0.04567508 0.51160462 1.00000000 0.46689441
## spread   0.01885192 0.82180371 0.46689441 1.00000000
##
## , , 2021-05-11
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03670119 0.04195735 0.01470095
## ESDE     0.03670119 1.00000000 0.51862500 0.82777470
## IEDE     0.04195735 0.51862500 1.00000000 0.47465469
## spread   0.01470095 0.82777470 0.47465469 1.00000000
##
## , , 2021-05-12
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m 1.000000000 0.0323267 0.03467938 0.009483525
## ESDE     0.032326704 1.0000000 0.51715717 0.826855045
## IEDE     0.034679377 0.5171572 1.00000000 0.473651756
## spread   0.009483525 0.8268550 0.47365176 1.000000000
##
## , , 2021-05-13
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m 1.000000000 0.03020218 0.0324719 0.007317193
## ESDE     0.030202181 1.00000000 0.5151916 0.825836664
## IEDE     0.032471902 0.51519160 1.0000000 0.471381852
## spread   0.007317193 0.82583666 0.4713819 1.000000000
##
## , , 2021-05-14
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01633746 0.01909098 -0.004462379
## ESDE      0.016337458 1.00000000 0.52683035  0.828920260
## IEDE      0.019090976 0.52683035 1.00000000  0.481693934
## spread   -0.004462379 0.82892026 0.48169393  1.000000000
##
## , , 2021-05-17
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.001004179 0.01275101 -0.02646674
## ESDE      0.001004179 1.000000000 0.52471620  0.82941203
## IEDE      0.012751015 0.524716195 1.00000000  0.47882241
## spread   -0.026466739 0.829412030 0.47882241  1.00000000
##
## , , 2021-05-18
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.005386982 0.005077018 -0.03198664
```

```
## ESDE     -0.005386982  1.000000000 0.530236497  0.83097235
## IEDE      0.005077018  0.530236497 1.000000000  0.48489242
## spread   -0.031986640  0.830972353 0.484892421  1.00000000
##
## , , 2021-05-19
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01193313 0.01519457 -0.01771559
## ESDE      0.01193313 1.00000000 0.53333898  0.83247456
## IEDE      0.01519457 0.53333898 1.00000000  0.48766326
## spread   -0.01771559 0.83247456 0.48766326  1.00000000
##
## , , 2021-05-20
##
##               eurois3m         ESDE       IEDE      spread
## eurois3m  1.0000000000 0.0008497415 0.01412725 -0.03083717
## ESDE      0.0008497415 1.0000000000 0.52646265  0.83436746
## IEDE      0.0141272490 0.5264626544 1.00000000  0.47876561
## spread   -0.0308371743 0.8343674606 0.47876561  1.00000000
##
## , , 2021-05-21
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02135973 -0.01228819 -0.05045164
## ESDE     -0.02135973  1.00000000  0.57410034  0.84881946
## IEDE     -0.01228819  0.57410034  1.00000000  0.53142524
## spread   -0.05045164  0.84881946  0.53142524  1.00000000
##
## , , 2021-05-24
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.01660778 -0.008491066 -0.04615108
## ESDE     -0.016607777  1.00000000  0.566449085  0.84535252
## IEDE     -0.008491066  0.56644908  1.000000000  0.52320657
## spread   -0.046151078  0.84535252  0.523206568  1.00000000
##
## , , 2021-05-25
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01358303 -0.00757871 -0.04311015
## ESDE     -0.01358303  1.00000000  0.55904921  0.84233873
## IEDE     -0.00757871  0.55904921  1.00000000  0.51661640
## spread   -0.04311015  0.84233873  0.51661640  1.00000000
##
## , , 2021-05-26
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.01987022 -0.004278165 -0.04681135
## ESDE     -0.019870217  1.00000000  0.538460304  0.84509208
## IEDE     -0.004278165  0.53846030  1.000000000  0.50012294
## spread   -0.046811350  0.84509208  0.500122935  1.00000000
##
```

```
## , , 2021-05-27
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01391804 0.00312183 -0.04108139
## ESDE     -0.01391804  1.00000000 0.53604274  0.84238580
## IEDE      0.00312183  0.53604274 1.00000000  0.49519903
## spread   -0.04108139  0.84238580 0.49519903  1.00000000
##
## , , 2021-05-28
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.01025305 0.005069451 -0.03631251
## ESDE     -0.010253051  1.00000000 0.532051865  0.83909395
## IEDE      0.005069451  0.53205187 1.000000000  0.49027269
## spread   -0.036312507  0.83909395 0.490272686  1.00000000
##
## , , 2021-05-31
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.01200153 0.001589887 -0.02620809
## ESDE     -0.012001534  1.00000000 0.528978170  0.83255291
## IEDE      0.001589887  0.52897817 1.000000000  0.48248603
## spread   -0.026208085  0.83255291 0.482486026  1.00000000
##
## , , 2021-06-01
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.003739053 0.005334511 -0.02195131
## ESDE     -0.003739053  1.000000000 0.526066538  0.83008950
## IEDE      0.005334511  0.526066538 1.000000000  0.47976705
## spread   -0.021951314  0.830089503 0.479767051  1.00000000
##
## , , 2021-06-02
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m 1.000000000 0.008703573 0.02199477 0.005750819
## ESDE     0.008703573 1.000000000 0.52555965 0.826340037
## IEDE     0.021994768 0.525559655 1.00000000 0.483841411
## spread   0.005750819 0.826340037 0.48384141 1.000000000
##
## , , 2021-06-03
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.01224776 0.02267085 0.004908163
## ESDE     0.012247755 1.00000000 0.52298846 0.824022387
## IEDE     0.022670848 0.52298846 1.00000000 0.480443155
## spread   0.004908163 0.82402239 0.48044315 1.000000000
##
## , , 2021-06-04
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m 1.000000000 0.0126845 0.02203813 0.001677973
```

```
## ESDE     0.012684499 1.0000000 0.51997558 0.821723054
## IEDE     0.022038132 0.5199756 1.00000000 0.476455726
## spread   0.001677973 0.8217231 0.47645573 1.000000000
##
## , , 2021-06-07
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 -0.01094412 0.001585143 -0.009080699
## ESDE     -0.010944123  1.00000000 0.526381821  0.819424864
## IEDE      0.001585143  0.52638182 1.000000000  0.476593634
## spread   -0.009080699  0.81942486 0.476593634  1.000000000
##
## , , 2021-06-08
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.00920262 0.002421542 -0.01195357
## ESDE     -0.009202620  1.00000000 0.526096120  0.81640070
## IEDE      0.002421542  0.52609612 1.000000000  0.47722830
## spread   -0.011953574  0.81640070 0.477228296  1.00000000
##
## , , 2021-06-09
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.00877622 0.00434958 -0.01357602
## ESDE     -0.00877622  1.00000000 0.51266257  0.82172771
## IEDE      0.00434958  0.51266257 1.00000000  0.46728644
## spread   -0.01357602  0.82172771 0.46728644  1.00000000
##
## , , 2021-06-10
##
##               eurois3m          ESDE        IEDE       spread
## eurois3m  1.0000000000 -0.0003389579 0.006689208 -0.008920956
## ESDE     -0.0003389579  1.0000000000 0.508141262  0.822940072
## IEDE      0.0066892085  0.5081412617 1.000000000  0.464673667
## spread   -0.0089209564  0.8229400724 0.464673667  1.000000000
##
## , , 2021-06-11
##
##              eurois3m        ESDE         IEDE     spread
## eurois3m  1.000000000 0.001563016 -0.001992617 0.00354811
## ESDE      0.001563016 1.000000000  0.501657695 0.81008262
## IEDE     -0.001992617 0.501657695  1.000000000 0.42738991
## spread    0.003548110 0.810082622  0.427389905 1.00000000
##
## , , 2021-06-14
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0262086 0.01154254 0.02737477
## ESDE     0.02620860 1.0000000 0.50753415 0.81845744
## IEDE     0.01154254 0.5075341 1.00000000 0.43819609
## spread   0.02737477 0.8184574 0.43819609 1.00000000
##
```

```
## , , 2021-06-15
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m 1.000000000 0.004057894 0.000261735 0.01425568
## ESDE     0.004057894 1.000000000 0.515089236 0.81633582
## IEDE     0.000261735 0.515089236 1.000000000 0.44449932
## spread   0.014255681 0.816335824 0.444499315 1.00000000
##
## , , 2021-06-16
##
##               eurois3m         ESDE        IEDE        spread
## eurois3m  1.0000000000 -0.001735449 0.004514918 -0.0009478966
## ESDE     -0.0017354487  1.000000000 0.511660338  0.8165430503
## IEDE      0.0045149185  0.511660338 1.000000000  0.4423519546
## spread   -0.0009478966  0.816543050 0.442351955  1.0000000000
##
## , , 2021-06-17
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01105336 -0.01076292 -0.01915234
## ESDE     -0.01105336  1.00000000  0.51177585  0.81700230
## IEDE     -0.01076292  0.51177585  1.00000000  0.44682140
## spread   -0.01915234  0.81700230  0.44682140  1.00000000
##
## , , 2021-06-18
##
##              eurois3m         ESDE         IEDE      spread
## eurois3m  1.000000000 -0.003103732 -0.003332704 -0.01419601
## ESDE     -0.003103732  1.000000000  0.547668150  0.82606178
## IEDE     -0.003332704  0.547668150  1.000000000  0.47805924
## spread   -0.014196015  0.826061779  0.478059236  1.00000000
##
## , , 2021-06-21
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m 1.000000000 0.01005369 0.004860997 0.002548143
## ESDE     0.010053691 1.00000000 0.554308519 0.837776778
## IEDE     0.004860997 0.55430852 1.000000000 0.488417965
## spread   0.002548143 0.83777678 0.488417965 1.000000000
##
## , , 2021-06-22
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.01746994 0.01407254 0.005094739
## ESDE     0.017469940 1.00000000 0.56983736 0.835162510
## IEDE     0.014072536 0.56983736 1.00000000 0.493572336
## spread   0.005094739 0.83516251 0.49357234 1.000000000
##
## , , 2021-06-23
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.02057122 0.02234703 -0.000530139
```

```
## ESDE      0.020571223 1.00000000 0.56280272  0.830620796
## IEDE      0.022347029 0.56280272 1.00000000  0.483612402
## spread   -0.000530139 0.83062080 0.48361240  1.000000000
##
## , , 2021-06-24
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.02650008 0.0259841 0.01139351
## ESDE     0.02650008 1.00000000 0.5565855 0.82911527
## IEDE     0.02598410 0.55658551 1.0000000 0.48087685
## spread   0.01139351 0.82911527 0.4808768 1.00000000
##
## , , 2021-06-25
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02791766 0.02579462 0.01218547
## ESDE     0.02791766 1.00000000 0.54941834 0.83064265
## IEDE     0.02579462 0.54941834 1.00000000 0.47722079
## spread   0.01218547 0.83064265 0.47722079 1.00000000
##
## , , 2021-06-28
##
##              eurois3m       ESDE        IEDE       spread
## eurois3m  1.000000000 0.01102019 0.002503904 -0.003828642
## ESDE      0.011020189 1.00000000 0.559492543  0.833147332
## IEDE      0.002503904 0.55949254 1.000000000  0.490429192
## spread   -0.003828642 0.83314733 0.490429192  1.000000000
##
## , , 2021-06-29
##
##              eurois3m      ESDE       IEDE       spread
## eurois3m  1.000000000 0.0156282 0.01083634 -0.001531426
## ESDE      0.015628195 1.0000000 0.55514333  0.831673723
## IEDE      0.010836343 0.5551433 1.00000000  0.487450243
## spread   -0.001531426 0.8316737 0.48745024  1.000000000
##
## , , 2021-07-01
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.02141317 0.0214513 0.00490498
## ESDE     0.02141317 1.00000000 0.5514013 0.83084118
## IEDE     0.02145130 0.55140128 1.0000000 0.48758838
## spread   0.00490498 0.83084118 0.4875884 1.00000000
##
## , , 2021-07-02
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.01971357 0.02039238 0.004525251
## ESDE     0.019713571 1.00000000 0.67851257 0.809113246
## IEDE     0.020392375 0.67851257 1.00000000 0.535788927
## spread   0.004525251 0.80911325 0.53578893 1.000000000
##
```

```
## , , 2021-07-05
##
##              eurois3m         ESDE        IEDE     spread
## eurois3m  1.000000000 -0.001221913 0.009038618 -0.0390736
## ESDE     -0.001221913  1.000000000 0.663889951  0.8085784
## IEDE      0.009038618  0.663889951 1.000000000  0.5269723
## spread   -0.039073600  0.808578408 0.526972325  1.0000000
##
## , , 2021-07-06
##
##               eurois3m         ESDE      IEDE      spread
## eurois3m  1.0000000000 0.0003849027 0.0107040 -0.03608931
## ESDE      0.0003849027 1.0000000000 0.6492630  0.80858737
## IEDE      0.0107040008 0.6492629625 1.0000000  0.51854174
## spread   -0.0360893102 0.8085873676 0.5185417  1.00000000
##
## , , 2021-07-07
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.004201636 0.01549326 -0.02750737
## ESDE      0.004201636 1.000000000 0.63691683  0.80824729
## IEDE      0.015493262 0.636916833 1.00000000  0.51645055
## spread   -0.027507369 0.808247288 0.51645055  1.00000000
##
## , , 2021-07-08
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.02703147 0.002397922 -0.06227754
## ESDE     -0.027031471  1.00000000 0.625093765  0.81418616
## IEDE      0.002397922  0.62509377 1.000000000  0.51174714
## spread   -0.062277542  0.81418616 0.511747136  1.00000000
##
## , , 2021-07-09
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.006806539 0.01382007 -0.02883624
## ESDE     -0.006806539  1.000000000 0.62129675  0.82000624
## IEDE      0.013820074  0.621296751 1.00000000  0.51591867
## spread   -0.028836240  0.820006245 0.51591867  1.00000000
##
## , , 2021-07-12
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.007497312 0.01288051 -0.03083629
## ESDE     -0.007497312  1.000000000 0.61012113  0.82069207
## IEDE      0.012880508  0.610121129 1.00000000  0.50999488
## spread   -0.030836286  0.820692070 0.50999488  1.00000000
##
## , , 2021-07-13
##
##               eurois3m          ESDE       IEDE      spread
## eurois3m  1.0000000000 -0.0003675031 0.01626275 -0.01811151
```

```
## ESDE     -0.0003675031  1.0000000000 0.59954608  0.81989774
## IEDE      0.0162627484  0.5995460834 1.00000000  0.50414458
## spread   -0.0181115067  0.8198977407 0.50414458  1.00000000
##
## , , 2021-07-14
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.06089445 -0.06499376 -0.05551911
## ESDE     -0.06089445  1.00000000  0.59754513  0.82067537
## IEDE     -0.06499376  0.59754513  1.00000000  0.50396171
## spread   -0.05551911  0.82067537  0.50396171  1.00000000
##
## , , 2021-07-15
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03635576 -0.05150272 -0.02544242
## ESDE     -0.03635576  1.00000000  0.58709853  0.82048740
## IEDE     -0.05150272  0.58709853  1.00000000  0.49772007
## spread   -0.02544242  0.82048740  0.49772007  1.00000000
##
## , , 2021-07-16
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04464264 -0.04470126 -0.04085197
## ESDE     -0.04464264  1.00000000  0.56837492  0.82620205
## IEDE     -0.04470126  0.56837492  1.00000000  0.47911508
## spread   -0.04085197  0.82620205  0.47911508  1.00000000
##
## , , 2021-07-19
##
##              eurois3m         ESDE        IEDE       spread
## eurois3m  1.000000000 -0.008870233 -0.02203426 -0.004797401
## ESDE     -0.008870233  1.000000000  0.56325485  0.827832146
## IEDE     -0.022034256  0.563254847  1.00000000  0.479275848
## spread   -0.004797401  0.827832146  0.47927585  1.000000000
##
## , , 2021-07-20
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03140171 -0.04006804 -0.02839715
## ESDE     -0.03140171  1.00000000  0.58473855  0.83824099
## IEDE     -0.04006804  0.58473855  1.00000000  0.50795968
## spread   -0.02839715  0.83824099  0.50795968  1.00000000
##
## , , 2021-07-21
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02662993 -0.03397823 -0.02623222
## ESDE     -0.02662993  1.00000000  0.57683454  0.83620917
## IEDE     -0.03397823  0.57683454  1.00000000  0.50272974
## spread   -0.02623222  0.83620917  0.50272974  1.00000000
##
```

```
## , , 2021-07-22
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02144805 -0.03277973 -0.01547211
## ESDE     -0.02144805  1.00000000  0.56799462  0.83304198
## IEDE     -0.03277973  0.56799462  1.00000000  0.49425054
## spread   -0.01547211  0.83304198  0.49425054  1.00000000
##
## , , 2021-07-23
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01122785 -0.02658414 -0.01322168
## ESDE     -0.01122785  1.00000000  0.55338114  0.82043409
## IEDE     -0.02658414  0.55338114  1.00000000  0.49053138
## spread   -0.01322168  0.82043409  0.49053138  1.00000000
##
## , , 2021-07-26
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.007876812 -0.02126405 -0.01221663
## ESDE     -0.007876812  1.000000000  0.56342038  0.82403742
## IEDE     -0.021264048  0.563420378  1.00000000  0.49935114
## spread   -0.012216630  0.824037419  0.49935114  1.00000000
##
## , , 2021-07-27
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.007697446 -0.0186911 -0.01311776
## ESDE     -0.007697446  1.000000000  0.5599590  0.82377932
## IEDE     -0.018691097  0.559958962  1.0000000  0.49669650
## spread   -0.013117755  0.823779317  0.4966965  1.00000000
##
## , , 2021-07-28
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.006579282 -0.01609093 -0.01323533
## ESDE     -0.006579282  1.000000000  0.56594400  0.82299601
## IEDE     -0.016090927  0.565943999  1.00000000  0.49653786
## spread   -0.013235334  0.822996014  0.49653786  1.00000000
##
## , , 2021-07-29
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.007028772 -0.01709097 -0.01278038
## ESDE     -0.007028772  1.000000000  0.56253589  0.81983702
## IEDE     -0.017090972  0.562535887  1.00000000  0.48707139
## spread   -0.012780380  0.819837017  0.48707139  1.00000000
##
## , , 2021-07-30
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.005711063 -0.01808046 -0.01207411
```

```
## ESDE     -0.005711063  1.000000000  0.55373881  0.81877738
## IEDE     -0.018080456  0.553738807  1.00000000  0.47663440
## spread   -0.012074107  0.818777384  0.47663440  1.00000000
##
## , , 2021-08-02
##
##              eurois3m         ESDE       IEDE     spread
## eurois3m  1.000000000 -0.003498236 -0.0142129 -0.0117877
## ESDE     -0.003498236  1.000000000  0.5487192  0.8141816
## IEDE     -0.014212897  0.548719204  1.0000000  0.4780624
## spread   -0.011787696  0.814181581  0.4780624  1.0000000
##
## , , 2021-08-03
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.002016981 -0.0112483 -0.01235171
## ESDE     -0.002016981  1.000000000  0.5489413  0.81390844
## IEDE     -0.011248299  0.548941293  1.0000000  0.48680034
## spread   -0.012351715  0.813908438  0.4868003  1.00000000
##
## , , 2021-08-04
##
##               eurois3m          ESDE         IEDE       spread
## eurois3m  1.0000000000 -0.0003589884 -0.009035937 -0.009752628
## ESDE     -0.0003589884  1.0000000000  0.543198307  0.813413873
## IEDE     -0.0090359365  0.5431983072  1.000000000  0.481621615
## spread   -0.0097526283  0.8134138735  0.481621615  1.000000000
##
## , , 2021-08-05
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.001748984 -0.01100011 -0.01062977
## ESDE     -0.001748984  1.000000000  0.54723534  0.81211064
## IEDE     -0.011000112  0.547235342  1.00000000  0.47649807
## spread   -0.010629774  0.812110640  0.47649807  1.00000000
##
## , , 2021-08-06
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04752105 -0.03632084 -0.06149624
## ESDE     -0.04752105  1.00000000  0.55659412  0.82544195
## IEDE     -0.03632084  0.55659412  1.00000000  0.49206950
## spread   -0.06149624  0.82544195  0.49206950  1.00000000
##
## , , 2021-08-09
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03135453 -0.03758025 -0.06075753
## ESDE     -0.03135453  1.00000000  0.54429511  0.81921926
## IEDE     -0.03758025  0.54429511  1.00000000  0.48874545
## spread   -0.06075753  0.81921926  0.48874545  1.00000000
##
```

```
## , , 2021-08-10
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.02424079 -0.02307096 -0.0462755
## ESDE     -0.02424079  1.00000000  0.53849546  0.8166575
## IEDE     -0.02307096  0.53849546  1.00000000  0.4900724
## spread   -0.04627550  0.81665749  0.49007236  1.0000000
##
## , , 2021-08-11
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.01462148 -0.009848097 -0.03893794
## ESDE     -0.014621476  1.00000000  0.536159335  0.81675528
## IEDE     -0.009848097  0.53615933  1.000000000  0.48638687
## spread   -0.038937945  0.81675528  0.486386871  1.00000000
##
## , , 2021-08-12
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03516322 -0.01206234 -0.06042393
## ESDE     -0.03516322  1.00000000  0.53037612  0.82158107
## IEDE     -0.01206234  0.53037612  1.00000000  0.48118385
## spread   -0.06042393  0.82158107  0.48118385  1.00000000
##
## , , 2021-08-13
##
##              eurois3m         ESDE         IEDE      spread
## eurois3m  1.000000000 -0.008911107 -0.008433801 -0.03916171
## ESDE     -0.008911107  1.000000000  0.512908357  0.82634114
## IEDE     -0.008433801  0.512908357  1.000000000  0.47118217
## spread   -0.039161712  0.826341141  0.471182170  1.00000000
##
## , , 2021-08-16
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.006563103 0.01079544 -0.04814383
## ESDE     -0.006563103  1.000000000 0.50914297  0.82357722
## IEDE      0.010795441  0.509142967 1.00000000  0.46073093
## spread   -0.048143826  0.823577219 0.46073093  1.00000000
##
## , , 2021-08-17
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.009052904 0.008305557 -0.04912205
## ESDE     -0.009052904  1.000000000 0.511659924  0.82540408
## IEDE      0.008305557  0.511659924 1.000000000  0.46435297
## spread   -0.049122049  0.825404081 0.464352966  1.00000000
##
## , , 2021-08-18
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.005809642 0.009672953 -0.04383619
```

```
## ESDE     -0.005809642  1.000000000 0.510920217  0.82560794
## IEDE      0.009672953  0.510920217 1.000000000  0.46391079
## spread   -0.043836194  0.825607944 0.463910790  1.00000000
##
## , , 2021-08-19
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.002685684 0.01163769 -0.03789256
## ESDE     -0.002685684  1.000000000 0.51023247  0.82390693
## IEDE      0.011637688  0.510232473 1.00000000  0.46476859
## spread   -0.037892559  0.823906934 0.46476859  1.00000000
##
## , , 2021-08-20
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.003818011 0.01236808 -0.03825799
## ESDE     -0.003818011  1.000000000 0.50348430  0.82699407
## IEDE      0.012368083  0.503484296 1.00000000  0.45384260
## spread   -0.038257985  0.826994074 0.45384260  1.00000000
##
## , , 2021-08-23
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.001467746 0.01141352 -0.03947069
## ESDE     -0.001467746  1.000000000 0.50233756  0.82201038
## IEDE      0.011413519  0.502337563 1.00000000  0.45415240
## spread   -0.039470689  0.822010382 0.45415240  1.00000000
##
## , , 2021-08-24
##
##               eurois3m         ESDE       IEDE      spread
## eurois3m  1.0000000000 0.0003924865 0.01233903 -0.03592632
## ESDE      0.0003924865 1.0000000000 0.50001666  0.81401693
## IEDE      0.0123390290 0.5000166576 1.00000000  0.44079791
## spread   -0.0359263156 0.8140169273 0.44079791  1.00000000
##
## , , 2021-08-25
##
##              eurois3m         ESDE      IEDE     spread
## eurois3m  1.000000000 -0.001832825 0.0151769 -0.0224626
## ESDE     -0.001832825  1.000000000 0.4988192  0.8095619
## IEDE      0.015176898  0.498819234 1.0000000  0.4418025
## spread   -0.022462602  0.809561917 0.4418025  1.0000000
##
## , , 2021-08-26
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 -0.1023420 -0.00511455 -0.05635891
## ESDE     -0.10234196  1.0000000  0.44346294  0.72511434
## IEDE     -0.00511455  0.4434629  1.00000000  0.46058707
## spread   -0.05635891  0.7251143  0.46058707  1.00000000
##
```

```
## , , 2021-08-27
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.08068973 0.002906698 -0.04933699
## ESDE     -0.080689730  1.00000000 0.445427551  0.72950505
## IEDE      0.002906698  0.44542755 1.000000000  0.45920857
## spread   -0.049336991  0.72950505 0.459208571  1.00000000
##
## , , 2021-08-30
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.06845074 0.004801591 -0.03143265
## ESDE     -0.068450743  1.00000000 0.447326593  0.73225094
## IEDE      0.004801591  0.44732659 1.000000000  0.45380889
## spread   -0.031432646  0.73225094 0.453808893  1.00000000
##
## , , 2021-08-31
##
##               eurois3m        ESDE         IEDE      spread
## eurois3m  1.0000000000 -0.06103777 0.0007811659 -0.03123786
## ESDE     -0.0610377728  1.00000000 0.4473140524  0.73674059
## IEDE      0.0007811659  0.44731405 1.0000000000  0.45503875
## spread   -0.0312378646  0.73674059 0.4550387484  1.00000000
##
## , , 2021-09-01
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.05361896 0.004691705 -0.02283406
## ESDE     -0.053618962  1.00000000 0.450671092  0.72298376
## IEDE      0.004691705  0.45067109 1.000000000  0.46360152
## spread   -0.022834059  0.72298376 0.463601521  1.00000000
##
## , , 2021-09-02
##
##               eurois3m        ESDE       IEDE        spread
## eurois3m  1.0000000000 -0.03991156 0.01297905 -0.0001849062
## ESDE     -0.0399115571  1.00000000 0.45699275  0.7267042966
## IEDE      0.0129790452  0.45699275 1.00000000  0.4657632620
## spread   -0.0001849062  0.72670430 0.46576326  1.0000000000
##
## , , 2021-09-03
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 -0.02730778 0.01789149 0.006668921
## ESDE     -0.027307781  1.00000000 0.46139923 0.735763998
## IEDE      0.017891490  0.46139923 1.00000000 0.465532773
## spread    0.006668921  0.73576400 0.46553277 1.000000000
##
## , , 2021-09-06
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m  1.000000000 -0.02230557 0.02230416 -0.002069543
```

```
## ESDE     -0.022305569  1.00000000 0.46381568  0.739677404
## IEDE      0.022304164  0.46381568 1.00000000  0.457551282
## spread   -0.002069543  0.73967740 0.45755128  1.000000000
##
## , , 2021-09-07
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m  1.000000000 -0.02091365 0.02225187 -0.003898878
## ESDE     -0.020913649  1.00000000 0.45967741  0.746818774
## IEDE      0.022251867  0.45967741 1.00000000  0.454402950
## spread   -0.003898878  0.74681877 0.45440295  1.000000000
##
## , , 2021-09-08
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m  1.000000000 -0.01864538 0.01941266 -0.006337535
## ESDE     -0.018645375  1.00000000 0.47186447  0.756949455
## IEDE      0.019412663  0.47186447 1.00000000  0.466337915
## spread   -0.006337535  0.75694946 0.46633791  1.000000000
##
## , , 2021-09-09
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m  1.000000000 -0.02041536 0.01435143 -0.009773115
## ESDE     -0.020415355  1.00000000 0.47447206  0.762675842
## IEDE      0.014351433  0.47447206 1.00000000  0.464760490
## spread   -0.009773115  0.76267584 0.46476049  1.000000000
##
## , , 2021-09-10
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0051677 0.03983425 0.01547078
## ESDE     0.00516770 1.0000000 0.51888359 0.78546841
## IEDE     0.03983425 0.5188836 1.00000000 0.51449791
## spread   0.01547078 0.7854684 0.51449791 1.00000000
##
## , , 2021-09-13
##
##             eurois3m        ESDE      IEDE     spread
## eurois3m 1.000000000 0.001041518 0.0323243 0.00298394
## ESDE     0.001041518 1.000000000 0.5174274 0.78853840
## IEDE     0.032324305 0.517427435 1.0000000 0.50911043
## spread   0.002983940 0.788538404 0.5091104 1.00000000
##
## , , 2021-09-14
##
##               eurois3m        ESDE       IEDE        spread
## eurois3m  1.0000000000 0.001227415 0.02926576 -0.0003017067
## ESDE      0.0012274153 1.000000000 0.51757615  0.7922500827
## IEDE      0.0292657550 0.517576150 1.00000000  0.5063615968
## spread   -0.0003017067 0.792250083 0.50636160  1.0000000000
##
```

```
## , , 2021-09-15
##
##            eurois3m       ESDE      IEDE    spread
## eurois3m 1.00000000 0.02549532 0.0437112 0.0262759
## ESDE     0.02549532 1.00000000 0.5220130 0.7995047
## IEDE     0.04371120 0.52201302 1.0000000 0.5079261
## spread   0.02627590 0.79950465 0.5079261 1.0000000
##
## , , 2021-09-16
##
##             eurois3m        ESDE      IEDE      spread
## eurois3m 1.000000000 0.008853304 0.0318199 0.006345569
## ESDE     0.008853304 1.000000000 0.5213846 0.803300796
## IEDE     0.031819897 0.521384587 1.0000000 0.503955173
## spread   0.006345569 0.803300796 0.5039552 1.000000000
##
## , , 2021-09-17
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.01054758 0.02903369 0.005836231
## ESDE     0.010547581 1.00000000 0.50942275 0.805735573
## IEDE     0.029033694 0.50942275 1.00000000 0.488057117
## spread   0.005836231 0.80573557 0.48805712 1.000000000
##
## , , 2021-09-20
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.03609098 0.01497587 0.003566338
## ESDE     0.036090977 1.00000000 0.48970403 0.797061969
## IEDE     0.014975867 0.48970403 1.00000000 0.482801139
## spread   0.003566338 0.79706197 0.48280114 1.000000000
##
## , , 2021-09-21
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.02044397 0.002671821 -0.01532579
## ESDE      0.020443968 1.00000000 0.504941370  0.80555475
## IEDE      0.002671821 0.50494137 1.000000000  0.49832832
## spread   -0.015325786 0.80555475 0.498328317  1.00000000
##
## , , 2021-09-22
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.01669769 0.003463775 -0.02180706
## ESDE      0.016697694 1.00000000 0.504011994  0.80621532
## IEDE      0.003463775 0.50401199 1.000000000  0.49190688
## spread   -0.021807061 0.80621532 0.491906882  1.00000000
##
## , , 2021-09-23
##
##              eurois3m      ESDE        IEDE      spread
## eurois3m  1.000000000 0.0185322 0.007973704 -0.01706642
```

```
## ESDE      0.018532202 1.0000000 0.517810729  0.81217692
## IEDE      0.007973704 0.5178107 1.000000000  0.50811456
## spread   -0.017066425 0.8121769 0.508114557  1.00000000
##
## , , 2021-09-24
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m  1.00000000 0.0161269 0.01123757 -0.01575747
## ESDE      0.01612690 1.0000000 0.51289394  0.81172008
## IEDE      0.01123757 0.5128939 1.00000000  0.50157861
## spread   -0.01575747 0.8117201 0.50157861  1.00000000
##
## , , 2021-09-27
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01884696 0.01824649 -0.003688236
## ESDE      0.018846959 1.00000000 0.51314635  0.804349748
## IEDE      0.018246492 0.51314635 1.00000000  0.506835955
## spread   -0.003688236 0.80434975 0.50683596  1.000000000
##
## , , 2021-09-28
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.02313809 0.01965573 -0.005880626
## ESDE      0.023138091 1.00000000 0.51142201  0.801055590
## IEDE      0.019655728 0.51142201 1.00000000  0.499669006
## spread   -0.005880626 0.80105559 0.49966901  1.000000000
##
## , , 2021-09-29
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.00953638 -0.003490878 -0.04530416
## ESDE      0.009536380 1.00000000  0.513343044  0.79600980
## IEDE     -0.003490878 0.51334304  1.000000000  0.50651794
## spread   -0.045304157 0.79600980  0.506517939  1.00000000
##
## , , 2021-09-30
##
##               eurois3m        ESDE          IEDE      spread
## eurois3m  1.0000000000 0.005587012 -0.0001096047 -0.03618986
## ESDE      0.0055870123 1.000000000  0.5056621368  0.77464090
## IEDE     -0.0001096047 0.505662137  1.0000000000  0.49925341
## spread   -0.0361898563 0.774640904  0.4992534080  1.00000000
##
## , , 2021-10-01
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 0.007031744 0.003507197 -0.03527287
## ESDE      0.007031744 1.000000000 0.504376417  0.77427524
## IEDE      0.003507197 0.504376417 1.000000000  0.48473978
## spread   -0.035272867 0.774275243 0.484739780  1.00000000
##
```

```
## , , 2021-10-04
##
##              eurois3m        ESDE      IEDE     spread
## eurois3m  1.000000000 0.006696852 0.0142383 -0.0240327
## ESDE      0.006696852 1.000000000 0.5023864  0.7777963
## IEDE      0.014238300 0.502386360 1.0000000  0.4820380
## spread   -0.024032705 0.777796319 0.4820380  1.0000000
##
## , , 2021-10-05
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.004111481 0.01524238 -0.02726593
## ESDE      0.004111481 1.000000000 0.49959486  0.78334455
## IEDE      0.015242376 0.499594856 1.00000000  0.47467122
## spread   -0.027265933 0.783344552 0.47467122  1.00000000
##
## , , 2021-10-06
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.03344575 0.06924056 -0.01654393
## ESDE      0.03344575 1.00000000 0.51614612  0.78384173
## IEDE      0.06924056 0.51614612 1.00000000  0.46340150
## spread   -0.01654393 0.78384173 0.46340150  1.00000000
##
## , , 2021-10-07
##
##              eurois3m        ESDE       IEDE     spread
## eurois3m  1.000000000 0.008324157 0.04029139 -0.0464879
## ESDE      0.008324157 1.000000000 0.52403576  0.7924878
## IEDE      0.040291390 0.524035756 1.00000000  0.4767308
## spread   -0.046487905 0.792487754 0.47673081  1.0000000
##
## , , 2021-10-08
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m  1.00000000 0.0102182 0.03927074 -0.03786419
## ESDE      0.01021820 1.0000000 0.52358971  0.78978711
## IEDE      0.03927074 0.5235897 1.00000000  0.48133257
## spread   -0.03786419 0.7897871 0.48133257  1.00000000
##
## , , 2021-10-11
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01606512 0.03610597 -0.03359706
## ESDE      0.01606512 1.00000000 0.51445477  0.78916218
## IEDE      0.03610597 0.51445477 1.00000000  0.47741100
## spread   -0.03359706 0.78916218 0.47741100  1.00000000
##
## , , 2021-10-12
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.03044856 0.005023811 -0.03507395
```

```
## ESDE      0.030448560 1.00000000 0.482675074  0.78492949
## IEDE      0.005023811 0.48267507 1.000000000  0.47169227
## spread   -0.035073951 0.78492949 0.471692267  1.00000000
##
## , , 2021-10-13
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.03529053 0.01435096 -0.02364204
## ESDE      0.03529053 1.00000000 0.49691407  0.79352566
## IEDE      0.01435096 0.49691407 1.00000000  0.48441573
## spread   -0.02364204 0.79352566 0.48441573  1.00000000
##
## , , 2021-10-14
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02927183 0.01931996 -0.03298648
## ESDE      0.02927183 1.00000000 0.49454332  0.79458698
## IEDE      0.01931996 0.49454332 1.00000000  0.47240243
## spread   -0.03298648 0.79458698 0.47240243  1.00000000
##
## , , 2021-10-15
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.03012425 0.01302768 -0.02776329
## ESDE      0.03012425 1.00000000 0.48416190  0.79752933
## IEDE      0.01302768 0.48416190 1.00000000  0.46008446
## spread   -0.02776329 0.79752933 0.46008446  1.00000000
##
## , , 2021-10-18
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02912419 0.01416734 -0.02378725
## ESDE      0.02912419 1.00000000 0.49220478  0.80668345
## IEDE      0.01416734 0.49220478 1.00000000  0.46624979
## spread   -0.02378725 0.80668345 0.46624979  1.00000000
##
## , , 2021-10-19
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.01892804 0.02232137 -0.0243027
## ESDE      0.01892804 1.00000000 0.48372084  0.8068136
## IEDE      0.02232137 0.48372084 1.00000000  0.4613168
## spread   -0.02430270 0.80681356 0.46131684  1.0000000
##
## , , 2021-10-20
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.01801287 0.02261459 -0.0224958
## ESDE      0.01801287 1.00000000 0.48408012  0.8077675
## IEDE      0.02261459 0.48408012 1.00000000  0.4597471
## spread   -0.02249580 0.80776746 0.45974711  1.0000000
##
```

```
## , , 2021-10-21
##
##              eurois3m        ESDE      IEDE      spread
## eurois3m  1.000000000 0.007711639 0.0114287 -0.02274883
## ESDE      0.007711639 1.000000000 0.4992656  0.80319810
## IEDE      0.011428697 0.499265585 1.0000000  0.45735269
## spread   -0.022748831 0.803198104 0.4573527  1.00000000
##
## , , 2021-10-22
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 0.006988696 0.001544561 -0.02855109
## ESDE      0.006988696 1.000000000 0.497447349  0.80308263
## IEDE      0.001544561 0.497447349 1.000000000  0.46220208
## spread   -0.028551093 0.803082625 0.462202084  1.00000000
##
## , , 2021-10-25
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 0.001275698 -0.007457074 -0.03117453
## ESDE      0.001275698 1.000000000  0.504459174  0.80593222
## IEDE     -0.007457074 0.504459174  1.000000000  0.46588179
## spread   -0.031174526 0.805932224  0.465881792  1.00000000
##
## , , 2021-10-26
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 0.004638597 -0.005195661 -0.02281892
## ESDE      0.004638597 1.000000000  0.499597096  0.80170322
## IEDE     -0.005195661 0.499597096  1.000000000  0.44587221
## spread   -0.022818922 0.801703222  0.445872205  1.00000000
##
## , , 2021-10-27
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.01349944 -0.006775371 -0.04343894
## ESDE     -0.013499444  1.00000000  0.498422008  0.80815386
## IEDE     -0.006775371  0.49842201  1.000000000  0.44511711
## spread   -0.043438943  0.80815386  0.445117110  1.00000000
##
## , , 2021-10-28
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03907293 -0.08182549 -0.05086423
## ESDE     -0.03907293  1.00000000  0.50758783  0.80832390
## IEDE     -0.08182549  0.50758783  1.00000000  0.43772736
## spread   -0.05086423  0.80832390  0.43772736  1.00000000
##
## , , 2021-10-29
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.06827654 -0.08593195 -0.1231302
```

```
## ESDE     -0.06827654  1.00000000  0.51401946  0.7881171
## IEDE     -0.08593195  0.51401946  1.00000000  0.4233713
## spread   -0.12313022  0.78811711  0.42337132  1.0000000
##
## , , 2021-11-01
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.07478656 -0.08383251 -0.1215394
## ESDE     -0.07478656  1.00000000  0.55221855  0.8257268
## IEDE     -0.08383251  0.55221855  1.00000000  0.4752564
## spread   -0.12153942  0.82572681  0.47525638  1.0000000
##
## , , 2021-11-02
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.07763857 -0.07052978 -0.1168006
## ESDE     -0.07763857  1.00000000  0.53193372  0.8243888
## IEDE     -0.07052978  0.53193372  1.00000000  0.4675508
## spread   -0.11680062  0.82438885  0.46755080  1.0000000
##
## , , 2021-11-03
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.008529511 0.008815598 -0.05338773
## ESDE     -0.008529511  1.000000000 0.569120612  0.83427881
## IEDE      0.008815598  0.569120612 1.000000000  0.50392501
## spread   -0.053387730  0.834278806 0.503925010  1.00000000
##
## , , 2021-11-04
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 0.001314392 -0.02327535 -0.03742284
## ESDE      0.001314392 1.000000000  0.55685068  0.83283702
## IEDE     -0.023275352 0.556850680  1.00000000  0.49306059
## spread   -0.037422835 0.832837016  0.49306059  1.00000000
##
## , , 2021-11-05
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.001941086 -0.02560351 -0.04199572
## ESDE     -0.001941086  1.000000000  0.55522316  0.83203419
## IEDE     -0.025603509  0.555223165  1.00000000  0.49748192
## spread   -0.041995722  0.832034187  0.49748192  1.00000000
##
## , , 2021-11-08
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 0.000303668 -0.02716463 -0.03781197
## ESDE      0.000303668 1.000000000  0.54871615  0.83058755
## IEDE     -0.027164634 0.548716146  1.00000000  0.49217157
## spread   -0.037811967 0.830587553  0.49217157  1.00000000
##
```

```
## , , 2021-11-09
##
##               eurois3m         ESDE        IEDE      spread
## eurois3m  1.0000000000 0.0009444093 -0.02705808 -0.02578511
## ESDE      0.0009444093 1.0000000000  0.54337100  0.82713952
## IEDE     -0.0270580773 0.5433710039  1.00000000  0.48449548
## spread   -0.0257851125 0.8271395178  0.48449548  1.00000000
##
## , , 2021-11-10
##
##              eurois3m         ESDE        IEDE       spread
## eurois3m  1.000000000 -0.001341772 0.007681608 -0.007429377
## ESDE     -0.001341772  1.000000000 0.517253928  0.815216691
## IEDE      0.007681608  0.517253928 1.000000000  0.493553437
## spread   -0.007429377  0.815216691 0.493553437  1.000000000
##
## , , 2021-11-11
##
##               eurois3m        ESDE         IEDE      spread
## eurois3m  1.0000000000 -0.02043415 0.0007860811 -0.01550552
## ESDE     -0.0204341538  1.00000000 0.5222820602  0.80756475
## IEDE      0.0007860811  0.52228206 1.0000000000  0.49586980
## spread   -0.0155055169  0.80756475 0.4958698013  1.00000000
##
## , , 2021-11-12
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.02611608 -0.002756784 -0.01249738
## ESDE     -0.026116076  1.00000000  0.521313857  0.80360582
## IEDE     -0.002756784  0.52131386  1.000000000  0.48967850
## spread   -0.012497385  0.80360582  0.489678502  1.00000000
##
## , , 2021-11-15
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.02816221 -0.001294513 -0.01868746
## ESDE     -0.028162209  1.00000000  0.513166356  0.81243782
## IEDE     -0.001294513  0.51316636  1.000000000  0.48055319
## spread   -0.018687455  0.81243782  0.480553189  1.00000000
##
## , , 2021-11-16
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.01710168 -0.008292998 -0.01165527
## ESDE     -0.017101682  1.00000000  0.508057407  0.81308542
## IEDE     -0.008292998  0.50805741  1.000000000  0.47434951
## spread   -0.011655273  0.81308542  0.474349506  1.00000000
##
## , , 2021-11-17
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 -0.01178928 -0.01819523 -0.009049369
```

```
## ESDE     -0.011789276  1.00000000  0.45797695  0.813914828
## IEDE     -0.018195233  0.45797695  1.00000000  0.421554135
## spread   -0.009049369  0.81391483  0.42155413  1.000000000
##
## , , 2021-11-18
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 -0.0332513 -0.04501339 -0.02204533
## ESDE     -0.03325130  1.0000000  0.47402589  0.81441656
## IEDE     -0.04501339  0.4740259  1.00000000  0.42957324
## spread   -0.02204533  0.8144166  0.42957324  1.00000000
##
## , , 2021-11-19
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03404382 -0.03886351 -0.02429199
## ESDE     -0.03404382  1.00000000  0.47306074  0.81565922
## IEDE     -0.03886351  0.47306074  1.00000000  0.43110435
## spread   -0.02429199  0.81565922  0.43110435  1.00000000
##
## , , 2021-11-22
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.02808931 -0.02920859 -0.0222942
## ESDE     -0.02808931  1.00000000  0.47412097  0.8154592
## IEDE     -0.02920859  0.47412097  1.00000000  0.4228242
## spread   -0.02229420  0.81545920  0.42282420  1.0000000
##
## , , 2021-11-23
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02325125 -0.02285514 -0.02048708
## ESDE     -0.02325125  1.00000000  0.46848056  0.79876153
## IEDE     -0.02285514  0.46848056  1.00000000  0.35196484
## spread   -0.02048708  0.79876153  0.35196484  1.00000000
##
## , , 2021-11-24
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.08938654 -0.04536836 -0.1270275
## ESDE     -0.08938654  1.00000000  0.47652923  0.8095713
## IEDE     -0.04536836  0.47652923  1.00000000  0.3677987
## spread   -0.12702750  0.80957132  0.36779868  1.0000000
##
## , , 2021-11-25
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 -0.1095193 -0.07749361 -0.1368193
## ESDE     -0.10951933  1.0000000  0.49369301  0.8131751
## IEDE     -0.07749361  0.4936930  1.00000000  0.3886737
## spread   -0.13681932  0.8131751  0.38867370  1.0000000
##
```

```
## , , 2021-11-26
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.09705864 -0.06669568 -0.1235944
## ESDE     -0.09705864  1.00000000  0.48891501  0.8126329
## IEDE     -0.06669568  0.48891501  1.00000000  0.3937631
## spread   -0.12359442  0.81263294  0.39376310  1.0000000
##
## , , 2021-11-29
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.09243896 -0.07321009 -0.1101407
## ESDE     -0.09243896  1.00000000  0.49535798  0.8091600
## IEDE     -0.07321009  0.49535798  1.00000000  0.3917002
## spread   -0.11014075  0.80916002  0.39170023  1.0000000
##
## , , 2021-11-30
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.09998116 -0.05398934 -0.08637792
## ESDE     -0.09998116  1.00000000  0.48129470  0.78919333
## IEDE     -0.05398934  0.48129470  1.00000000  0.40028265
## spread   -0.08637792  0.78919333  0.40028265  1.00000000
##
## , , 2021-12-01
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.08565932 -0.03550005 -0.07853408
## ESDE     -0.08565932  1.00000000  0.47745224  0.79012865
## IEDE     -0.03550005  0.47745224  1.00000000  0.38491760
## spread   -0.07853408  0.79012865  0.38491760  1.00000000
##
## , , 2021-12-02
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.09468212 -0.04406654 -0.09706159
## ESDE     -0.09468212  1.00000000  0.50231045  0.81086198
## IEDE     -0.04406654  0.50231045  1.00000000  0.42347712
## spread   -0.09706159  0.81086198  0.42347712  1.00000000
##
## , , 2021-12-03
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 -0.1082470 -0.07224682 -0.1099058
## ESDE     -0.10824704  1.0000000  0.51716159  0.8153091
## IEDE     -0.07224682  0.5171616  1.00000000  0.4412630
## spread   -0.10990579  0.8153091  0.44126304  1.0000000
##
## , , 2021-12-06
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.08301687 -0.05808015 -0.09365173
```

```
## ESDE     -0.08301687  1.00000000  0.52031627  0.81505250
## IEDE     -0.05808015  0.52031627  1.00000000  0.44680941
## spread   -0.09365173  0.81505250  0.44680941  1.00000000
##
## , , 2021-12-07
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.07435305 -0.04855485 -0.09310523
## ESDE     -0.07435305  1.00000000  0.51663922  0.80973415
## IEDE     -0.04855485  0.51663922  1.00000000  0.43743198
## spread   -0.09310523  0.80973415  0.43743198  1.00000000
##
## , , 2021-12-08
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 -0.0761642 -0.05576096 -0.07448895
## ESDE     -0.07616420  1.0000000  0.52291427  0.79118822
## IEDE     -0.05576096  0.5229143  1.00000000  0.42005255
## spread   -0.07448895  0.7911882  0.42005255  1.00000000
##
## , , 2021-12-09
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.09228648 -0.01897916 -0.1636010
## ESDE     -0.09228648  1.00000000  0.51105250  0.7824653
## IEDE     -0.01897916  0.51105250  1.00000000  0.3847230
## spread   -0.16360105  0.78246533  0.38472297  1.0000000
##
## , , 2021-12-10
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.08803237 0.01530576 -0.1537526
## ESDE     -0.08803237  1.00000000 0.49947457  0.7862003
## IEDE      0.01530576  0.49947457 1.00000000  0.3820928
## spread   -0.15375262  0.78620030 0.38209276  1.0000000
##
## , , 2021-12-13
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 -0.0291397 0.03834398 -0.1073436
## ESDE     -0.02913970  1.0000000 0.51924697  0.7970026
## IEDE      0.03834398  0.5192470 1.00000000  0.4088930
## spread   -0.10734365  0.7970026 0.40889297  1.0000000
##
## , , 2021-12-14
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.007728242 0.04645872 -0.06355102
## ESDE      0.007728242 1.000000000 0.51771642  0.80024199
## IEDE      0.046458717 0.517716419 1.00000000  0.41431782
## spread   -0.063551020 0.800241986 0.41431782  1.00000000
##
```

```
## , , 2021-12-15
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.003027024 0.04314914 -0.05779734
## ESDE      0.003027024 1.000000000 0.51507389  0.80111683
## IEDE      0.043149143 0.515073895 1.00000000  0.41805003
## spread   -0.057797336 0.801116830 0.41805003  1.00000000
##
## , , 2021-12-16
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.008249088 0.04220738 -0.04988154
## ESDE     -0.008249088  1.000000000 0.50102693  0.78286001
## IEDE      0.042207378  0.501026930 1.00000000  0.42195754
## spread   -0.049881543  0.782860014 0.42195754  1.00000000
##
## , , 2021-12-17
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01174661 0.04891254 -0.02593283
## ESDE      0.01174661 1.00000000 0.51185625  0.79777615
## IEDE      0.04891254 0.51185625 1.00000000  0.43913204
## spread   -0.02593283 0.79777615 0.43913204  1.00000000
##
## , , 2021-12-20
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.005099577 0.03291167 -0.04063235
## ESDE     -0.005099577  1.000000000 0.51759707  0.80414456
## IEDE      0.032911674  0.517597066 1.00000000  0.44885343
## spread   -0.040632349  0.804144556 0.44885343  1.00000000
##
## , , 2021-12-21
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.008782796 0.01616343 -0.04035654
## ESDE     -0.008782796  1.000000000 0.52352032  0.80568923
## IEDE      0.016163426  0.523520316 1.00000000  0.44792954
## spread   -0.040356540  0.805689229 0.44792954  1.00000000
##
## , , 2021-12-22
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.006252032 0.01622591 -0.03698014
## ESDE     -0.006252032  1.000000000 0.52896424  0.80302328
## IEDE      0.016225911  0.528964240 1.00000000  0.45042312
## spread   -0.036980144  0.803023278 0.45042312  1.00000000
##
## , , 2021-12-23
##
##              eurois3m         ESDE        IEDE     spread
## eurois3m  1.000000000 -0.004866786 0.003589649 -0.0412770
```

```
## ESDE     -0.004866786  1.000000000 0.516163679  0.7989629
## IEDE      0.003589649  0.516163679 1.000000000  0.4654289
## spread   -0.041276997  0.798962942 0.465428876  1.0000000
##
## , , 2021-12-27
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.01027827 0.005656667 -0.05432345
## ESDE     -0.010278266  1.00000000 0.511769206  0.79925582
## IEDE      0.005656667  0.51176921 1.000000000  0.45395963
## spread   -0.054323452  0.79925582 0.453959627  1.00000000
##
## , , 2021-12-28
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.008531087 0.01009287 -0.05409673
## ESDE     -0.008531087  1.000000000 0.50660124  0.79781957
## IEDE      0.010092868  0.506601240 1.00000000  0.43944747
## spread   -0.054096727  0.797819573 0.43944747  1.00000000
##
## , , 2021-12-29
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.006851377 0.009010268 -0.05077174
## ESDE     -0.006851377  1.000000000 0.509881472  0.79414713
## IEDE      0.009010268  0.509881472 1.000000000  0.45727926
## spread   -0.050771735  0.794147132 0.457279261  1.00000000
##
## , , 2021-12-30
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.001757105 0.02645453 -0.04434065
## ESDE     -0.001757105  1.000000000 0.50408772  0.79658008
## IEDE      0.026454526  0.504087724 1.00000000  0.45122925
## spread   -0.044340647  0.796580081 0.45122925  1.00000000
##
## , , 2022-01-03
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.002185555 0.02982081 -0.03734883
## ESDE     -0.002185555  1.000000000 0.49452808  0.79225679
## IEDE      0.029820809  0.494528075 1.00000000  0.45447323
## spread   -0.037348833  0.792256791 0.45447323  1.00000000
##
## , , 2022-01-04
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.004862053 0.01700228 -0.03579701
## ESDE     -0.004862053  1.000000000 0.58901141  0.79218213
## IEDE      0.017002281  0.589011408 1.00000000  0.47196940
## spread   -0.035797010  0.792182125 0.47196940  1.00000000
##
```

```
## , , 2022-01-05
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.008895024 0.01546648 -0.05033524
## ESDE     -0.008895024  1.000000000 0.58011598  0.79191268
## IEDE      0.015466481  0.580115975 1.00000000  0.46529110
## spread   -0.050335240  0.791912684 0.46529110  1.00000000
##
## , , 2022-01-06
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.004196404 0.01656463 -0.04156496
## ESDE     -0.004196404  1.000000000 0.57295122  0.79592831
## IEDE      0.016564628  0.572951223 1.00000000  0.46365089
## spread   -0.041564959  0.795928305 0.46365089  1.00000000
##
## , , 2022-01-07
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.00336796 0.005579274 -0.04154596
## ESDE     -0.003367960  1.00000000 0.564386526  0.79800945
## IEDE      0.005579274  0.56438653 1.000000000  0.46284561
## spread   -0.041545961  0.79800945 0.462845614  1.00000000
##
## , , 2022-01-10
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.01600433 -0.009432888 -0.05493613
## ESDE      0.016004330 1.00000000  0.551349827  0.79177818
## IEDE     -0.009432888 0.55134983  1.000000000  0.46368748
## spread   -0.054936130 0.79177818  0.463687481  1.00000000
##
## , , 2022-01-11
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.01437486 -0.008912678 -0.06469906
## ESDE      0.014374863 1.00000000  0.545879383  0.78204992
## IEDE     -0.008912678 0.54587938  1.000000000  0.45839509
## spread   -0.064699061 0.78204992  0.458395093  1.00000000
##
## , , 2022-01-12
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.02582253 0.004122072 -0.03508398
## ESDE      0.025822525 1.00000000 0.542627944  0.78618681
## IEDE      0.004122072 0.54262794 1.000000000  0.46081740
## spread   -0.035083977 0.78618681 0.460817401  1.00000000
##
## , , 2022-01-13
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.01918818 0.003776544 -0.03675277
```

```
## ESDE      0.019188182 1.00000000 0.537947892  0.79087151
## IEDE      0.003776544 0.53794789 1.000000000  0.46039447
## spread   -0.036752774 0.79087151 0.460394469  1.00000000
##
## , , 2022-01-14
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02489273 0.02594433 -0.02515008
## ESDE      0.02489273 1.00000000 0.53372078  0.79475213
## IEDE      0.02594433 0.53372078 1.00000000  0.46559573
## spread   -0.02515008 0.79475213 0.46559573  1.00000000
##
## , , 2022-01-17
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01368586 0.03961329 -0.03243787
## ESDE      0.01368586 1.00000000 0.51278737  0.79855781
## IEDE      0.03961329 0.51278737 1.00000000  0.44855003
## spread   -0.03243787 0.79855781 0.44855003  1.00000000
##
## , , 2022-01-18
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.00407964 0.02330779 -0.02812948
## ESDE      0.00407964 1.00000000 0.51595904  0.79624528
## IEDE      0.02330779 0.51595904 1.00000000  0.44364116
## spread   -0.02812948 0.79624528 0.44364116  1.00000000
##
## , , 2022-01-19
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.002804718 0.02101956 -0.02633528
## ESDE      0.002804718 1.000000000 0.55002714  0.78893419
## IEDE      0.021019563 0.550027143 1.00000000  0.46689924
## spread   -0.026335281 0.788934186 0.46689924  1.00000000
##
## , , 2022-01-20
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.01591183 0.007228811 -0.04291797
## ESDE     -0.015911834  1.00000000 0.551831076  0.79686271
## IEDE      0.007228811  0.55183108 1.000000000  0.47344464
## spread   -0.042917975  0.79686271 0.473444643  1.00000000
##
## , , 2022-01-21
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.03999952 -0.01324375 -0.0718451
## ESDE     -0.03999952  1.00000000  0.55530899  0.8048093
## IEDE     -0.01324375  0.55530899  1.00000000  0.4830679
## spread   -0.07184510  0.80480933  0.48306791  1.0000000
##
```

```
## , , 2022-01-24
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.03199565 -0.009021645 -0.06183666
## ESDE     -0.031995649  1.00000000  0.550761492  0.80878289
## IEDE     -0.009021645  0.55076149  1.000000000  0.48120897
## spread   -0.061836660  0.80878289  0.481208973  1.00000000
##
## , , 2022-01-25
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01529807 0.02594435 -0.02090541
## ESDE      0.01529807 1.00000000 0.55765214  0.81458569
## IEDE      0.02594435 0.55765214 1.00000000  0.48989879
## spread   -0.02090541 0.81458569 0.48989879  1.00000000
##
## , , 2022-01-26
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m  1.00000000 0.0162036 0.02561812 -0.01998241
## ESDE      0.01620360 1.0000000 0.55192154  0.81186908
## IEDE      0.02561812 0.5519215 1.00000000  0.48444311
## spread   -0.01998241 0.8118691 0.48444311  1.00000000
##
## , , 2022-01-27
##
##              eurois3m        ESDE       IEDE     spread
## eurois3m  1.000000000 0.004503781 0.02140404 -0.0286216
## ESDE      0.004503781 1.000000000 0.54600590  0.8272002
## IEDE      0.021404043 0.546005898 1.00000000  0.4839291
## spread   -0.028621604 0.827200179 0.48392908  1.0000000
##
## , , 2022-01-28
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02491184 0.02744268 -0.06200647
## ESDE     -0.02491184  1.00000000 0.50483833  0.83966537
## IEDE      0.02744268  0.50483833 1.00000000  0.43404975
## spread   -0.06200647  0.83966537 0.43404975  1.00000000
##
## , , 2022-01-31
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.04797267 0.002746222 -0.05009172
## ESDE     -0.047972673  1.00000000 0.512374741  0.82422852
## IEDE      0.002746222  0.51237474 1.000000000  0.42902920
## spread   -0.050091719  0.82422852 0.429029204  1.00000000
##
## , , 2022-02-01
##
##              eurois3m        ESDE        IEDE     spread
## eurois3m  1.000000000 -0.04073489 0.006635609 -0.0392510
```

```
## ESDE     -0.040734895  1.00000000 0.507374008  0.7868758
## IEDE      0.006635609  0.50737401 1.000000000  0.4906765
## spread   -0.039250999  0.78687578 0.490676470  1.0000000
##
## , , 2022-02-02
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.03371525 0.008117651 -0.02880682
## ESDE     -0.033715249  1.00000000 0.505951130  0.75390676
## IEDE      0.008117651  0.50595113 1.000000000  0.45162465
## spread   -0.028806823  0.75390676 0.451624654  1.00000000
##
## , , 2022-02-03
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.03698248 0.01376388 -0.02724216
## ESDE     -0.03698248  1.00000000 0.49742505  0.75599962
## IEDE      0.01376388  0.49742505 1.00000000  0.45010053
## spread   -0.02724216  0.75599962 0.45010053  1.00000000
##
## , , 2022-02-04
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 -0.0894172 -0.04390346 -0.06331703
## ESDE     -0.08941720  1.0000000  0.58088198  0.77830215
## IEDE     -0.04390346  0.5808820  1.00000000  0.50446529
## spread   -0.06331703  0.7783021  0.50446529  1.00000000
##
## , , 2022-02-07
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.08884584 -0.03963292 -0.06120791
## ESDE     -0.08884584  1.00000000  0.55970995  0.76414226
## IEDE     -0.03963292  0.55970995  1.00000000  0.50145995
## spread   -0.06120791  0.76414226  0.50145995  1.00000000
##
## , , 2022-02-08
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.02645429 0.04722697 0.04849787
## ESDE     -0.02645429  1.00000000 0.55771514 0.76741721
## IEDE      0.04722697  0.55771514 1.00000000 0.51003065
## spread    0.04849787  0.76741721 0.51003065 1.00000000
##
## , , 2022-02-09
##
##             eurois3m       ESDE       IEDE    spread
## eurois3m  1.00000000 -0.0222447 0.05569956 0.0422872
## ESDE     -0.02224470  1.0000000 0.54555649 0.7713671
## IEDE      0.05569956  0.5455565 1.00000000 0.4948930
## spread    0.04228720  0.7713671 0.49489304 1.0000000
##
```

```
## , , 2022-02-10
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 -0.0245463 0.03029351 0.02365858
## ESDE     -0.02454630  1.0000000 0.54072786 0.77516385
## IEDE      0.03029351  0.5407279 1.00000000 0.49696284
## spread    0.02365858  0.7751638 0.49696284 1.00000000
##
## , , 2022-02-11
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.06134639 0.009620142 -0.01018857
## ESDE     -0.061346393  1.00000000 0.544698273  0.78735856
## IEDE      0.009620142  0.54469827 1.000000000  0.50205712
## spread   -0.010188566  0.78735856 0.502057124  1.00000000
##
## , , 2022-02-14
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 -0.03053459 0.02337067 0.009507358
## ESDE     -0.030534588  1.00000000 0.54690895 0.795463756
## IEDE      0.023370666  0.54690895 1.00000000 0.504334646
## spread    0.009507358  0.79546376 0.50433465 1.000000000
##
## , , 2022-02-15
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.01381004 0.03875112 0.01751415
## ESDE     -0.01381004  1.00000000 0.55299970 0.80021232
## IEDE      0.03875112  0.55299970 1.00000000 0.50792579
## spread    0.01751415  0.80021232 0.50792579 1.00000000
##
## , , 2022-02-16
##
##              eurois3m         ESDE       IEDE     spread
## eurois3m  1.000000000 -0.002909169 0.04532201 0.02267263
## ESDE     -0.002909169  1.000000000 0.55719211 0.80542494
## IEDE      0.045322006  0.557192109 1.00000000 0.51272238
## spread    0.022672627  0.805424939 0.51272238 1.00000000
##
## , , 2022-02-17
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m 1.000000000 0.004398265 0.04717697 0.02719284
## ESDE     0.004398265 1.000000000 0.55281433 0.80734224
## IEDE     0.047176972 0.552814326 1.00000000 0.50876464
## spread   0.027192842 0.807342244 0.50876464 1.00000000
##
## , , 2022-02-18
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m 1.000000000 0.004761152 0.03293976 0.02472803
```

```
## ESDE     0.004761152 1.000000000 0.53576680 0.80734478
## IEDE     0.032939757 0.535766805 1.00000000 0.48208337
## spread   0.024728029 0.807344783 0.48208337 1.00000000
##
## , , 2022-02-21
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02260433 0.06350651 0.03424197
## ESDE     0.02260433 1.00000000 0.54671871 0.81124910
## IEDE     0.06350651 0.54671871 1.00000000 0.48898386
## spread   0.03424197 0.81124910 0.48898386 1.00000000
##
## , , 2022-02-22
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02398163 0.06006082 0.03174854
## ESDE     0.02398163 1.00000000 0.55454144 0.82074883
## IEDE     0.06006082 0.55454144 1.00000000 0.49996038
## spread   0.03174854 0.82074883 0.49996038 1.00000000
##
## , , 2022-02-23
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01137419 0.04880612 0.02040564
## ESDE     0.01137419 1.00000000 0.55604464 0.82196946
## IEDE     0.04880612 0.55604464 1.00000000 0.50116075
## spread   0.02040564 0.82196946 0.50116075 1.00000000
##
## , , 2022-02-24
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m 1.000000000 0.001135398 0.02427764 0.007441955
## ESDE     0.001135398 1.000000000 0.55748018 0.823316813
## IEDE     0.024277644 0.557480184 1.00000000 0.503585301
## spread   0.007441955 0.823316813 0.50358530 1.000000000
##
## , , 2022-02-25
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01377854 0.04873619 -0.01792434
## ESDE     -0.01377854  1.00000000 0.46871255  0.82818171
## IEDE      0.04873619  0.46871255 1.00000000  0.38815716
## spread   -0.01792434  0.82818171 0.38815716  1.00000000
##
## , , 2022-02-28
##
##               eurois3m          ESDE       IEDE      spread
## eurois3m  1.0000000000 -0.0000126268 0.06125543 -0.01040004
## ESDE     -0.0000126268  1.0000000000 0.48229242  0.82759309
## IEDE      0.0612554309  0.4822924162 1.00000000  0.39976452
## spread   -0.0104000378  0.8275930900 0.39976452  1.00000000
##
```

```
## , , 2022-03-01
##
##              eurois3m         ESDE       IEDE       spread
## eurois3m  1.000000000 -0.005163941 0.05738356 -0.007020321
## ESDE     -0.005163941  1.000000000 0.47816132  0.811103764
## IEDE      0.057383556  0.478161321 1.00000000  0.404659554
## spread   -0.007020321  0.811103764 0.40465955  1.000000000
##
## , , 2022-03-02
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.00118504 0.05332441 -0.00236366
## ESDE      0.00118504 1.00000000 0.53988092  0.85469514
## IEDE      0.05332441 0.53988092 1.00000000  0.48113254
## spread   -0.00236366 0.85469514 0.48113254  1.00000000
##
## , , 2022-03-03
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02328309 0.03865641 -0.01890068
## ESDE     -0.02328309  1.00000000 0.54348677  0.85383419
## IEDE      0.03865641  0.54348677 1.00000000  0.48836568
## spread   -0.01890068  0.85383419 0.48836568  1.00000000
##
## , , 2022-03-04
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01987995 0.02091446 -0.01983446
## ESDE     -0.01987995  1.00000000 0.52491239  0.85044804
## IEDE      0.02091446  0.52491239 1.00000000  0.47858850
## spread   -0.01983446  0.85044804 0.47858850  1.00000000
##
## , , 2022-03-07
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.1062972 0.1058038 0.04493691
## ESDE     0.10629723 1.0000000 0.5492264 0.84519023
## IEDE     0.10580378 0.5492264 1.0000000 0.49115022
## spread   0.04493691 0.8451902 0.4911502 1.00000000
##
## , , 2022-03-08
##
##              eurois3m       ESDE      IEDE       spread
## eurois3m  1.000000000 0.01019022 0.1203359 -0.001941848
## ESDE      0.010190222 1.00000000 0.5240072  0.841784375
## IEDE      0.120335931 0.52400720 1.0000000  0.478702109
## spread   -0.001941848 0.84178438 0.4787021  1.000000000
##
## , , 2022-03-09
##
##              eurois3m        ESDE      IEDE       spread
## eurois3m  1.000000000 0.004562819 0.1054349 -0.009206799
```

```
## ESDE      0.004562819 1.000000000 0.5442502  0.852622127
## IEDE      0.105434932 0.544250203 1.0000000  0.503647890
## spread   -0.009206799 0.852622127 0.5036479  1.000000000
##
## , , 2022-03-10
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.002102845 0.08969931 -0.01246286
## ESDE     -0.002102845  1.000000000 0.54092385  0.84942989
## IEDE      0.089699307  0.540923845 1.00000000  0.49918346
## spread   -0.012462862  0.849429894 0.49918346  1.00000000
##
## , , 2022-03-11
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.004789797 0.08240555 -0.01630942
## ESDE     -0.004789797  1.000000000 0.51894500  0.86518492
## IEDE      0.082405550  0.518944995 1.00000000  0.45980875
## spread   -0.016309417  0.865184917 0.45980875  1.00000000
##
## , , 2022-03-14
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m  1.000000000 0.003736874 0.08769453 -0.005707717
## ESDE      0.003736874 1.000000000 0.52105300  0.862172776
## IEDE      0.087694528 0.521053001 1.00000000  0.467406703
## spread   -0.005707717 0.862172776 0.46740670  1.000000000
##
## , , 2022-03-15
##
##              eurois3m        ESDE      IEDE       spread
## eurois3m  1.000000000 0.008133995 0.0881329 -0.004170439
## ESDE      0.008133995 1.000000000 0.5200962  0.858434833
## IEDE      0.088132899 0.520096243 1.0000000  0.465164509
## spread   -0.004170439 0.858434833 0.4651645  1.000000000
##
## , , 2022-03-16
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01455174 0.04344797 -0.009468169
## ESDE      0.014551745 1.00000000 0.50037899  0.853122010
## IEDE      0.043447971 0.50037899 1.00000000  0.459769445
## spread   -0.009468169 0.85312201 0.45976945  1.000000000
##
## , , 2022-03-17
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.008820044 0.005711372 -0.02207443
## ESDE     -0.008820044  1.000000000 0.524411992  0.84984276
## IEDE      0.005711372  0.524411992 1.000000000  0.46777914
## spread   -0.022074434  0.849842756 0.467779139  1.00000000
##
```

```
## , , 2022-03-18
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.005938274 0.01875962 -0.02380409
## ESDE     -0.005938274  1.000000000 0.51887949  0.84630816
## IEDE      0.018759618  0.518879486 1.00000000  0.45920865
## spread   -0.023804091  0.846308162 0.45920865  1.00000000
##
## , , 2022-03-21
##
##              eurois3m         ESDE       IEDE     spread
## eurois3m  1.000000000 -0.009401643 0.02352754 -0.0247310
## ESDE     -0.009401643  1.000000000 0.51163437  0.8433547
## IEDE      0.023527540  0.511634374 1.00000000  0.4559655
## spread   -0.024731001  0.843354655 0.45596547  1.0000000
##
## , , 2022-03-22
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.009002286 0.01580426 -0.02461523
## ESDE     -0.009002286  1.000000000 0.51027423  0.84085219
## IEDE      0.015804264  0.510274229 1.00000000  0.45535223
## spread   -0.024615226  0.840852192 0.45535223  1.00000000
##
## , , 2022-03-23
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01839622 0.02525581 -0.01907793
## ESDE      0.01839622 1.00000000 0.51352362  0.82297428
## IEDE      0.02525581 0.51352362 1.00000000  0.45589728
## spread   -0.01907793 0.82297428 0.45589728  1.00000000
##
## , , 2022-03-24
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02281585 0.01927789 -0.01701828
## ESDE      0.02281585 1.00000000 0.50343228  0.82108308
## IEDE      0.01927789 0.50343228 1.00000000  0.45292339
## spread   -0.01701828 0.82108308 0.45292339  1.00000000
##
## , , 2022-03-25
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.003266361 0.003483936 -0.01399395
## ESDE     -0.003266361  1.000000000 0.507489573  0.81195823
## IEDE      0.003483936  0.507489573 1.000000000  0.45030627
## spread   -0.013993946  0.811958231 0.450306267  1.00000000
##
## , , 2022-03-28
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01181935 -0.01041637 -0.01761351
```

```
## ESDE     -0.01181935  1.00000000  0.51687067  0.81135984
## IEDE     -0.01041637  0.51687067  1.00000000  0.45153447
## spread   -0.01761351  0.81135984  0.45153447  1.00000000
##
## , , 2022-03-29
##
##              eurois3m         ESDE         IEDE      spread
## eurois3m  1.000000000 -0.008829297 -0.007417537 -0.01641466
## ESDE     -0.008829297  1.000000000  0.510577569  0.81245108
## IEDE     -0.007417537  0.510577569  1.000000000  0.44612721
## spread   -0.016414663  0.812451083  0.446127212  1.00000000
##
## , , 2022-03-30
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.02137895 0.007378447 -0.03861573
## ESDE     -0.021378951  1.00000000 0.498972318  0.81429449
## IEDE      0.007378447  0.49897232 1.000000000  0.43076453
## spread   -0.038615731  0.81429449 0.430764530  1.00000000
##
## , , 2022-03-31
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.001920838 0.01752047 -0.03772159
## ESDE      0.001920838 1.000000000 0.50694036  0.78761016
## IEDE      0.017520468 0.506940365 1.00000000  0.42715119
## spread   -0.037721594 0.787610160 0.42715119  1.00000000
##
## , , 2022-04-01
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.008999636 -0.09448409 -0.02331942
## ESDE     -0.008999636  1.000000000  0.48749092  0.78860331
## IEDE     -0.094484090  0.487490917  1.00000000  0.39292370
## spread   -0.023319420  0.788603308  0.39292370  1.00000000
##
## , , 2022-04-04
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.002908662 -0.06899448 -0.01429953
## ESDE     -0.002908662  1.000000000  0.53124724  0.80030138
## IEDE     -0.068994476  0.531247238  1.00000000  0.50215349
## spread   -0.014299532  0.800301377  0.50215349  1.00000000
##
## , , 2022-04-05
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.05906116 -0.01606455 -0.01782705
## ESDE      0.05906116 1.00000000  0.53687515  0.78666008
## IEDE     -0.01606455 0.53687515  1.00000000  0.49194207
## spread   -0.01782705 0.78666008  0.49194207  1.00000000
##
```

```
## , , 2022-04-06
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01358127 -0.03471769 -0.03904416
## ESDE     -0.01358127  1.00000000  0.55896875  0.75433291
## IEDE     -0.03471769  0.55896875  1.00000000  0.52123942
## spread   -0.03904416  0.75433291  0.52123942  1.00000000
##
## , , 2022-04-07
##
##               eurois3m         ESDE        IEDE       spread
## eurois3m  1.000000e+00 3.941443e-05 -0.03003709 -0.006467681
## ESDE      3.941443e-05 1.000000e+00  0.55087903  0.755109115
## IEDE     -3.003709e-02 5.508790e-01  1.00000000  0.497968236
## spread   -6.467681e-03 7.551091e-01  0.49796824  1.000000000
##
## , , 2022-04-08
##
##               eurois3m         ESDE        IEDE       spread
## eurois3m  1.0000000000 0.0006651503 -0.02628686 -0.007107648
## ESDE      0.0006651503 1.0000000000  0.54861081  0.759869408
## IEDE     -0.0262868553 0.5486108114  1.00000000  0.495261580
## spread   -0.0071076475 0.7598694083  0.49526158  1.000000000
##
## , , 2022-04-11
##
##              eurois3m        ESDE         IEDE       spread
## eurois3m  1.000000000 0.007549455 -0.004652426 -0.001487646
## ESDE      0.007549455 1.000000000  0.545012839  0.764864578
## IEDE     -0.004652426 0.545012839  1.000000000  0.493212316
## spread   -0.001487646 0.764864578  0.493212316  1.000000000
##
## , , 2022-04-12
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.07390068 -0.1247930 -0.04391478
## ESDE     -0.07390068  1.00000000  0.5889661  0.77061565
## IEDE     -0.12479296  0.58896614  1.0000000  0.49954181
## spread   -0.04391478  0.77061565  0.4995418  1.00000000
##
## , , 2022-04-13
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.06477808 -0.1136542 -0.03459661
## ESDE     -0.06477808  1.00000000  0.5800680  0.77148606
## IEDE     -0.11365420  0.58006801  1.0000000  0.48666871
## spread   -0.03459661  0.77148606  0.4866687  1.00000000
##
## , , 2022-04-14
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.06179641 -0.09892172 -0.02293742
```

```
## ESDE     -0.06179641  1.00000000  0.57082292  0.76391969
## IEDE     -0.09892172  0.57082292  1.00000000  0.48294035
## spread   -0.02293742  0.76391969  0.48294035  1.00000000
##
## , , 2022-04-19
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.05951837 -0.06929959 -0.05306816
## ESDE     -0.05951837  1.00000000  0.55643910  0.75783538
## IEDE     -0.06929959  0.55643910  1.00000000  0.44163808
## spread   -0.05306816  0.75783538  0.44163808  1.00000000
##
## , , 2022-04-20
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 -0.0431487 -0.06487743 -0.04393307
## ESDE     -0.04314870  1.0000000  0.54615530  0.76312481
## IEDE     -0.06487743  0.5461553  1.00000000  0.44035377
## spread   -0.04393307  0.7631248  0.44035377  1.00000000
##
## , , 2022-04-21
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04121468 -0.05733974 -0.04422271
## ESDE     -0.04121468  1.00000000  0.54148280  0.76860905
## IEDE     -0.05733974  0.54148280  1.00000000  0.44123959
## spread   -0.04422271  0.76860905  0.44123959  1.00000000
##
## , , 2022-04-22
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03502744 -0.04998294 -0.04007313
## ESDE     -0.03502744  1.00000000  0.52774156  0.77357997
## IEDE     -0.04998294  0.52774156  1.00000000  0.43635243
## spread   -0.04007313  0.77357997  0.43635243  1.00000000
##
## , , 2022-04-25
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.01964701 -0.002565117 0.08829386
## ESDE      0.019647012 1.00000000  0.528887911 0.77260813
## IEDE     -0.002565117 0.52888791  1.000000000 0.44232079
## spread    0.088293864 0.77260813  0.442320794 1.00000000
##
## , , 2022-04-26
##
##            eurois3m       ESDE      IEDE    spread
## eurois3m 1.00000000 0.07157463 0.1021877 0.1458213
## ESDE     0.07157463 1.00000000 0.5425835 0.7832542
## IEDE     0.10218769 0.54258349 1.0000000 0.4777996
## spread   0.14582134 0.78325416 0.4777996 1.0000000
##
```

```
## , , 2022-04-27
##
##            eurois3m       ESDE      IEDE    spread
## eurois3m 1.00000000 0.06550374 0.0992361 0.1311304
## ESDE     0.06550374 1.00000000 0.5245026 0.7864920
## IEDE     0.09923610 0.52450262 1.0000000 0.4657671
## spread   0.13113039 0.78649198 0.4657671 1.0000000
##
## , , 2022-04-28
##
##            eurois3m       ESDE       IEDE    spread
## eurois3m 1.00000000 0.06111424 0.09034099 0.1134320
## ESDE     0.06111424 1.00000000 0.52073683 0.7831587
## IEDE     0.09034099 0.52073683 1.00000000 0.4655785
## spread   0.11343196 0.78315872 0.46557851 1.0000000
##
## , , 2022-04-29
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.04244213 0.0819774 0.07616843
## ESDE     0.04244213 1.00000000 0.5114290 0.79528864
## IEDE     0.08197740 0.51142900 1.0000000 0.44669975
## spread   0.07616843 0.79528864 0.4466998 1.00000000
##
## , , 2022-05-02
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03945297 0.07564005 0.06812135
## ESDE     0.03945297 1.00000000 0.51784315 0.78900628
## IEDE     0.07564005 0.51784315 1.00000000 0.45056823
## spread   0.06812135 0.78900628 0.45056823 1.00000000
##
## , , 2022-05-03
##
##            eurois3m       ESDE      IEDE    spread
## eurois3m 1.00000000 0.06567607 0.1238801 0.1036859
## ESDE     0.06567607 1.00000000 0.5329037 0.7966495
## IEDE     0.12388013 0.53290366 1.0000000 0.4909722
## spread   0.10368592 0.79664950 0.4909722 1.0000000
##
## , , 2022-05-04
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.06259446 0.1166201 0.09226039
## ESDE     0.06259446 1.00000000 0.5309473 0.79706679
## IEDE     0.11662010 0.53094732 1.0000000 0.48428205
## spread   0.09226039 0.79706679 0.4842820 1.00000000
##
## , , 2022-05-05
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01657785 0.09350078 0.02739876
```

```
## ESDE     0.01657785 1.00000000 0.53024848 0.82469369
## IEDE     0.09350078 0.53024848 1.00000000 0.47940272
## spread   0.02739876 0.82469369 0.47940272 1.00000000
##
## , , 2022-05-06
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.01268095 0.1090251 0.02045018
## ESDE     0.01268095 1.00000000 0.5174708 0.82395411
## IEDE     0.10902505 0.51747079 1.0000000 0.46797801
## spread   0.02045018 0.82395411 0.4679780 1.00000000
##
## , , 2022-05-09
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 -0.00524785 0.05861634 0.001181212
## ESDE     -0.005247850  1.00000000 0.51975800 0.824154168
## IEDE      0.058616344  0.51975800 1.00000000 0.470363653
## spread    0.001181212  0.82415417 0.47036365 1.000000000
##
## , , 2022-05-10
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02100942 0.05744385 0.02790408
## ESDE     0.02100942 1.00000000 0.51460426 0.82799495
## IEDE     0.05744385 0.51460426 1.00000000 0.46489239
## spread   0.02790408 0.82799495 0.46489239 1.00000000
##
## , , 2022-05-11
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02370375 0.05535292 0.02626967
## ESDE     0.02370375 1.00000000 0.51846069 0.82717899
## IEDE     0.05535292 0.51846069 1.00000000 0.46741094
## spread   0.02626967 0.82717899 0.46741094 1.00000000
##
## , , 2022-05-12
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06369942 0.06264516 0.05860388
## ESDE     0.06369942 1.00000000 0.51314107 0.85280014
## IEDE     0.06264516 0.51314107 1.00000000 0.47277242
## spread   0.05860388 0.85280014 0.47277242 1.00000000
##
## , , 2022-05-13
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05081373 0.02870102 0.03486238
## ESDE     0.05081373 1.00000000 0.51275512 0.84807874
## IEDE     0.02870102 0.51275512 1.00000000 0.49869389
## spread   0.03486238 0.84807874 0.49869389 1.00000000
##
```

```
## , , 2022-05-16
##
##            eurois3m       ESDE       IEDE    spread
## eurois3m 1.00000000 0.06686287 0.03315139 0.0510102
## ESDE     0.06686287 1.00000000 0.51125843 0.8497480
## IEDE     0.03315139 0.51125843 1.00000000 0.4943245
## spread   0.05101020 0.84974801 0.49432452 1.0000000
##
## , , 2022-05-17
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04658876 0.08660599 0.03420364
## ESDE     0.04658876 1.00000000 0.45467068 0.84730979
## IEDE     0.08660599 0.45467068 1.00000000 0.44378072
## spread   0.03420364 0.84730979 0.44378072 1.00000000
##
## , , 2022-05-18
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04528889 0.08307881 0.02821155
## ESDE     0.04528889 1.00000000 0.46041332 0.84071168
## IEDE     0.08307881 0.46041332 1.00000000 0.43795374
## spread   0.02821155 0.84071168 0.43795374 1.00000000
##
## , , 2022-05-19
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.03776027 0.0560301 0.02591498
## ESDE     0.03776027 1.00000000 0.4632778 0.83711942
## IEDE     0.05603010 0.46327784 1.0000000 0.42982948
## spread   0.02591498 0.83711942 0.4298295 1.00000000
##
## , , 2022-05-20
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06921594 0.04934154 0.04278974
## ESDE     0.06921594 1.00000000 0.46055500 0.83579312
## IEDE     0.04934154 0.46055500 1.00000000 0.42961845
## spread   0.04278974 0.83579312 0.42961845 1.00000000
##
## , , 2022-05-23
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.08700173 0.04189296 0.06459029
## ESDE     0.08700173 1.00000000 0.42507238 0.85152713
## IEDE     0.04189296 0.42507238 1.00000000 0.38786260
## spread   0.06459029 0.85152713 0.38786260 1.00000000
##
## , , 2022-05-24
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.08331376 0.03766194 0.06011676
```

```
## ESDE     0.08331376 1.00000000 0.42347204 0.84934039
## IEDE     0.03766194 0.42347204 1.00000000 0.38811179
## spread   0.06011676 0.84934039 0.38811179 1.00000000
##
## , , 2022-05-25
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06451917 0.02670019 0.05398061
## ESDE     0.06451917 1.00000000 0.43422245 0.84188958
## IEDE     0.02670019 0.43422245 1.00000000 0.39257280
## spread   0.05398061 0.84188958 0.39257280 1.00000000
##
## , , 2022-05-26
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.0235072 0.0074893 0.02706322
## ESDE     0.02350720 1.0000000 0.4514660 0.84012048
## IEDE     0.00748930 0.4514660 1.0000000 0.40658496
## spread   0.02706322 0.8401205 0.4065850 1.00000000
##
## , , 2022-05-27
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03809547 0.02069821 0.03968387
## ESDE     0.03809547 1.00000000 0.47444382 0.84565529
## IEDE     0.02069821 0.47444382 1.00000000 0.43207422
## spread   0.03968387 0.84565529 0.43207422 1.00000000
##
## , , 2022-05-30
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m 1.000000000 0.0184441 0.02724594 0.003121299
## ESDE     0.018444100 1.0000000 0.47372533 0.842726939
## IEDE     0.027245940 0.4737253 1.00000000 0.427457057
## spread   0.003121299 0.8427269 0.42745706 1.000000000
##
## , , 2022-05-31
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.01752864 0.03541555 0.002550272
## ESDE     0.017528642 1.00000000 0.45114563 0.839645815
## IEDE     0.035415546 0.45114563 1.00000000 0.416351459
## spread   0.002550272 0.83964581 0.41635146 1.000000000
##
## , , 2022-06-01
##
##            eurois3m       ESDE       IEDE    spread
## eurois3m 1.00000000 0.02926809 0.04145738 0.0146483
## ESDE     0.02926809 1.00000000 0.47598788 0.8475673
## IEDE     0.04145738 0.47598788 1.00000000 0.4424529
## spread   0.01464830 0.84756732 0.44245292 1.0000000
##
```

```
## , , 2022-06-02
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.02787713 0.02403809 0.008840339
## ESDE     0.027877126 1.00000000 0.46747939 0.843405704
## IEDE     0.024038093 0.46747939 1.00000000 0.443518943
## spread   0.008840339 0.84340570 0.44351894 1.000000000
##
## , , 2022-06-03
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0361093 0.02396125 0.01774126
## ESDE     0.03610930 1.0000000 0.46303447 0.84752318
## IEDE     0.02396125 0.4630345 1.00000000 0.43609441
## spread   0.01774126 0.8475232 0.43609441 1.00000000
##
## , , 2022-06-06
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.02683913 0.02071816 0.007709267
## ESDE     0.026839126 1.00000000 0.47684032 0.856467105
## IEDE     0.020718164 0.47684032 1.00000000 0.448619627
## spread   0.007709267 0.85646710 0.44861963 1.000000000
##
## , , 2022-06-07
##
##              eurois3m       ESDE        IEDE       spread
## eurois3m  1.000000000 0.01892564 -0.01515357 -0.003941604
## ESDE      0.018925640 1.00000000  0.46906667  0.853307537
## IEDE     -0.015153568 0.46906667  1.00000000  0.455127165
## spread   -0.003941604 0.85330754  0.45512716  1.000000000
##
## , , 2022-06-08
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.03639471 0.01032184 -0.05626993
## ESDE     -0.03639471  1.00000000 0.40510801  0.86390303
## IEDE      0.01032184  0.40510801 1.00000000  0.39164215
## spread   -0.05626993  0.86390303 0.39164215  1.00000000
##
## , , 2022-06-09
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.05260849 -0.005664636 -0.07169395
## ESDE     -0.052608495  1.00000000  0.418542494  0.86116586
## IEDE     -0.005664636  0.41854249  1.000000000  0.40197492
## spread   -0.071693948  0.86116586  0.401974917  1.00000000
##
## , , 2022-06-10
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1892680 0.1868581 0.1781883
```

```
## ESDE     0.1892680 1.0000000 0.4764927 0.8719660
## IEDE     0.1868581 0.4764927 1.0000000 0.4592785
## spread   0.1781883 0.8719660 0.4592785 1.0000000
##
## , , 2022-06-13
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1356464 0.1452364 0.1476411
## ESDE     0.1356464 1.0000000 0.5647167 0.8608824
## IEDE     0.1452364 0.5647167 1.0000000 0.5058081
## spread   0.1476411 0.8608824 0.5058081 1.0000000
##
## , , 2022-06-14
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1505363 0.1472070 0.1530905
## ESDE     0.1505363 1.0000000 0.5682292 0.8619912
## IEDE     0.1472070 0.5682292 1.0000000 0.5121400
## spread   0.1530905 0.8619912 0.5121400 1.0000000
##
## , , 2022-06-15
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1400145 0.1364408 0.1372602
## ESDE     0.1400145 1.0000000 0.5623479 0.8579294
## IEDE     0.1364408 0.5623479 1.0000000 0.5066459
## spread   0.1372602 0.8579294 0.5066459 1.0000000
##
## , , 2022-06-16
##
##              eurois3m        ESDE      IEDE       spread
## eurois3m  1.000000000 0.003644641 0.0584233 -0.009390478
## ESDE      0.003644641 1.000000000 0.5798286  0.874937849
## IEDE      0.058423299 0.579828558 1.0000000  0.529452773
## spread   -0.009390478 0.874937849 0.5294528  1.000000000
##
## , , 2022-06-17
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.001483289 0.05574818 -0.01199242
## ESDE      0.001483289 1.000000000 0.56421733  0.87245049
## IEDE      0.055748180 0.564217333 1.00000000  0.51534625
## spread   -0.011992424 0.872450487 0.51534625  1.00000000
##
## , , 2022-06-20
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 -0.0130482 0.04860536 -0.02464258
## ESDE     -0.01304820  1.0000000 0.55386970  0.87590421
## IEDE      0.04860536  0.5538697 1.00000000  0.50979965
## spread   -0.02464258  0.8759042 0.50979965  1.00000000
##
```

```
## , , 2022-06-21
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 -0.0099939 0.0364561 -0.01829549
## ESDE     -0.00999390  1.0000000 0.5474091  0.87094472
## IEDE      0.03645610  0.5474091 1.0000000  0.50163372
## spread   -0.01829549  0.8709447 0.5016337  1.00000000
##
## , , 2022-06-22
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.009927383 0.02631309 -0.01802203
## ESDE     -0.009927383  1.000000000 0.54184420  0.86628190
## IEDE      0.026313085  0.541844195 1.00000000  0.49411248
## spread   -0.018022032  0.866281900 0.49411248  1.00000000
##
## , , 2022-06-23
##
##              eurois3m         ESDE       IEDE     spread
## eurois3m  1.000000000 -0.004420767 0.03009987 -0.0108440
## ESDE     -0.004420767  1.000000000 0.53787426  0.8619242
## IEDE      0.030099875  0.537874262 1.00000000  0.4908177
## spread   -0.010844005  0.861924222 0.49081773  1.0000000
##
## , , 2022-06-24
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.008932164 0.04232154 -0.02218791
## ESDE     -0.008932164  1.000000000 0.52067238  0.85761371
## IEDE      0.042321540  0.520672381 1.00000000  0.46541760
## spread   -0.022187912  0.857613715 0.46541760  1.00000000
##
## , , 2022-06-27
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.02939966 0.009698009 -0.04133262
## ESDE     -0.029399661  1.00000000 0.523120329  0.85492982
## IEDE      0.009698009  0.52312033 1.000000000  0.46890241
## spread   -0.041332624  0.85492982 0.468902414  1.00000000
##
## , , 2022-06-28
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02286376 0.08670785 -0.03073285
## ESDE     -0.02286376  1.00000000 0.47510009  0.85104574
## IEDE      0.08670785  0.47510009 1.00000000  0.43792676
## spread   -0.03073285  0.85104574 0.43792676  1.00000000
##
## , , 2022-06-29
##
##              eurois3m         ESDE       IEDE       spread
## eurois3m  1.000000000 -0.009741292 0.07549987 -0.008440695
```

```
## ESDE     -0.009741292  1.000000000 0.47449444  0.846963537
## IEDE      0.075499867  0.474494436 1.00000000  0.432181612
## spread   -0.008440695  0.846963537 0.43218161  1.000000000
##
## , , 2022-06-30
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.00127297 0.06567485 0.001904241
## ESDE     0.001272970 1.00000000 0.46578025 0.846235024
## IEDE     0.065674846 0.46578025 1.00000000 0.420938880
## spread   0.001904241 0.84623502 0.42093888 1.000000000
##
## , , 2022-07-01
##
##              eurois3m         ESDE       IEDE       spread
## eurois3m  1.000000000 -0.004348462 0.04336904 -0.009433044
## ESDE     -0.004348462  1.000000000 0.48234793  0.845432478
## IEDE      0.043369039  0.482347934 1.00000000  0.453023575
## spread   -0.009433044  0.845432478 0.45302358  1.000000000
##
## , , 2022-07-04
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.03205676 0.007292154 -0.03030018
## ESDE     -0.032056763  1.00000000 0.536231865  0.84983987
## IEDE      0.007292154  0.53623186 1.000000000  0.48986222
## spread   -0.030300182  0.84983987 0.489862218  1.00000000
##
## , , 2022-07-05
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02458833 0.00964044 -0.02650471
## ESDE     -0.02458833  1.00000000 0.54121227  0.84736603
## IEDE      0.00964044  0.54121227 1.00000000  0.49285421
## spread   -0.02650471  0.84736603 0.49285421  1.00000000
##
## , , 2022-07-06
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03099914 0.03375877 0.04282696
## ESDE     0.03099914 1.00000000 0.54144954 0.84925476
## IEDE     0.03375877 0.54144954 1.00000000 0.49301554
## spread   0.04282696 0.84925476 0.49301554 1.00000000
##
## , , 2022-07-07
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02731292 0.03475971 0.03836302
## ESDE     0.02731292 1.00000000 0.53652553 0.84576839
## IEDE     0.03475971 0.53652553 1.00000000 0.48894046
## spread   0.03836302 0.84576839 0.48894046 1.00000000
##
```

```
## , , 2022-07-08
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m 1.000000000 0.000206631 0.05343942 0.01402016
## ESDE     0.000206631 1.000000000 0.51470595 0.84409599
## IEDE     0.053439424 0.514705949 1.00000000 0.47216300
## spread   0.014020165 0.844095990 0.47216300 1.00000000
##
## , , 2022-07-11
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01930797 0.03674973 -0.01240264
## ESDE     -0.01930797  1.00000000 0.51561897  0.84323368
## IEDE      0.03674973  0.51561897 1.00000000  0.47309322
## spread   -0.01240264  0.84323368 0.47309322  1.00000000
##
## , , 2022-07-12
##
##               eurois3m         ESDE       IEDE       spread
## eurois3m  1.0000000000 -0.001605716 0.03447082 0.0001633598
## ESDE     -0.0016057158  1.000000000 0.50691105 0.8435533192
## IEDE      0.0344708237  0.506911047 1.00000000 0.4661061322
## spread    0.0001633598  0.843553319 0.46610613 1.0000000000
##
## , , 2022-07-13
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m 1.000000000 0.001186864 0.03203259 0.005078203
## ESDE     0.001186864 1.000000000 0.50578514 0.840112069
## IEDE     0.032032592 0.505785145 1.00000000 0.463604953
## spread   0.005078203 0.840112069 0.46360495 1.000000000
##
## , , 2022-07-14
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.04053336 -0.0247848 -0.04276387
## ESDE     -0.04053336  1.00000000  0.5195841  0.84183988
## IEDE     -0.02478480  0.51958409  1.0000000  0.47971101
## spread   -0.04276387  0.84183988  0.4797110  1.00000000
##
## , , 2022-07-15
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 -0.0395019 -0.02592471 -0.04371902
## ESDE     -0.03950190  1.0000000  0.57722637  0.86131928
## IEDE     -0.02592471  0.5772264  1.00000000  0.55167652
## spread   -0.04371902  0.8613193  0.55167652  1.00000000
##
## , , 2022-07-18
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 -0.0747592 -0.06979583 -0.04187422
```

```
## ESDE     -0.07475920  1.0000000  0.57525360  0.85348035
## IEDE     -0.06979583  0.5752536  1.00000000  0.53963492
## spread   -0.04187422  0.8534803  0.53963492  1.00000000
##
## , , 2022-07-19
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.06638286 -0.06152514 -0.03846361
## ESDE     -0.06638286  1.00000000  0.57169061  0.84885868
## IEDE     -0.06152514  0.57169061  1.00000000  0.53929443
## spread   -0.03846361  0.84885868  0.53929443  1.00000000
##
## , , 2022-07-20
##
##               eurois3m        ESDE         IEDE      spread
## eurois3m  1.0000000000 -0.00278024 0.0003244001 -0.01505475
## ESDE     -0.0027802397  1.00000000 0.5765549233  0.84226932
## IEDE      0.0003244001  0.57655492 1.0000000000  0.53251717
## spread   -0.0150547510  0.84226932 0.5325171690  1.00000000
##
## , , 2022-07-21
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01525384 0.01516178 0.01466286
## ESDE     0.01525384 1.00000000 0.57550381 0.84046123
## IEDE     0.01516178 0.57550381 1.00000000 0.53321271
## spread   0.01466286 0.84046123 0.53321271 1.00000000
##
## , , 2022-07-22
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.03372799 -0.06683841 -0.1024847
## ESDE     -0.03372799  1.00000000  0.58632068  0.8233521
## IEDE     -0.06683841  0.58632068  1.00000000  0.5805703
## spread   -0.10248468  0.82335206  0.58057026  1.0000000
##
## , , 2022-07-25
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03968451 -0.05644458 -0.09275144
## ESDE     -0.03968451  1.00000000  0.55313839  0.79639773
## IEDE     -0.05644458  0.55313839  1.00000000  0.56953686
## spread   -0.09275144  0.79639773  0.56953686  1.00000000
##
## , , 2022-07-26
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03818392 -0.04399045 -0.08369325
## ESDE     -0.03818392  1.00000000  0.54477104  0.79707205
## IEDE     -0.04399045  0.54477104  1.00000000  0.55837062
## spread   -0.08369325  0.79707205  0.55837062  1.00000000
##
```

```
## , , 2022-07-27
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03017109 -0.02897658 -0.07666951
## ESDE     -0.03017109  1.00000000  0.68098099  0.74748942
## IEDE     -0.02897658  0.68098099  1.00000000  0.46710910
## spread   -0.07666951  0.74748942  0.46710910  1.00000000
##
## , , 2022-07-28
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02619833 -0.0258508 -0.07199093
## ESDE     -0.02619833  1.00000000  0.6677822  0.74459663
## IEDE     -0.02585080  0.66778220  1.0000000  0.46231271
## spread   -0.07199093  0.74459663  0.4623127  1.00000000
##
## , , 2022-07-29
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04441921 -0.02560821 -0.07554587
## ESDE     -0.04441921  1.00000000  0.64937315  0.75103298
## IEDE     -0.02560821  0.64937315  1.00000000  0.46085395
## spread   -0.07554587  0.75103298  0.46085395  1.00000000
##
## , , 2022-08-01
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03204611 -0.02183915 -0.06324401
## ESDE     -0.03204611  1.00000000  0.62418528  0.76519086
## IEDE     -0.02183915  0.62418528  1.00000000  0.45378582
## spread   -0.06324401  0.76519086  0.45378582  1.00000000
##
## , , 2022-08-02
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02900521 -0.0176855 -0.05883263
## ESDE     -0.02900521  1.00000000  0.6036391  0.74787203
## IEDE     -0.01768550  0.60363911  1.0000000  0.39599655
## spread   -0.05883263  0.74787203  0.3959965  1.00000000
##
## , , 2022-08-03
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02227442 -0.01463752 -0.05371114
## ESDE     -0.02227442  1.00000000  0.53125853  0.72639650
## IEDE     -0.01463752  0.53125853  1.00000000  0.39408427
## spread   -0.05371114  0.72639650  0.39408427  1.00000000
##
## , , 2022-08-04
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 -0.0039850 -0.001941261 -0.03150904
```

```
## ESDE     -0.003985000  1.0000000  0.536965944  0.74272385
## IEDE     -0.001941261  0.5369659  1.000000000  0.41079796
## spread   -0.031509036  0.7427238  0.410797963  1.00000000
##
## , , 2022-08-05
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.002991406 0.01704597 -0.02398998
## ESDE      0.002991406 1.000000000 0.53386126  0.74941174
## IEDE      0.017045967 0.533861259 1.00000000  0.41536972
## spread   -0.023989983 0.749411741 0.41536972  1.00000000
##
## , , 2022-08-08
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01539602 0.03458772 -0.001451704
## ESDE      0.015396018 1.00000000 0.53761386  0.757291340
## IEDE      0.034587718 0.53761386 1.00000000  0.435816025
## spread   -0.001451704 0.75729134 0.43581603  1.000000000
##
## , , 2022-08-09
##
##               eurois3m       ESDE       IEDE        spread
## eurois3m  1.000000e+00 0.01697834 0.03435896 -4.903073e-05
## ESDE      1.697834e-02 1.00000000 0.54172867  7.672814e-01
## IEDE      3.435896e-02 0.54172867 1.00000000  4.479018e-01
## spread   -4.903073e-05 0.76728139 0.44790177  1.000000e+00
##
## , , 2022-08-10
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01785087 0.03425784 -0.001907324
## ESDE      0.017850870 1.00000000 0.53796840  0.771193024
## IEDE      0.034257842 0.53796840 1.00000000  0.447312570
## spread   -0.001907324 0.77119302 0.44731257  1.000000000
##
## , , 2022-08-11
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02950435 0.03021053 0.02293421
## ESDE     0.02950435 1.00000000 0.53103439 0.77548242
## IEDE     0.03021053 0.53103439 1.00000000 0.43800716
## spread   0.02293421 0.77548242 0.43800716 1.00000000
##
## , , 2022-08-12
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.03269372 0.0301686 0.02729043
## ESDE     0.03269372 1.00000000 0.5277745 0.77986237
## IEDE     0.03016860 0.52777447 1.0000000 0.43890547
## spread   0.02729043 0.77986237 0.4389055 1.00000000
##
```

```
## , , 2022-08-15
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03928699 0.03126149 0.02923666
## ESDE     0.03928699 1.00000000 0.52446724 0.78592264
## IEDE     0.03126149 0.52446724 1.00000000 0.44166320
## spread   0.02923666 0.78592264 0.44166320 1.00000000
##
## , , 2022-08-16
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03109269 0.02949969 0.01543625
## ESDE     0.03109269 1.00000000 0.52142119 0.78913753
## IEDE     0.02949969 0.52142119 1.00000000 0.44171956
## spread   0.01543625 0.78913753 0.44171956 1.00000000
##
## , , 2022-08-17
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.01169986 0.01472844 -0.0643922
## ESDE     -0.01169986  1.00000000 0.51963766  0.7931412
## IEDE      0.01472844  0.51963766 1.00000000  0.4365606
## spread   -0.06439220  0.79314115 0.43656063  1.0000000
##
## , , 2022-08-18
##
##             eurois3m      ESDE      IEDE      spread
## eurois3m  1.00000000 0.1334787 0.3100463 -0.04182109
## ESDE      0.13347871 1.0000000 0.5791897  0.75931352
## IEDE      0.31004628 0.5791897 1.0000000  0.32999202
## spread   -0.04182109 0.7593135 0.3299920  1.00000000
##
## , , 2022-08-19
##
##             eurois3m      ESDE      IEDE      spread
## eurois3m  1.00000000 0.1089048 0.2843196 -0.04234881
## ESDE      0.10890483 1.0000000 0.5702106  0.76439868
## IEDE      0.28431960 0.5702106 1.0000000  0.33796400
## spread   -0.04234881 0.7643987 0.3379640  1.00000000
##
## , , 2022-08-22
##
##             eurois3m      ESDE      IEDE      spread
## eurois3m  1.00000000 0.1017703 0.2634089 -0.03516803
## ESDE      0.10177033 1.0000000 0.5612704  0.77051479
## IEDE      0.26340886 0.5612704 1.0000000  0.34232058
## spread   -0.03516803 0.7705148 0.3423206  1.00000000
##
## , , 2022-08-23
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.09004193 0.2430500 -0.03581571
```

```
## ESDE      0.09004193 1.00000000 0.5537417  0.77714191
## IEDE      0.24305003 0.55374171 1.0000000  0.34949725
## spread   -0.03581571 0.77714191 0.3494972  1.00000000
##
## , , 2022-08-24
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.07917938 0.2233220 -0.03648641
## ESDE      0.07917938 1.00000000 0.5468145  0.78468176
## IEDE      0.22332196 0.54681450 1.0000000  0.35509645
## spread   -0.03648641 0.78468176 0.3550964  1.00000000
##
## , , 2022-08-25
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.07202247 0.2028784 -0.02891498
## ESDE      0.07202247 1.00000000 0.5417090  0.78385314
## IEDE      0.20287839 0.54170901 1.0000000  0.35777198
## spread   -0.02891498 0.78385314 0.3577720  1.00000000
##
## , , 2022-08-26
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.02996095 0.1974540 -0.08733266
## ESDE      0.02996095 1.00000000 0.5083009  0.79048047
## IEDE      0.19745396 0.50830088 1.0000000  0.30939519
## spread   -0.08733266 0.79048047 0.3093952  1.00000000
##
## , , 2022-08-29
##
##             eurois3m        ESDE      IEDE     spread
## eurois3m  1.00000000 -0.09036705 0.1261666 -0.1917837
## ESDE     -0.09036705  1.00000000 0.5114584  0.8117872
## IEDE      0.12616657  0.51145840 1.0000000  0.3353248
## spread   -0.19178367  0.81178724 0.3353248  1.0000000
##
## , , 2022-08-30
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.04092029 0.06671296 -0.1515262
## ESDE     -0.04092029  1.00000000 0.46961845  0.8118771
## IEDE      0.06671296  0.46961845 1.00000000  0.3255756
## spread   -0.15152622  0.81187714 0.32557561  1.0000000
##
## , , 2022-08-31
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.03485009 0.06037386 -0.1404333
## ESDE     -0.03485009  1.00000000 0.47077590  0.8104140
## IEDE      0.06037386  0.47077590 1.00000000  0.3380600
## spread   -0.14043326  0.81041402 0.33806002  1.0000000
##
```

```
## , , 2022-09-01
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.03827282 0.06333835 -0.1407279
## ESDE     -0.03827282  1.00000000 0.44609739  0.7950852
## IEDE      0.06333835  0.44609739 1.00000000  0.2686400
## spread   -0.14072791  0.79508517 0.26864001  1.0000000
##
## , , 2022-09-02
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.03314626 0.05949197 -0.1290116
## ESDE     -0.03314626  1.00000000 0.44827954  0.7962095
## IEDE      0.05949197  0.44827954 1.00000000  0.2787322
## spread   -0.12901163  0.79620954 0.27873220  1.0000000
##
## , , 2022-09-05
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.02487452 0.04966474 -0.1370959
## ESDE     -0.02487452  1.00000000 0.45286366  0.7941264
## IEDE      0.04966474  0.45286366 1.00000000  0.2942470
## spread   -0.13709592  0.79412641 0.29424697  1.0000000
##
## , , 2022-09-06
##
##              eurois3m         ESDE       IEDE     spread
## eurois3m  1.000000000 -0.009266232 0.07647054 -0.1120558
## ESDE     -0.009266232  1.000000000 0.46611455  0.7986173
## IEDE      0.076470541  0.466114550 1.00000000  0.3274286
## spread   -0.112055752  0.798617285 0.32742862  1.0000000
##
## , , 2022-09-07
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.06753022 -0.01127959 -0.1212075
## ESDE     -0.06753022  1.00000000  0.53453556  0.7891817
## IEDE     -0.01127959  0.53453556  1.00000000  0.3536420
## spread   -0.12120746  0.78918171  0.35364200  1.0000000
##
## , , 2022-09-08
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 -0.0675765 -0.01474599 -0.1134271
## ESDE     -0.06757650  1.0000000  0.53470681  0.7910101
## IEDE     -0.01474599  0.5347068  1.00000000  0.3640382
## spread   -0.11342707  0.7910101  0.36403815  1.0000000
##
## , , 2022-09-09
##
##             eurois3m      ESDE        IEDE      spread
## eurois3m 1.000000000 0.1088506 0.004954739 0.004866564
```

```
## ESDE     0.108850592 1.0000000 0.517276579 0.796666298
## IEDE     0.004954739 0.5172766 1.000000000 0.371093164
## spread   0.004866564 0.7966663 0.371093164 1.000000000
##
## , , 2022-09-12
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1055947 0.00623055 0.01099129
## ESDE     0.10559468 1.0000000 0.51225637 0.80172685
## IEDE     0.00623055 0.5122564 1.00000000 0.37439188
## spread   0.01099129 0.8017268 0.37439188 1.00000000
##
## , , 2022-09-13
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.08871391 -0.002761394 0.004998322
## ESDE      0.088713913 1.00000000  0.514383412 0.803258081
## IEDE     -0.002761394 0.51438341  1.000000000 0.383477031
## spread    0.004998322 0.80325808  0.383477031 1.000000000
##
## , , 2022-09-14
##
##               eurois3m       ESDE          IEDE     spread
## eurois3m  1.0000000000 0.08134215 -0.0009692716 0.00353278
## ESDE      0.0813421455 1.00000000  0.5106306073 0.80601954
## IEDE     -0.0009692716 0.51063061  1.0000000000 0.38930133
## spread    0.0035327797 0.80601954  0.3893013256 1.00000000
##
## , , 2022-09-15
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m 1.000000000 0.07816692 0.000576292 0.003419608
## ESDE     0.078166923 1.00000000 0.503877511 0.804227620
## IEDE     0.000576292 0.50387751 1.000000000 0.394918324
## spread   0.003419608 0.80422762 0.394918324 1.000000000
##
## , , 2022-09-16
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m 1.0000000000 0.07252298 0.0004207909 0.001398845
## ESDE     0.0725229826 1.00000000 0.4993992860 0.803734992
## IEDE     0.0004207909 0.49939929 1.0000000000 0.402828062
## spread   0.0013988447 0.80373499 0.4028280618 1.000000000
##
## , , 2022-09-19
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m 1.0000000000 0.06782346 0.0002077231 0.002632825
## ESDE     0.0678234610 1.00000000 0.4950548270 0.804804821
## IEDE     0.0002077231 0.49505483 1.0000000000 0.399724501
## spread   0.0026328249 0.80480482 0.3997245011 1.000000000
##
```

```
## , , 2022-09-20
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m 1.000000000 0.06207656 0.001972695 0.001403058
## ESDE     0.062076564 1.00000000 0.494229761 0.808163473
## IEDE     0.001972695 0.49422976 1.000000000 0.404737749
## spread   0.001403058 0.80816347 0.404737749 1.000000000
##
## , , 2022-09-21
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m 1.000000000 0.05835622 0.007214146 0.001918625
## ESDE     0.058356215 1.00000000 0.494639820 0.809146765
## IEDE     0.007214146 0.49463982 1.000000000 0.410159836
## spread   0.001918625 0.80914677 0.410159836 1.000000000
##
## , , 2022-09-22
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.04263147 -0.02454386 -0.01428763
## ESDE      0.04263147 1.00000000  0.50059156  0.81247976
## IEDE     -0.02454386 0.50059156  1.00000000  0.43936266
## spread   -0.01428763 0.81247976  0.43936266  1.00000000
##
## , , 2022-09-23
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.07200434 -0.03525279 0.01127789
## ESDE      0.07200434 1.00000000  0.47607325 0.81628510
## IEDE     -0.03525279 0.47607325  1.00000000 0.42471391
## spread    0.01127789 0.81628510  0.42471391 1.00000000
##
## , , 2022-09-26
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.06031407 -0.03179374 0.003088263
## ESDE      0.060314066 1.00000000  0.49348364 0.826414209
## IEDE     -0.031793742 0.49348364  1.00000000 0.445785788
## spread    0.003088263 0.82641421  0.44578579 1.000000000
##
## , , 2022-09-27
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.03633824 -0.03716034 -0.02086575
## ESDE      0.03633824 1.00000000  0.53110221  0.84372722
## IEDE     -0.03716034 0.53110221  1.00000000  0.49356584
## spread   -0.02086575 0.84372722  0.49356584  1.00000000
##
## , , 2022-09-28
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.03861211 -0.02698482 -0.01124007
```

```
## ESDE      0.03861211 1.00000000  0.53463120  0.84191386
## IEDE     -0.02698482 0.53463120  1.00000000  0.49973673
## spread   -0.01124007 0.84191386  0.49973673  1.00000000
##
## , , 2022-09-29
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.03606084 -0.02223536 -0.01124802
## ESDE      0.03606084 1.00000000  0.53054149  0.83925932
## IEDE     -0.02223536 0.53054149  1.00000000  0.49417864
## spread   -0.01124802 0.83925932  0.49417864  1.00000000
##
## , , 2022-09-30
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.03732121 0.04907786 -0.01362742
## ESDE      0.03732121 1.00000000 0.52637048  0.83652184
## IEDE      0.04907786 0.52637048 1.00000000  0.48692055
## spread   -0.01362742 0.83652184 0.48692055  1.00000000
##
## , , 2022-10-03
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.03891097 0.06809355 -0.003146744
## ESDE      0.038910975 1.00000000 0.51982348  0.833741785
## IEDE      0.068093553 0.51982348 1.00000000  0.487938547
## spread   -0.003146744 0.83374179 0.48793855  1.000000000
##
## , , 2022-10-04
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.04135404 0.07163691 0.006025002
## ESDE     0.041354041 1.00000000 0.52795931 0.830774513
## IEDE     0.071636906 0.52795931 1.00000000 0.516734797
## spread   0.006025002 0.83077451 0.51673480 1.000000000
##
## , , 2022-10-05
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m 1.000000000 0.04760095 0.0759008 0.009082757
## ESDE     0.047600948 1.00000000 0.5307879 0.828625628
## IEDE     0.075900804 0.53078793 1.0000000 0.512230080
## spread   0.009082757 0.82862563 0.5122301 1.000000000
##
## , , 2022-10-06
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01644985 0.07921798 -0.03305388
## ESDE     -0.01644985  1.00000000 0.41041187  0.85023664
## IEDE      0.07921798  0.41041187 1.00000000  0.43713971
## spread   -0.03305388  0.85023664 0.43713971  1.00000000
##
```

```
## , , 2022-10-07
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.01356098 0.07367585 -0.0309011
## ESDE     -0.01356098  1.00000000 0.42077231  0.8471534
## IEDE      0.07367585  0.42077231 1.00000000  0.4373168
## spread   -0.03090110  0.84715339 0.43731676  1.0000000
##
## , , 2022-10-10
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.004562512 0.08101603 -0.01917046
## ESDE     -0.004562512  1.000000000 0.44325786  0.84683440
## IEDE      0.081016033  0.443257864 1.00000000  0.46904583
## spread   -0.019170463  0.846834396 0.46904583  1.00000000
##
## , , 2022-10-11
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01295924 0.06143224 -0.02668753
## ESDE     -0.01295924  1.00000000 0.51770524  0.86396814
## IEDE      0.06143224  0.51770524 1.00000000  0.52976555
## spread   -0.02668753  0.86396814 0.52976555  1.00000000
##
## , , 2022-10-12
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01005472 0.06143639 -0.02310578
## ESDE     -0.01005472  1.00000000 0.51456256  0.85968143
## IEDE      0.06143639  0.51456256 1.00000000  0.52419934
## spread   -0.02310578  0.85968143 0.52419934  1.00000000
##
## , , 2022-10-13
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m 1.0000000000 0.008228504 0.06436954 0.0004515259
## ESDE     0.0082285042 1.000000000 0.51575115 0.8584801132
## IEDE     0.0643695357 0.515751151 1.00000000 0.5193920511
## spread   0.0004515259 0.858480113 0.51939205 1.0000000000
##
## , , 2022-10-14
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03496604 0.04841976 0.01445327
## ESDE     0.03496604 1.00000000 0.49678936 0.85478653
## IEDE     0.04841976 0.49678936 1.00000000 0.50457741
## spread   0.01445327 0.85478653 0.50457741 1.00000000
##
## , , 2022-10-17
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.02549884 0.05200654 0.006635176
```

```
## ESDE     0.025498835 1.00000000 0.49227362 0.851653452
## IEDE     0.052006539 0.49227362 1.00000000 0.496060321
## spread   0.006635176 0.85165345 0.49606032 1.000000000
##
## , , 2022-10-18
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.00300068 0.03861545 -0.01330953
## ESDE      0.00300068 1.00000000 0.49324799  0.84910906
## IEDE      0.03861545 0.49324799 1.00000000  0.49227986
## spread   -0.01330953 0.84910906 0.49227986  1.00000000
##
## , , 2022-10-19
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.005725299 0.03733747 -0.01108132
## ESDE      0.005725299 1.000000000 0.49418976  0.84661586
## IEDE      0.037337466 0.494189763 1.00000000  0.48875280
## spread   -0.011081324 0.846615862 0.48875280  1.00000000
##
## , , 2022-10-20
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01422931 0.03985441 -0.02179916
## ESDE      0.01422931 1.00000000 0.49479337  0.84032332
## IEDE      0.03985441 0.49479337 1.00000000  0.48286082
## spread   -0.02179916 0.84032332 0.48286082  1.00000000
##
## , , 2022-10-21
##
##              eurois3m      ESDE       IEDE       spread
## eurois3m  1.000000000 0.0294320 0.06328078 -0.001485415
## ESDE      0.029431996 1.0000000 0.52738541  0.845700740
## IEDE      0.063280777 0.5273854 1.00000000  0.524549661
## spread   -0.001485415 0.8457007 0.52454966  1.000000000
##
## , , 2022-10-24
##
##              eurois3m        ESDE         IEDE     spread
## eurois3m  1.000000000 0.002901399 -0.007109453 0.01237732
## ESDE      0.002901399 1.000000000  0.530466924 0.83001250
## IEDE     -0.007109453 0.530466924  1.000000000 0.48052671
## spread    0.012377321 0.830012497  0.480526709 1.00000000
##
## , , 2022-10-25
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 0.002430427 -0.005951606 0.006884757
## ESDE      0.002430427 1.000000000  0.532779947 0.828568874
## IEDE     -0.005951606 0.532779947  1.000000000 0.484408148
## spread    0.006884757 0.828568874  0.484408148 1.000000000
##
```

```
## , , 2022-10-26
##
##               eurois3m        ESDE         IEDE       spread
## eurois3m  1.0000000000 0.001113474 -0.003760531 0.0007089793
## ESDE      0.0011134744 1.000000000  0.526342573 0.8282002755
## IEDE     -0.0037605310 0.526342573  1.000000000 0.4723748024
## spread    0.0007089793 0.828200276  0.472374802 1.0000000000
##
## , , 2022-10-27
##
##              eurois3m        ESDE         IEDE    spread
## eurois3m  1.000000000 0.001411284 -0.002877005 0.0019447
## ESDE      0.001411284 1.000000000  0.523538115 0.8256814
## IEDE     -0.002877005 0.523538115  1.000000000 0.4691424
## spread    0.001944700 0.825681431  0.469142380 1.0000000
##
## , , 2022-10-28
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03468809 0.01600676 0.03450985
## ESDE     0.03468809 1.00000000 0.55217446 0.85602269
## IEDE     0.01600676 0.55217446 1.00000000 0.50468435
## spread   0.03450985 0.85602269 0.50468435 1.00000000
##
## , , 2022-10-31
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02953288 0.01469277 0.03021493
## ESDE     0.02953288 1.00000000 0.54786238 0.85011723
## IEDE     0.01469277 0.54786238 1.00000000 0.49943887
## spread   0.03021493 0.85011723 0.49943887 1.00000000
##
## , , 2022-11-01
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.08086009 0.03867976 0.08893942
## ESDE     0.08086009 1.00000000 0.54645292 0.85201056
## IEDE     0.03867976 0.54645292 1.00000000 0.49864953
## spread   0.08893942 0.85201056 0.49864953 1.00000000
##
## , , 2022-11-02
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07113535 0.09288594 0.06925285
## ESDE     0.07113535 1.00000000 0.49110791 0.84798408
## IEDE     0.09288594 0.49110791 1.00000000 0.42845281
## spread   0.06925285 0.84798408 0.42845281 1.00000000
##
## , , 2022-11-03
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06186119 0.08288506 0.06066618
```

```
## ESDE     0.06186119 1.00000000 0.49299381 0.84433374
## IEDE     0.08288506 0.49299381 1.00000000 0.43077368
## spread   0.06066618 0.84433374 0.43077368 1.00000000
##
## , , 2022-11-04
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07165044 0.07956019 0.05789116
## ESDE     0.07165044 1.00000000 0.48989262 0.83018871
## IEDE     0.07956019 0.48989262 1.00000000 0.43433893
## spread   0.05789116 0.83018871 0.43433893 1.00000000
##
## , , 2022-11-07
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0630117 0.06964099 0.03802216
## ESDE     0.06301170 1.0000000 0.49098234 0.82531429
## IEDE     0.06964099 0.4909823 1.00000000 0.43641183
## spread   0.03802216 0.8253143 0.43641183 1.00000000
##
## , , 2022-11-08
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05818562 0.06443375 0.03344943
## ESDE     0.05818562 1.00000000 0.49155380 0.82470624
## IEDE     0.06443375 0.49155380 1.00000000 0.43852323
## spread   0.03344943 0.82470624 0.43852323 1.00000000
##
## , , 2022-11-09
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06266274 0.07004835 0.03929191
## ESDE     0.06266274 1.00000000 0.50326636 0.82744995
## IEDE     0.07004835 0.50326636 1.00000000 0.45358719
## spread   0.03929191 0.82744995 0.45358719 1.00000000
##
## , , 2022-11-10
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05797706 0.06467336 0.03428356
## ESDE     0.05797706 1.00000000 0.50239455 0.82308318
## IEDE     0.06467336 0.50239455 1.00000000 0.45320104
## spread   0.03428356 0.82308318 0.45320104 1.00000000
##
## , , 2022-11-11
##
##              eurois3m       ESDE      IEDE       spread
## eurois3m 1.0000000000 0.03356898 0.0484227 0.0006884246
## ESDE     0.0335689781 1.00000000 0.5121664 0.8293498659
## IEDE     0.0484227041 0.51216643 1.0000000 0.4645589961
## spread   0.0006884246 0.82934987 0.4645590 1.0000000000
##
```

```
## , , 2022-11-14
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.09999464 -0.2260289 -0.0616552
## ESDE     -0.09999464  1.00000000  0.5348850  0.8273142
## IEDE     -0.22602885  0.53488498  1.0000000  0.4608845
## spread   -0.06165520  0.82731417  0.4608845  1.0000000
##
## , , 2022-11-15
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.08952728 -0.2079942 -0.05550821
## ESDE     -0.08952728  1.00000000  0.5278991  0.82599200
## IEDE     -0.20799415  0.52789911  1.0000000  0.45397084
## spread   -0.05550821  0.82599200  0.4539708  1.00000000
##
## , , 2022-11-16
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.07001878 -0.1834509 -0.03944341
## ESDE     -0.07001878  1.00000000  0.5394265  0.84114719
## IEDE     -0.18345092  0.53942653  1.0000000  0.47154346
## spread   -0.03944341  0.84114719  0.4715435  1.00000000
##
## , , 2022-11-17
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.06033839 -0.1662397 -0.03754938
## ESDE     -0.06033839  1.00000000  0.5320751  0.83252108
## IEDE     -0.16623966  0.53207514  1.0000000  0.46972038
## spread   -0.03754938  0.83252108  0.4697204  1.00000000
##
## , , 2022-11-18
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.06227282 -0.06675362 -0.04133584
## ESDE     -0.06227282  1.00000000  0.44561586  0.83127329
## IEDE     -0.06675362  0.44561586  1.00000000  0.39962991
## spread   -0.04133584  0.83127329  0.39962991  1.00000000
##
## , , 2022-11-21
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03370614 -0.06095979 -0.02162665
## ESDE     -0.03370614  1.00000000  0.42171402  0.84145174
## IEDE     -0.06095979  0.42171402  1.00000000  0.38526017
## spread   -0.02162665  0.84145174  0.38526017  1.00000000
##
## , , 2022-11-22
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04177731 -0.04407603 -0.04909574
```

```
## ESDE     -0.04177731  1.00000000  0.42067907  0.83670582
## IEDE     -0.04407603  0.42067907  1.00000000  0.37057888
## spread   -0.04909574  0.83670582  0.37057888  1.00000000
##
## , , 2022-11-23
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04746696 -0.05320483 -0.05252519
## ESDE     -0.04746696  1.00000000  0.43516860  0.83533154
## IEDE     -0.05320483  0.43516860  1.00000000  0.38257193
## spread   -0.05252519  0.83533154  0.38257193  1.00000000
##
## , , 2022-11-24
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.04309192 -0.04820263 -0.0498588
## ESDE     -0.04309192  1.00000000  0.46924976  0.8418865
## IEDE     -0.04820263  0.46924976  1.00000000  0.4221212
## spread   -0.04985880  0.84188650  0.42212117  1.0000000
##
## , , 2022-11-25
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04020926 -0.04288583 -0.04742377
## ESDE     -0.04020926  1.00000000  0.47222256  0.83937125
## IEDE     -0.04288583  0.47222256  1.00000000  0.42555394
## spread   -0.04742377  0.83937125  0.42555394  1.00000000
##
## , , 2022-11-28
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.08216511 -0.06290599 -0.07679088
## ESDE     -0.08216511  1.00000000  0.49034049  0.84218222
## IEDE     -0.06290599  0.49034049  1.00000000  0.44227265
## spread   -0.07679088  0.84218222  0.44227265  1.00000000
##
## , , 2022-11-29
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.05667741 -0.0284924 -0.0489141
## ESDE     -0.05667741  1.00000000  0.4990102  0.8414844
## IEDE     -0.02849240  0.49901015  1.0000000  0.4564035
## spread   -0.04891410  0.84148435  0.4564035  1.0000000
##
## , , 2022-11-30
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.08677406 -0.03534922 -0.1007777
## ESDE     -0.08677406  1.00000000  0.49921241  0.8403008
## IEDE     -0.03534922  0.49921241  1.00000000  0.4556529
## spread   -0.10077770  0.84030085  0.45565291  1.0000000
##
```

```
## , , 2022-12-01
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.06411458 -0.02501645 -0.0813841
## ESDE     -0.06411458  1.00000000  0.50724864  0.8439678
## IEDE     -0.02501645  0.50724864  1.00000000  0.4643564
## spread   -0.08138410  0.84396775  0.46435641  1.0000000
##
## , , 2022-12-02
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.05228866 -0.01285545 -0.06595924
## ESDE     -0.05228866  1.00000000  0.51479634  0.84164820
## IEDE     -0.01285545  0.51479634  1.00000000  0.48368578
## spread   -0.06595924  0.84164820  0.48368578  1.00000000
##
## , , 2022-12-05
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.05134606 -0.01701688 -0.06801007
## ESDE     -0.05134606  1.00000000  0.51484123  0.83964431
## IEDE     -0.01701688  0.51484123  1.00000000  0.48317940
## spread   -0.06801007  0.83964431  0.48317940  1.00000000
##
## , , 2022-12-06
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04405018 -0.01316935 -0.06112204
## ESDE     -0.04405018  1.00000000  0.51719415  0.83942910
## IEDE     -0.01316935  0.51719415  1.00000000  0.48483379
## spread   -0.06112204  0.83942910  0.48483379  1.00000000
##
## , , 2022-12-07
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.03651601 -0.00550367 -0.0556137
## ESDE     -0.03651601  1.00000000  0.51748133  0.8357652
## IEDE     -0.00550367  0.51748133  1.00000000  0.4735091
## spread   -0.05561370  0.83576525  0.47350912  1.0000000
##
## , , 2022-12-08
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.02495573 -0.007074885 0.01456205
## ESDE      0.024955729 1.00000000  0.500021499 0.84194793
## IEDE     -0.007074885 0.50002150  1.000000000 0.45301999
## spread    0.014562055 0.84194793  0.453019985 1.00000000
##
## , , 2022-12-09
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.05541028 -0.01245413 -0.04649971
```

```
## ESDE     -0.05541028  1.00000000  0.49017800  0.84510423
## IEDE     -0.01245413  0.49017800  1.00000000  0.44950238
## spread   -0.04649971  0.84510423  0.44950238  1.00000000
##
## , , 2022-12-12
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.05213122 -0.007350089 -0.04366627
## ESDE     -0.052131218  1.00000000  0.477135459  0.83865254
## IEDE     -0.007350089  0.47713546  1.000000000  0.44646912
## spread   -0.043666269  0.83865254  0.446469117  1.00000000
##
## , , 2022-12-13
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.03840972 0.005198787 -0.03074482
## ESDE     -0.038409725  1.00000000 0.480137338  0.83660291
## IEDE      0.005198787  0.48013734 1.000000000  0.44842841
## spread   -0.030744825  0.83660291 0.448428409  1.00000000
##
## , , 2022-12-14
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.03340324 0.004754983 -0.02887584
## ESDE     -0.033403242  1.00000000 0.481377763  0.83455064
## IEDE      0.004754983  0.48137776 1.000000000  0.44856572
## spread   -0.028875839  0.83455064 0.448565722  1.00000000
##
## , , 2022-12-15
##
##              eurois3m        ESDE         IEDE       spread
## eurois3m  1.000000000 -0.01924534 -0.003402861 -0.004766631
## ESDE     -0.019245336  1.00000000  0.470140100  0.829266491
## IEDE     -0.003402861  0.47014010  1.000000000  0.415623496
## spread   -0.004766631  0.82926649  0.415623496  1.000000000
##
## , , 2022-12-16
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.07282794 -0.07793379 -0.07924072
## ESDE     -0.07282794  1.00000000  0.60439284  0.86329491
## IEDE     -0.07793379  0.60439284  1.00000000  0.61213939
## spread   -0.07924072  0.86329491  0.61213939  1.00000000
##
## , , 2022-12-19
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.07520125 -0.0841322 -0.09260149
## ESDE     -0.07520125  1.00000000  0.5974603  0.85907752
## IEDE     -0.08413220  0.59746027  1.0000000  0.60371108
## spread   -0.09260149  0.85907752  0.6037111  1.00000000
##
```

```
## , , 2022-12-20
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.06857073 -0.06773252 -0.09798174
## ESDE     -0.06857073  1.00000000  0.58671718  0.85221358
## IEDE     -0.06773252  0.58671718  1.00000000  0.58203563
## spread   -0.09798174  0.85221358  0.58203563  1.00000000
##
## , , 2022-12-21
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.06661271 -0.0663222 -0.0920031
## ESDE     -0.06661271  1.00000000  0.5857152  0.8466742
## IEDE     -0.06632220  0.58571516  1.0000000  0.5713994
## spread   -0.09200310  0.84667424  0.5713994  1.0000000
##
## , , 2022-12-22
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.05931511 -0.05803588 -0.08498161
## ESDE     -0.05931511  1.00000000  0.57368283  0.84494621
## IEDE     -0.05803588  0.57368283  1.00000000  0.55708110
## spread   -0.08498161  0.84494621  0.55708110  1.00000000
##
## , , 2022-12-23
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01949971 -0.1032845 -0.06656363
## ESDE     -0.01949971  1.00000000  0.5027602  0.83878631
## IEDE     -0.10328450  0.50276018  1.0000000  0.51635543
## spread   -0.06656363  0.83878631  0.5163554  1.00000000
##
## , , 2022-12-27
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02400776 -0.0826038 -0.06846064
## ESDE     -0.02400776  1.00000000  0.4896187  0.83787312
## IEDE     -0.08260380  0.48961870  1.0000000  0.49810583
## spread   -0.06846064  0.83787312  0.4981058  1.00000000
##
## , , 2022-12-28
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04138161 -0.06267094 -0.05786116
## ESDE     -0.04138161  1.00000000  0.47544504  0.82290515
## IEDE     -0.06267094  0.47544504  1.00000000  0.49415508
## spread   -0.05786116  0.82290515  0.49415508  1.00000000
##
## , , 2022-12-29
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.009334387 -0.0530889 -0.01883941
```

```
## ESDE      0.009334387 1.000000000  0.4612617  0.82986925
## IEDE     -0.053088900 0.461261716  1.0000000  0.48093903
## spread   -0.018839405 0.829869251  0.4809390  1.00000000
##
## , , 2022-12-30
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.03693982 -0.04919704 0.04690194
## ESDE      0.03693982 1.00000000  0.45878681 0.82325474
## IEDE     -0.04919704 0.45878681  1.00000000 0.45285757
## spread    0.04690194 0.82325474  0.45285757 1.00000000
##
## , , 2023-01-02
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.00191956 -0.1987820 -0.02527746
## ESDE      0.00191956 1.00000000  0.4553320  0.82201290
## IEDE     -0.19878197 0.45533197  1.0000000  0.46244121
## spread   -0.02527746 0.82201290  0.4624412  1.00000000
##
## , , 2023-01-03
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.01489091 -0.1582368 -0.0171415
## ESDE      0.01489091 1.00000000  0.4770519  0.8217460
## IEDE     -0.15823678 0.47705189  1.0000000  0.4671593
## spread   -0.01714150 0.82174602  0.4671593  1.0000000
##
## , , 2023-01-04
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.01522018 -0.1386451 -0.0188856
## ESDE      0.01522018 1.00000000  0.4771109  0.8177782
## IEDE     -0.13864510 0.47711087  1.0000000  0.4529374
## spread   -0.01888560 0.81777819  0.4529374  1.0000000
##
## , , 2023-01-05
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.02505727 -0.1255001 0.01381211
## ESDE      0.02505727 1.00000000  0.4742890 0.80226329
## IEDE     -0.12550009 0.47428899  1.0000000 0.42093461
## spread    0.01381211 0.80226329  0.4209346 1.00000000
##
## , , 2023-01-06
##
##              eurois3m        ESDE        IEDE     spread
## eurois3m  1.000000000 0.001780186 -0.08664004 0.01195264
## ESDE      0.001780186 1.000000000  0.46288159 0.79960992
## IEDE     -0.086640035 0.462881594  1.00000000 0.42188714
## spread    0.011952642 0.799609923  0.42188714 1.00000000
##
```

```
## , , 2023-01-09
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01408094 -0.09606741 -0.01573664
## ESDE     -0.01408094  1.00000000  0.47375921  0.80344125
## IEDE     -0.09606741  0.47375921  1.00000000  0.43659826
## spread   -0.01573664  0.80344125  0.43659826  1.00000000
##
## , , 2023-01-10
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 0.001172189 -0.08959303 -0.009808359
## ESDE      0.001172189 1.000000000  0.46614457  0.803808042
## IEDE     -0.089593033 0.466144569  1.00000000  0.434372188
## spread   -0.009808359 0.803808042  0.43437219  1.000000000
##
## , , 2023-01-11
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 -0.0124051 -0.08877764 -0.02491503
## ESDE     -0.01240510  1.0000000  0.50788918  0.82349878
## IEDE     -0.08877764  0.5078892  1.00000000  0.47896152
## spread   -0.02491503  0.8234988  0.47896152  1.00000000
##
## , , 2023-01-12
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01855063 -0.09109313 -0.03468746
## ESDE     -0.01855063  1.00000000  0.52152150  0.82670763
## IEDE     -0.09109313  0.52152150  1.00000000  0.49957683
## spread   -0.03468746  0.82670763  0.49957683  1.00000000
##
## , , 2023-01-13
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01505191 -0.06505891 -0.03290746
## ESDE     -0.01505191  1.00000000  0.51694007  0.82557468
## IEDE     -0.06505891  0.51694007  1.00000000  0.49233764
## spread   -0.03290746  0.82557468  0.49233764  1.00000000
##
## , , 2023-01-16
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01078585 -0.05461809 -0.03194369
## ESDE     -0.01078585  1.00000000  0.51575889  0.82138251
## IEDE     -0.05461809  0.51575889  1.00000000  0.48350777
## spread   -0.03194369  0.82138251  0.48350777  1.00000000
##
## , , 2023-01-17
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.004616358 -0.0486514 -0.03046714
```

```
## ESDE     -0.004616358  1.000000000  0.5082548  0.81554766
## IEDE     -0.048651397  0.508254789  1.0000000  0.48059995
## spread   -0.030467142  0.815547658  0.4806000  1.00000000
##
## , , 2023-01-18
##
##               eurois3m         ESDE        IEDE      spread
## eurois3m  1.0000000000 0.0005576465 -0.02695665 -0.02139465
## ESDE      0.0005576465 1.0000000000  0.50381735  0.81496694
## IEDE     -0.0269566517 0.5038173531  1.00000000  0.49293721
## spread   -0.0213946514 0.8149669405  0.49293721  1.00000000
##
## , , 2023-01-19
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.03771377 -0.004659514 0.01671293
## ESDE      0.037713772 1.00000000  0.530281667 0.84524743
## IEDE     -0.004659514 0.53028167  1.000000000 0.51758946
## spread    0.016712935 0.84524743  0.517589464 1.00000000
##
## , , 2023-01-20
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.03554852 -0.002059659 0.01460857
## ESDE      0.035548519 1.00000000  0.526979812 0.84077438
## IEDE     -0.002059659 0.52697981  1.000000000 0.51307193
## spread    0.014608569 0.84077438  0.513071926 1.00000000
##
## , , 2023-01-23
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05134944 0.02518944 0.03708198
## ESDE     0.05134944 1.00000000 0.54328204 0.84361051
## IEDE     0.02518944 0.54328204 1.00000000 0.53562353
## spread   0.03708198 0.84361051 0.53562353 1.00000000
##
## , , 2023-01-24
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05319362 0.03039775 0.03225566
## ESDE     0.05319362 1.00000000 0.53997256 0.83980896
## IEDE     0.03039775 0.53997256 1.00000000 0.52673786
## spread   0.03225566 0.83980896 0.52673786 1.00000000
##
## , , 2023-01-25
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 -0.1631842 -0.1265865 -0.08844605
## ESDE     -0.16318418  1.0000000  0.6077187  0.83661515
## IEDE     -0.12658648  0.6077187  1.0000000  0.56130886
## spread   -0.08844605  0.8366151  0.5613089  1.00000000
##
```

```
## , , 2023-01-26
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m  1.0000000 -0.1888459 -0.1202229 -0.1564748
## ESDE     -0.1888459  1.0000000  0.5877433  0.8355523
## IEDE     -0.1202229  0.5877433  1.0000000  0.5052005
## spread   -0.1564748  0.8355523  0.5052005  1.0000000
##
## , , 2023-01-27
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m  1.0000000 -0.1793218 -0.1257693 -0.1454226
## ESDE     -0.1793218  1.0000000  0.5821767  0.8333050
## IEDE     -0.1257693  0.5821767  1.0000000  0.4984836
## spread   -0.1454226  0.8333050  0.4984836  1.0000000
##
## , , 2023-01-30
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m  1.0000000 -0.1631890 -0.1141450 -0.1352251
## ESDE     -0.1631890  1.0000000  0.5755983  0.8323016
## IEDE     -0.1141450  0.5755983  1.0000000  0.4962802
## spread   -0.1352251  0.8323016  0.4962802  1.0000000
##
## , , 2023-01-31
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m  1.0000000 -0.1719539 -0.1426435 -0.1311065
## ESDE     -0.1719539  1.0000000  0.5768168  0.8293285
## IEDE     -0.1426435  0.5768168  1.0000000  0.4913731
## spread   -0.1311065  0.8293285  0.4913731  1.0000000
##
## , , 2023-02-01
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m  1.0000000 -0.1555220 -0.1248997 -0.1200131
## ESDE     -0.1555220  1.0000000  0.5531033  0.8232511
## IEDE     -0.1248997  0.5531033  1.0000000  0.4853485
## spread   -0.1200131  0.8232511  0.4853485  1.0000000
##
## , , 2023-02-02
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 -0.1361022 -0.07768814 -0.1260305
## ESDE     -0.13610222  1.0000000  0.54186364  0.8187111
## IEDE     -0.07768814  0.5418636  1.00000000  0.4604300
## spread   -0.12603047  0.8187111  0.46043000  1.0000000
##
## , , 2023-02-03
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.09084037 -0.05914057 -0.08852641
```

```
## ESDE     -0.09084037  1.00000000  0.57218502  0.85950733
## IEDE     -0.05914057  0.57218502  1.00000000  0.50668531
## spread   -0.08852641  0.85950733  0.50668531  1.00000000
##
## , , 2023-02-06
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04537803 -0.01059705 -0.04588354
## ESDE     -0.04537803  1.00000000  0.56674459  0.85664501
## IEDE     -0.01059705  0.56674459  1.00000000  0.50317520
## spread   -0.04588354  0.85664501  0.50317520  1.00000000
##
## , , 2023-02-07
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.04572405 -0.01430526 -0.0495235
## ESDE     -0.04572405  1.00000000  0.56072082  0.8536563
## IEDE     -0.01430526  0.56072082  1.00000000  0.4990461
## spread   -0.04952350  0.85365633  0.49904605  1.0000000
##
## , , 2023-02-08
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03924895 -0.01084726 -0.04514728
## ESDE     -0.03924895  1.00000000  0.55446354  0.85067609
## IEDE     -0.01084726  0.55446354  1.00000000  0.49440606
## spread   -0.04514728  0.85067609  0.49440606  1.00000000
##
## , , 2023-02-09
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.03425955 -0.008021334 -0.04192892
## ESDE     -0.034259549  1.00000000  0.548136511  0.84557693
## IEDE     -0.008021334  0.54813651  1.000000000  0.48985933
## spread   -0.041928916  0.84557693  0.489859325  1.00000000
##
## , , 2023-02-10
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02089383 -0.02353415 -0.02088178
## ESDE     -0.02089383  1.00000000  0.52732637  0.84239194
## IEDE     -0.02353415  0.52732637  1.00000000  0.45845212
## spread   -0.02088178  0.84239194  0.45845212  1.00000000
##
## , , 2023-02-13
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.03420945 -0.03681458 -0.0215316
## ESDE     -0.03420945  1.00000000  0.53429388  0.8335974
## IEDE     -0.03681458  0.53429388  1.00000000  0.4547564
## spread   -0.02153160  0.83359736  0.45475639  1.0000000
##
```

```
## , , 2023-02-14
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04459876 -0.03416218 -0.02476174
## ESDE     -0.04459876  1.00000000  0.52877619  0.82876605
## IEDE     -0.03416218  0.52877619  1.00000000  0.45485292
## spread   -0.02476174  0.82876605  0.45485292  1.00000000
##
## , , 2023-02-15
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03774039 -0.03025877 -0.02125378
## ESDE     -0.03774039  1.00000000  0.51963776  0.82769144
## IEDE     -0.03025877  0.51963776  1.00000000  0.44500043
## spread   -0.02125378  0.82769144  0.44500043  1.00000000
##
## , , 2023-02-16
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 -0.01766344 -0.01943434 -0.008701344
## ESDE     -0.017663444  1.00000000  0.52672840  0.837481194
## IEDE     -0.019434337  0.52672840  1.00000000  0.458944183
## spread   -0.008701344  0.83748119  0.45894418  1.000000000
##
## , , 2023-02-17
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01183746 -0.01099346 -0.00900321
## ESDE     -0.01183746  1.00000000  0.52492373  0.83417633
## IEDE     -0.01099346  0.52492373  1.00000000  0.45588783
## spread   -0.00900321  0.83417633  0.45588783  1.00000000
##
## , , 2023-02-20
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.01846259 -0.01068854 0.009923816
## ESDE     -0.018462591  1.00000000  0.52183365 0.828944271
## IEDE     -0.010688542  0.52183365  1.00000000 0.453895817
## spread    0.009923816  0.82894427  0.45389582 1.000000000
##
## , , 2023-02-21
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 -0.03219554 -0.02857581 -0.004717785
## ESDE     -0.032195543  1.00000000  0.52382628  0.828468730
## IEDE     -0.028575806  0.52382628  1.00000000  0.457079173
## spread   -0.004717785  0.82846873  0.45707917  1.000000000
##
## , , 2023-02-22
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03314674 -0.04172786 -0.01922012
```

```
## ESDE     -0.03314674  1.00000000  0.52150984  0.82137219
## IEDE     -0.04172786  0.52150984  1.00000000  0.50956731
## spread   -0.01922012  0.82137219  0.50956731  1.00000000
##
## , , 2023-02-23
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02993173 -0.04226772 -0.02670214
## ESDE     -0.02993173  1.00000000  0.51896562  0.81957087
## IEDE     -0.04226772  0.51896562  1.00000000  0.50590568
## spread   -0.02670214  0.81957087  0.50590568  1.00000000
##
## , , 2023-02-24
##
##              eurois3m         ESDE        IEDE     spread
## eurois3m  1.000000000 -0.003375038 -0.01929661 0.01498687
## ESDE     -0.003375038  1.000000000  0.52029985 0.81968423
## IEDE     -0.019296610  0.520299848  1.00000000 0.50431945
## spread    0.014986867  0.819684235  0.50431945 1.00000000
##
## , , 2023-02-27
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.000166952 -0.0253449 0.004627551
## ESDE      0.000166952 1.000000000  0.5142412 0.816064733
## IEDE     -0.025344904 0.514241171  1.0000000 0.501976920
## spread    0.004627551 0.816064733  0.5019769 1.000000000
##
## , , 2023-02-28
##
##              eurois3m        ESDE        IEDE     spread
## eurois3m  1.000000000 0.001732807 -0.01769998 0.01496689
## ESDE      0.001732807 1.000000000  0.51142989 0.79565505
## IEDE     -0.017699980 0.511429885  1.00000000 0.49615111
## spread    0.014966885 0.795655052  0.49615111 1.00000000
##
## , , 2023-03-01
##
##              eurois3m        ESDE        IEDE     spread
## eurois3m  1.000000000 0.003098419 -0.01415225 0.01274841
## ESDE      0.003098419 1.000000000  0.51070199 0.79431959
## IEDE     -0.014152255 0.510701995  1.00000000 0.48251493
## spread    0.012748407 0.794319588  0.48251493 1.00000000
##
## , , 2023-03-02
##
##               eurois3m         ESDE         IEDE     spread
## eurois3m  1.0000000000 0.0009078614 -0.008209811 0.01134808
## ESDE      0.0009078614 1.0000000000  0.493125294 0.78686855
## IEDE     -0.0082098114 0.4931252938  1.000000000 0.47952443
## spread    0.0113480768 0.7868685531  0.479524426 1.00000000
##
```

```
## , , 2023-03-03
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.005213584 -0.01214275 0.002294293
## ESDE     -0.005213584  1.000000000  0.50475313 0.794629613
## IEDE     -0.012142747  0.504753129  1.00000000 0.486771236
## spread    0.002294293  0.794629613  0.48677124 1.000000000
##
## , , 2023-03-06
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.07883166 0.03818524 -0.06564253
## ESDE     -0.07883166  1.00000000 0.43281380  0.80970126
## IEDE      0.03818524  0.43281380 1.00000000  0.42269513
## spread   -0.06564253  0.80970126 0.42269513  1.00000000
##
## , , 2023-03-07
##
##             eurois3m        ESDE      IEDE      spread
## eurois3m  1.00000000 -0.06596706 0.0432972 -0.06768208
## ESDE     -0.06596706  1.00000000 0.4401989  0.80516980
## IEDE      0.04329720  0.44019886 1.0000000  0.41676356
## spread   -0.06768208  0.80516980 0.4167636  1.00000000
##
## , , 2023-03-08
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.02231983 -0.006498312 -0.02204445
## ESDE     -0.022319833  1.00000000  0.414829652  0.80964529
## IEDE     -0.006498312  0.41482965  1.000000000  0.38584457
## spread   -0.022044453  0.80964529  0.385844567  1.00000000
##
## , , 2023-03-09
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03159093 -0.01804938 -0.05802021
## ESDE     -0.03159093  1.00000000  0.42966048  0.80168650
## IEDE     -0.01804938  0.42966048  1.00000000  0.40413347
## spread   -0.05802021  0.80168650  0.40413347  1.00000000
##
## , , 2023-03-10
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02651881 -0.01101128 -0.05292863
## ESDE     -0.02651881  1.00000000  0.28514827  0.79392856
## IEDE     -0.01101128  0.28514827  1.00000000  0.34887515
## spread   -0.05292863  0.79392856  0.34887515  1.00000000
##
## , , 2023-03-13
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.08116719 0.01880866 0.08421411
```

```
## ESDE     0.08116719 1.00000000 0.30871997 0.80664458
## IEDE     0.01880866 0.30871997 1.00000000 0.35901149
## spread   0.08421411 0.80664458 0.35901149 1.00000000
##
## , , 2023-03-14
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.3110576 0.1385349 0.2094903
## ESDE     0.3110576 1.0000000 0.4296614 0.8094447
## IEDE     0.1385349 0.4296614 1.0000000 0.4332595
## spread   0.2094903 0.8094447 0.4332595 1.0000000
##
## , , 2023-03-15
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.3527001 0.1689149 0.2539743
## ESDE     0.3527001 1.0000000 0.4507457 0.8193789
## IEDE     0.1689149 0.4507457 1.0000000 0.4498320
## spread   0.2539743 0.8193789 0.4498320 1.0000000
##
## , , 2023-03-16
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.3524347 0.1880338 0.2561866
## ESDE     0.3524347 1.0000000 0.4877497 0.8269810
## IEDE     0.1880338 0.4877497 1.0000000 0.4774727
## spread   0.2561866 0.8269810 0.4774727 1.0000000
##
## , , 2023-03-17
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.3224556 0.1684381 0.2438810
## ESDE     0.3224556 1.0000000 0.4881980 0.8252842
## IEDE     0.1684381 0.4881980 1.0000000 0.4742735
## spread   0.2438810 0.8252842 0.4742735 1.0000000
##
## , , 2023-03-20
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.3001795 0.1585040 0.2209925
## ESDE     0.3001795 1.0000000 0.4881937 0.8232133
## IEDE     0.1585040 0.4881937 1.0000000 0.4689588
## spread   0.2209925 0.8232133 0.4689588 1.0000000
##
## , , 2023-03-21
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.2729454 0.1418322 0.1902541
## ESDE     0.2729454 1.0000000 0.4903051 0.8217715
## IEDE     0.1418322 0.4903051 1.0000000 0.4698104
## spread   0.1902541 0.8217715 0.4698104 1.0000000
##
```

```
## , , 2023-03-22
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.2419165 0.1266639 0.1708547
## ESDE     0.2419165 1.0000000 0.4991996 0.8155083
## IEDE     0.1266639 0.4991996 1.0000000 0.4698321
## spread   0.1708547 0.8155083 0.4698321 1.0000000
##
## , , 2023-03-23
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.2235147 0.1046502 0.1544909
## ESDE     0.2235147 1.0000000 0.4683930 0.8151201
## IEDE     0.1046502 0.4683930 1.0000000 0.4441329
## spread   0.1544909 0.8151201 0.4441329 1.0000000
##
## , , 2023-03-24
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.2076655 0.0954601 0.1400399
## ESDE     0.2076655 1.0000000 0.4687348 0.8148097
## IEDE     0.0954601 0.4687348 1.0000000 0.4437097
## spread   0.1400399 0.8148097 0.4437097 1.0000000
##
## , , 2023-03-27
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.1932545 0.07937108 0.1358906
## ESDE     0.19325454 1.0000000 0.46792593 0.8140070
## IEDE     0.07937108 0.4679259 1.00000000 0.4359821
## spread   0.13589063 0.8140070 0.43598206 1.0000000
##
## , , 2023-03-28
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.1779363 0.07532151 0.1232355
## ESDE     0.17793625 1.0000000 0.47452682 0.8108008
## IEDE     0.07532151 0.4745268 1.00000000 0.4456548
## spread   0.12323555 0.8108008 0.44565478 1.0000000
##
## , , 2023-03-29
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.1686133 0.02755268 0.1067527
## ESDE     0.16861327 1.0000000 0.32656997 0.7957946
## IEDE     0.02755268 0.3265700 1.00000000 0.4040821
## spread   0.10675274 0.7957946 0.40408206 1.0000000
##
## , , 2023-03-30
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1475724 0.02600969 0.08924397
```

```
## ESDE     0.14757244 1.0000000 0.34068623 0.80378224
## IEDE     0.02600969 0.3406862 1.00000000 0.40615275
## spread   0.08924397 0.8037822 0.40615275 1.00000000
##
## , , 2023-03-31
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1341627 0.02741039 0.08369674
## ESDE     0.13416268 1.0000000 0.35343569 0.80053885
## IEDE     0.02741039 0.3534357 1.00000000 0.41150935
## spread   0.08369674 0.8005389 0.41150935 1.00000000
##
## , , 2023-04-03
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1176952 0.02596969 0.07822008
## ESDE     0.11769524 1.0000000 0.36588267 0.79143100
## IEDE     0.02596969 0.3658827 1.00000000 0.41243927
## spread   0.07822008 0.7914310 0.41243927 1.00000000
##
## , , 2023-04-04
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1070865 0.02622043 0.06961541
## ESDE     0.10708645 1.0000000 0.37452120 0.78300420
## IEDE     0.02622043 0.3745212 1.00000000 0.42747071
## spread   0.06961541 0.7830042 0.42747071 1.00000000
##
## , , 2023-04-05
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.09560802 0.02586237 0.05963072
## ESDE     0.09560802 1.00000000 0.38555852 0.78676759
## IEDE     0.02586237 0.38555852 1.00000000 0.42897489
## spread   0.05963072 0.78676759 0.42897489 1.00000000
##
## , , 2023-04-06
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07689059 0.01605353 0.05031152
## ESDE     0.07689059 1.00000000 0.40652058 0.78620660
## IEDE     0.01605353 0.40652058 1.00000000 0.43050625
## spread   0.05031152 0.78620660 0.43050625 1.00000000
##
## , , 2023-04-11
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06996724 0.01669489 0.04599757
## ESDE     0.06996724 1.00000000 0.41524798 0.78676089
## IEDE     0.01669489 0.41524798 1.00000000 0.43209068
## spread   0.04599757 0.78676089 0.43209068 1.00000000
##
```

```
## , , 2023-04-12
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06150124 0.02038434 0.03765599
## ESDE     0.06150124 1.00000000 0.35561873 0.79389716
## IEDE     0.02038434 0.35561873 1.00000000 0.36468166
## spread   0.03765599 0.79389716 0.36468166 1.00000000
##
## , , 2023-04-13
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06478388 0.02697079 0.02625431
## ESDE     0.06478388 1.00000000 0.37314553 0.78679513
## IEDE     0.02697079 0.37314553 1.00000000 0.36748877
## spread   0.02625431 0.78679513 0.36748877 1.00000000
##
## , , 2023-04-14
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07060552 0.01942886 0.02235894
## ESDE     0.07060552 1.00000000 0.37581961 0.78537063
## IEDE     0.01942886 0.37581961 1.00000000 0.37505709
## spread   0.02235894 0.78537063 0.37505709 1.00000000
##
## , , 2023-04-17
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06949584 0.01913677 0.01546192
## ESDE     0.06949584 1.00000000 0.38789714 0.78727013
## IEDE     0.01913677 0.38789714 1.00000000 0.38270080
## spread   0.01546192 0.78727013 0.38270080 1.00000000
##
## , , 2023-04-18
##
##              eurois3m      ESDE        IEDE       spread
## eurois3m  1.000000000 0.0394120 -0.01284647 -0.008325706
## ESDE      0.039412002 1.0000000  0.43275272  0.798676452
## IEDE     -0.012846472 0.4327527  1.00000000  0.420975756
## spread   -0.008325706 0.7986765  0.42097576  1.000000000
##
## , , 2023-04-19
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.04841544 -0.001763353 0.003582632
## ESDE      0.048415440 1.00000000  0.447570431 0.805618198
## IEDE     -0.001763353 0.44757043  1.000000000 0.432321440
## spread    0.003582632 0.80561820  0.432321440 1.000000000
##
## , , 2023-04-20
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.01281355 -0.02553309 -0.04438786
```

```
## ESDE      0.01281355 1.00000000  0.46269226  0.81079486
## IEDE     -0.02553309 0.46269226  1.00000000  0.44802360
## spread   -0.04438786 0.81079486  0.44802360  1.00000000
##
## , , 2023-04-21
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 0.006463695 -0.03482734 -0.04238949
## ESDE      0.006463695 1.000000000  0.46828131  0.81117461
## IEDE     -0.034827339 0.468281306  1.00000000  0.44742013
## spread   -0.042389492 0.811174613  0.44742013  1.00000000
##
## , , 2023-04-24
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.003065506 -0.04310527 -0.02053075
## ESDE     -0.003065506  1.000000000  0.47298754  0.80373467
## IEDE     -0.043105270  0.472987545  1.00000000  0.43909188
## spread   -0.020530755  0.803734667  0.43909188  1.00000000
##
## , , 2023-04-25
##
##              eurois3m         ESDE        IEDE       spread
## eurois3m  1.000000000 -0.004032361 -0.03017625 -0.009208992
## ESDE     -0.004032361  1.000000000  0.47188536  0.796703090
## IEDE     -0.030176254  0.471885361  1.00000000  0.443907041
## spread   -0.009208992  0.796703090  0.44390704  1.000000000
##
## , , 2023-04-26
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 0.003362168 -0.02221071 -0.004397176
## ESDE      0.003362168 1.000000000  0.48012481  0.803481761
## IEDE     -0.022210707 0.480124814  1.00000000  0.450404206
## spread   -0.004397176 0.803481761  0.45040421  1.000000000
##
## , , 2023-04-27
##
##               eurois3m          ESDE        IEDE       spread
## eurois3m  1.0000000000 -0.0001445323 -0.01192318 -0.001830891
## ESDE     -0.0001445323  1.0000000000  0.47461481  0.800163301
## IEDE     -0.0119231781  0.4746148074  1.00000000  0.451710419
## spread   -0.0018308906  0.8001633012  0.45171042  1.000000000
##
## , , 2023-04-28
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.007652324 0.06604161 -0.02911271
## ESDE     -0.007652324  1.000000000 0.44272530  0.80079800
## IEDE      0.066041615  0.442725303 1.00000000  0.39454955
## spread   -0.029112709  0.800798001 0.39454955  1.00000000
##
```

```
## , , 2023-05-02
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.03021864 0.03985534 -0.05123224
## ESDE     -0.03021864  1.00000000 0.46026763  0.80807597
## IEDE      0.03985534  0.46026763 1.00000000  0.41332014
## spread   -0.05123224  0.80807597 0.41332014  1.00000000
##
## , , 2023-05-03
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m  1.00000000 0.0096763 0.07331181 -0.01252764
## ESDE      0.00967630 1.0000000 0.53330612  0.82900350
## IEDE      0.07331181 0.5333061 1.00000000  0.48773703
## spread   -0.01252764 0.8290035 0.48773703  1.00000000
##
## , , 2023-05-04
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01191267 0.02037542 0.02076148
## ESDE     0.01191267 1.00000000 0.51159600 0.81636289
## IEDE     0.02037542 0.51159600 1.00000000 0.41963108
## spread   0.02076148 0.81636289 0.41963108 1.00000000
##
## , , 2023-05-05
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07992177 0.05313154 0.02631771
## ESDE     0.07992177 1.00000000 0.51965492 0.80067069
## IEDE     0.05313154 0.51965492 1.00000000 0.42309176
## spread   0.02631771 0.80067069 0.42309176 1.00000000
##
## , , 2023-05-08
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06240074 0.04512391 0.02780305
## ESDE     0.06240074 1.00000000 0.52003930 0.78132176
## IEDE     0.04512391 0.52003930 1.00000000 0.41893937
## spread   0.02780305 0.78132176 0.41893937 1.00000000
##
## , , 2023-05-09
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05830219 0.04555343 0.03322774
## ESDE     0.05830219 1.00000000 0.51768299 0.78163710
## IEDE     0.04555343 0.51768299 1.00000000 0.42324634
## spread   0.03322774 0.78163710 0.42324634 1.00000000
##
## , , 2023-05-10
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.008551746 0.03361061 -0.02175824
```

```
## ESDE     -0.008551746  1.000000000 0.51038589  0.79129302
## IEDE      0.033610607  0.510385890 1.00000000  0.42413688
## spread   -0.021758235  0.791293020 0.42413688  1.00000000
##
## , , 2023-05-11
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.009873613 0.02956516 -0.02710173
## ESDE     -0.009873613  1.000000000 0.51275743  0.79479300
## IEDE      0.029565155  0.512757430 1.00000000  0.43150172
## spread   -0.027101733  0.794793003 0.43150172  1.00000000
##
## , , 2023-05-12
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.002532106 0.02700464 -0.02692009
## ESDE     -0.002532106  1.000000000 0.50295288  0.78504442
## IEDE      0.027004638  0.502952878 1.00000000  0.43437016
## spread   -0.026920089  0.785044424 0.43437016  1.00000000
##
## , , 2023-05-15
##
##              eurois3m         ESDE       IEDE       spread
## eurois3m  1.000000000 -0.005439375 0.09164763 -0.005331978
## ESDE     -0.005439375  1.000000000 0.46112174  0.781812877
## IEDE      0.091647628  0.461121737 1.00000000  0.440972864
## spread   -0.005331978  0.781812877 0.44097286  1.000000000
##
## , , 2023-05-16
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01095231 0.08419555 -0.01630377
## ESDE     -0.01095231  1.00000000 0.45843227  0.79228318
## IEDE      0.08419555  0.45843227 1.00000000  0.43016208
## spread   -0.01630377  0.79228318 0.43016208  1.00000000
##
## , , 2023-05-17
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 -0.0106905 0.08642148 -0.02386685
## ESDE     -0.01069050  1.0000000 0.46082267  0.79461085
## IEDE      0.08642148  0.4608227 1.00000000  0.42870827
## spread   -0.02386685  0.7946108 0.42870827  1.00000000
##
## , , 2023-05-18
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01369997 0.07022521 -0.02671347
## ESDE     -0.01369997  1.00000000 0.48182788  0.80074972
## IEDE      0.07022521  0.48182788 1.00000000  0.44461055
## spread   -0.02671347  0.80074972 0.44461055  1.00000000
##
```

```
## , , 2023-05-19
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01555705 0.08036049 -0.03043986
## ESDE     -0.01555705  1.00000000 0.47200513  0.80287280
## IEDE      0.08036049  0.47200513 1.00000000  0.43316797
## spread   -0.03043986  0.80287280 0.43316797  1.00000000
##
## , , 2023-05-22
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.03594758 0.06441902 -0.05190606
## ESDE     -0.03594758  1.00000000 0.47741109  0.81069233
## IEDE      0.06441902  0.47741109 1.00000000  0.43846561
## spread   -0.05190606  0.81069233 0.43846561  1.00000000
##
## , , 2023-05-23
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.03374013 0.07061112 -0.05390563
## ESDE     -0.03374013  1.00000000 0.43582851  0.81078342
## IEDE      0.07061112  0.43582851 1.00000000  0.37020297
## spread   -0.05390563  0.81078342 0.37020297  1.00000000
##
## , , 2023-05-24
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.03128967 0.06497559 -0.0489749
## ESDE     -0.03128967  1.00000000 0.43955511  0.8051530
## IEDE      0.06497559  0.43955511 1.00000000  0.3783630
## spread   -0.04897490  0.80515305 0.37836299  1.0000000
##
## , , 2023-05-25
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02063732 0.05958806 -0.06175087
## ESDE     -0.02063732  1.00000000 0.44549774  0.80169850
## IEDE      0.05958806  0.44549774 1.00000000  0.38415168
## spread   -0.06175087  0.80169850 0.38415168  1.00000000
##
## , , 2023-05-26
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m  1.00000000 0.0176498 0.09406017 -0.08331203
## ESDE      0.01764980 1.0000000 0.47070974  0.75634375
## IEDE      0.09406017 0.4707097 1.00000000  0.34645213
## spread   -0.08331203 0.7563437 0.34645213  1.00000000
##
## , , 2023-05-29
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.003351036 0.09625851 -0.06986145
```

```
## ESDE      0.003351036 1.000000000 0.44133674  0.74188073
## IEDE      0.096258514 0.441336739 1.00000000  0.36136849
## spread   -0.069861450 0.741880734 0.36136849  1.00000000
##
## , , 2023-05-30
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01693843 0.08266905 -0.08327805
## ESDE      0.01693843 1.00000000 0.44393889  0.73214059
## IEDE      0.08266905 0.44393889 1.00000000  0.37084975
## spread   -0.08327805 0.73214059 0.37084975  1.00000000
##
## , , 2023-05-31
##
##              eurois3m        ESDE      IEDE      spread
## eurois3m  1.000000000 0.004967285 0.0846868 -0.09383736
## ESDE      0.004967285 1.000000000 0.4309454  0.74496002
## IEDE      0.084686799 0.430945377 1.0000000  0.34453721
## spread   -0.093837365 0.744960019 0.3445372  1.00000000
##
## , , 2023-06-01
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.001516333 0.07395472 -0.08790377
## ESDE      0.001516333 1.000000000 0.43961491  0.75355595
## IEDE      0.073954720 0.439614909 1.00000000  0.35736337
## spread   -0.087903770 0.753555948 0.35736337  1.00000000
##
## , , 2023-06-02
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.03698237 0.07897324 -0.01474994
## ESDE      0.03698237 1.00000000 0.45309608  0.77671860
## IEDE      0.07897324 0.45309608 1.00000000  0.37009497
## spread   -0.01474994 0.77671860 0.37009497  1.00000000
##
## , , 2023-06-05
##
##              eurois3m        ESDE      IEDE      spread
## eurois3m  1.000000000 0.009002497 0.0518239 -0.04565489
## ESDE      0.009002497 1.000000000 0.4903177  0.79995747
## IEDE      0.051823901 0.490317727 1.0000000  0.42442079
## spread   -0.045654892 0.799957471 0.4244208  1.00000000
##
## , , 2023-06-06
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01083092 0.03468079 -0.04403105
## ESDE      0.01083092 1.00000000 0.45482440  0.79706098
## IEDE      0.03468079 0.45482440 1.00000000  0.42322423
## spread   -0.04403105 0.79706098 0.42322423  1.00000000
##
```

```
## , , 2023-06-07
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.007055426 0.09887993 -0.06636846
## ESDE      0.007055426 1.000000000 0.44221821  0.79593003
## IEDE      0.098879926 0.442218208 1.00000000  0.38748851
## spread   -0.066368463 0.795930026 0.38748851  1.00000000
##
## , , 2023-06-08
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.03404747 0.1437140 -0.04302108
## ESDE      0.03404747 1.00000000 0.4514282  0.79747620
## IEDE      0.14371404 0.45142816 1.0000000  0.39310415
## spread   -0.04302108 0.79747620 0.3931042  1.00000000
##
## , , 2023-06-09
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.03510202 0.1339644 -0.05125569
## ESDE      0.03510202 1.00000000 0.4581233  0.73349323
## IEDE      0.13396441 0.45812333 1.0000000  0.35917294
## spread   -0.05125569 0.73349323 0.3591729  1.00000000
##
## , , 2023-06-12
##
##              eurois3m       ESDE      IEDE       spread
## eurois3m  1.000000000 0.07238971 0.1135532 -0.006342389
## ESDE      0.072389708 1.00000000 0.4432033  0.752519930
## IEDE      0.113553181 0.44320331 1.0000000  0.351857229
## spread   -0.006342389 0.75251993 0.3518572  1.000000000
##
## , , 2023-06-13
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.1005470 0.1325094 0.01448194
## ESDE     0.10054699 1.0000000 0.5023620 0.76364194
## IEDE     0.13250935 0.5023620 1.0000000 0.39282631
## spread   0.01448194 0.7636419 0.3928263 1.00000000
##
## , , 2023-06-14
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.08228527 0.1533567 -0.01166979
## ESDE      0.08228527 1.00000000 0.4928294  0.76735740
## IEDE      0.15335670 0.49282940 1.0000000  0.37954765
## spread   -0.01166979 0.76735740 0.3795476  1.00000000
##
## , , 2023-06-15
##
##              eurois3m       ESDE      IEDE       spread
## eurois3m  1.000000000 0.08072257 0.1356587 -0.008229877
```

```
## ESDE      0.080722572 1.00000000 0.4710562  0.772668172
## IEDE      0.135658658 0.47105622 1.0000000  0.375438375
## spread   -0.008229877 0.77266817 0.3754384  1.000000000
##
## , , 2023-06-16
##
##              eurois3m       ESDE      IEDE       spread
## eurois3m  1.000000000 0.07337424 0.1241037 -0.007952028
## ESDE      0.073374243 1.00000000 0.4766579  0.769649112
## IEDE      0.124103652 0.47665795 1.0000000  0.385264864
## spread   -0.007952028 0.76964911 0.3852649  1.000000000
##
## , , 2023-06-19
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.04153051 0.06426471 -0.0346727
## ESDE      0.04153051 1.00000000 0.52586891  0.7900819
## IEDE      0.06426471 0.52586891 1.00000000  0.4606609
## spread   -0.03467270 0.79008190 0.46066089  1.0000000
##
## , , 2023-06-20
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.03042439 0.05470464 -0.04511402
## ESDE      0.03042439 1.00000000 0.52572396  0.79439115
## IEDE      0.05470464 0.52572396 1.00000000  0.46266623
## spread   -0.04511402 0.79439115 0.46266623  1.00000000
##
## , , 2023-06-21
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.03671966 0.05676922 -0.0360900
## ESDE      0.03671966 1.00000000 0.52483004  0.7968935
## IEDE      0.05676922 0.52483004 1.00000000  0.4629638
## spread   -0.03609000 0.79689349 0.46296378  1.0000000
##
## , , 2023-06-22
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02672457 0.02217777 -0.01233358
## ESDE      0.02672457 1.00000000 0.52148581  0.79380074
## IEDE      0.02217777 0.52148581 1.00000000  0.44800197
## spread   -0.01233358 0.79380074 0.44800197  1.00000000
##
## , , 2023-06-23
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.03267383 0.02069487 -0.009191088
## ESDE      0.032673827 1.00000000 0.51114874  0.794393930
## IEDE      0.020694872 0.51114874 1.00000000  0.446577594
## spread   -0.009191088 0.79439393 0.44657759  1.000000000
##
```

```
## , , 2023-06-26
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.04582098 0.04195877 -0.001056067
## ESDE      0.045820977 1.00000000 0.51428083  0.796685384
## IEDE      0.041958772 0.51428083 1.00000000  0.448456098
## spread   -0.001056067 0.79668538 0.44845610  1.000000000
##
## , , 2023-06-27
##
##              eurois3m       ESDE         IEDE       spread
## eurois3m  1.000000000 0.03315347 -0.004094316 -0.001617002
## ESDE      0.033153470 1.00000000  0.510201067  0.798236305
## IEDE     -0.004094316 0.51020107  1.000000000  0.441339030
## spread   -0.001617002 0.79823630  0.441339030  1.000000000
##
## , , 2023-06-28
##
##              eurois3m      ESDE        IEDE      spread
## eurois3m  1.000000000 0.0475606 -0.02730438 0.005820389
## ESDE      0.047560597 1.0000000  0.50254739 0.800079550
## IEDE     -0.027304379 0.5025474  1.00000000 0.439115002
## spread    0.005820389 0.8000796  0.43911500 1.000000000
##
## , , 2023-06-29
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.05630842 -0.01786057 0.01564009
## ESDE      0.05630842 1.00000000  0.50273073 0.80344295
## IEDE     -0.01786057 0.50273073  1.00000000 0.44149606
## spread    0.01564009 0.80344295  0.44149606 1.00000000
##
## , , 2023-06-30
##
##              eurois3m       ESDE        IEDE       spread
## eurois3m  1.000000000 0.02916141 -0.01356599 -0.002704814
## ESDE      0.029161411 1.00000000  0.48176842  0.813253021
## IEDE     -0.013565989 0.48176842  1.00000000  0.432000420
## spread   -0.002704814 0.81325302  0.43200042  1.000000000
##
## , , 2023-07-03
##
##              eurois3m       ESDE        IEDE       spread
## eurois3m  1.000000000 0.01909507 -0.03488161 -0.005779495
## ESDE      0.019095068 1.00000000  0.48480015  0.813260184
## IEDE     -0.034881607 0.48480015  1.00000000  0.431344800
## spread   -0.005779495 0.81326018  0.43134480  1.000000000
##
## , , 2023-07-04
##
##              eurois3m       ESDE        IEDE       spread
## eurois3m  1.000000000 0.01840333 -0.02912751 -0.007055117
```

```
## ESDE      0.018403327 1.00000000  0.48446865  0.811838439
## IEDE     -0.029127511 0.48446865  1.00000000  0.426928483
## spread   -0.007055117 0.81183844  0.42692848  1.000000000
##
## , , 2023-07-05
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.04399046 -0.007896282 0.03236497
## ESDE      0.043990459 1.00000000  0.498752719 0.82089090
## IEDE     -0.007896282 0.49875272  1.000000000 0.44549769
## spread    0.032364974 0.82089090  0.445497689 1.00000000
##
## , , 2023-07-06
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1027111 0.03076444 0.09509026
## ESDE     0.10271113 1.0000000 0.51671397 0.83569809
## IEDE     0.03076444 0.5167140 1.00000000 0.46739797
## spread   0.09509026 0.8356981 0.46739797 1.00000000
##
## , , 2023-07-07
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.08241748 0.01325385 0.07239821
## ESDE     0.08241748 1.00000000 0.53041691 0.83806467
## IEDE     0.01325385 0.53041691 1.00000000 0.48567376
## spread   0.07239821 0.83806467 0.48567376 1.00000000
##
## , , 2023-07-10
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05990392 0.00273412 0.06716125
## ESDE     0.05990392 1.00000000 0.52962831 0.83035749
## IEDE     0.00273412 0.52962831 1.00000000 0.47937337
## spread   0.06716125 0.83035749 0.47937337 1.00000000
##
## , , 2023-07-11
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.05021858 0.005719093 0.06386812
## ESDE     0.050218577 1.00000000 0.524572132 0.82416398
## IEDE     0.005719093 0.52457213 1.000000000 0.47644377
## spread   0.063868116 0.82416398 0.476443770 1.00000000
##
## , , 2023-07-12
##
##              eurois3m       ESDE         IEDE    spread
## eurois3m 1.0000000000 0.05488766 0.0009979731 0.0642007
## ESDE     0.0548876574 1.00000000 0.4852074827 0.8292212
## IEDE     0.0009979731 0.48520748 1.0000000000 0.4406411
## spread   0.0642007006 0.82922124 0.4406410522 1.0000000
##
```

```
## , , 2023-07-13
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.06350911 0.007161752 0.07607703
## ESDE     0.063509108 1.00000000 0.487469752 0.83072903
## IEDE     0.007161752 0.48746975 1.000000000 0.44017436
## spread   0.076077030 0.83072903 0.440174364 1.00000000
##
## , , 2023-07-14
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07078632 0.02235977 0.08015102
## ESDE     0.07078632 1.00000000 0.50449241 0.83455775
## IEDE     0.02235977 0.50449241 1.00000000 0.45961110
## spread   0.08015102 0.83455775 0.45961110 1.00000000
##
## , , 2023-07-17
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03089874 0.01573959 0.05199175
## ESDE     0.03089874 1.00000000 0.50070562 0.83296012
## IEDE     0.01573959 0.50070562 1.00000000 0.45945678
## spread   0.05199175 0.83296012 0.45945678 1.00000000
##
## , , 2023-07-18
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03684524 0.02993793 0.05501314
## ESDE     0.03684524 1.00000000 0.50601123 0.83273209
## IEDE     0.02993793 0.50601123 1.00000000 0.46752522
## spread   0.05501314 0.83273209 0.46752522 1.00000000
##
## , , 2023-07-19
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.04002517 -0.01406055 -0.0263162
## ESDE     -0.04002517  1.00000000  0.52762621  0.8469869
## IEDE     -0.01406055  0.52762621  1.00000000  0.4907187
## spread   -0.02631620  0.84698685  0.49071869  1.0000000
##
## , , 2023-07-20
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.05352458 -0.02569309 -0.04639657
## ESDE     -0.05352458  1.00000000  0.52507183  0.84443194
## IEDE     -0.02569309  0.52507183  1.00000000  0.48732161
## spread   -0.04639657  0.84443194  0.48732161  1.00000000
##
## , , 2023-07-21
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04998672 -0.01943369 -0.04685343
```

```
## ESDE     -0.04998672  1.00000000  0.51853127  0.84220977
## IEDE     -0.01943369  0.51853127  1.00000000  0.47842325
## spread   -0.04685343  0.84220977  0.47842325  1.00000000
##
## , , 2023-07-24
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04085377 -0.01290909 -0.04887292
## ESDE     -0.04085377  1.00000000  0.51632528  0.83728346
## IEDE     -0.01290909  0.51632528  1.00000000  0.47353656
## spread   -0.04887292  0.83728346  0.47353656  1.00000000
##
## , , 2023-07-25
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03123921 -0.01296935 -0.04536562
## ESDE     -0.03123921  1.00000000  0.48116113  0.82193368
## IEDE     -0.01296935  0.48116113  1.00000000  0.46901715
## spread   -0.04536562  0.82193368  0.46901715  1.00000000
##
## , , 2023-07-26
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.02801414 -0.007435882 -0.05014928
## ESDE     -0.028014138  1.00000000  0.481559312  0.81642562
## IEDE     -0.007435882  0.48155931  1.000000000  0.45725882
## spread   -0.050149278  0.81642562  0.457258816  1.00000000
##
## , , 2023-07-27
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02492178 -0.00856716 -0.04386029
## ESDE     -0.02492178  1.00000000  0.48442948  0.81196840
## IEDE     -0.00856716  0.48442948  1.00000000  0.44601632
## spread   -0.04386029  0.81196840  0.44601632  1.00000000
##
## , , 2023-07-28
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.02053138 -0.005723015 -0.03986528
## ESDE     -0.020531383  1.00000000  0.492014714  0.81628323
## IEDE     -0.005723015  0.49201471  1.000000000  0.45286100
## spread   -0.039865282  0.81628323  0.452861001  1.00000000
##
## , , 2023-07-31
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 0.002251827 -0.02775047 -0.02264566
## ESDE      0.002251827 1.000000000  0.45056296  0.81837152
## IEDE     -0.027750474 0.450562962  1.00000000  0.42188268
## spread   -0.022645658 0.818371516  0.42188268  1.00000000
##
```

```
## , , 2023-08-01
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.00431889 -0.05628575 -0.02246331
## ESDE     -0.00431889  1.00000000  0.45388843  0.81687923
## IEDE     -0.05628575  0.45388843  1.00000000  0.40657757
## spread   -0.02246331  0.81687923  0.40657757  1.00000000
##
## , , 2023-08-02
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.001520437 -0.04813555 -0.02085616
## ESDE     -0.001520437  1.000000000  0.45966399  0.81490567
## IEDE     -0.048135550  0.459663992  1.00000000  0.41116773
## spread   -0.020856156  0.814905675  0.41116773  1.00000000
##
## , , 2023-08-03
##
##              eurois3m       ESDE        IEDE       spread
## eurois3m  1.000000000 0.01806074 -0.03516479 -0.005364603
## ESDE      0.018060745 1.00000000  0.46565725  0.819176499
## IEDE     -0.035164793 0.46565725  1.00000000  0.418730555
## spread   -0.005364603 0.81917650  0.41873056  1.000000000
##
## , , 2023-08-04
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.01327936 -0.00146234 -0.03642296
## ESDE      0.01327936 1.00000000  0.45162207  0.80292369
## IEDE     -0.00146234 0.45162207  1.00000000  0.35188566
## spread   -0.03642296 0.80292369  0.35188566  1.00000000
##
## , , 2023-08-07
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 0.005510187 -0.01199565 -0.04115416
## ESDE      0.005510187 1.000000000  0.47411152  0.80899868
## IEDE     -0.011995652 0.474111517  1.00000000  0.38527818
## spread   -0.041154157 0.808998677  0.38527818  1.00000000
##
## , , 2023-08-08
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.02425369 -0.004082242 -0.02495118
## ESDE      0.024253690 1.00000000  0.476609297  0.81249957
## IEDE     -0.004082242 0.47660930  1.000000000  0.39433013
## spread   -0.024951183 0.81249957  0.394330132  1.00000000
##
## , , 2023-08-09
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01104083 0.01889099 -0.01613104
```

```
## ESDE      0.01104083 1.00000000 0.47304826  0.81076707
## IEDE      0.01889099 0.47304826 1.00000000  0.40085208
## spread   -0.01613104 0.81076707 0.40085208  1.00000000
##
## , , 2023-08-10
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.02924132 0.07259984 0.05713669
## ESDE     -0.02924132  1.00000000 0.37573934 0.64306953
## IEDE      0.07259984  0.37573934 1.00000000 0.47400280
## spread    0.05713669  0.64306953 0.47400280 1.00000000
##
## , , 2023-08-11
##
##             eurois3m        ESDE      IEDE     spread
## eurois3m  1.00000000 -0.06464133 0.1333097 0.01739143
## ESDE     -0.06464133  1.00000000 0.3507219 0.66399857
## IEDE      0.13330971  0.35072195 1.0000000 0.44072242
## spread    0.01739143  0.66399857 0.4407224 1.00000000
##
## , , 2023-08-14
##
##             eurois3m       ESDE      IEDE     spread
## eurois3m  1.00000000 -0.0549404 0.1244976 0.01324089
## ESDE     -0.05494040  1.0000000 0.3691627 0.67385856
## IEDE      0.12449764  0.3691627 1.0000000 0.42495051
## spread    0.01324089  0.6738586 0.4249505 1.00000000
##
## , , 2023-08-15
##
##             eurois3m        ESDE      IEDE     spread
## eurois3m  1.00000000 -0.04968167 0.1250141 0.01158368
## ESDE     -0.04968167  1.00000000 0.3761294 0.68856123
## IEDE      0.12501409  0.37612942 1.0000000 0.42473222
## spread    0.01158368  0.68856123 0.4247322 1.00000000
##
## , , 2023-08-16
##
##             eurois3m        ESDE      IEDE      spread
## eurois3m  1.00000000 -0.09212623 0.1028613 -0.04185276
## ESDE     -0.09212623  1.00000000 0.3868303  0.72052644
## IEDE      0.10286132  0.38683026 1.0000000  0.42433245
## spread   -0.04185276  0.72052644 0.4243324  1.00000000
##
## , , 2023-08-17
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.08397086 0.09469712 -0.03919363
## ESDE     -0.08397086  1.00000000 0.39384936  0.72812973
## IEDE      0.09469712  0.39384936 1.00000000  0.42644832
## spread   -0.03919363  0.72812973 0.42644832  1.00000000
##
```

```
## , , 2023-08-18
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.07438592 0.08686051 -0.03658948
## ESDE     -0.07438592  1.00000000 0.40290092  0.73790901
## IEDE      0.08686051  0.40290092 1.00000000  0.42819072
## spread   -0.03658948  0.73790901 0.42819072  1.00000000
##
## , , 2023-08-21
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.05471844 0.07264847 -0.02240139
## ESDE     -0.05471844  1.00000000 0.41022744  0.74724069
## IEDE      0.07264847  0.41022744 1.00000000  0.42705500
## spread   -0.02240139  0.74724069 0.42705500  1.00000000
##
## , , 2023-08-22
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m  1.000000000 -0.03294117 0.06000489 -0.009033168
## ESDE     -0.032941166  1.00000000 0.41782395  0.755456592
## IEDE      0.060004888  0.41782395 1.00000000  0.427931303
## spread   -0.009033168  0.75545659 0.42793130  1.000000000
##
## , , 2023-08-23
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 -0.01485538 0.05794382 0.004590209
## ESDE     -0.014855382  1.00000000 0.42597293 0.772282751
## IEDE      0.057943823  0.42597293 1.00000000 0.428822798
## spread    0.004590209  0.77228275 0.42882280 1.000000000
##
## , , 2023-08-24
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.07226251 0.002712143 -0.03571261
## ESDE     -0.072262512  1.00000000 0.439596667  0.77830697
## IEDE      0.002712143  0.43959667 1.000000000  0.43491202
## spread   -0.035712613  0.77830697 0.434912021  1.00000000
##
## , , 2023-08-25
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 -0.0709135 -0.001975829 -0.04574635
## ESDE     -0.070913498  1.0000000  0.448214764  0.78335303
## IEDE     -0.001975829  0.4482148  1.000000000  0.43965824
## spread   -0.045746353  0.7833530  0.439658244  1.00000000
##
## , , 2023-08-28
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.06073502 -0.004281134 -0.04372875
```

```
## ESDE     -0.060735020  1.00000000  0.451880800  0.78623396
## IEDE     -0.004281134  0.45188080  1.000000000  0.44132098
## spread   -0.043728745  0.78623396  0.441320985  1.00000000
##
## , , 2023-08-29
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02854045 -0.04922101 -0.01719107
## ESDE     -0.02854045  1.00000000  0.41852921  0.79197633
## IEDE     -0.04922101  0.41852921  1.00000000  0.40575171
## spread   -0.01719107  0.79197633  0.40575171  1.00000000
##
## , , 2023-08-30
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 -0.01837206 -0.03638256 -0.006700657
## ESDE     -0.018372059  1.00000000  0.43476312  0.796724697
## IEDE     -0.036382557  0.43476312  1.00000000  0.422476248
## spread   -0.006700657  0.79672470  0.42247625  1.000000000
##
## , , 2023-08-31
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 -0.01646962 -0.05368109 -0.008672552
## ESDE     -0.016469623  1.00000000  0.42080672  0.799044989
## IEDE     -0.053681095  0.42080672  1.00000000  0.406858232
## spread   -0.008672552  0.79904499  0.40685823  1.000000000
##
## , , 2023-09-01
##
##               eurois3m         ESDE        IEDE     spread
## eurois3m  1.0000000000 0.0008773652 -0.08019687 0.01250317
## ESDE      0.0008773652 1.0000000000  0.40568485 0.80208055
## IEDE     -0.0801968741 0.4056848522  1.00000000 0.37970632
## spread    0.0125031682 0.8020805499  0.37970632 1.00000000
##
## , , 2023-09-04
##
##              eurois3m        ESDE       IEDE     spread
## eurois3m  1.000000000 0.005002175 -0.0695205 0.01425203
## ESDE      0.005002175 1.000000000  0.4183341 0.80721955
## IEDE     -0.069520505 0.418334137  1.0000000 0.38874472
## spread    0.014252025 0.807219554  0.3887447 1.00000000
##
## , , 2023-09-05
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.01769401 -0.05265628 0.03118278
## ESDE      0.01769401 1.00000000  0.43520191 0.81120830
## IEDE     -0.05265628 0.43520191  1.00000000 0.40606995
## spread    0.03118278 0.81120830  0.40606995 1.00000000
##
```

```
## , , 2023-09-06
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.02264372 -0.06316253 0.02709652
## ESDE      0.02264372 1.00000000  0.42827935 0.81085133
## IEDE     -0.06316253 0.42827935  1.00000000 0.40521934
## spread    0.02709652 0.81085133  0.40521934 1.00000000
##
## , , 2023-09-07
##
##              eurois3m         ESDE        IEDE     spread
## eurois3m  1.000000000 -0.002274227 -0.06987412 -0.0124560
## ESDE     -0.002274227  1.000000000  0.44450697  0.8158522
## IEDE     -0.069874117  0.444506971  1.00000000  0.4211401
## spread   -0.012455999  0.815852242  0.42114012  1.0000000
##
## , , 2023-09-08
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.02186092 -0.04875478 0.001874623
## ESDE      0.021860922 1.00000000  0.46521677 0.819253640
## IEDE     -0.048754777 0.46521677  1.00000000 0.436226436
## spread    0.001874623 0.81925364  0.43622644 1.000000000
##
## , , 2023-09-11
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01511565 -0.0730783 -0.01020461
## ESDE      0.01511565 1.00000000  0.4652367  0.81896386
## IEDE     -0.07307830 0.46523670  1.0000000  0.44224477
## spread   -0.01020461 0.81896386  0.4422448  1.00000000
##
## , , 2023-09-12
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01954504 -0.0648333 -0.00513928
## ESDE      0.01954504 1.00000000  0.4592887  0.82187736
## IEDE     -0.06483330 0.45928869  1.0000000  0.43339255
## spread   -0.00513928 0.82187736  0.4333925  1.00000000
##
## , , 2023-09-13
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.004964165 -0.1042950 -0.00661013
## ESDE     -0.004964165  1.000000000  0.4672976  0.81962138
## IEDE     -0.104295047  0.467297601  1.0000000  0.43052210
## spread   -0.006610130  0.819621379  0.4305221  1.00000000
##
## , , 2023-09-14
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02574303 -0.07252005 -0.04069488
```

```
## ESDE     -0.02574303  1.00000000  0.43834947  0.82396462
## IEDE     -0.07252005  0.43834947  1.00000000  0.38339447
## spread   -0.04069488  0.82396462  0.38339447  1.00000000
##
## , , 2023-09-15
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.06237247 -0.01015666 0.05426048
## ESDE      0.06237247 1.00000000  0.46211354 0.83433051
## IEDE     -0.01015666 0.46211354  1.00000000 0.41091516
## spread    0.05426048 0.83433051  0.41091516 1.00000000
##
## , , 2023-09-18
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.02801318 -0.02382276 0.01766347
## ESDE      0.02801318 1.00000000  0.48200128 0.84244162
## IEDE     -0.02382276 0.48200128  1.00000000 0.43356339
## spread    0.01766347 0.84244162  0.43356339 1.00000000
##
## , , 2023-09-19
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.02830929 -0.02132445 0.01840602
## ESDE      0.02830929 1.00000000  0.48193956 0.83966007
## IEDE     -0.02132445 0.48193956  1.00000000 0.43183823
## spread    0.01840602 0.83966007  0.43183823 1.00000000
##
## , , 2023-09-20
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.01797933 -0.01428079 0.006875884
## ESDE      0.017979333 1.00000000  0.48108103 0.837925741
## IEDE     -0.014280791 0.48108103  1.00000000 0.431402179
## spread    0.006875884 0.83792574  0.43140218 1.000000000
##
## , , 2023-09-21
##
##              eurois3m       ESDE        IEDE       spread
## eurois3m  1.000000000 0.01247824 -0.01800965 -0.001585364
## ESDE      0.012478240 1.00000000  0.48852699  0.837098504
## IEDE     -0.018009653 0.48852699  1.00000000  0.442309577
## spread   -0.001585364 0.83709850  0.44230958  1.000000000
##
## , , 2023-09-22
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02401687 0.01198448 0.03776837
## ESDE     0.02401687 1.00000000 0.49234074 0.82698217
## IEDE     0.01198448 0.49234074 1.00000000 0.45860182
## spread   0.03776837 0.82698217 0.45860182 1.00000000
##
```

```
## , , 2023-09-25
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.03726549 0.0274626 0.05246822
## ESDE     0.03726549 1.00000000 0.5058568 0.83016108
## IEDE     0.02746260 0.50585677 1.0000000 0.47639818
## spread   0.05246822 0.83016108 0.4763982 1.00000000
##
## , , 2023-09-26
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03362618 0.02334805 0.06351757
## ESDE     0.03362618 1.00000000 0.50485471 0.82327726
## IEDE     0.02334805 0.50485471 1.00000000 0.46905966
## spread   0.06351757 0.82327726 0.46905966 1.00000000
##
## , , 2023-09-27
##
##              eurois3m         ESDE       IEDE     spread
## eurois3m  1.000000000 -0.004133526 0.03371658 0.01100598
## ESDE     -0.004133526  1.000000000 0.48312560 0.82914604
## IEDE      0.033716583  0.483125604 1.00000000 0.44008300
## spread    0.011005980  0.829146038 0.44008300 1.00000000
##
## , , 2023-09-28
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01346036 0.02853128 0.03584256
## ESDE     0.01346036 1.00000000 0.48005096 0.82836687
## IEDE     0.02853128 0.48005096 1.00000000 0.43063468
## spread   0.03584256 0.82836687 0.43063468 1.00000000
##
## , , 2023-09-29
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01338778 0.02697228 0.02639508
## ESDE     0.01338778 1.00000000 0.48191957 0.82508103
## IEDE     0.02697228 0.48191957 1.00000000 0.43187178
## spread   0.02639508 0.82508103 0.43187178 1.00000000
##
## , , 2023-10-02
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02015531 0.02450707 0.02748807
## ESDE     0.02015531 1.00000000 0.47912497 0.82490016
## IEDE     0.02450707 0.47912497 1.00000000 0.43113514
## spread   0.02748807 0.82490016 0.43113514 1.00000000
##
## , , 2023-10-03
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02659451 0.02473024 0.03413084
```

```
## ESDE     0.02659451 1.00000000 0.47912491 0.82658172
## IEDE     0.02473024 0.47912491 1.00000000 0.42762703
## spread   0.03413084 0.82658172 0.42762703 1.00000000
##
## , , 2023-10-04
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.04800578 0.009516945 0.06478347
## ESDE     0.048005777 1.00000000 0.450283123 0.82969714
## IEDE     0.009516945 0.45028312 1.000000000 0.37821109
## spread   0.064783468 0.82969714 0.378211091 1.00000000
##
## , , 2023-10-05
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.04277174 0.009638211 0.05544042
## ESDE     0.042771739 1.00000000 0.465049063 0.83252971
## IEDE     0.009638211 0.46504906 1.000000000 0.39661692
## spread   0.055440421 0.83252971 0.396616925 1.00000000
##
## , , 2023-10-06
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05225676 0.02581642 0.05735186
## ESDE     0.05225676 1.00000000 0.47256478 0.83121759
## IEDE     0.02581642 0.47256478 1.00000000 0.40504105
## spread   0.05735186 0.83121759 0.40504105 1.00000000
##
## , , 2023-10-09
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.06804716 0.1118429 0.08578934
## ESDE     0.06804716 1.00000000 0.4644464 0.83016384
## IEDE     0.11184288 0.46444636 1.0000000 0.45434268
## spread   0.08578934 0.83016384 0.4543427 1.00000000
##
## , , 2023-10-10
##
##            eurois3m       ESDE       IEDE    spread
## eurois3m 1.00000000 0.09754607 0.08308044 0.0922433
## ESDE     0.09754607 1.00000000 0.41468572 0.8252525
## IEDE     0.08308044 0.41468572 1.00000000 0.4321832
## spread   0.09224330 0.82525246 0.43218321 1.0000000
##
## , , 2023-10-11
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04455398 0.06079579 0.03114644
## ESDE     0.04455398 1.00000000 0.43777951 0.84581166
## IEDE     0.06079579 0.43777951 1.00000000 0.44656160
## spread   0.03114644 0.84581166 0.44656160 1.00000000
##
```

```
## , , 2023-10-12
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.09548022 0.1115720 0.06170342
## ESDE     0.09548022 1.00000000 0.4621325 0.84437424
## IEDE     0.11157196 0.46213251 1.0000000 0.45755554
## spread   0.06170342 0.84437424 0.4575555 1.00000000
##
## , , 2023-10-13
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.0880085 0.1029598 0.05573087
## ESDE     0.08800850 1.0000000 0.4710853 0.84435796
## IEDE     0.10295977 0.4710853 1.0000000 0.46232069
## spread   0.05573087 0.8443580 0.4623207 1.00000000
##
## , , 2023-10-16
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.1115137 0.1198648 0.07740112
## ESDE     0.11151375 1.0000000 0.4978854 0.84904193
## IEDE     0.11986484 0.4978854 1.0000000 0.48320407
## spread   0.07740112 0.8490419 0.4832041 1.00000000
##
## , , 2023-10-17
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.1064047 0.1110053 0.07448919
## ESDE     0.10640473 1.0000000 0.4967661 0.85096118
## IEDE     0.11100534 0.4967661 1.0000000 0.47830829
## spread   0.07448919 0.8509612 0.4783083 1.00000000
##
## , , 2023-10-18
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.09158566 0.1065490 0.05374116
## ESDE     0.09158566 1.00000000 0.4949989 0.84711016
## IEDE     0.10654904 0.49499891 1.0000000 0.47092513
## spread   0.05374116 0.84711016 0.4709251 1.00000000
##
## , , 2023-10-19
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.1044708 0.1188511 0.08646323
## ESDE     0.10447082 1.0000000 0.5009951 0.84437896
## IEDE     0.11885112 0.5009951 1.0000000 0.47840198
## spread   0.08646323 0.8443790 0.4784020 1.00000000
##
## , , 2023-10-20
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0698084 0.09559095 0.07057319
```

```
## ESDE     0.06980840 1.0000000 0.50605107 0.83547806
## IEDE     0.09559095 0.5060511 1.00000000 0.47708183
## spread   0.07057319 0.8354781 0.47708183 1.00000000
##
## , , 2023-10-23
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07504559 0.06579839 0.07534382
## ESDE     0.07504559 1.00000000 0.50080717 0.83397809
## IEDE     0.06579839 0.50080717 1.00000000 0.46964549
## spread   0.07534382 0.83397809 0.46964549 1.00000000
##
## , , 2023-10-24
##
##            eurois3m       ESDE       IEDE    spread
## eurois3m 1.00000000 0.08976517 0.06455376 0.1044617
## ESDE     0.08976517 1.00000000 0.49956158 0.8326486
## IEDE     0.06455376 0.49956158 1.00000000 0.4616850
## spread   0.10446172 0.83264858 0.46168503 1.0000000
##
## , , 2023-10-25
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06252882 0.08492779 0.07337364
## ESDE     0.06252882 1.00000000 0.47057914 0.83311829
## IEDE     0.08492779 0.47057914 1.00000000 0.43253797
## spread   0.07337364 0.83311829 0.43253797 1.00000000
##
## , , 2023-10-26
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.0465628 0.0373962 0.04695045
## ESDE     0.04656280 1.0000000 0.4753952 0.83208206
## IEDE     0.03739620 0.4753952 1.0000000 0.45050196
## spread   0.04695045 0.8320821 0.4505020 1.00000000
##
## , , 2023-10-27
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03860013 0.02910798 0.03627494
## ESDE     0.03860013 1.00000000 0.48503997 0.83242678
## IEDE     0.02910798 0.48503997 1.00000000 0.45908923
## spread   0.03627494 0.83242678 0.45908923 1.00000000
##
## , , 2023-10-30
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03676722 0.02871921 0.03232648
## ESDE     0.03676722 1.00000000 0.49940959 0.83429702
## IEDE     0.02871921 0.49940959 1.00000000 0.47103954
## spread   0.03232648 0.83429702 0.47103954 1.00000000
##
```

```
## , , 2023-10-31
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.002938088 0.03736383 -0.02046427
## ESDE     -0.002938088  1.000000000 0.47821180  0.83910882
## IEDE      0.037363828  0.478211798 1.00000000  0.43911102
## spread   -0.020464273  0.839108817 0.43911102  1.00000000
##
## , , 2023-11-01
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01096643 0.03936936 -0.01074251
## ESDE      0.01096643 1.00000000 0.48062823  0.83703931
## IEDE      0.03936936 0.48062823 1.00000000  0.44081967
## spread   -0.01074251 0.83703931 0.44081967  1.00000000
##
## , , 2023-11-02
##
##              eurois3m      ESDE       IEDE       spread
## eurois3m  1.000000000 0.0153168 0.03130615 -0.001185269
## ESDE      0.015316800 1.0000000 0.47751576  0.834007450
## IEDE      0.031306151 0.4775158 1.00000000  0.430595866
## spread   -0.001185269 0.8340074 0.43059587  1.000000000
##
## , , 2023-11-03
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.01245267 0.004086858 -0.02505404
## ESDE     -0.012452670  1.00000000 0.501141968  0.83812065
## IEDE      0.004086858  0.50114197 1.000000000  0.45265176
## spread   -0.025054035  0.83812065 0.452651758  1.00000000
##
## , , 2023-11-06
##
##              eurois3m         ESDE       IEDE       spread
## eurois3m 1.0000000000 0.0007794187 0.02277767 0.0006109193
## ESDE     0.0007794187 1.0000000000 0.51226544 0.8307138474
## IEDE     0.0227776656 0.5122654370 1.00000000 0.4851746226
## spread   0.0006109193 0.8307138474 0.48517462 1.0000000000
##
## , , 2023-11-07
##
##             eurois3m        ESDE      IEDE      spread
## eurois3m 1.000000000 0.005207223 0.0246460 0.001648617
## ESDE     0.005207223 1.000000000 0.5260236 0.831501552
## IEDE     0.024646003 0.526023593 1.0000000 0.492091982
## spread   0.001648617 0.831501552 0.4920920 1.000000000
##
## , , 2023-11-08
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m  1.000000000 0.004823961 0.02187506 -0.001213457
```

```
## ESDE      0.004823961 1.000000000 0.52372547  0.830150574
## IEDE      0.021875064 0.523725471 1.00000000  0.488949989
## spread   -0.001213457 0.830150574 0.48894999  1.000000000
##
## , , 2023-11-09
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m  1.000000000 0.001184883 0.01759209 -0.006759757
## ESDE      0.001184883 1.000000000 0.52940526  0.832791848
## IEDE      0.017592094 0.529405260 1.00000000  0.494329419
## spread   -0.006759757 0.832791848 0.49432942  1.000000000
##
## , , 2023-11-10
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01236805 0.01154497 -0.008025083
## ESDE      0.012368053 1.00000000 0.52348192  0.829662364
## IEDE      0.011544968 0.52348192 1.00000000  0.489693163
## spread   -0.008025083 0.82966236 0.48969316  1.000000000
##
## , , 2023-11-13
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.02319655 0.01099979 -0.004434025
## ESDE      0.023196547 1.00000000 0.51945169  0.827815318
## IEDE      0.010999788 0.51945169 1.00000000  0.485312382
## spread   -0.004434025 0.82781532 0.48531238  1.000000000
##
## , , 2023-11-14
##
##               eurois3m       ESDE          IEDE      spread
## eurois3m  1.0000000000 0.01400069 -0.0009321502 -0.02328067
## ESDE      0.0140006886 1.00000000  0.5192893281  0.82553247
## IEDE     -0.0009321502 0.51928933  1.0000000000  0.48754131
## spread   -0.0232806679 0.82553247  0.4875413051  1.00000000
##
## , , 2023-11-15
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 0.001799315 -0.01853792 -0.03718216
## ESDE      0.001799315 1.000000000  0.52521197  0.82688199
## IEDE     -0.018537925 0.525211968  1.00000000  0.49666988
## spread   -0.037182158 0.826881993  0.49666988  1.00000000
##
## , , 2023-11-16
##
##               eurois3m          ESDE        IEDE      spread
## eurois3m  1.0000000000 -0.0003600196 -0.01562961 -0.03579051
## ESDE     -0.0003600196  1.0000000000  0.52311652  0.82573187
## IEDE     -0.0156296109  0.5231165224  1.00000000  0.49425163
## spread   -0.0357905064  0.8257318732  0.49425163  1.00000000
##
```

```
## , , 2023-11-17
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03353336 -0.00430632 -0.05747259
## ESDE     -0.03353336  1.00000000  0.47590203  0.83367244
## IEDE     -0.00430632  0.47590203  1.00000000  0.46205338
## spread   -0.05747259  0.83367244  0.46205338  1.00000000
##
## , , 2023-11-20
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.03476689 -0.002655153 -0.05615834
## ESDE     -0.034766895  1.00000000  0.477959421  0.83198547
## IEDE     -0.002655153  0.47795942  1.000000000  0.46071900
## spread   -0.056158341  0.83198547  0.460718995  1.00000000
##
## , , 2023-11-21
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.04692485 0.1043595 0.05362645
## ESDE     0.04692485 1.00000000 0.5129358 0.83867365
## IEDE     0.10435949 0.51293578 1.0000000 0.50859805
## spread   0.05362645 0.83867365 0.5085981 1.00000000
##
## , , 2023-11-22
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04088834 0.09324605 0.04154653
## ESDE     0.04088834 1.00000000 0.51389621 0.83598301
## IEDE     0.09324605 0.51389621 1.00000000 0.50535298
## spread   0.04154653 0.83598301 0.50535298 1.00000000
##
## , , 2023-11-23
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03362228 0.08789167 0.03241626
## ESDE     0.03362228 1.00000000 0.51058127 0.83413791
## IEDE     0.08789167 0.51058127 1.00000000 0.49832777
## spread   0.03241626 0.83413791 0.49832777 1.00000000
##
## , , 2023-11-24
##
##            eurois3m       ESDE       IEDE    spread
## eurois3m 1.00000000 0.03350549 0.08695374 0.0305209
## ESDE     0.03350549 1.00000000 0.50909134 0.8322934
## IEDE     0.08695374 0.50909134 1.00000000 0.4933257
## spread   0.03052090 0.83229337 0.49332569 1.0000000
##
## , , 2023-11-27
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.06555356 0.1319337 0.06004347
```

```
## ESDE     0.06555356 1.00000000 0.5161549 0.83259364
## IEDE     0.13193372 0.51615490 1.0000000 0.49723429
## spread   0.06004347 0.83259364 0.4972343 1.00000000
##
## , , 2023-11-28
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.06345147 0.1233553 0.05541898
## ESDE     0.06345147 1.00000000 0.5253643 0.83221743
## IEDE     0.12335531 0.52536431 1.0000000 0.49821527
## spread   0.05541898 0.83221743 0.4982153 1.00000000
##
## , , 2023-11-29
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.06190745 0.1129366 0.05176421
## ESDE     0.06190745 1.00000000 0.5216632 0.83111221
## IEDE     0.11293655 0.52166325 1.0000000 0.49279875
## spread   0.05176421 0.83111221 0.4927988 1.00000000
##
## , , 2023-11-30
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.0516173 0.1055874 0.02708031
## ESDE     0.05161730 1.0000000 0.5181858 0.82813446
## IEDE     0.10558736 0.5181858 1.0000000 0.48442303
## spread   0.02708031 0.8281345 0.4844230 1.00000000
##
## , , 2023-12-01
##
##            eurois3m       ESDE      IEDE    spread
## eurois3m 1.00000000 0.06170248 0.0850306 0.0347132
## ESDE     0.06170248 1.00000000 0.4386741 0.8348751
## IEDE     0.08503060 0.43867406 1.0000000 0.4263843
## spread   0.03471320 0.83487507 0.4263843 1.0000000
##
## , , 2023-12-04
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.006630363 0.06227808 -0.01479736
## ESDE      0.006630363 1.000000000 0.44718589  0.84034007
## IEDE      0.062278083 0.447185889 1.00000000  0.43221232
## spread   -0.014797359 0.840340070 0.43221232  1.00000000
##
## , , 2023-12-05
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.004781342 0.06161405 -0.01777309
## ESDE      0.004781342 1.000000000 0.43847295  0.83877441
## IEDE      0.061614054 0.438472951 1.00000000  0.41588696
## spread   -0.017773091 0.838774414 0.41588696  1.00000000
##
```

```
## , , 2023-12-06
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.003862478 0.04612937 -0.02027887
## ESDE      0.003862478 1.000000000 0.44421654  0.83652812
## IEDE      0.046129368 0.444216539 1.00000000  0.42972215
## spread   -0.020278874 0.836528121 0.42972215  1.00000000
##
## , , 2023-12-07
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.00562671 0.05627624 -0.02058697
## ESDE      0.00562671 1.00000000 0.44876683  0.83427860
## IEDE      0.05627624 0.44876683 1.00000000  0.42984418
## spread   -0.02058697 0.83427860 0.42984418  1.00000000
##
## , , 2023-12-08
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.00634885 0.04742866 -0.0217992
## ESDE      0.00634885 1.00000000 0.45377670  0.8321641
## IEDE      0.04742866 0.45377670 1.00000000  0.4325396
## spread   -0.02179920 0.83216410 0.43253956  1.0000000
##
## , , 2023-12-11
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 -0.0364757 0.03224604 -0.08416747
## ESDE     -0.03647570  1.0000000 0.45875824  0.83619520
## IEDE      0.03224604  0.4587582 1.00000000  0.42904486
## spread   -0.08416747  0.8361952 0.42904486  1.00000000
##
## , , 2023-12-12
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.03223655 0.03071331 -0.0760476
## ESDE     -0.03223655  1.00000000 0.46214487  0.8319258
## IEDE      0.03071331  0.46214487 1.00000000  0.4313108
## spread   -0.07604760  0.83192584 0.43131079  1.0000000
##
## , , 2023-12-13
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.01346242 0.1621745 -0.04477888
## ESDE      0.01346242 1.00000000 0.4704567  0.83021252
## IEDE      0.16217452 0.47045669 1.0000000  0.41549587
## spread   -0.04477888 0.83021252 0.4154959  1.00000000
##
## , , 2023-12-14
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.04141477 0.1497805 -0.02476212
```

```
## ESDE      0.04141477 1.00000000 0.4697360  0.82894044
## IEDE      0.14978054 0.46973602 1.0000000  0.41894127
## spread   -0.02476212 0.82894044 0.4189413  1.00000000
##
## , , 2023-12-15
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.02277865 0.1350240 -0.03836318
## ESDE      0.02277865 1.00000000 0.4563505  0.84863644
## IEDE      0.13502398 0.45635052 1.0000000  0.38605524
## spread   -0.03836318 0.84863644 0.3860552  1.00000000
##
## , , 2023-12-18
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.01201033 0.1076654 -0.04656844
## ESDE      0.01201033 1.00000000 0.4689330  0.84749962
## IEDE      0.10766541 0.46893298 1.0000000  0.40362335
## spread   -0.04656844 0.84749962 0.4036234  1.00000000
##
## , , 2023-12-19
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.006998024 0.09923231 -0.05368631
## ESDE      0.006998024 1.000000000 0.47042041  0.84382848
## IEDE      0.099232310 0.470420415 1.00000000  0.40406894
## spread   -0.053686307 0.843828484 0.40406894  1.00000000
##
## , , 2023-12-20
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.04427539 0.1225448 -0.01474324
## ESDE      0.04427539 1.00000000 0.5104136  0.85088450
## IEDE      0.12254478 0.51041363 1.0000000  0.44551077
## spread   -0.01474324 0.85088450 0.4455108  1.00000000
##
## , , 2023-12-21
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.04459443 0.1037540 -0.01821168
## ESDE      0.04459443 1.00000000 0.5054696  0.84521066
## IEDE      0.10375396 0.50546958 1.0000000  0.44718692
## spread   -0.01821168 0.84521066 0.4471869  1.00000000
##
## , , 2023-12-22
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.03750729 0.09416081 -0.01972442
## ESDE      0.03750729 1.00000000 0.50369333  0.84290774
## IEDE      0.09416081 0.50369333 1.00000000  0.44750169
## spread   -0.01972442 0.84290774 0.44750169  1.00000000
##
```

```
## , , 2023-12-27
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.005238634 0.06779127 -0.06106206
## ESDE      0.005238634 1.000000000 0.50679294  0.84244634
## IEDE      0.067791270 0.506792936 1.00000000  0.45208445
## spread   -0.061062056 0.842446343 0.45208445  1.00000000
##
## , , 2023-12-28
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01421174 0.08114038 -0.05072203
## ESDE      0.01421174 1.00000000 0.47253541  0.84035351
## IEDE      0.08114038 0.47253541 1.00000000  0.39368119
## spread   -0.05072203 0.84035351 0.39368119  1.00000000
##
## , , 2023-12-29
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02152223 0.04503221 -0.08473382
## ESDE     -0.02152223  1.00000000 0.48736378  0.84398494
## IEDE      0.04503221  0.48736378 1.00000000  0.41547733
## spread   -0.08473382  0.84398494 0.41547733  1.00000000
##
## , , 2024-01-02
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01949187 0.02518826 -0.07115556
## ESDE     -0.01949187  1.00000000 0.47966499  0.83611830
## IEDE      0.02518826  0.47966499 1.00000000  0.38622997
## spread   -0.07115556  0.83611830 0.38622997  1.00000000
##
## , , 2024-01-03
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02344875 0.02506853 -0.06403072
## ESDE     -0.02344875  1.00000000 0.47441189  0.82579215
## IEDE      0.02506853  0.47441189 1.00000000  0.39071906
## spread   -0.06403072  0.82579215 0.39071906  1.00000000
##
## , , 2024-01-04
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.09808365 0.03890341 0.04582683
## ESDE     0.09808365 1.00000000 0.46635119 0.83671053
## IEDE     0.03890341 0.46635119 1.00000000 0.39278311
## spread   0.04582683 0.83671053 0.39278311 1.00000000
##
## , , 2024-01-05
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.08961518 0.03944111 0.04294357
```

```
## ESDE     0.08961518 1.00000000 0.46666717 0.83193631
## IEDE     0.03944111 0.46666717 1.00000000 0.40022306
## spread   0.04294357 0.83193631 0.40022306 1.00000000
##
## , , 2024-01-08
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07883584 0.03850796 0.03547212
## ESDE     0.07883584 1.00000000 0.46323542 0.83142209
## IEDE     0.03850796 0.46323542 1.00000000 0.40047464
## spread   0.03547212 0.83142209 0.40047464 1.00000000
##
## , , 2024-01-09
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07213019 0.03373631 0.03265194
## ESDE     0.07213019 1.00000000 0.46437188 0.82803316
## IEDE     0.03373631 0.46437188 1.00000000 0.39647404
## spread   0.03265194 0.82803316 0.39647404 1.00000000
##
## , , 2024-01-10
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06796087 0.03156552 0.03063104
## ESDE     0.06796087 1.00000000 0.45452218 0.83104697
## IEDE     0.03156552 0.45452218 1.00000000 0.38978554
## spread   0.03063104 0.83104697 0.38978554 1.00000000
##
## , , 2024-01-11
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0665636 0.02610813 0.03815744
## ESDE     0.06656360 1.0000000 0.45373369 0.82686098
## IEDE     0.02610813 0.4537337 1.00000000 0.38119477
## spread   0.03815744 0.8268610 0.38119477 1.00000000
##
## , , 2024-01-12
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.03232774 0.01253763 0.006927646
## ESDE     0.032327741 1.00000000 0.46221818 0.829747308
## IEDE     0.012537632 0.46221818 1.00000000 0.393203967
## spread   0.006927646 0.82974731 0.39320397 1.000000000
##
## , , 2024-01-15
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03577339 0.02269688 0.01666212
## ESDE     0.03577339 1.00000000 0.47049654 0.82737421
## IEDE     0.02269688 0.47049654 1.00000000 0.41262587
## spread   0.01666212 0.82737421 0.41262587 1.00000000
##
```

```
## , , 2024-01-16
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m 1.0000000000 0.01989001 0.0008569488 0.004934423
## ESDE     0.0198900132 1.00000000 0.4759506775 0.826483644
## IEDE     0.0008569488 0.47595068 1.0000000000 0.418113719
## spread   0.0049344229 0.82648364 0.4181137187 1.000000000
##
## , , 2024-01-17
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.01916699 0.005593187 -0.01038638
## ESDE      0.019166993 1.00000000 0.478011746  0.82494197
## IEDE      0.005593187 0.47801175 1.000000000  0.42102323
## spread   -0.010386382 0.82494197 0.421023226  1.00000000
##
## , , 2024-01-18
##
##              eurois3m         ESDE         IEDE      spread
## eurois3m  1.000000000 -0.006224968 -0.009420564 -0.03722167
## ESDE     -0.006224968  1.000000000  0.495021125  0.83322980
## IEDE     -0.009420564  0.495021125  1.000000000  0.44229822
## spread   -0.037221666  0.833229802  0.442298217  1.00000000
##
## , , 2024-01-19
##
##             eurois3m      ESDE        IEDE     spread
## eurois3m  1.00000000 0.1752860 -0.09275243 -0.1075734
## ESDE      0.17528598 1.0000000  0.33954083  0.6490042
## IEDE     -0.09275243 0.3395408  1.00000000  0.4655800
## spread   -0.10757335 0.6490042  0.46558003  1.0000000
##
## , , 2024-01-22
##
##             eurois3m      ESDE        IEDE      spread
## eurois3m  1.00000000 0.1526385 -0.09202764 -0.08576397
## ESDE      0.15263850 1.0000000  0.35977976  0.64861420
## IEDE     -0.09202764 0.3597798  1.00000000  0.44610246
## spread   -0.08576397 0.6486142  0.44610246  1.00000000
##
## , , 2024-01-23
##
##             eurois3m      ESDE        IEDE      spread
## eurois3m  1.00000000 0.1378675 -0.08761314 -0.07972703
## ESDE      0.13786751 1.0000000  0.37507804  0.66541461
## IEDE     -0.08761314 0.3750780  1.00000000  0.44633265
## spread   -0.07972703 0.6654146  0.44633265  1.00000000
##
## , , 2024-01-24
##
##             eurois3m      ESDE        IEDE      spread
## eurois3m  1.00000000 0.1278418 -0.07705212 -0.06800182
```

```
## ESDE      0.12784184 1.0000000  0.38741184  0.67988074
## IEDE     -0.07705212 0.3874118  1.00000000  0.44572262
## spread   -0.06800182 0.6798807  0.44572262  1.00000000
##
## , , 2024-01-25
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m  1.00000000 0.1203966 -0.0698829 -0.05503616
## ESDE      0.12039657 1.0000000  0.3966219  0.69473136
## IEDE     -0.06988290 0.3966219  1.0000000  0.44172443
## spread   -0.05503616 0.6947314  0.4417244  1.00000000
##
## , , 2024-01-26
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m  1.0000000 0.1086378 -0.0616516 -0.0513661
## ESDE      0.1086378 1.0000000  0.4172826  0.7085140
## IEDE     -0.0616516 0.4172826  1.0000000  0.4765876
## spread   -0.0513661 0.7085140  0.4765876  1.0000000
##
## , , 2024-01-29
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.09834775 -0.04856081 -0.04437084
## ESDE      0.09834775 1.00000000  0.42553539  0.71989459
## IEDE     -0.04856081 0.42553539  1.00000000  0.47387830
## spread   -0.04437084 0.71989459  0.47387830  1.00000000
##
## , , 2024-01-30
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.07722748 -0.05472349 -0.05450958
## ESDE      0.07722748 1.00000000  0.43811274  0.73274469
## IEDE     -0.05472349 0.43811274  1.00000000  0.47717232
## spread   -0.05450958 0.73274469  0.47717232  1.00000000
##
## , , 2024-01-31
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.05856968 -0.05616625 -0.06603837
## ESDE      0.05856968 1.00000000  0.45214358  0.74700784
## IEDE     -0.05616625 0.45214358  1.00000000  0.48342906
## spread   -0.06603837 0.74700784  0.48342906  1.00000000
##
## , , 2024-02-01
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.03870245 -0.08053781 -0.05502812
## ESDE      0.03870245 1.00000000  0.46947273  0.74365017
## IEDE     -0.08053781 0.46947273  1.00000000  0.44865055
## spread   -0.05502812 0.74365017  0.44865055  1.00000000
##
```

```
## , , 2024-02-02
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.02064007 -0.08747005 -0.05565252
## ESDE      0.02064007 1.00000000  0.48010790  0.75111946
## IEDE     -0.08747005 0.48010790  1.00000000  0.45028715
## spread   -0.05565252 0.75111946  0.45028715  1.00000000
##
## , , 2024-02-05
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.02678725 -0.07094826 -0.04766896
## ESDE      0.02678725 1.00000000  0.48804954  0.75781519
## IEDE     -0.07094826 0.48804954  1.00000000  0.45258455
## spread   -0.04766896 0.75781519  0.45258455  1.00000000
##
## , , 2024-02-06
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.00314674 -0.07060575 -0.0504185
## ESDE      0.00314674 1.00000000  0.49546415  0.7601474
## IEDE     -0.07060575 0.49546415  1.00000000  0.4558985
## spread   -0.05041850 0.76014738  0.45589848  1.0000000
##
## , , 2024-02-07
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.01564017 -0.04440606 -0.05150938
## ESDE      0.01564017 1.00000000  0.49830898  0.76190335
## IEDE     -0.04440606 0.49830898  1.00000000  0.44987498
## spread   -0.05150938 0.76190335  0.44987498  1.00000000
##
## , , 2024-02-08
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.008571098 -0.05452074 -0.07968541
## ESDE     -0.008571098  1.000000000  0.50152105  0.77034440
## IEDE     -0.054520737  0.501521050  1.00000000  0.45453172
## spread   -0.079685415  0.770344403  0.45453172  1.00000000
##
## , , 2024-02-09
##
##              eurois3m         ESDE        IEDE      spread
## eurois3m  1.000000000 -0.003463031 -0.04757633 -0.06679107
## ESDE     -0.003463031  1.000000000  0.50066122  0.77556952
## IEDE     -0.047576328  0.500661221  1.00000000  0.45332326
## spread   -0.066791071  0.775569515  0.45332326  1.00000000
##
## , , 2024-02-12
##
##              eurois3m         ESDE       IEDE     spread
## eurois3m  1.000000000 -0.008109155 -0.0477265 -0.0676899
```

```
## ESDE     -0.008109155  1.000000000  0.5016434  0.7808444
## IEDE     -0.047726498  0.501643390  1.0000000  0.4548495
## spread   -0.067689898  0.780844447  0.4548495  1.0000000
##
## , , 2024-02-13
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 -0.0426891 -0.0713426 -0.09363913
## ESDE     -0.04268910  1.0000000  0.5270142  0.79709271
## IEDE     -0.07134260  0.5270142  1.0000000  0.48100592
## spread   -0.09363913  0.7970927  0.4810059  1.00000000
##
## , , 2024-02-14
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02294948 -0.06372453 -0.09263769
## ESDE     -0.02294948  1.00000000  0.51966851  0.78789556
## IEDE     -0.06372453  0.51966851  1.00000000  0.47769318
## spread   -0.09263769  0.78789556  0.47769318  1.00000000
##
## , , 2024-02-15
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.05034049 -0.07478826 -0.1120376
## ESDE     -0.05034049  1.00000000  0.53671527  0.8054536
## IEDE     -0.07478826  0.53671527  1.00000000  0.4962517
## spread   -0.11203765  0.80545361  0.49625174  1.0000000
##
## , , 2024-02-16
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04212166 -0.06467216 -0.09853471
## ESDE     -0.04212166  1.00000000  0.53026395  0.80963032
## IEDE     -0.06467216  0.53026395  1.00000000  0.48764571
## spread   -0.09853471  0.80963032  0.48764571  1.00000000
##
## , , 2024-02-19
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 -0.0545762 -0.07101348 -0.1017856
## ESDE     -0.05457620  1.0000000  0.54096932  0.8143897
## IEDE     -0.07101348  0.5409693  1.00000000  0.4951720
## spread   -0.10178558  0.8143897  0.49517196  1.0000000
##
## , , 2024-02-20
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.04731042 -0.06147208 -0.09184127
## ESDE     -0.04731042  1.00000000  0.53713957  0.81491777
## IEDE     -0.06147208  0.53713957  1.00000000  0.49144577
## spread   -0.09184127  0.81491777  0.49144577  1.00000000
##
```

```
## , , 2024-02-21
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.02597002 -0.05236268 -0.08337495
## ESDE     -0.02597002  1.00000000  0.53109931  0.81219682
## IEDE     -0.05236268  0.53109931  1.00000000  0.48690378
## spread   -0.08337495  0.81219682  0.48690378  1.00000000
##
## , , 2024-02-22
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01661913 -0.04743415 -0.06675138
## ESDE     -0.01661913  1.00000000  0.52203251  0.81340869
## IEDE     -0.04743415  0.52203251  1.00000000  0.47366785
## spread   -0.06675138  0.81340869  0.47366785  1.00000000
##
## , , 2024-02-23
##
##              eurois3m         ESDE        IEDE     spread
## eurois3m  1.000000000 -0.002964292 -0.03398194 -0.0476457
## ESDE     -0.002964292  1.000000000  0.52471547  0.8185859
## IEDE     -0.033981945  0.524715472  1.00000000  0.4776179
## spread   -0.047645702  0.818585857  0.47761792  1.0000000
##
## , , 2024-02-26
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.01968442 -0.01522144 -0.0175780
## ESDE      0.01968442 1.00000000  0.53014614  0.8253069
## IEDE     -0.01522144 0.53014614  1.00000000  0.4847780
## spread   -0.01757800 0.82530692  0.48477802  1.0000000
##
## , , 2024-02-27
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.02230183 -0.009205015 -0.01125881
## ESDE      0.022301829 1.00000000  0.526674614  0.82449175
## IEDE     -0.009205015 0.52667461  1.000000000  0.48128999
## spread   -0.011258805 0.82449175  0.481289992  1.00000000
##
## , , 2024-02-28
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.02091198 -0.002675452 -0.01738226
## ESDE      0.020911976 1.00000000  0.522313885  0.82106340
## IEDE     -0.002675452 0.52231388  1.000000000  0.47136416
## spread   -0.017382258 0.82106340  0.471364161  1.00000000
##
## , , 2024-02-29
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.02517931 -0.006332646 -0.0128716
```

```
## ESDE      0.025179307 1.00000000  0.502183675  0.8213146
## IEDE     -0.006332646 0.50218368  1.000000000  0.4570257
## spread   -0.012871600 0.82131460  0.457025725  1.0000000
##
## , , 2024-03-01
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.04570812 -0.05196588 -0.00481059
## ESDE      0.04570812 1.00000000  0.47585028  0.81979256
## IEDE     -0.05196588 0.47585028  1.00000000  0.44357457
## spread   -0.00481059 0.81979256  0.44357457  1.00000000
##
## , , 2024-03-04
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.02358117 -0.04686111 -0.0276156
## ESDE      0.02358117 1.00000000  0.47678842  0.8233318
## IEDE     -0.04686111 0.47678842  1.00000000  0.4416335
## spread   -0.02761560 0.82333176  0.44163351  1.0000000
##
## , , 2024-03-05
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.05462144 -0.02145654 0.01039343
## ESDE      0.05462144 1.00000000  0.49053978 0.83225849
## IEDE     -0.02145654 0.49053978  1.00000000 0.45554073
## spread    0.01039343 0.83225849  0.45554073 1.00000000
##
## , , 2024-03-06
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.07205465 0.0174877 0.04320847
## ESDE     0.07205465 1.00000000 0.5023540 0.83278379
## IEDE     0.01748770 0.50235398 1.0000000 0.47476964
## spread   0.04320847 0.83278379 0.4747696 1.00000000
##
## , , 2024-03-07
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.01633899 0.009470875 -0.01947203
## ESDE      0.016338993 1.00000000 0.494044111  0.84633608
## IEDE      0.009470875 0.49404411 1.000000000  0.45999870
## spread   -0.019472026 0.84633608 0.459998700  1.00000000
##
## , , 2024-03-08
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05496046 0.04222002 0.01120111
## ESDE     0.05496046 1.00000000 0.50766089 0.84612971
## IEDE     0.04222002 0.50766089 1.00000000 0.46952354
## spread   0.01120111 0.84612971 0.46952354 1.00000000
##
```

```
## , , 2024-03-11
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.04764302 0.04569693 0.007057092
## ESDE     0.047643019 1.00000000 0.49844695 0.843314104
## IEDE     0.045696927 0.49844695 1.00000000 0.461371706
## spread   0.007057092 0.84331410 0.46137171 1.000000000
##
## , , 2024-03-12
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01373676 0.02809455 -0.009651446
## ESDE      0.013736763 1.00000000 0.50314343  0.839989904
## IEDE      0.028094550 0.50314343 1.00000000  0.463978501
## spread   -0.009651446 0.83998990 0.46397850  1.000000000
##
## , , 2024-03-13
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01670133 0.02742886 -0.003942286
## ESDE      0.016701331 1.00000000 0.49935687  0.838522577
## IEDE      0.027428856 0.49935687 1.00000000  0.449895624
## spread   -0.003942286 0.83852258 0.44989562  1.000000000
##
## , , 2024-03-14
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.007049348 0.02218883 -0.03396003
## ESDE     -0.007049348  1.000000000 0.49526635  0.84380150
## IEDE      0.022188827  0.495266352 1.00000000  0.44279866
## spread   -0.033960030  0.843801499 0.44279866  1.00000000
##
## , , 2024-03-15
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.009961219 0.01884289 -0.03793135
## ESDE     -0.009961219  1.000000000 0.49710182  0.84249354
## IEDE      0.018842885  0.497101822 1.00000000  0.44545531
## spread   -0.037931347  0.842493544 0.44545531  1.00000000
##
## , , 2024-03-18
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01134122 0.01550247 -0.03917314
## ESDE     -0.01134122  1.00000000 0.49942151  0.84095868
## IEDE      0.01550247  0.49942151 1.00000000  0.44863730
## spread   -0.03917314  0.84095868 0.44863730  1.00000000
##
## , , 2024-03-19
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m  1.00000000 0.0134456 0.02312456 -0.01121855
```

```
## ESDE      0.01344560 1.0000000 0.50127725  0.84467192
## IEDE      0.02312456 0.5012773 1.00000000  0.45143330
## spread   -0.01121855 0.8446719 0.45143330  1.00000000
##
## , , 2024-03-20
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02644413 0.02117128 -0.00129565
## ESDE      0.02644413 1.00000000 0.49198167  0.84382831
## IEDE      0.02117128 0.49198167 1.00000000  0.44624872
## spread   -0.00129565 0.84382831 0.44624872  1.00000000
##
## , , 2024-03-21
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01600297 0.02299909 -0.008579182
## ESDE      0.016002970 1.00000000 0.48330265  0.842546429
## IEDE      0.022999093 0.48330265 1.00000000  0.440861997
## spread   -0.008579182 0.84254643 0.44086200  1.000000000
##
## , , 2024-03-22
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.004507231 0.02756276 -0.01053079
## ESDE      0.004507231 1.000000000 0.47819763  0.83773207
## IEDE      0.027562764 0.478197633 1.00000000  0.44049524
## spread   -0.010530795 0.837732072 0.44049524  1.00000000
##
## , , 2024-03-25
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 -0.0463211 -0.04556728 -0.05617322
## ESDE     -0.04632110  1.0000000  0.52086194  0.84415645
## IEDE     -0.04556728  0.5208619  1.00000000  0.47950497
## spread   -0.05617322  0.8441565  0.47950497  1.00000000
##
## , , 2024-03-26
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.05121689 -0.07238645 -0.05453398
## ESDE     -0.05121689  1.00000000  0.51947219  0.84115850
## IEDE     -0.07238645  0.51947219  1.00000000  0.47484866
## spread   -0.05453398  0.84115850  0.47484866  1.00000000
##
## , , 2024-03-27
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.05648878 -0.05990588 -0.06307523
## ESDE     -0.05648878  1.00000000  0.51111031  0.84086804
## IEDE     -0.05990588  0.51111031  1.00000000  0.46578980
## spread   -0.06307523  0.84086804  0.46578980  1.00000000
##
```

```
## , , 2024-03-28
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 0.005736148 -0.003433323 0.003580095
## ESDE      0.005736148 1.000000000  0.520636928 0.842882281
## IEDE     -0.003433323 0.520636928  1.000000000 0.477362529
## spread    0.003580095 0.842882281  0.477362529 1.000000000
##
## , , 2024-04-02
##
##              eurois3m         ESDE         IEDE       spread
## eurois3m  1.000000000 -0.001156512 -0.008514515 -0.007275796
## ESDE     -0.001156512  1.000000000  0.545128264  0.850212555
## IEDE     -0.008514515  0.545128264  1.000000000  0.509975725
## spread   -0.007275796  0.850212555  0.509975725  1.000000000
##
## , , 2024-04-03
##
##              eurois3m        ESDE         IEDE       spread
## eurois3m  1.000000000 0.007086607 -0.009743676 -0.008468255
## ESDE      0.007086607 1.000000000  0.525034822  0.837659759
## IEDE     -0.009743676 0.525034822  1.000000000  0.503830624
## spread   -0.008468255 0.837659759  0.503830624  1.000000000
##
## , , 2024-04-04
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.05569939 0.01112327 -0.1092296
## ESDE     -0.05569939  1.00000000 0.50843297  0.8379257
## IEDE      0.01112327  0.50843297 1.00000000  0.4707918
## spread   -0.10922961  0.83792574 0.47079176  1.0000000
##
## , , 2024-04-05
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 -0.1066692 -0.01844963 -0.1529061
## ESDE     -0.10666915  1.0000000  0.52526129  0.8529222
## IEDE     -0.01844963  0.5252613  1.00000000  0.4892714
## spread   -0.15290607  0.8529222  0.48927139  1.0000000
##
## , , 2024-04-08
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 -0.0957049 -0.01602634 -0.1393285
## ESDE     -0.09570490  1.0000000  0.53018833  0.8512287
## IEDE     -0.01602634  0.5301883  1.00000000  0.4981018
## spread   -0.13932847  0.8512287  0.49810179  1.0000000
##
## , , 2024-04-09
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 -0.1053421 -0.05032704 -0.1379184
```

```
## ESDE     -0.10534207  1.0000000  0.54662120  0.8479249
## IEDE     -0.05032704  0.5466212  1.00000000  0.4972560
## spread   -0.13791845  0.8479249  0.49725602  1.0000000
##
## , , 2024-04-10
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.09787097 -0.04790007 -0.1275683
## ESDE     -0.09787097  1.00000000  0.54234522  0.8449298
## IEDE     -0.04790007  0.54234522  1.00000000  0.4930365
## spread   -0.12756833  0.84492976  0.49303651  1.0000000
##
## , , 2024-04-11
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.08843504 0.02664325 -0.1169456
## ESDE     -0.08843504  1.00000000 0.36601082  0.8421515
## IEDE      0.02664325  0.36601082 1.00000000  0.3372257
## spread   -0.11694559  0.84215146 0.33722570  1.0000000
##
## , , 2024-04-12
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.07726366 0.02475294 -0.1085190
## ESDE     -0.07726366  1.00000000 0.37418728  0.8357818
## IEDE      0.02475294  0.37418728 1.00000000  0.3460331
## spread   -0.10851904  0.83578179 0.34603311  1.0000000
##
## , , 2024-04-15
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.04209496 0.01818737 -0.09338686
## ESDE     -0.04209496  1.00000000 0.38090185  0.83169291
## IEDE      0.01818737  0.38090185 1.00000000  0.35340654
## spread   -0.09338686  0.83169291 0.35340654  1.00000000
##
## , , 2024-04-16
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.06998626 0.01106503 -0.1193763
## ESDE     -0.06998626  1.00000000 0.38921454  0.8393071
## IEDE      0.01106503  0.38921454 1.00000000  0.3601580
## spread   -0.11937633  0.83930709 0.36015799  1.0000000
##
## , , 2024-04-17
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.06127295 0.01376672 -0.1027645
## ESDE     -0.06127295  1.00000000 0.39761788  0.8360717
## IEDE      0.01376672  0.39761788 1.00000000  0.3676898
## spread   -0.10276453  0.83607165 0.36768984  1.0000000
##
```

```
## , , 2024-04-18
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.04682095 0.01906841 -0.08353229
## ESDE     -0.04682095  1.00000000 0.40761672  0.83459911
## IEDE      0.01906841  0.40761672 1.00000000  0.37693736
## spread   -0.08353229  0.83459911 0.37693736  1.00000000
##
## , , 2024-04-19
##
##             eurois3m        ESDE      IEDE      spread
## eurois3m  1.00000000 -0.05053058 0.0126062 -0.07762798
## ESDE     -0.05053058  1.00000000 0.4241234  0.82535952
## IEDE      0.01260620  0.42412338 1.0000000  0.38438780
## spread   -0.07762798  0.82535952 0.3843878  1.00000000
##
## , , 2024-04-22
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 -0.0396185 0.01470129 -0.07247109
## ESDE     -0.03961850  1.0000000 0.43101880  0.82096743
## IEDE      0.01470129  0.4310188 1.00000000  0.38913037
## spread   -0.07247109  0.8209674 0.38913037  1.00000000
##
## , , 2024-04-23
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m  1.00000000 -0.04209582 0.01093715 -0.0716929
## ESDE     -0.04209582  1.00000000 0.45239166  0.8451598
## IEDE      0.01093715  0.45239166 1.00000000  0.4134659
## spread   -0.07169290  0.84515981 0.41346594  1.0000000
##
## , , 2024-04-24
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 -0.0302772 0.01822981 -0.04821478
## ESDE     -0.03027720  1.0000000 0.45775248  0.84029732
## IEDE      0.01822981  0.4577525 1.00000000  0.41975574
## spread   -0.04821478  0.8402973 0.41975574  1.00000000
##
## , , 2024-04-25
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02838958 0.01561844 -0.04675964
## ESDE     -0.02838958  1.00000000 0.47754798  0.84611130
## IEDE      0.01561844  0.47754798 1.00000000  0.44211380
## spread   -0.04675964  0.84611130 0.44211380  1.00000000
##
## , , 2024-04-26
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02392698 0.03681861 -0.03867862
```

```
## ESDE     -0.02392698  1.00000000 0.47251947  0.84280856
## IEDE      0.03681861  0.47251947 1.00000000  0.44143462
## spread   -0.03867862  0.84280856 0.44143462  1.00000000
##
## , , 2024-04-29
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02451755 0.03228823 -0.04344132
## ESDE     -0.02451755  1.00000000 0.47737137  0.84057744
## IEDE      0.03228823  0.47737137 1.00000000  0.44420826
## spread   -0.04344132  0.84057744 0.44420826  1.00000000
##
## , , 2024-04-30
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01533547 0.08237743 -0.00918869
## ESDE      0.01533547 1.00000000 0.49275385  0.84042332
## IEDE      0.08237743 0.49275385 1.00000000  0.45592704
## spread   -0.00918869 0.84042332 0.45592704  1.00000000
##
## , , 2024-05-02
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.02158881 0.0793329 -0.01576314
## ESDE      0.02158881 1.00000000 0.4935583  0.83535577
## IEDE      0.07933290 0.49355826 1.0000000  0.45385426
## spread   -0.01576314 0.83535577 0.4538543  1.00000000
##
## , , 2024-05-03
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02795782 0.07583174 -0.04152114
## ESDE      0.02795782 1.00000000 0.49408569  0.82876637
## IEDE      0.07583174 0.49408569 1.00000000  0.45037239
## spread   -0.04152114 0.82876637 0.45037239  1.00000000
##
## , , 2024-05-06
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02458951 0.06328373 -0.03913786
## ESDE      0.02458951 1.00000000 0.49646835  0.82737534
## IEDE      0.06328373 0.49646835 1.00000000  0.44865112
## spread   -0.03913786 0.82737534 0.44865112  1.00000000
##
## , , 2024-05-07
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02282928 0.06258288 -0.03377504
## ESDE      0.02282928 1.00000000 0.49157709  0.82386093
## IEDE      0.06258288 0.49157709 1.00000000  0.45118934
## spread   -0.03377504 0.82386093 0.45118934  1.00000000
##
```

```
## , , 2024-05-08
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01229565 0.04303594 -0.07792349
## ESDE     -0.01229565  1.00000000 0.49290009  0.82476755
## IEDE      0.04303594  0.49290009 1.00000000  0.45114784
## spread   -0.07792349  0.82476755 0.45114784  1.00000000
##
## , , 2024-05-09
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01045667 0.02595194 -0.07951608
## ESDE     -0.01045667  1.00000000 0.49127728  0.82223870
## IEDE      0.02595194  0.49127728 1.00000000  0.45516332
## spread   -0.07951608  0.82223870 0.45516332  1.00000000
##
## , , 2024-05-10
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.004627613 0.02853671 -0.07004358
## ESDE     -0.004627613  1.000000000 0.49703575  0.82212153
## IEDE      0.028536712  0.497035749 1.00000000  0.45770143
## spread   -0.070043577  0.822121529 0.45770143  1.00000000
##
## , , 2024-05-13
##
##              eurois3m         ESDE       IEDE     spread
## eurois3m  1.000000000 -0.001524603 0.03496366 -0.0686313
## ESDE     -0.001524603  1.000000000 0.49635505  0.8200848
## IEDE      0.034963658  0.496355045 1.00000000  0.4520289
## spread   -0.068631303  0.820084829 0.45202893  1.0000000
##
## , , 2024-05-14
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.007025981 0.02108493 -0.05452171
## ESDE     -0.007025981  1.000000000 0.49727814  0.81762294
## IEDE      0.021084934  0.497278136 1.00000000  0.44918773
## spread   -0.054521709  0.817622941 0.44918773  1.00000000
##
## , , 2024-05-15
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.007133205 0.02708996 -0.04408175
## ESDE     -0.007133205  1.000000000 0.49256007  0.81307884
## IEDE      0.027089965  0.492560073 1.00000000  0.45225628
## spread   -0.044081749  0.813078842 0.45225628  1.00000000
##
## , , 2024-05-16
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.03558745 0.02347953 -0.07624916
```

```
## ESDE     -0.03558745  1.00000000 0.48040128  0.82848258
## IEDE      0.02347953  0.48040128 1.00000000  0.43305001
## spread   -0.07624916  0.82848258 0.43305001  1.00000000
##
## , , 2024-05-17
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.04808739 0.01044318 -0.07331269
## ESDE     -0.04808739  1.00000000 0.48502440  0.82569394
## IEDE      0.01044318  0.48502440 1.00000000  0.43565411
## spread   -0.07331269  0.82569394 0.43565411  1.00000000
##
## , , 2024-05-20
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.05067043 -0.001199897 -0.07557522
## ESDE     -0.050670434  1.00000000  0.489996179  0.82568587
## IEDE     -0.001199897  0.48999618  1.000000000  0.44127095
## spread   -0.075575215  0.82568587  0.441270955  1.00000000
##
## , , 2024-05-21
##
##               eurois3m        ESDE         IEDE      spread
## eurois3m  1.0000000000 -0.04353997 0.0009404534 -0.06912716
## ESDE     -0.0435399659  1.00000000 0.4907445432  0.82293107
## IEDE      0.0009404534  0.49074454 1.0000000000  0.44115566
## spread   -0.0691271644  0.82293107 0.4411556563  1.00000000
##
## , , 2024-05-22
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.02861704 0.002662671 -0.05761186
## ESDE     -0.028617040  1.00000000 0.487340737  0.82261895
## IEDE      0.002662671  0.48734074 1.000000000  0.44086466
## spread   -0.057611862  0.82261895 0.440864663  1.00000000
##
## , , 2024-05-23
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.02269024 0.003344253 -0.05110087
## ESDE     -0.022690236  1.00000000 0.486932494  0.82209473
## IEDE      0.003344253  0.48693249 1.000000000  0.44044150
## spread   -0.051100866  0.82209473 0.440441499  1.00000000
##
## , , 2024-05-24
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.008099744 0.01743458 -0.05218155
## ESDE      0.008099744 1.000000000 0.49059377  0.81152865
## IEDE      0.017434585 0.490593769 1.00000000  0.43882007
## spread   -0.052181553 0.811528654 0.43882007  1.00000000
##
```

```
## , , 2024-05-27
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.002825111 0.02478837 -0.05160852
## ESDE      0.002825111 1.000000000 0.48300290  0.81263448
## IEDE      0.024788369 0.483002905 1.00000000  0.43397173
## spread   -0.051608515 0.812634483 0.43397173  1.00000000
##
## , , 2024-05-28
##
##              eurois3m         ESDE       IEDE     spread
## eurois3m  1.000000000 -0.002990816 0.01642477 -0.0592052
## ESDE     -0.002990816  1.000000000 0.48770281  0.8134748
## IEDE      0.016424774  0.487702809 1.00000000  0.4412671
## spread   -0.059205202  0.813474783 0.44126707  1.0000000
##
## , , 2024-05-29
##
##               eurois3m        ESDE         IEDE      spread
## eurois3m  1.0000000000 -0.01212351 0.0009772046 -0.08942187
## ESDE     -0.0121235076  1.00000000 0.4921577459  0.80485484
## IEDE      0.0009772046  0.49215775 1.0000000000  0.45076405
## spread   -0.0894218652  0.80485484 0.4507640519  1.00000000
##
## , , 2024-05-30
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 -0.02037748 -0.001707859 -0.08078864
## ESDE     -0.020377481  1.00000000  0.496610389  0.79339628
## IEDE     -0.001707859  0.49661039  1.000000000  0.44993002
## spread   -0.080788645  0.79339628  0.449930021  1.00000000
##
## , , 2024-05-31
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.03250349 0.01414362 -0.08118064
## ESDE     -0.03250349  1.00000000 0.46792480  0.79680236
## IEDE      0.01414362  0.46792480 1.00000000  0.43423862
## spread   -0.08118064  0.79680236 0.43423862  1.00000000
##
## , , 2024-06-03
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01452002 0.02776064 -0.08745699
## ESDE     -0.01452002  1.00000000 0.47559121  0.78145226
## IEDE      0.02776064  0.47559121 1.00000000  0.42088161
## spread   -0.08745699  0.78145226 0.42088161  1.00000000
##
## , , 2024-06-04
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01844789 0.02707938 -0.08430171
```

```
## ESDE     -0.01844789  1.00000000 0.47547168  0.78687820
## IEDE      0.02707938  0.47547168 1.00000000  0.42257583
## spread   -0.08430171  0.78687820 0.42257583  1.00000000
##
## , , 2024-06-05
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.007049758 0.03053393 -0.06706843
## ESDE     -0.007049758  1.000000000 0.49844336  0.80208702
## IEDE      0.030533929  0.498443360 1.00000000  0.44750054
## spread   -0.067068426  0.802087021 0.44750054  1.00000000
##
## , , 2024-06-06
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01355908 0.01695347 -0.02612743
## ESDE      0.01355908 1.00000000 0.48489251  0.80487973
## IEDE      0.01695347 0.48489251 1.00000000  0.42143276
## spread   -0.02612743 0.80487973 0.42143276  1.00000000
##
## , , 2024-06-07
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m  1.00000000 0.01379788 0.0213292 -0.03360019
## ESDE      0.01379788 1.00000000 0.4858616  0.80299973
## IEDE      0.02132920 0.48586156 1.0000000  0.41929744
## spread   -0.03360019 0.80299973 0.4192974  1.00000000
##
## , , 2024-06-10
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.01213074 0.03406102 -0.009394715
## ESDE      0.012130744 1.00000000 0.48201398  0.788780003
## IEDE      0.034061017 0.48201398 1.00000000  0.433776209
## spread   -0.009394715 0.78878000 0.43377621  1.000000000
##
## , , 2024-06-11
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.09664621 -0.08276829 -0.1091039
## ESDE     -0.09664621  1.00000000  0.63402459  0.8475894
## IEDE     -0.08276829  0.63402459  1.00000000  0.5906810
## spread   -0.10910388  0.84758937  0.59068103  1.0000000
##
## , , 2024-06-12
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.08884496 -0.07482077 -0.1018520
## ESDE     -0.08884496  1.00000000  0.62394387  0.8460725
## IEDE     -0.07482077  0.62394387  1.00000000  0.5805956
## spread   -0.10185202  0.84607247  0.58059561  1.0000000
##
```

```
## , , 2024-06-13
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 -0.04760368 -0.02331806 -0.002303472
## ESDE     -0.047603683  1.00000000  0.61498142  0.835567429
## IEDE     -0.023318062  0.61498142  1.00000000  0.574571870
## spread   -0.002303472  0.83556743  0.57457187  1.000000000
##
## , , 2024-06-14
##
##              eurois3m        ESDE         IEDE    spread
## eurois3m  1.000000000 -0.02726659 -0.008680396 0.0046149
## ESDE     -0.027266586  1.00000000  0.666829989 0.8477567
## IEDE     -0.008680396  0.66682999  1.000000000 0.6112067
## spread    0.004614900  0.84775671  0.611206678 1.0000000
##
## , , 2024-06-17
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06580313 0.03516759 0.05803091
## ESDE     0.06580313 1.00000000 0.66815527 0.85305203
## IEDE     0.03516759 0.66815527 1.00000000 0.62518967
## spread   0.05803091 0.85305203 0.62518967 1.00000000
##
## , , 2024-06-18
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06887283 0.04329749 0.05815884
## ESDE     0.06887283 1.00000000 0.66759384 0.85323991
## IEDE     0.04329749 0.66759384 1.00000000 0.62093709
## spread   0.05815884 0.85323991 0.62093709 1.00000000
##
## , , 2024-06-19
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07087515 0.04050569 0.06125106
## ESDE     0.07087515 1.00000000 0.65427881 0.85040111
## IEDE     0.04050569 0.65427881 1.00000000 0.60647772
## spread   0.06125106 0.85040111 0.60647772 1.00000000
##
## , , 2024-06-20
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.08124099 0.05355372 0.07727711
## ESDE     0.08124099 1.00000000 0.64688501 0.84940673
## IEDE     0.05355372 0.64688501 1.00000000 0.60069227
## spread   0.07727711 0.84940673 0.60069227 1.00000000
##
## , , 2024-06-21
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.09681103 0.06845501 0.08371595
```

```
## ESDE     0.09681103 1.00000000 0.64355252 0.84831137
## IEDE     0.06845501 0.64355252 1.00000000 0.59441401
## spread   0.08371595 0.84831137 0.59441401 1.00000000
##
## , , 2024-06-24
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.04499467 0.001630815 0.04664433
## ESDE     0.044994674 1.00000000 0.635123173 0.84567554
## IEDE     0.001630815 0.63512317 1.000000000 0.58257560
## spread   0.046644328 0.84567554 0.582575604 1.00000000
##
## , , 2024-06-25
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.04154453 0.002052046 0.04076697
## ESDE     0.041544527 1.00000000 0.628227696 0.84454276
## IEDE     0.002052046 0.62822770 1.000000000 0.57588623
## spread   0.040766974 0.84454276 0.575886231 1.00000000
##
## , , 2024-06-26
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.04490125 0.004108464 0.04405347
## ESDE     0.044901247 1.00000000 0.615332054 0.84273620
## IEDE     0.004108464 0.61533205 1.000000000 0.56176337
## spread   0.044053473 0.84273620 0.561763374 1.00000000
##
## , , 2024-06-27
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06501955 0.02551597 0.05836998
## ESDE     0.06501955 1.00000000 0.60629821 0.84099499
## IEDE     0.02551597 0.60629821 1.00000000 0.55290905
## spread   0.05836998 0.84099499 0.55290905 1.00000000
##
## , , 2024-06-28
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05772848 0.02285837 0.04853403
## ESDE     0.05772848 1.00000000 0.60412894 0.84299080
## IEDE     0.02285837 0.60412894 1.00000000 0.55336330
## spread   0.04853403 0.84299080 0.55336330 1.00000000
##
## , , 2024-07-01
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03795615 0.02051043 0.03747158
## ESDE     0.03795615 1.00000000 0.59063188 0.83833786
## IEDE     0.02051043 0.59063188 1.00000000 0.54317584
## spread   0.03747158 0.83833786 0.54317584 1.00000000
##
```

```
## , , 2024-07-02
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02839681 0.01130787 0.02759651
## ESDE     0.02839681 1.00000000 0.66193707 0.85406257
## IEDE     0.01130787 0.66193707 1.00000000 0.59810409
## spread   0.02759651 0.85406257 0.59810409 1.00000000
##
## , , 2024-07-03
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.02603381 0.009560108 0.02281599
## ESDE     0.026033808 1.00000000 0.651864297 0.85194118
## IEDE     0.009560108 0.65186430 1.000000000 0.59101260
## spread   0.022815991 0.85194118 0.591012596 1.00000000
##
## , , 2024-07-04
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.02545194 0.009679171 0.02056483
## ESDE     0.025451938 1.00000000 0.636225031 0.84836170
## IEDE     0.009679171 0.63622503 1.000000000 0.57293956
## spread   0.020564828 0.84836170 0.572939561 1.00000000
##
## , , 2024-07-05
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.02359755 0.0101212 0.01688855
## ESDE     0.02359755 1.00000000 0.6251725 0.84691894
## IEDE     0.01012120 0.62517251 1.0000000 0.56315561
## spread   0.01688855 0.84691894 0.5631556 1.00000000
##
## , , 2024-07-08
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02255458 0.01084678 0.01391021
## ESDE     0.02255458 1.00000000 0.61583533 0.84569984
## IEDE     0.01084678 0.61583533 1.00000000 0.55364447
## spread   0.01391021 0.84569984 0.55364447 1.00000000
##
## , , 2024-07-09
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02649177 0.02399645 0.01494933
## ESDE     0.02649177 1.00000000 0.60535640 0.84299523
## IEDE     0.02399645 0.60535640 1.00000000 0.54340356
## spread   0.01494933 0.84299523 0.54340356 1.00000000
##
## , , 2024-07-10
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04432857 0.06914413 0.02975065
```

```
## ESDE     0.04432857 1.00000000 0.60151189 0.84297658
## IEDE     0.06914413 0.60151189 1.00000000 0.54098881
## spread   0.02975065 0.84297658 0.54098881 1.00000000
##
## , , 2024-07-11
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.02567666 0.05570317 0.003121988
## ESDE     0.025676664 1.00000000 0.59713723 0.845216305
## IEDE     0.055703172 0.59713723 1.00000000 0.536766125
## spread   0.003121988 0.84521631 0.53676612 1.000000000
##
## , , 2024-07-12
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.02803764 0.0463375 0.01055488
## ESDE     0.02803764 1.00000000 0.5864472 0.84217454
## IEDE     0.04633750 0.58644725 1.0000000 0.52559595
## spread   0.01055488 0.84217454 0.5255960 1.00000000
##
## , , 2024-07-15
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01696227 0.02053901 0.03170911
## ESDE     0.01696227 1.00000000 0.57810656 0.83441368
## IEDE     0.02053901 0.57810656 1.00000000 0.50992511
## spread   0.03170911 0.83441368 0.50992511 1.00000000
##
## , , 2024-07-16
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02052058 0.02704957 0.03347347
## ESDE     0.02052058 1.00000000 0.57251549 0.83304215
## IEDE     0.02704957 0.57251549 1.00000000 0.51024722
## spread   0.03347347 0.83304215 0.51024722 1.00000000
##
## , , 2024-07-17
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02304612 0.02968108 0.02955652
## ESDE     0.02304612 1.00000000 0.56936682 0.82880990
## IEDE     0.02968108 0.56936682 1.00000000 0.50252258
## spread   0.02955652 0.82880990 0.50252258 1.00000000
##
## , , 2024-07-18
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0451221 0.04905777 0.05270686
## ESDE     0.04512210 1.0000000 0.56692221 0.82993337
## IEDE     0.04905777 0.5669222 1.00000000 0.50380768
## spread   0.05270686 0.8299334 0.50380768 1.00000000
##
```

```
## , , 2024-07-19
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02952328 0.03964826 0.02772434
## ESDE     0.02952328 1.00000000 0.56034616 0.82938363
## IEDE     0.03964826 0.56034616 1.00000000 0.49879039
## spread   0.02772434 0.82938363 0.49879039 1.00000000
##
## , , 2024-07-22
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04471072 0.05006272 0.04137671
## ESDE     0.04471072 1.00000000 0.55535898 0.82893495
## IEDE     0.05006272 0.55535898 1.00000000 0.49576299
## spread   0.04137671 0.82893495 0.49576299 1.00000000
##
## , , 2024-07-23
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01018223 0.03370019 -0.01398065
## ESDE      0.01018223 1.00000000 0.54979814  0.82895149
## IEDE      0.03370019 0.54979814 1.00000000  0.49003111
## spread   -0.01398065 0.82895149 0.49003111  1.00000000
##
## , , 2024-07-24
##
##               eurois3m        ESDE          IEDE      spread
## eurois3m  1.0000000000 -0.01371459 -0.0007189245 -0.03203128
## ESDE     -0.0137145900  1.00000000  0.6119378482  0.84285563
## IEDE     -0.0007189245  0.61193785  1.0000000000  0.54349673
## spread   -0.0320312773  0.84285563  0.5434967266  1.00000000
##
## , , 2024-07-25
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.03213496 -0.01136033 0.00397042
## ESDE      0.03213496 1.00000000  0.58396138 0.84285156
## IEDE     -0.01136033 0.58396138  1.00000000 0.52345490
## spread    0.00397042 0.84285156  0.52345490 1.00000000
##
## , , 2024-07-26
##
##              eurois3m      ESDE         IEDE     spread
## eurois3m  1.000000000 0.0401501 -0.002114698 0.01242536
## ESDE      0.040150103 1.0000000  0.577130610 0.84127692
## IEDE     -0.002114698 0.5771306  1.000000000 0.51814076
## spread    0.012425357 0.8412769  0.518140763 1.00000000
##
## , , 2024-07-29
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.04164721 -0.01939327 0.01728909
```

```
## ESDE      0.04164721 1.00000000  0.56285792 0.83877930
## IEDE     -0.01939327 0.56285792  1.00000000 0.50360010
## spread    0.01728909 0.83877930  0.50360010 1.00000000
##
## , , 2024-07-30
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.04595539 -0.02522264 0.03198916
## ESDE      0.04595539 1.00000000  0.55220800 0.83573154
## IEDE     -0.02522264 0.55220800  1.00000000 0.48897235
## spread    0.03198916 0.83573154  0.48897235 1.00000000
##
## , , 2024-07-31
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.04299262 -0.003384321 0.02270606
## ESDE      0.042992616 1.00000000  0.525075948 0.83031317
## IEDE     -0.003384321 0.52507595  1.000000000 0.43923735
## spread    0.022706056 0.83031317  0.439237349 1.00000000
##
## , , 2024-08-01
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.02414094 -0.0314913 0.01765296
## ESDE      0.02414094 1.00000000  0.5270430 0.82648787
## IEDE     -0.03149130 0.52704298  1.0000000 0.43531547
## spread    0.01765296 0.82648787  0.4353155 1.00000000
##
## , , 2024-08-02
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.04467266 -0.007322475 0.03473875
## ESDE      0.044672660 1.00000000  0.580397843 0.84972316
## IEDE     -0.007322475 0.58039784  1.000000000 0.49852630
## spread    0.034738745 0.84972316  0.498526299 1.00000000
##
## , , 2024-08-05
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.1385773 0.03042774 0.1155672
## ESDE     0.13857728 1.0000000 0.57449352 0.8586150
## IEDE     0.03042774 0.5744935 1.00000000 0.5018160
## spread   0.11556723 0.8586150 0.50181604 1.0000000
##
## , , 2024-08-06
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1129645 0.0346369 0.0867777
## ESDE     0.1129645 1.0000000 0.5488272 0.8583304
## IEDE     0.0346369 0.5488272 1.0000000 0.4746045
## spread   0.0867777 0.8583304 0.4746045 1.0000000
##
```

```
## , , 2024-08-07
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.2186695 0.06409454 0.1584966
## ESDE     0.21866946 1.0000000 0.54177127 0.8590178
## IEDE     0.06409454 0.5417713 1.00000000 0.4764074
## spread   0.15849656 0.8590178 0.47640738 1.0000000
##
## , , 2024-08-08
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.1941449 0.05878035 0.1365268
## ESDE     0.19414488 1.0000000 0.53636971 0.8568424
## IEDE     0.05878035 0.5363697 1.00000000 0.4730335
## spread   0.13652681 0.8568424 0.47303352 1.0000000
##
## , , 2024-08-09
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.1857079 0.04953882 0.1294966
## ESDE     0.18570788 1.0000000 0.52270925 0.8540068
## IEDE     0.04953882 0.5227093 1.00000000 0.4635432
## spread   0.12949665 0.8540068 0.46354324 1.0000000
##
## , , 2024-08-12
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.1743703 0.04017312 0.1234874
## ESDE     0.17437029 1.0000000 0.51304747 0.8506301
## IEDE     0.04017312 0.5130475 1.00000000 0.4513406
## spread   0.12348736 0.8506301 0.45134062 1.0000000
##
## , , 2024-08-13
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.1649139 0.04436898 0.1195791
## ESDE     0.16491387 1.0000000 0.51334298 0.8476904
## IEDE     0.04436898 0.5133430 1.00000000 0.4544578
## spread   0.11957906 0.8476904 0.45445777 1.0000000
##
## , , 2024-08-14
##
##            eurois3m      ESDE       IEDE    spread
## eurois3m 1.00000000 0.1476509 0.03987682 0.1041765
## ESDE     0.14765094 1.0000000 0.51310752 0.8452558
## IEDE     0.03987682 0.5131075 1.00000000 0.4557827
## spread   0.10417647 0.8452558 0.45578269 1.0000000
##
## , , 2024-08-15
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1373400 0.02127752 0.09415969
```

```
## ESDE     0.13734001 1.0000000 0.39188502 0.83963187
## IEDE     0.02127752 0.3918850 1.00000000 0.38692019
## spread   0.09415969 0.8396319 0.38692019 1.00000000
##
## , , 2024-08-16
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1235458 0.02085216 0.08151284
## ESDE     0.12354577 1.0000000 0.40170471 0.83759872
## IEDE     0.02085216 0.4017047 1.00000000 0.39254232
## spread   0.08151284 0.8375987 0.39254232 1.00000000
##
## , , 2024-08-19
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1299492 0.02352588 0.08408866
## ESDE     0.12994916 1.0000000 0.41039989 0.83622904
## IEDE     0.02352588 0.4103999 1.00000000 0.39854346
## spread   0.08408866 0.8362290 0.39854346 1.00000000
##
## , , 2024-08-20
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.09951314 0.01762046 0.05757091
## ESDE     0.09951314 1.00000000 0.42303922 0.84038361
## IEDE     0.01762046 0.42303922 1.00000000 0.40824133
## spread   0.05757091 0.84038361 0.40824133 1.00000000
##
## , , 2024-08-21
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0898063 0.01360873 0.04261847
## ESDE     0.08980630 1.0000000 0.43044225 0.83548009
## IEDE     0.01360873 0.4304423 1.00000000 0.41444770
## spread   0.04261847 0.8354801 0.41444770 1.00000000
##
## , , 2024-08-22
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05978064 0.00049449 0.01850736
## ESDE     0.05978064 1.00000000 0.44411835 0.83618655
## IEDE     0.00049449 0.44411835 1.00000000 0.42465134
## spread   0.01850736 0.83618655 0.42465134 1.00000000
##
## , , 2024-08-23
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.04419006 0.002053124 0.01476944
## ESDE     0.044190062 1.00000000 0.446940656 0.82928174
## IEDE     0.002053124 0.44694066 1.000000000 0.42742386
## spread   0.014769441 0.82928174 0.427423863 1.00000000
##
```

```
## , , 2024-08-26
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.04646676 0.003530048 0.02446696
## ESDE     0.046466760 1.00000000 0.450700549 0.82639714
## IEDE     0.003530048 0.45070055 1.000000000 0.42314036
## spread   0.024466964 0.82639714 0.423140358 1.00000000
##
## , , 2024-08-27
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04325042 0.00488206 0.02108652
## ESDE     0.04325042 1.00000000 0.45404946 0.82540824
## IEDE     0.00488206 0.45404946 1.00000000 0.42390302
## spread   0.02108652 0.82540824 0.42390302 1.00000000
##
## , , 2024-08-28
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.01698505 0.02792054 -0.0080353
## ESDE      0.01698505 1.00000000 0.38723203  0.8339486
## IEDE      0.02792054 0.38723203 1.00000000  0.3465114
## spread   -0.00803530 0.83394864 0.34651140  1.0000000
##
## , , 2024-08-29
##
##              eurois3m        ESDE      IEDE      spread
## eurois3m  1.000000000 0.009086029 0.0277493 -0.01007566
## ESDE      0.009086029 1.000000000 0.3898727  0.82713239
## IEDE      0.027749298 0.389872660 1.0000000  0.35632415
## spread   -0.010075664 0.827132386 0.3563241  1.00000000
##
## , , 2024-08-30
##
##              eurois3m         ESDE         IEDE      spread
## eurois3m  1.000000000 -0.006461057 -0.005675857 -0.01489412
## ESDE     -0.006461057  1.000000000  0.409567092  0.82488328
## IEDE     -0.005675857  0.409567092  1.000000000  0.36586596
## spread   -0.014894123  0.824883282  0.365865961  1.00000000
##
## , , 2024-09-02
##
##             eurois3m        ESDE      IEDE      spread
## eurois3m  1.00000000 -0.02441899 0.0123043 -0.05808282
## ESDE     -0.02441899  1.00000000 0.4141680  0.82308702
## IEDE      0.01230430  0.41416796 1.0000000  0.36332472
## spread   -0.05808282  0.82308702 0.3633247  1.00000000
##
## , , 2024-09-03
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.07365056 0.01955193 0.05483881
```

```
## ESDE     0.07365056 1.00000000 0.41487748 0.83430152
## IEDE     0.01955193 0.41487748 1.00000000 0.36394819
## spread   0.05483881 0.83430152 0.36394819 1.00000000
##
## , , 2024-09-04
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m 1.000000000 0.03265821 0.005484054 0.01492772
## ESDE     0.032658211 1.00000000 0.428518744 0.83724986
## IEDE     0.005484054 0.42851874 1.000000000 0.37892532
## spread   0.014927717 0.83724986 0.378925319 1.00000000
##
## , , 2024-09-05
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03827918 0.01232967 0.02692495
## ESDE     0.03827918 1.00000000 0.44042951 0.83451016
## IEDE     0.01232967 0.44042951 1.00000000 0.39335369
## spread   0.02692495 0.83451016 0.39335369 1.00000000
##
## , , 2024-09-06
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02638713 0.03255932 0.02489213
## ESDE     0.02638713 1.00000000 0.43877184 0.83121479
## IEDE     0.03255932 0.43877184 1.00000000 0.39829849
## spread   0.02489213 0.83121479 0.39829849 1.00000000
##
## , , 2024-09-09
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03420902 0.05034358 0.02785483
## ESDE     0.03420902 1.00000000 0.46482886 0.83142398
## IEDE     0.05034358 0.46482886 1.00000000 0.41309492
## spread   0.02785483 0.83142398 0.41309492 1.00000000
##
## , , 2024-09-10
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03269261 0.04983824 0.02407004
## ESDE     0.03269261 1.00000000 0.46778655 0.82976914
## IEDE     0.04983824 0.46778655 1.00000000 0.41599216
## spread   0.02407004 0.82976914 0.41599216 1.00000000
##
## , , 2024-09-11
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03075374 0.04819342 0.01875768
## ESDE     0.03075374 1.00000000 0.47061273 0.82790554
## IEDE     0.04819342 0.47061273 1.00000000 0.41821612
## spread   0.01875768 0.82790554 0.41821612 1.00000000
##
```

```
## , , 2024-09-12
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m 1.000000000 0.02875079 0.03632342 0.006813796
## ESDE     0.028750793 1.00000000 0.47225340 0.824081284
## IEDE     0.036323424 0.47225340 1.00000000 0.425255860
## spread   0.006813796 0.82408128 0.42525586 1.000000000
##
## , , 2024-09-13
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01190165 0.01792721 -0.01610946
## ESDE      0.01190165 1.00000000 0.49272676  0.82884632
## IEDE      0.01792721 0.49272676 1.00000000  0.45459252
## spread   -0.01610946 0.82884632 0.45459252  1.00000000
##
## , , 2024-09-16
##
##              eurois3m         ESDE         IEDE      spread
## eurois3m  1.000000000 -0.004845439 -0.002091966 -0.04063837
## ESDE     -0.004845439  1.000000000  0.502315204  0.82979155
## IEDE     -0.002091966  0.502315204  1.000000000  0.46887751
## spread   -0.040638371  0.829791552  0.468877515  1.00000000
##
## , , 2024-09-17
##
##              eurois3m        ESDE       IEDE     spread
## eurois3m  1.000000000 0.004872441 0.01332877 -0.0420026
## ESDE      0.004872441 1.000000000 0.50221706  0.8271745
## IEDE      0.013328767 0.502217057 1.00000000  0.4651312
## spread   -0.042002597 0.827174542 0.46513117  1.0000000
##
## , , 2024-09-18
##
##               eurois3m        ESDE         IEDE      spread
## eurois3m  1.0000000000 0.003851298 0.0003252525 -0.04233167
## ESDE      0.0038512978 1.000000000 0.4988131553  0.82597586
## IEDE      0.0003252525 0.498813155 1.0000000000  0.46498601
## spread   -0.0423316683 0.825975861 0.4649860099  1.00000000
##
## , , 2024-09-19
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01790242 0.01506506 -0.06054784
## ESDE     -0.01790242  1.00000000 0.48670416  0.82721775
## IEDE      0.01506506  0.48670416 1.00000000  0.45188029
## spread   -0.06054784  0.82721775 0.45188029  1.00000000
##
## , , 2024-09-20
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.01804609 0.02050555 -0.06016761
```

```
## ESDE     -0.01804609  1.00000000 0.46362082  0.82768614
## IEDE      0.02050555  0.46362082 1.00000000  0.41566261
## spread   -0.06016761  0.82768614 0.41566261  1.00000000
##
## , , 2024-09-23
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.01581108 0.05255875 -0.03505519
## ESDE      0.01581108 1.00000000 0.49103678  0.82833555
## IEDE      0.05255875 0.49103678 1.00000000  0.43359243
## spread   -0.03505519 0.82833555 0.43359243  1.00000000
##
## , , 2024-09-24
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.03116567 0.05277018 -0.03829248
## ESDE      0.03116567 1.00000000 0.49188251  0.82245728
## IEDE      0.05277018 0.49188251 1.00000000  0.43439734
## spread   -0.03829248 0.82245728 0.43439734  1.00000000
##
## , , 2024-09-25
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02916354 0.02292862 -0.04339602
## ESDE      0.02916354 1.00000000 0.48944201  0.82105029
## IEDE      0.02292862 0.48944201 1.00000000  0.43760136
## spread   -0.04339602 0.82105029 0.43760136  1.00000000
##
## , , 2024-09-26
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02587832 -0.0497843 -0.06937475
## ESDE     -0.02587832  1.00000000  0.5121429  0.82202284
## IEDE     -0.04978430  0.51214288  1.0000000  0.45018280
## spread   -0.06937475  0.82202284  0.4501828  1.00000000
##
## , , 2024-09-27
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01748363 -0.04219014 -0.05022657
## ESDE     -0.01748363  1.00000000  0.50545874  0.83122031
## IEDE     -0.04219014  0.50545874  1.00000000  0.42508533
## spread   -0.05022657  0.83122031  0.42508533  1.00000000
##
## , , 2024-09-30
##
##               eurois3m         ESDE        IEDE      spread
## eurois3m  1.0000000000 0.0004827469 -0.05025204 -0.02886963
## ESDE      0.0004827469 1.0000000000  0.49545524  0.83060420
## IEDE     -0.0502520444 0.4954552405  1.00000000  0.41756336
## spread   -0.0288696283 0.8306042038  0.41756336  1.00000000
##
```

```
## , , 2024-10-01
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.01579814 -0.02738823 -0.03835188
## ESDE     -0.01579814  1.00000000  0.46873147  0.83036412
## IEDE     -0.02738823  0.46873147  1.00000000  0.40375405
## spread   -0.03835188  0.83036412  0.40375405  1.00000000
##
## , , 2024-10-02
##
##             eurois3m      ESDE        IEDE      spread
## eurois3m  1.00000000 0.0042016 -0.00997568 -0.03047792
## ESDE      0.00420160 1.0000000  0.47547924  0.82706255
## IEDE     -0.00997568 0.4754792  1.00000000  0.40913385
## spread   -0.03047792 0.8270625  0.40913385  1.00000000
##
## , , 2024-10-03
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 0.005380319 -0.006729866 -0.02826104
## ESDE      0.005380319 1.000000000  0.473336980  0.82586850
## IEDE     -0.006729866 0.473336980  1.000000000  0.41006825
## spread   -0.028261041 0.825868502  0.410068248  1.00000000
##
## , , 2024-10-04
##
##              eurois3m        ESDE         IEDE      spread
## eurois3m  1.000000000 0.005199697 -0.001449961 -0.02505773
## ESDE      0.005199697 1.000000000  0.461886242  0.81794962
## IEDE     -0.001449961 0.461886242  1.000000000  0.41753283
## spread   -0.025057728 0.817949622  0.417532835  1.00000000
##
## , , 2024-10-07
##
##               eurois3m        ESDE          IEDE      spread
## eurois3m  1.0000000000 0.002129455 -0.0007704921 -0.02507259
## ESDE      0.0021294554 1.000000000  0.4811941641  0.82793388
## IEDE     -0.0007704921 0.481194164  1.0000000000  0.43925412
## spread   -0.0250725918 0.827933882  0.4392541226  1.00000000
##
## , , 2024-10-08
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m  1.00000000 0.0190792 0.04178194 -0.01038977
## ESDE      0.01907920 1.0000000 0.48725845  0.82755863
## IEDE      0.04178194 0.4872584 1.00000000  0.44307664
## spread   -0.01038977 0.8275586 0.44307664  1.00000000
##
## , , 2024-10-09
##
##              eurois3m         ESDE       IEDE      spread
## eurois3m  1.000000000 -0.006110032 0.04703084 -0.01231276
```

```
## ESDE     -0.006110032  1.000000000 0.47839791  0.82139035
## IEDE      0.047030844  0.478397911 1.00000000  0.44284168
## spread   -0.012312760  0.821390350 0.44284168  1.00000000
##
## , , 2024-10-10
##
##              eurois3m         ESDE       IEDE       spread
## eurois3m  1.000000000 -0.009971623 0.03426877 -0.007729656
## ESDE     -0.009971623  1.000000000 0.48070733  0.820170785
## IEDE      0.034268768  0.480707327 1.00000000  0.442873461
## spread   -0.007729656  0.820170785 0.44287346  1.000000000
##
## , , 2024-10-11
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05103604 0.03172887 0.04945925
## ESDE     0.05103604 1.00000000 0.47079969 0.82834691
## IEDE     0.03172887 0.47079969 1.00000000 0.43347423
## spread   0.04945925 0.82834691 0.43347423 1.00000000
##
## , , 2024-10-14
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0408066 0.02274189 0.03903125
## ESDE     0.04080660 1.0000000 0.47603725 0.82744490
## IEDE     0.02274189 0.4760373 1.00000000 0.43696078
## spread   0.03903125 0.8274449 0.43696078 1.00000000
##
## , , 2024-10-15
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02561989 0.01542437 0.02188668
## ESDE     0.02561989 1.00000000 0.48019748 0.82770221
## IEDE     0.01542437 0.48019748 1.00000000 0.44058113
## spread   0.02188668 0.82770221 0.44058113 1.00000000
##
## , , 2024-10-16
##
##              eurois3m       ESDE        IEDE       spread
## eurois3m  1.000000000 0.00995753 0.003922106 -0.002862394
## ESDE      0.009957530 1.00000000 0.486742468  0.828333967
## IEDE      0.003922106 0.48674247 1.000000000  0.448289626
## spread   -0.002862394 0.82833397 0.448289626  1.000000000
##
## , , 2024-10-17
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.01021219 -0.002175877 -0.01713505
## ESDE      0.010212192 1.00000000  0.487059888  0.82248475
## IEDE     -0.002175877 0.48705989  1.000000000  0.45149340
## spread   -0.017135050 0.82248475  0.451493396  1.00000000
##
```

```
## , , 2024-10-18
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.07541115 -0.01190666 -0.1184222
## ESDE     -0.07541115  1.00000000  0.48349785  0.8300248
## IEDE     -0.01190666  0.48349785  1.00000000  0.4439944
## spread   -0.11842215  0.83002480  0.44399442  1.0000000
##
## , , 2024-10-21
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.08426279 -0.02748383 -0.1225777
## ESDE     -0.08426279  1.00000000  0.50101170  0.8331202
## IEDE     -0.02748383  0.50101170  1.00000000  0.4596542
## spread   -0.12257770  0.83312020  0.45965416  1.0000000
##
## , , 2024-10-22
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m  1.0000000 -0.1279465 -0.0494891 -0.1665029
## ESDE     -0.1279465  1.0000000  0.5125789  0.8445838
## IEDE     -0.0494891  0.5125789  1.0000000  0.4724827
## spread   -0.1665029  0.8445838  0.4724827  1.0000000
##
## , , 2024-10-23
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.09036213 -0.04578595 -0.2085380
## ESDE     -0.09036213  1.00000000  0.50541563  0.7435873
## IEDE     -0.04578595  0.50541563  1.00000000  0.4464233
## spread   -0.20853803  0.74358730  0.44642328  1.0000000
##
## , , 2024-10-24
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.06195151 -0.02454532 -0.1794171
## ESDE     -0.06195151  1.00000000  0.51899970  0.7528071
## IEDE     -0.02454532  0.51899970  1.00000000  0.4551088
## spread   -0.17941706  0.75280706  0.45510881  1.0000000
##
## , , 2024-10-25
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.05592691 -0.01894436 -0.1653253
## ESDE     -0.05592691  1.00000000  0.51536390  0.7591839
## IEDE     -0.01894436  0.51536390  1.00000000  0.4528816
## spread   -0.16532533  0.75918387  0.45288161  1.0000000
##
## , , 2024-10-28
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m  1.00000000 -0.06195994 -0.02730029 -0.1559012
```

```
## ESDE     -0.06195994  1.00000000  0.52024783  0.7650787
## IEDE     -0.02730029  0.52024783  1.00000000  0.4556645
## spread   -0.15590124  0.76507873  0.45566453  1.0000000
##
## , , 2024-10-29
##
##              eurois3m        ESDE         IEDE     spread
## eurois3m  1.000000000 -0.05104425 -0.004996389 -0.1387284
## ESDE     -0.051044254  1.00000000  0.506504163  0.7708916
## IEDE     -0.004996389  0.50650416  1.000000000  0.4442124
## spread   -0.138728401  0.77089157  0.444212357  1.0000000
##
## , , 2024-10-30
##
##              eurois3m        ESDE         IEDE     spread
## eurois3m  1.000000000 -0.04410813 -0.002626305 -0.1244725
## ESDE     -0.044108130  1.00000000  0.505433059  0.7682533
## IEDE     -0.002626305  0.50543306  1.000000000  0.4435178
## spread   -0.124472541  0.76825332  0.443517836  1.0000000
##
## , , 2024-10-31
##
##              eurois3m       ESDE        IEDE     spread
## eurois3m  1.000000000 -0.0568676 0.002487124 -0.1290281
## ESDE     -0.056867597  1.0000000 0.484953223  0.7838492
## IEDE      0.002487124  0.4849532 1.000000000  0.4271699
## spread   -0.129028080  0.7838492 0.427169867  1.0000000
##
## , , 2024-11-01
##
##              eurois3m        ESDE        IEDE     spread
## eurois3m  1.000000000 -0.04878642 0.004464774 -0.1175107
## ESDE     -0.048786416  1.00000000 0.486450029  0.7788894
## IEDE      0.004464774  0.48645003 1.000000000  0.4245742
## spread   -0.117510697  0.77888944 0.424574164  1.0000000
##
## , , 2024-11-04
##
##               eurois3m        ESDE          IEDE     spread
## eurois3m  1.0000000000 -0.04298134 -0.0003266341 -0.1110089
## ESDE     -0.0429813351  1.00000000  0.4875993170  0.7825791
## IEDE     -0.0003266341  0.48759932  1.0000000000  0.4280379
## spread   -0.1110089061  0.78257906  0.4280379195  1.0000000
##
## , , 2024-11-05
##
##               eurois3m        ESDE         IEDE     spread
## eurois3m  1.0000000000 -0.03683316 0.0009124425 -0.0995850
## ESDE     -0.0368331608  1.00000000 0.4886207911  0.7861065
## IEDE      0.0009124425  0.48862079 1.0000000000  0.4302561
## spread   -0.0995849965  0.78610652 0.4302560714  1.0000000
##
```

```
## , , 2024-11-06
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m  1.00000000 -0.02444681 0.01048555 -0.07287042
## ESDE     -0.02444681  1.00000000 0.49125279  0.78848720
## IEDE      0.01048555  0.49125279 1.00000000  0.43628222
## spread   -0.07287042  0.78848720 0.43628222  1.00000000
##
## , , 2024-11-07
##
##             eurois3m      ESDE       IEDE     spread
## eurois3m  1.00000000 0.0257513 0.02565174 -0.0180498
## ESDE      0.02575130 1.0000000 0.50162395  0.8312703
## IEDE      0.02565174 0.5016239 1.00000000  0.4553286
## spread   -0.01804980 0.8312703 0.45532862  1.0000000
##
## , , 2024-11-08
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02892137 0.04303536 -0.01026828
## ESDE      0.02892137 1.00000000 0.49756850  0.83042219
## IEDE      0.04303536 0.49756850 1.00000000  0.46112587
## spread   -0.01026828 0.83042219 0.46112587  1.00000000
##
## , , 2024-11-11
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.02791662 0.03667785 -0.0107625
## ESDE      0.02791662 1.00000000 0.49165088  0.8282890
## IEDE      0.03667785 0.49165088 1.00000000  0.4595911
## spread   -0.01076250 0.82828903 0.45959106  1.0000000
##
## , , 2024-11-12
##
##              eurois3m        ESDE      IEDE      spread
## eurois3m  1.000000000 0.005421413 0.0431254 -0.04239463
## ESDE      0.005421413 1.000000000 0.4878706  0.82778067
## IEDE      0.043125402 0.487870555 1.0000000  0.45086317
## spread   -0.042394627 0.827780667 0.4508632  1.00000000
##
## , , 2024-11-13
##
##             eurois3m      ESDE       IEDE      spread
## eurois3m  1.00000000 0.0453228 0.07305359 -0.01838727
## ESDE      0.04532280 1.0000000 0.50772524  0.82583788
## IEDE      0.07305359 0.5077252 1.00000000  0.45969469
## spread   -0.01838727 0.8258379 0.45969469  1.00000000
##
## , , 2024-11-14
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02289509 0.09556549 -0.03678971
```

```
## ESDE      0.02289509 1.00000000 0.42681611  0.83034987
## IEDE      0.09556549 0.42681611 1.00000000  0.37709813
## spread   -0.03678971 0.83034987 0.37709813  1.00000000
##
## , , 2024-11-15
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.00690734 0.07772589 -0.0465936
## ESDE      0.00690734 1.00000000 0.45324811  0.8394568
## IEDE      0.07772589 0.45324811 1.00000000  0.4054261
## spread   -0.04659360 0.83945685 0.40542609  1.0000000
##
## , , 2024-11-18
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.007321108 0.07003881 -0.04296461
## ESDE      0.007321108 1.000000000 0.46243272  0.83821412
## IEDE      0.070038806 0.462432718 1.00000000  0.41533855
## spread   -0.042964607 0.838214124 0.41533855  1.00000000
##
## , , 2024-11-19
##
##              eurois3m        ESDE       IEDE      spread
## eurois3m  1.000000000 0.008073255 0.06509982 -0.04015952
## ESDE      0.008073255 1.000000000 0.46497689  0.83480893
## IEDE      0.065099816 0.464976895 1.00000000  0.41453059
## spread   -0.040159516 0.834808935 0.41453059  1.00000000
##
## , , 2024-11-20
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02347777 0.04480227 -0.02407573
## ESDE      0.02347777 1.00000000 0.45877350  0.83376320
## IEDE      0.04480227 0.45877350 1.00000000  0.41079588
## spread   -0.02407573 0.83376320 0.41079588  1.00000000
##
## , , 2024-11-21
##
##             eurois3m       ESDE       IEDE      spread
## eurois3m  1.00000000 0.02238921 0.04490968 -0.02139599
## ESDE      0.02238921 1.00000000 0.46185229  0.83179256
## IEDE      0.04490968 0.46185229 1.00000000  0.41519638
## spread   -0.02139599 0.83179256 0.41519638  1.00000000
##
## , , 2024-11-22
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.05690284 0.0621936 0.01253237
## ESDE     0.05690284 1.00000000 0.4705717 0.83260307
## IEDE     0.06219360 0.47057171 1.0000000 0.42404107
## spread   0.01253237 0.83260307 0.4240411 1.00000000
##
```

```
## , , 2024-11-25
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.03300822 0.1275552 0.01771223
## ESDE     0.03300822 1.00000000 0.4697420 0.83076233
## IEDE     0.12755517 0.46974204 1.0000000 0.42553883
## spread   0.01771223 0.83076233 0.4255388 1.00000000
##
## , , 2024-11-26
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.03082704 0.1144975 0.01487809
## ESDE     0.03082704 1.00000000 0.4744658 0.82983482
## IEDE     0.11449751 0.47446581 1.0000000 0.43274357
## spread   0.01487809 0.82983482 0.4327436 1.00000000
##
## , , 2024-11-27
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.03510516 0.1007868 0.04129899
## ESDE     0.03510516 1.00000000 0.4762924 0.82485861
## IEDE     0.10078675 0.47629240 1.0000000 0.42937973
## spread   0.04129899 0.82485861 0.4293797 1.00000000
##
## , , 2024-11-28
##
##            eurois3m      ESDE      IEDE     spread
## eurois3m 1.00000000 0.0436966 0.1029458 0.05059901
## ESDE     0.04369660 1.0000000 0.4840117 0.82603230
## IEDE     0.10294578 0.4840117 1.0000000 0.43917330
## spread   0.05059901 0.8260323 0.4391733 1.00000000
##
## , , 2024-11-29
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03327439 0.08568591 0.03938332
## ESDE     0.03327439 1.00000000 0.49323871 0.82652877
## IEDE     0.08568591 0.49323871 1.00000000 0.44676265
## spread   0.03938332 0.82652877 0.44676265 1.00000000
##
## , , 2024-12-02
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03677373 0.07972466 0.03553024
## ESDE     0.03677373 1.00000000 0.48773345 0.81634971
## IEDE     0.07972466 0.48773345 1.00000000 0.44717825
## spread   0.03553024 0.81634971 0.44717825 1.00000000
##
## , , 2024-12-03
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02778779 0.07096811 0.02566271
```

```
## ESDE     0.02778779 1.00000000 0.49470679 0.82241682
## IEDE     0.07096811 0.49470679 1.00000000 0.45440838
## spread   0.02566271 0.82241682 0.45440838 1.00000000
##
## , , 2024-12-04
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06786338 0.08297025 0.06297909
## ESDE     0.06786338 1.00000000 0.50723336 0.83840921
## IEDE     0.08297025 0.50723336 1.00000000 0.46874107
## spread   0.06297909 0.83840921 0.46874107 1.00000000
##
## , , 2024-12-05
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.08043698 0.08612909 0.08595402
## ESDE     0.08043698 1.00000000 0.51273861 0.83734147
## IEDE     0.08612909 0.51273861 1.00000000 0.47151937
## spread   0.08595402 0.83734147 0.47151937 1.00000000
##
## , , 2024-12-06
##
##            eurois3m       ESDE       IEDE    spread
## eurois3m 1.00000000 0.05818944 0.06881753 0.0653075
## ESDE     0.05818944 1.00000000 0.55576510 0.8508938
## IEDE     0.06881753 0.55576510 1.00000000 0.5106699
## spread   0.06530750 0.85089376 0.51066990 1.0000000
##
## , , 2024-12-09
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.05149888 0.0628644 0.05631129
## ESDE     0.05149888 1.00000000 0.5498450 0.84820465
## IEDE     0.06286440 0.54984495 1.0000000 0.50480376
## spread   0.05631129 0.84820465 0.5048038 1.00000000
##
## , , 2024-12-10
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05448291 0.05331525 0.05544846
## ESDE     0.05448291 1.00000000 0.51759786 0.84768345
## IEDE     0.05331525 0.51759786 1.00000000 0.47911740
## spread   0.05544846 0.84768345 0.47911740 1.00000000
##
## , , 2024-12-11
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.05118688 0.0447359 0.05038249
## ESDE     0.05118688 1.00000000 0.5037311 0.84446136
## IEDE     0.04473590 0.50373107 1.0000000 0.45981785
## spread   0.05038249 0.84446136 0.4598178 1.00000000
##
```

```
## , , 2024-12-12
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04608088 0.04399489 0.03290563
## ESDE     0.04608088 1.00000000 0.50279532 0.84056420
## IEDE     0.04399489 0.50279532 1.00000000 0.45718757
## spread   0.03290563 0.84056420 0.45718757 1.00000000
##
## , , 2024-12-13
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06898919 0.02609542 0.07488453
## ESDE     0.06898919 1.00000000 0.48314945 0.83889667
## IEDE     0.02609542 0.48314945 1.00000000 0.41859355
## spread   0.07488453 0.83889667 0.41859355 1.00000000
##
## , , 2024-12-16
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04125753 0.02501087 0.05573806
## ESDE     0.04125753 1.00000000 0.47996290 0.83602672
## IEDE     0.02501087 0.47996290 1.00000000 0.42090709
## spread   0.05573806 0.83602672 0.42090709 1.00000000
##
## , , 2024-12-17
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05226875 0.03259904 0.05268479
## ESDE     0.05226875 1.00000000 0.48852169 0.82892210
## IEDE     0.03259904 0.48852169 1.00000000 0.42537588
## spread   0.05268479 0.82892210 0.42537588 1.00000000
##
## , , 2024-12-18
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.05492241 0.03095682 0.05202308
## ESDE     0.05492241 1.00000000 0.48846781 0.82815296
## IEDE     0.03095682 0.48846781 1.00000000 0.42749764
## spread   0.05202308 0.82815296 0.42749764 1.00000000
##
## , , 2024-12-19
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0508982 0.02910288 0.04603897
## ESDE     0.05089820 1.0000000 0.48812111 0.82574507
## IEDE     0.02910288 0.4881211 1.00000000 0.42405388
## spread   0.04603897 0.8257451 0.42405388 1.00000000
##
## , , 2024-12-20
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.04688411 0.02569893 0.04088504
```

```
## ESDE     0.04688411 1.00000000 0.48880991 0.82463762
## IEDE     0.02569893 0.48880991 1.00000000 0.42610728
## spread   0.04088504 0.82463762 0.42610728 1.00000000
##
## , , 2024-12-23
##
##             eurois3m       ESDE      IEDE      spread
## eurois3m 1.000000000 0.01631827 0.0210611 0.007832359
## ESDE     0.016318274 1.00000000 0.4889625 0.824548595
## IEDE     0.021061101 0.48896251 1.0000000 0.428208436
## spread   0.007832359 0.82454859 0.4282084 1.000000000
##
## , , 2024-12-24
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m 1.000000000 0.01623932 0.008414129 0.007644095
## ESDE     0.016239321 1.00000000 0.489179269 0.823656420
## IEDE     0.008414129 0.48917927 1.000000000 0.429955460
## spread   0.007644095 0.82365642 0.429955460 1.000000000
##
## , , 2024-12-27
##
##               eurois3m      ESDE       IEDE        spread
## eurois3m  1.0000000000 0.0281226 0.01179377 -0.0008795413
## ESDE      0.0281225984 1.0000000 0.48945074  0.8115880515
## IEDE      0.0117937728 0.4894507 1.00000000  0.4303764358
## spread   -0.0008795413 0.8115881 0.43037644  1.0000000000
##
## , , 2024-12-30
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.03552022 0.0153513 0.02983304
## ESDE     0.03552022 1.00000000 0.4905430 0.80302383
## IEDE     0.01535130 0.49054296 1.0000000 0.42542689
## spread   0.02983304 0.80302383 0.4254269 1.00000000
##
## , , 2025-01-02
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02171799 0.01730581 0.01537578
## ESDE     0.02171799 1.00000000 0.47186894 0.81294203
## IEDE     0.01730581 0.47186894 1.00000000 0.41256680
## spread   0.01537578 0.81294203 0.41256680 1.00000000
##
## , , 2025-01-03
##
##              eurois3m        ESDE       IEDE       spread
## eurois3m  1.000000000 -0.01862684 -0.0494584 -0.002438207
## ESDE     -0.018626840  1.00000000  0.4834773  0.812354142
## IEDE     -0.049458396  0.48347730  1.0000000  0.416863079
## spread   -0.002438207  0.81235414  0.4168631  1.000000000
##
```

```
## , , 2025-01-06
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.01240907 -0.02740936 0.004925131
## ESDE     -0.012409065  1.00000000  0.48186661 0.810850283
## IEDE     -0.027409364  0.48186661  1.00000000 0.436727445
## spread    0.004925131  0.81085028  0.43672744 1.000000000
##
## , , 2025-01-07
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m  1.00000000 -0.03695809 -0.02250501 -0.02440004
## ESDE     -0.03695809  1.00000000  0.44594363  0.83866087
## IEDE     -0.02250501  0.44594363  1.00000000  0.40516612
## spread   -0.02440004  0.83866087  0.40516612  1.00000000
##
## , , 2025-01-08
##
##              eurois3m        ESDE        IEDE       spread
## eurois3m  1.000000000 -0.02040262 -0.03882527 -0.004775316
## ESDE     -0.020402623  1.00000000  0.44097908  0.836832754
## IEDE     -0.038825269  0.44097908  1.00000000  0.394664031
## spread   -0.004775316  0.83683275  0.39466403  1.000000000
##
## , , 2025-01-09
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 -0.00858429 -0.02968333 0.005462878
## ESDE     -0.008584290  1.00000000  0.44734428 0.835536902
## IEDE     -0.029683329  0.44734428  1.00000000 0.400920712
## spread    0.005462878  0.83553690  0.40092071 1.000000000
##
## , , 2025-01-10
##
##              eurois3m        ESDE       IEDE     spread
## eurois3m  1.000000000 0.009222374 -0.0145373 0.01126269
## ESDE      0.009222374 1.000000000  0.4569217 0.83305389
## IEDE     -0.014537299 0.456921710  1.0000000 0.40847711
## spread    0.011262694 0.833053890  0.4084771 1.00000000
##
## , , 2025-01-13
##
##              eurois3m        ESDE        IEDE      spread
## eurois3m  1.000000000 0.006478111 -0.01012131 0.002847738
## ESDE      0.006478111 1.000000000  0.45658400 0.831218323
## IEDE     -0.010121306 0.456583996  1.00000000 0.401610445
## spread    0.002847738 0.831218323  0.40161045 1.000000000
##
## , , 2025-01-14
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.01874478 -0.002827015 0.01221358
```

```
## ESDE      0.018744781 1.00000000  0.463833770 0.83396609
## IEDE     -0.002827015 0.46383377  1.000000000 0.41065164
## spread    0.012213581 0.83396609  0.410651641 1.00000000
##
## , , 2025-01-15
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.01992275 -0.001035301 0.01125546
## ESDE      0.019922751 1.00000000  0.460395247 0.83318426
## IEDE     -0.001035301 0.46039525  1.000000000 0.41048569
## spread    0.011255459 0.83318426  0.410485689 1.00000000
##
## , , 2025-01-16
##
##              eurois3m       ESDE         IEDE     spread
## eurois3m  1.000000000 0.03561016 -0.005441394 0.02668033
## ESDE      0.035610159 1.00000000  0.386334032 0.85242904
## IEDE     -0.005441394 0.38633403  1.000000000 0.33609930
## spread    0.026680327 0.85242904  0.336099299 1.00000000
##
## , , 2025-01-17
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.02855573 -0.02331189 0.01518122
## ESDE      0.02855573 1.00000000  0.39934395 0.84926750
## IEDE     -0.02331189 0.39934395  1.00000000 0.35925911
## spread    0.01518122 0.84926750  0.35925911 1.00000000
##
## , , 2025-01-20
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06164762 0.03006602 0.02583293
## ESDE     0.06164762 1.00000000 0.44551222 0.84080622
## IEDE     0.03006602 0.44551222 1.00000000 0.37511154
## spread   0.02583293 0.84080622 0.37511154 1.00000000
##
## , , 2025-01-21
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.05511574 -0.04354054 0.04620906
## ESDE      0.05511574 1.00000000  0.43898326 0.83613188
## IEDE     -0.04354054 0.43898326  1.00000000 0.35536960
## spread    0.04620906 0.83613188  0.35536960 1.00000000
##
## , , 2025-01-22
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.05299889 -0.0384846 0.04384995
## ESDE      0.05299889 1.00000000  0.4396787 0.83449767
## IEDE     -0.03848460 0.43967865  1.0000000 0.35573365
## spread    0.04384995 0.83449767  0.3557337 1.00000000
##
```

```
## , , 2025-01-23
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.05914031 -0.03149967 0.04402929
## ESDE      0.05914031 1.00000000  0.44131399 0.83247962
## IEDE     -0.03149967 0.44131399  1.00000000 0.36546719
## spread    0.04402929 0.83247962  0.36546719 1.00000000
##
## , , 2025-01-24
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.07030548 -0.02385337 0.05412141
## ESDE      0.07030548 1.00000000  0.44278847 0.83801199
## IEDE     -0.02385337 0.44278847  1.00000000 0.37217646
## spread    0.05412141 0.83801199  0.37217646 1.00000000
##
## , , 2025-01-27
##
##              eurois3m      ESDE         IEDE     spread
## eurois3m  1.000000000 0.0904006 -0.001819813 0.05863411
## ESDE      0.090400600 1.0000000  0.454540605 0.83455441
## IEDE     -0.001819813 0.4545406  1.000000000 0.38366890
## spread    0.058634113 0.8345544  0.383668904 1.00000000
##
## , , 2025-01-28
##
##             eurois3m      ESDE        IEDE    spread
## eurois3m 1.000000000 0.0811662 0.003566656 0.0495550
## ESDE     0.081166200 1.0000000 0.456830468 0.8327686
## IEDE     0.003566656 0.4568305 1.000000000 0.3881380
## spread   0.049554997 0.8327686 0.388138014 1.0000000
##
## , , 2025-01-29
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1080741 0.0248601 0.0692377
## ESDE     0.1080741 1.0000000 0.4736759 0.8359847
## IEDE     0.0248601 0.4736759 1.0000000 0.4071367
## spread   0.0692377 0.8359847 0.4071367 1.0000000
##
## , , 2025-01-30
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.09409932 0.02749486 0.05247077
## ESDE     0.09409932 1.00000000 0.47419819 0.83369289
## IEDE     0.02749486 0.47419819 1.00000000 0.40766669
## spread   0.05247077 0.83369289 0.40766669 1.00000000
##
## , , 2025-01-31
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0900401 0.02922059 0.05211937
```

```
## ESDE     0.09004010 1.0000000 0.47677181 0.83201066
## IEDE     0.02922059 0.4767718 1.00000000 0.41263135
## spread   0.05211937 0.8320107 0.41263135 1.00000000
##
## , , 2025-02-03
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.08381894 0.02316829 0.05641219
## ESDE     0.08381894 1.00000000 0.47764153 0.82795454
## IEDE     0.02316829 0.47764153 1.00000000 0.41035535
## spread   0.05641219 0.82795454 0.41035535 1.00000000
##
## , , 2025-02-04
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0753248 0.02382485 0.04835432
## ESDE     0.07532480 1.0000000 0.46400743 0.82787274
## IEDE     0.02382485 0.4640074 1.00000000 0.40013428
## spread   0.04835432 0.8278727 0.40013428 1.00000000
##
## , , 2025-02-05
##
##              eurois3m      ESDE         IEDE     spread
## eurois3m  1.000000000 0.0516113 -0.003397366 0.04192761
## ESDE      0.051611301 1.0000000  0.472828069 0.82362808
## IEDE     -0.003397366 0.4728281  1.000000000 0.40177131
## spread    0.041927613 0.8236281  0.401771309 1.00000000
##
## , , 2025-02-06
##
##               eurois3m       ESDE          IEDE     spread
## eurois3m  1.0000000000 0.05249356 -0.0003694352 0.05873585
## ESDE      0.0524935577 1.00000000  0.4752998296 0.81560971
## IEDE     -0.0003694352 0.47529983  1.0000000000 0.40121063
## spread    0.0587358506 0.81560971  0.4012106299 1.00000000
##
## , , 2025-02-07
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.03584319 0.0184536 0.03400331
## ESDE     0.03584319 1.00000000 0.4617901 0.81718475
## IEDE     0.01845360 0.46179011 1.0000000 0.38618741
## spread   0.03400331 0.81718475 0.3861874 1.00000000
##
## , , 2025-02-10
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.03029963 0.01684538 0.02547631
## ESDE     0.03029963 1.00000000 0.47333610 0.82346277
## IEDE     0.01684538 0.47333610 1.00000000 0.40312583
## spread   0.02547631 0.82346277 0.40312583 1.00000000
##
```

```
## , , 2025-02-11
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0297572 0.03203558 0.01961463
## ESDE     0.02975720 1.0000000 0.46470063 0.82138788
## IEDE     0.03203558 0.4647006 1.00000000 0.38544510
## spread   0.01961463 0.8213879 0.38544510 1.00000000
##
## , , 2025-02-12
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.1099614 0.02927894 0.03238634
## ESDE     0.10996143 1.0000000 0.38650527 0.73646206
## IEDE     0.02927894 0.3865053 1.00000000 0.38998991
## spread   0.03238634 0.7364621 0.38998991 1.00000000
##
## , , 2025-02-13
##
##              eurois3m       ESDE        IEDE      spread
## eurois3m  1.000000000 0.06454182 0.009617271 -0.03850274
## ESDE      0.064541823 1.00000000 0.399121014  0.74380033
## IEDE      0.009617271 0.39912101 1.000000000  0.39860572
## spread   -0.038502741 0.74380033 0.398605720  1.00000000
##
## , , 2025-02-14
##
##             eurois3m      ESDE        IEDE      spread
## eurois3m  1.00000000 0.0465454 -0.01424301 -0.05278459
## ESDE      0.04654540 1.0000000  0.41203718  0.75130516
## IEDE     -0.01424301 0.4120372  1.00000000  0.41109305
## spread   -0.05278459 0.7513052  0.41109305  1.00000000
##
## , , 2025-02-17
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.02946465 -0.007592184 -0.06506787
## ESDE      0.029464648 1.00000000  0.407311504  0.76515893
## IEDE     -0.007592184 0.40731150  1.000000000  0.39482478
## spread   -0.065067866 0.76515893  0.394824785  1.00000000
##
## , , 2025-02-18
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.05171627 -0.01190348 -0.0182651
## ESDE      0.05171627 1.00000000  0.40283845  0.7756767
## IEDE     -0.01190348 0.40283845  1.00000000  0.3759327
## spread   -0.01826510 0.77567674  0.37593270  1.0000000
##
## , , 2025-02-19
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.04841835 -0.008932056 -0.01628774
```

```
## ESDE      0.048418351 1.00000000  0.401573167  0.78060705
## IEDE     -0.008932056 0.40157317  1.000000000  0.36413278
## spread   -0.016287742 0.78060705  0.364132777  1.00000000
##
## , , 2025-02-20
##
##               eurois3m       ESDE          IEDE      spread
## eurois3m  1.0000000000 0.05844332 -0.0009404283 0.008429226
## ESDE      0.0584433247 1.00000000  0.4130673825 0.786664023
## IEDE     -0.0009404283 0.41306738  1.0000000000 0.374070356
## spread    0.0084292265 0.78666402  0.3740703560 1.000000000
##
## , , 2025-02-21
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06416979 0.02907968 0.01157838
## ESDE     0.06416979 1.00000000 0.42402129 0.78949337
## IEDE     0.02907968 0.42402129 1.00000000 0.38206427
## spread   0.01157838 0.78949337 0.38206427 1.00000000
##
## , , 2025-02-24
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.06850983 0.03950928 0.01395518
## ESDE     0.06850983 1.00000000 0.43766323 0.79200059
## IEDE     0.03950928 0.43766323 1.00000000 0.39152260
## spread   0.01395518 0.79200059 0.39152260 1.00000000
##
## , , 2025-02-25
##
##             eurois3m      ESDE      IEDE      spread
## eurois3m 1.000000000 0.0676917 0.0285190 0.007347476
## ESDE     0.067691700 1.0000000 0.4401097 0.792167201
## IEDE     0.028519000 0.4401097 1.0000000 0.399084776
## spread   0.007347476 0.7921672 0.3990848 1.000000000
##
## , , 2025-02-26
##
##              eurois3m       ESDE         IEDE      spread
## eurois3m  1.000000000 0.03679443 -0.001257553 -0.02010726
## ESDE      0.036794428 1.00000000  0.452047647  0.79671011
## IEDE     -0.001257553 0.45204765  1.000000000  0.41179427
## spread   -0.020107261 0.79671011  0.411794270  1.00000000
##
## , , 2025-02-27
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.05014494 0.01158699 -0.006505879
## ESDE      0.050144936 1.00000000 0.46582005  0.802151455
## IEDE      0.011586987 0.46582005 1.00000000  0.423844813
## spread   -0.006505879 0.80215145 0.42384481  1.000000000
##
```

```
## , , 2025-02-28
##
##              eurois3m       ESDE       IEDE       spread
## eurois3m  1.000000000 0.04170691 0.01532017 -0.009217484
## ESDE      0.041706914 1.00000000 0.42301388  0.806903544
## IEDE      0.015320166 0.42301388 1.00000000  0.397237071
## spread   -0.009217484 0.80690354 0.39723707  1.000000000
##
## , , 2025-03-03
##
##             eurois3m       ESDE        IEDE      spread
## eurois3m  1.00000000 0.03099226 -0.04419643 -0.04190167
## ESDE      0.03099226 1.00000000  0.38044310  0.77321135
## IEDE     -0.04419643 0.38044310  1.00000000  0.54458103
## spread   -0.04190167 0.77321135  0.54458103  1.00000000
##
## , , 2025-03-04
##
##               eurois3m       ESDE       IEDE        spread
## eurois3m  1.0000000000 0.08094219 -0.0465020 -0.0002366502
## ESDE      0.0809421899 1.00000000  0.3565482  0.7877503192
## IEDE     -0.0465020039 0.35654821  1.0000000  0.5141313931
## spread   -0.0002366502 0.78775032  0.5141314  1.0000000000
##
## , , 2025-03-05
##
##             eurois3m      ESDE        IEDE    spread
## eurois3m  1.00000000 0.1220260 -0.02292525 0.0305743
## ESDE      0.12202604 1.0000000  0.37044270 0.7921093
## IEDE     -0.02292525 0.3704427  1.00000000 0.5101239
## spread    0.03057430 0.7921093  0.51012389 1.0000000
##
## , , 2025-03-06
##
##             eurois3m      ESDE        IEDE     spread
## eurois3m  1.00000000 0.1049925 -0.01184457 0.03890852
## ESDE      0.10499249 1.0000000  0.37846785 0.78773893
## IEDE     -0.01184457 0.3784678  1.00000000 0.50649592
## spread    0.03890852 0.7877389  0.50649592 1.00000000
##
## , , 2025-03-07
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.07960553 -0.0170227 0.01674906
## ESDE      0.07960553 1.00000000  0.3919839 0.79315989
## IEDE     -0.01702270 0.39198388  1.0000000 0.50355621
## spread    0.01674906 0.79315989  0.5035562 1.00000000
##
## , , 2025-03-10
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.08004408 -0.03777556 0.02879273
```

```
## ESDE      0.08004408 1.00000000  0.39521828 0.79578547
## IEDE     -0.03777556 0.39521828  1.00000000 0.48852456
## spread    0.02879273 0.79578547  0.48852456 1.00000000
##
## , , 2025-03-11
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.05106453 -0.01781948 0.06861949
## ESDE      0.05106453 1.00000000  0.40149769 0.78120105
## IEDE     -0.01781948 0.40149769  1.00000000 0.48520901
## spread    0.06861949 0.78120105  0.48520901 1.00000000
##
## , , 2025-03-12
##
##             eurois3m       ESDE       IEDE     spread
## eurois3m  1.00000000 0.01039674 -0.0230759 0.04483244
## ESDE      0.01039674 1.00000000  0.4128397 0.78229337
## IEDE     -0.02307590 0.41283971  1.0000000 0.48555377
## spread    0.04483244 0.78229337  0.4855538 1.00000000
##
## , , 2025-03-13
##
##             eurois3m       ESDE        IEDE     spread
## eurois3m  1.00000000 0.01842398 -0.01830135 0.04333788
## ESDE      0.01842398 1.00000000  0.41827395 0.78617667
## IEDE     -0.01830135 0.41827395  1.00000000 0.48182943
## spread    0.04333788 0.78617667  0.48182943 1.00000000
##
## , , 2025-03-14
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m 1.000000000 0.006201074 0.02852075 0.02776833
## ESDE     0.006201074 1.000000000 0.15990281 0.79630029
## IEDE     0.028520749 0.159902808 1.00000000 0.22117051
## spread   0.027768325 0.796300290 0.22117051 1.00000000
##
## , , 2025-03-17
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.01226818 0.0268737 0.02658705
## ESDE     0.01226818 1.00000000 0.1821610 0.79832291
## IEDE     0.02687370 0.18216096 1.0000000 0.23666144
## spread   0.02658705 0.79832291 0.2366614 1.00000000
##
## , , 2025-03-18
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02728331 0.02739906 0.05037442
## ESDE     0.02728331 1.00000000 0.20600256 0.80050256
## IEDE     0.02739906 0.20600256 1.00000000 0.25172503
## spread   0.05037442 0.80050256 0.25172503 1.00000000
##
```

```
## , , 2025-03-19
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.02004535 0.02396265 0.04827501
## ESDE     0.02004535 1.00000000 0.22909383 0.80117843
## IEDE     0.02396265 0.22909383 1.00000000 0.26697122
## spread   0.04827501 0.80117843 0.26697122 1.00000000
##
## , , 2025-03-20
##
##            eurois3m       ESDE      IEDE     spread
## eurois3m 1.00000000 0.02267849 0.0257511 0.04252006
## ESDE     0.02267849 1.00000000 0.2513916 0.80274855
## IEDE     0.02575110 0.25139159 1.0000000 0.28199248
## spread   0.04252006 0.80274855 0.2819925 1.00000000
##
## , , 2025-03-21
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m 1.000000000 0.003079699 0.02691087 0.02967331
## ESDE     0.003079699 1.000000000 0.26651989 0.80246558
## IEDE     0.026910873 0.266519886 1.00000000 0.29473822
## spread   0.029673307 0.802465580 0.29473822 1.00000000
##
## , , 2025-03-24
##
##             eurois3m        ESDE       IEDE     spread
## eurois3m 1.000000000 0.002422331 0.02274857 0.02513135
## ESDE     0.002422331 1.000000000 0.29469471 0.80281774
## IEDE     0.022748569 0.294694709 1.00000000 0.30860329
## spread   0.025131352 0.802817741 0.30860329 1.00000000
##
## , , 2025-03-25
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01311501 0.01989999 0.02896905
## ESDE     0.01311501 1.00000000 0.30267128 0.80791970
## IEDE     0.01989999 0.30267128 1.00000000 0.31345638
## spread   0.02896905 0.80791970 0.31345638 1.00000000
##
## , , 2025-03-26
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01435238 0.02080821 0.02566153
## ESDE     0.01435238 1.00000000 0.32490380 0.80958506
## IEDE     0.02080821 0.32490380 1.00000000 0.32946362
## spread   0.02566153 0.80958506 0.32946362 1.00000000
##
## , , 2025-03-27
##
##            eurois3m       ESDE       IEDE    spread
## eurois3m 1.00000000 0.02189779 0.02077616 0.0345897
```

```
## ESDE     0.02189779 1.00000000 0.34100598 0.8106452
## IEDE     0.02077616 0.34100598 1.00000000 0.3400415
## spread   0.03458970 0.81064523 0.34004150 1.0000000
##
## , , 2025-03-28
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01631981 0.02137065 0.03056945
## ESDE     0.01631981 1.00000000 0.35547014 0.81000056
## IEDE     0.02137065 0.35547014 1.00000000 0.35097295
## spread   0.03056945 0.81000056 0.35097295 1.00000000
##
## , , 2025-03-31
##
##            eurois3m       ESDE       IEDE     spread
## eurois3m 1.00000000 0.01310009 0.02284093 0.02390735
## ESDE     0.01310009 1.00000000 0.36230724 0.81169440
## IEDE     0.02284093 0.36230724 1.00000000 0.35584177
## spread   0.02390735 0.81169440 0.35584177 1.00000000
##
## , , 2025-04-01
##
##             eurois3m        ESDE        IEDE      spread
## eurois3m 1.000000000 0.003796768 0.007192249 0.009283878
## ESDE     0.003796768 1.000000000 0.387764095 0.814861054
## IEDE     0.007192249 0.387764095 1.000000000 0.379496289
## spread   0.009283878 0.814861054 0.379496289 1.000000000
##
## , , 2025-04-02
##
##             eurois3m        ESDE        IEDE     spread
## eurois3m 1.000000000 0.007215517 0.005801268 0.01021661
## ESDE     0.007215517 1.000000000 0.349468570 0.82075348
## IEDE     0.005801268 0.349468570 1.000000000 0.31396863
## spread   0.010216611 0.820753484 0.313968632 1.00000000
##
## , , 2025-04-03
##
##             eurois3m        ESDE       IEDE      spread
## eurois3m 1.000000000 0.007289699 0.06973908 0.003626899
## ESDE     0.007289699 1.000000000 0.35726717 0.820234216
## IEDE     0.069739085 0.357267173 1.00000000 0.319074100
## spread   0.003626899 0.820234216 0.31907410 1.000000000
##
## , , 2025-04-04
##
##            eurois3m      ESDE       IEDE     spread
## eurois3m 1.00000000 0.0472233 0.07783792 0.02768048
## ESDE     0.04722330 1.0000000 0.39013858 0.82731741
## IEDE     0.07783792 0.3901386 1.00000000 0.35061458
## spread   0.02768048 0.8273174 0.35061458 1.00000000
##
```

```
## , , 2025-04-07
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1502823 0.1371176 0.1363921
## ESDE     0.1502823 1.0000000 0.4667730 0.8605069
## IEDE     0.1371176 0.4667730 1.0000000 0.4337806
## spread   0.1363921 0.8605069 0.4337806 1.0000000
##
## , , 2025-04-08
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1451508 0.2693284 0.1503218
## ESDE     0.1451508 1.0000000 0.4601408 0.8567225
## IEDE     0.2693284 0.4601408 1.0000000 0.4387328
## spread   0.1503218 0.8567225 0.4387328 1.0000000
##
## , , 2025-04-09
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1379388 0.2534651 0.1402332
## ESDE     0.1379388 1.0000000 0.4649728 0.8533503
## IEDE     0.2534651 0.4649728 1.0000000 0.4413717
## spread   0.1402332 0.8533503 0.4413717 1.0000000
##
## , , 2025-04-10
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1641080 0.2477503 0.1638619
## ESDE     0.1641080 1.0000000 0.4862767 0.8728247
## IEDE     0.2477503 0.4862767 1.0000000 0.4629695
## spread   0.1638619 0.8728247 0.4629695 1.0000000
##
## , , 2025-04-11
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1595008 0.2294436 0.1575060
## ESDE     0.1595008 1.0000000 0.5263698 0.8755308
## IEDE     0.2294436 0.5263698 1.0000000 0.5108589
## spread   0.1575060 0.8755308 0.5108589 1.0000000
##
## , , 2025-04-14
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1441210 0.2091601 0.1406304
## ESDE     0.1441210 1.0000000 0.5332976 0.8738281
## IEDE     0.2091601 0.5332976 1.0000000 0.5150218
## spread   0.1406304 0.8738281 0.5150218 1.0000000
##
## , , 2025-04-15
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1552128 0.1992206 0.1519315
```

```
## ESDE     0.1552128 1.0000000 0.5308175 0.8792697
## IEDE     0.1992206 0.5308175 1.0000000 0.5089912
## spread   0.1519315 0.8792697 0.5089912 1.0000000
##
## , , 2025-04-16
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1433210 0.1839459 0.1386093
## ESDE     0.1433210 1.0000000 0.5280002 0.8741132
## IEDE     0.1839459 0.5280002 1.0000000 0.5046572
## spread   0.1386093 0.8741132 0.5046572 1.0000000
##
## , , 2025-04-17
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1315541 0.1684839 0.1249881
## ESDE     0.1315541 1.0000000 0.5249953 0.8691730
## IEDE     0.1684839 0.5249953 1.0000000 0.4993871
## spread   0.1249881 0.8691730 0.4993871 1.0000000
##
## , , 2025-04-22
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1225163 0.1550344 0.1162696
## ESDE     0.1225163 1.0000000 0.5226111 0.8644699
## IEDE     0.1550344 0.5226111 1.0000000 0.4951631
## spread   0.1162696 0.8644699 0.4951631 1.0000000
##
## , , 2025-04-23
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1121124 0.1409684 0.1039576
## ESDE     0.1121124 1.0000000 0.5198747 0.8591939
## IEDE     0.1409684 0.5198747 1.0000000 0.4938874
## spread   0.1039576 0.8591939 0.4938874 1.0000000
##
## , , 2025-04-24
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1535208 0.1339723 0.1269108
## ESDE     0.1535208 1.0000000 0.5130029 0.8575302
## IEDE     0.1339723 0.5130029 1.0000000 0.4896040
## spread   0.1269108 0.8575302 0.4896040 1.0000000
##
## , , 2025-04-25
##
##           eurois3m      ESDE      IEDE    spread
## eurois3m 1.0000000 0.1328528 0.1198751 0.1083284
## ESDE     0.1328528 1.0000000 0.5145040 0.8566855
## IEDE     0.1198751 0.5145040 1.0000000 0.4895170
## spread   0.1083284 0.8566855 0.4895170 1.0000000
```

Check for residuals ARCH effect on DCC standard residuals

```
std_res <- residuals(dcc_fit) / sigma(dcc_fit)
MTS::MarchTest(std_res, lag = 10)
```

```
## Q(m) of squared series(LM test):
## Test statistic:  8.769307  p-value:  0.5541283
## Rank-based Test:
## Test statistic:  33.78083  p-value:  0.000201166
## Q_k(m) of squared series:
## Test statistic:  151.1795  p-value:  0.6790314
## Robust Test(5%) :  154.015  p-value:  0.6184608
```

Fail to reject the null, the series do not have any volatility clustering anymore.

Check for distributional assumptions.

```
apply(std_res, 2, check_dist)
```

```
## $eurois3m
##        aic      bic
## t 2892.279 2907.044
## n 3021.676 3031.520
##
## $ESDE
##        aic      bic
## t 2802.259 2817.024
## n 2880.573 2890.417
##
## $IEDE
##        aic      bic
## t 2568.722 2583.487
## n 2782.640 2792.483
##
## $spread
##        aic      bic
## t 2849.046 2863.811
## n 2870.983 2880.826
```

They appear to better follow a t-distribution

No ARCH effects and right distributional assumptions -> Model is correctly specified.

Validation

Validate the model via rolling forecast on test data, the window dimension is set to the same window used for the diagnostic, we keep the same structure we tested but re-estimating the coefficients at every iteration, we can do that by fixing the parameter when restricting the VAR model. Then transform back to spread and evaluate

```
set.seed(42)
start_time <- proc.time()

invert_diff <- function(d, last) last + d

window_size <- nrow(d_train)
len         <- nrow(d_test)

df <- data.frame(
  date = rep(as.Date(NA), len),
  act_spread  = numeric(len),act_eurois3m   = numeric(len),act_esde = numeric(len), act_iede = n
umeric(len),
  fc_spread   = numeric(len), fc_spread_lo   = numeric(len), fc_spread_up   = numeric(len),
  fc_eurois3m = numeric(len), fc_eurois3m_lo = numeric(len), fc_eurois3m_up = numeric(len),
  fc_esde = numeric(len), fc_esde_lo = numeric(len), fc_esde_up = numeric(len),
  fc_iede = numeric(len), fc_iede_lo = numeric(len), fc_iede_up = numeric(len)
)

last  <- xts::last
first <- xts::first

data_mat <- coredata(data)   # plain matrix, numeric indexing is safe

for (i in 1:len) {
  start <- n_train - window_size + i - 1
  end   <- start + window_size - 1
  window_data <- d_data[start:end, ]

  # Last value, d_data, end_data lost one row
  orig_idx <- end + 1

  # Guard: skip if out of bounds
  if (orig_idx + 1 > nrow(data)) break

  #last_spread  <- as.numeric(coredata(data)[orig_idx, "spread"])


  last_spread <- data_mat[orig_idx, "spread"]
  last_esde   <- data_mat[orig_idx, "ESDE"]
  last_iede   <- data_mat[orig_idx, "IEDE"]

  df$act_spread[i]   <- data_mat[orig_idx + 1, "spread"]
  df$act_eurois3m[i] <- data_mat[orig_idx + 1, "eurois3m"]
  df$act_esde[i]     <- data_mat[orig_idx + 1, "ESDE"]
  df$act_iede[i]     <- data_mat[orig_idx + 1, "IEDE"]

  #df$act_spread[i]  <- as.numeric(coredata(data)[orig_idx + 1, "spread"])
  #df$act_eurois3m[i] <- as.numeric(coredata(data)[orig_idx + 1, "eurois3m"])

  # VAR
  roll_var <- vars::VAR(window_data, p = fixed_p, type = "const")
  roll_var_s <- vars::restrict(roll_var, method = "manual", resmat = fixed_params)
```

```
  fc <- predict(roll_var_s, n.ahead = 1)

  # DCC
  roll_res <- residuals(roll_var_s)
  roll_dcc_fit <- dccfit(dcc_spec, data = roll_res, solver = c("hybrid", "gosolnp"),
                         fit.control = list(eval.se = FALSE))

  if (roll_dcc_fit@mfit$convergence != 0 || is.null(roll_dcc_fit@mfit$coef)) {
    warning(paste("DCC did not converge at i =", i, "- skipping"))
    next
  }

  dcc_fc  <- dccforecast(roll_dcc_fit, n.ahead = 1)
  dof  <- roll_dcc_fit@mfit$coef["[Joint]mshape"]
  t_sc <- qt(p = 0.975, df = dof)

  # Extract
  d_spread  <- fc$fcst$spread[1,  "fcst"]
  d_eurois3m  <- fc$fcst$eurois3m[1,  "fcst"]
  d_esde     <- fc$fcst$ESDE[1, "fcst"]
  d_iede     <- fc$fcst$IEDE[1, "fcst"]

  sigma <- rcov(dcc_fc)
  se    <- sqrt(diag(sigma[[1]][, , 1]))

  # Point forecasts
  df$date[i]       <- index(d_test)[i]
  df$fc_spread[i]  <- invert_diff(d_spread, last_spread)
  df$fc_eurois3m[i]  <- d_eurois3m
  df$fc_esde[i]     <- invert_diff(d_esde, last_esde)
  df$fc_iede[i]     <- invert_diff(d_iede, last_iede)

  # Confidence intervals
  df$fc_spread_lo[i]  <- invert_diff(d_spread  - t_sc * se[4], last_spread)
  df$fc_spread_up[i]  <- invert_diff(d_spread  + t_sc * se[4], last_spread)

  df$fc_eurois3m_lo[i]  <- d_eurois3m  - t_sc * se[1]
  df$fc_eurois3m_up[i]  <- d_eurois3m  + t_sc * se[1]

  df$fc_esde_lo[i]   <- invert_diff(d_spread - t_sc * se[2], last_spread)
  df$fc_esde_up[i]   <- invert_diff(d_spread + t_sc * se[2], last_spread)

  df$fc_iede_lo[i]     <- invert_diff(d_iede - t_sc * se[3], last_iede)
  df$fc_iede_up[i]     <- invert_diff(d_iede + t_sc * se[3], last_iede)



  if (i %% 25 == 0 || i == len) cat("Step", i, "/", len, "\n")
}
```

```
## Step 25 / 255
## Step 50 / 255
## Step 75 / 255
## Step 100 / 255
## Step 125 / 255
## Step 150 / 255
## Step 175 / 255
## Step 200 / 255
## Step 225 / 255
## Step 250 / 255
## Step 255 / 255
```

```
end_time <- proc.time()
execution_time <- end_time - start_time
cat("\n Total execution time: ", execution_time[3], "seconds")
```

```
##
##  Total execution time:  322.41 seconds
```

Metrics

```
rmse <- function(actual, forecast) sqrt(mean((actual - forecast)^2, na.rm = TRUE))

acc_dir <- function(actual, forecast) {
  d_actual = diff(actual)
  d_forecast = forecast[-1] - actual[-length(actual)]
  dir = mean(sign(d_forecast) == sign(d_actual), na.rm = TRUE) * 100
}

ci_coverage <- function(actual, lo, up) mean(actual >= lo & actual <= up, na.rm = TRUE) * 100

results <- data.frame(
  variable    = var_names,
  rmse        = c(
    rmse(df$act_spread,  df$fc_spread),
    rmse(df$act_eurois3m, df$fc_eurois3m),
    rmse(df$act_esde,     df$fc_esde),
    rmse(df$act_iede,     df$fc_iede)

  ),
  acc_dir    = c(
    acc_dir(df$act_spread, df$fc_spread),
    acc_dir(df$act_eurois3m, df$fc_eurois3m),
    acc_dir(df$act_esde,     df$fc_esde),
    acc_dir(df$act_iede,     df$fc_iede)
  ),
  ci_coverage_0.95 = c(
    ci_coverage(df$act_spread,  df$fc_spread_lo,  df$fc_spread_up),
    ci_coverage(df$act_eurois3m, df$fc_eurois3m_lo, df$fc_eurois3m_up),
    ci_coverage(df$act_esde,     df$fc_esde_lo,     df$fc_esde_up),
    ci_coverage(df$act_iede,     df$fc_iede_lo,     df$fc_iede_up)
  )
)

print(results)
```

```
##   variable       rmse  acc_dir ci_coverage_0.95
## 1 eurois3m 2.36356561 47.24409        98.823529
## 2     ESDE 0.02514003 49.21260        96.862745
## 3     IEDE 6.81861873 53.54331         3.529412
## 4   spread 1.23815691 51.57480        98.431373
```

Compare

I compare the forecast evaluation metrics of the model with a Random Walk model (tomorrow's value = today's value).

```
spread_rw <- numeric(len)

for (i in 1:len) {
  end <- n_train + i
  spread_rw[i] <- as.numeric(data[n_train + i - 1, "spread"])
}

df$spread_rw <- spread_rw

rmse_rw = rmse(df$act_spread, df$spread_rw)
acc_dir_rw = acc_dir(df$act_spread, df$spread_rw)
cat("\nRMSE RW: ", rmse_rw)
```

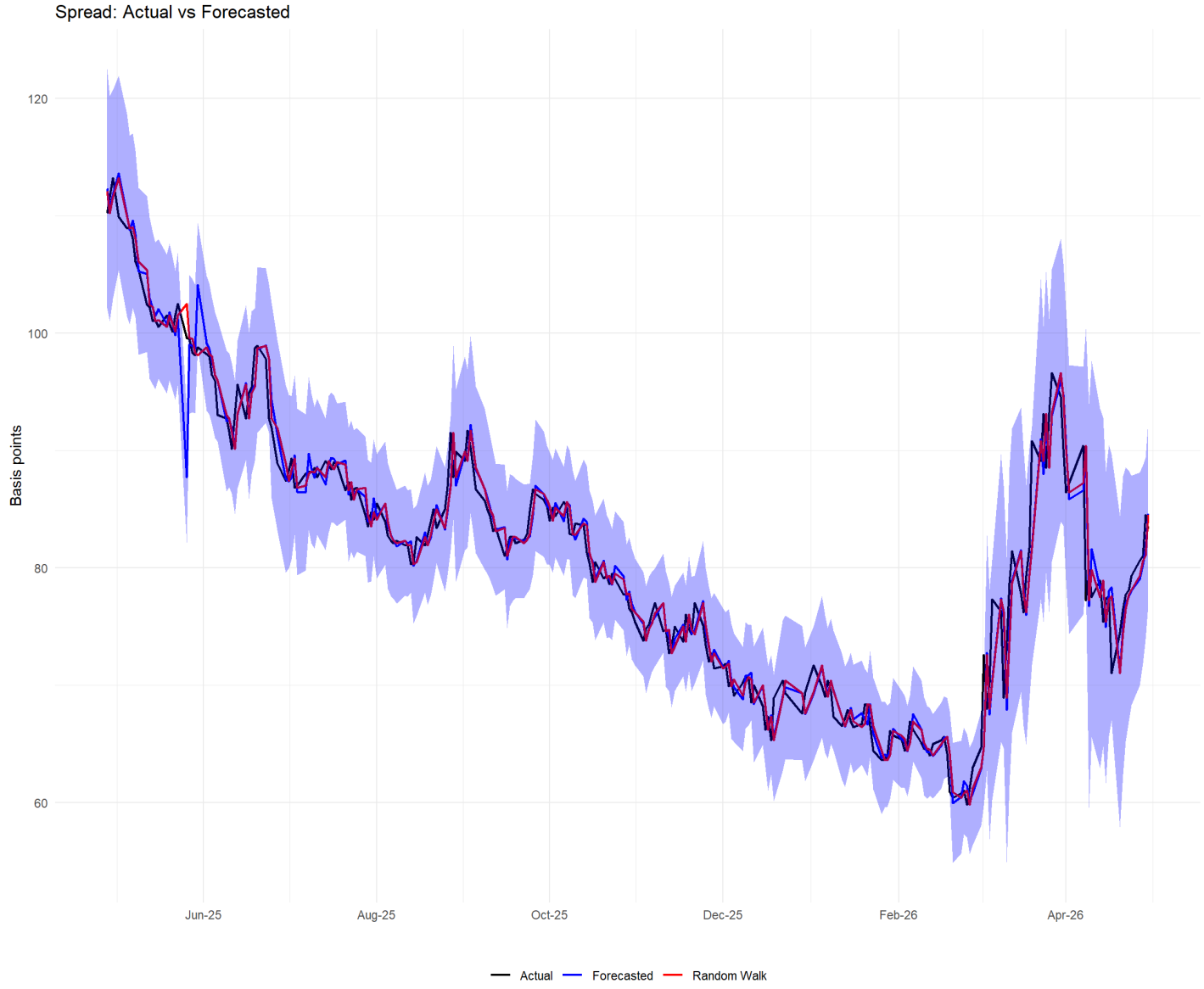
```
##
## RMSE RW:  2.200463
```

```
cat("\nDirection accuracy RW: ", acc_dir_rw)
```

```
##
## Direction accuracy RW:  1.181102
```

Model predict the spread like a Random Walk

```
p_spreads  <- ggplot(df, aes(x = date)) +
  geom_line(aes(y = act_spread, color = "Actual"), linewidth = 0.8) +
  geom_line(aes(y = fc_spread, color = "Forecasted"), linewidth = 0.8) +
  geom_line(aes(y = spread_rw, color = "Random Walk"), linewidth = 0.8) +
  geom_ribbon(aes(ymin = fc_spread_lo, ymax = fc_spread_up), fill = "blue", alpha = 0.3) +
  scale_color_manual(values = c("Actual" = "black", "Forecasted" = "blue", "Random Walk" = "re
d")) +
  labs(title = "Spread: Actual vs Forecasted", x = "", y = "Basis points") +
  scale_x_date(date_breaks = "2 month", date_labels = "%b-%y") +
  scale_y_continuous(breaks = scales::breaks_width(20)) +
  theme_minimal() +
  theme(legend.title = element_blank(), legend.position = "bottom")

print(p_spreads)
```



Forecast

Next week forecast

```
data_fit = tail(d_data, window_size)
last_date  <- last(index(data_fit))
last_spread <- last(data$spread)

fc_var <- vars::VAR(window_data, p = fixed_p, type = "const")
fc_var_s <- vars::restrict(roll_var, method = "manual", resmat = fixed_params)
fc <- predict(roll_var_s, n.ahead = 1)

fc_res <- residuals(roll_var_s)
fc_dcc_fit <- dccfit(dcc_spec, data = roll_res)
fc_dcc  <- dccforecast(roll_dcc_fit, n.ahead = 1)
dof  <- fc_dcc_fit@mfit$coef["[Joint]mshape"]
t_sc <- qt(p = 0.975, df = dof)
sigma <- rcov(fc_dcc)
se    <- sqrt(diag(sigma[[1]][, , 1]))

d_spread  <- fc$fcst$spread[1,  "fcst"]
fc_spread  <- invert_diff(d_spread, last_spread)
fc_spread_lo  <- invert_diff(d_spread  - t_sc * se[1], last_spread)
fc_spread_up  <- invert_diff(d_spread  + t_sc * se[1], last_spread)

forecast <- data.frame(
  spread = fc_spread |> as.numeric(),
  q_05 = fc_spread_lo |> as.numeric(),
  q_95 = fc_spread_up |> as.numeric()
)

cat("\nForecasted values:\n")
```

```
##
## Forecasted values:
```

```
cat("\nSpread on last date: ", as.character(last_date), ": ", last_spread)
```

```
##
## Spread on last date:  2026-04-30 :  83.3
```

```
cat("\nSpread forecasted for tomorrow:", as.character(last_date + 1), ":\n")
```

```
##
## Spread forecasted for tomorrow: 2026-05-01 :
```

```
print(forecast)
```

```
##     spread     q_05     q_95
## 1 83.39003 83.27353 83.50653
```